Kube-OVN Document

v1.17.0

Kube-OVN Team

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9.2 [illegible] 341
9.3 [illegible] 341
9.4 [illegible] 341

1. Kube-OVN

1.1 What is Kube-OVN?

Kube-OVN [illegible] CNCF [illegible] SDN [illegible]

Kube-OVN [illegible] Open Virtual Network[illegible]OVN[illegible] Open vSwitch [illegible] Kubernetes [illegible] OVN [illegible] OVS [illegible] Kubernetes [illegible] SDN [illegible]Kube-OVN [illegible] Kubernetes [illegible] Kubernetes [illegible]

1.2 Why Kube-OVN?

[illegible] Kubernetes [illegible]OVN [illegible] OVS [illegible]

[illegible] Kubernetes [illegible] KubeVirt, [illegible] Kube-OVN [illegible]Kube-OVN [illegible] SDN [illegible]

[illegible]

[illegible] SDN [illegible] Kube-OVN [illegible]

[illegible] OVS/OVN [illegible] SDN [illegible]Kube-OVN [illegible] IP [illegible]/[illegible]Underlay/Overlay [illegible] VPC [illegible]VPC EIP/FIP/SNAT[illegible]VPC [illegible] KubeVirt [illegible]Non-Primary CNI [illegible]BGP [illegible] OVN IPSec[illegible]QoS [illegible] ACL [illegible] ARM/x86 [illegible]

[illegible]

[illegible] Kube-OVN [illegible]

[illegible] eBPF[illegible]DPDK[illegible] Kube-OVN [illegible] OVN [illegible]Kube-OVN [illegible] Pod [illegible]

[illegible] Kube-OVN [illegible] CPU [illegible]

[illegible]

[illegible]Kube-OVN [illegible]

Kube-OVN [illegible] Grafana [illegible] Cilium [illegible] eBPF [illegible] NPM [illegible]

1.3 CNI [illegible]

Kubernetes [illegible] CNI [illegible]

1. [illegible]
2. [illegible]

[illegible] Kube-OVN [illegible] CNI [illegible] Kube-OVN [illegible] Kube-OVN [illegible]

1.3.1 [illegible] eBPF [illegible]

[illegible] Cilium [illegible] Calico eBPF[illegible]

Kube-OVN [illegible] Open vSwitch [illegible]

1.3.2 [illegible] CNI, Ingress, Service Mesh [illegible] Observability All in One [illegible]

[illegible] Cilium[illegible]

Kube-OVN [illegible] CNI [illegible]

1.3.3 [illegible] OpenShift [illegible]

[illegible] ovn-kubernetes[illegible]

[illegible] OpenShift [illegible] CNI [illegible] Cluster Network Operator [illegible]Kube-OVN [illegible] RedHat [illegible] Kubernetes [illegible]

1.3.4 [illegible] Kubernetes[illegible]EKS/AKS/GKE [illegible]

[illegible] Kubernetes [illegible] CNI[illegible]

1.3.5 [illegible] AI [illegible]

[illegible] Hostnetwork [illegible] host-device [illegible]

AI [illegible]

1.4 [illegible]OVN/ovn-kubernetes/Kube-OVN

[illegible]

1.4.1 OVN

OVN [illegible] Open vSwitch [illegible] OpenStack, Kubernetes [illegible] CMS [illegible]ovn-kubernetes [illegible] Kube-OVN [illegible] OVN [illegible] Kubernetes [illegible]

1.4.2 ovn-kubernetes

ovn-kubernetes [illegible] OVN [illegible] OVN [illegible] Kubernetes [illegible] CNI [illegible] OpenShift [illegible] OpenShift [illegible] UDN, Multihoming, Hardware Acceleration [illegible]

1.4.3 Kube-OVN

Kube-OVN [illegible] IP [illegible] Namespace [illegible] OVN [illegible] ovn-kubernetes [illegible] annotation [illegible] Pod [illegible] join [illegible] Underlay[illegible]VPC[illegible]KubeVirt [illegible]

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1.5 □□

2. □□□□

2.1 □□□□

Kube-OVN □□□□□ CNI □□□□□□□□□□□□□□□□ Kubernetes □□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

2.1.1 □□□□

- Kubernetes >= 1.29□
- Docker >= 1.12.6, Containerd >= 1.3.4□
- □□□□: CentOS 7/8, Ubuntu 16.04/18.04/20.04□
- □□ Linux □□□□□□□□□□□□□□□□□□□ `geneve`, `openvswitch`, `ip_tables` □ `iptable_nat`□Kube-OVN □□□□□□□□□□□□

□□□□□

1. □□□□□□□ 3.10.0-862□□□ `netfilter` □□□□ bug □□□ Kube-OVN □□□□□□□□□□□□□□□□□□□□□□□□□□ CentOS □□□□□□□□□□□□□□□□□□□□□□□□ bug □□ Floating IPs broken after kernel upgrade to Centos/RHEL 7.5 - DNAT not working□
2. Rocky Linux 8.6 □□□ 4.18.0-372.9.1.el8.x86_64 □□ TCP □□□□ TCP connection failed in Rocky Linux 8.6□□□□□□□ 4.18.0-372.13.1.el8_6.x86_64 □□□□□□
3. □□□□□□□ 4.4□□□□□□□ `openvswitch` □□□□□□□□□□□□□□□□□ `openvswitch` □□□□□□□□□□□□
4. Geneve □□□□□□□□□ IPv6□□□□ `cat /proc/cmdline` □□□□□□□□□□ □□□□ bug □□□ Geneve tunnels don't work when ipv6 is disabled□

2.1.2 □□□□

- □□□□□□□□□ IPv6□□□□□□□□□□□□ `ipv6.disable=1` □□□□□□□□ 0□
- `kube-proxy` □□□□□Kube-OVN □□□□ Service ClusterIP □□□ `kube-apiserver`□
- □□ kubelet □□□□□□□□ CNI□□□□□□□□□□□, kubelet □□□□□□□□□□□ `--network-plugin=cni --cni-bin-dir=/opt/cni/bin --cni-conf-dir=/etc/cni/net.d`□
- □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ `/etc/cni/net.d/` □□

2.1.3 □□□□

□□	□□	□□
ovn-central	6641/tcp	ovn nb db server □□□□
ovn-central	6642/tcp	ovn sb db server □□□□
ovn-central	6643/tcp	ovn northd server □□□□
ovn-central	6644/tcp	ovn raft server □□□□
ovn-ic	6645/tcp	ovn ic nb db server □□□□
ovn-ic	6646/tcp	ovn ic sb db server □□□□
ovs-ovn	Geneve 6081/udp, STT 7471/tcp, Vxlan 4789/udp	□□□□
kube-ovn-controller	10660/tcp	□□□□□□
kube-ovn-daemon	10665/tcp	□□□□□□
kube-ovn-monitor	10661/tcp	□□□□□□

□□□□□□□□□ firewalld□□□□□□□□ Packet Forwarding □□ Masquerade□

```
#    Packet Forwarding
firewall-cmd --add-forward --permanent
```

```
#    IPv4 Masquerade
firewall-cmd --add-masquerade --permanent
#   Kube-OVN IPv6/        Masquerade
firewall-cmd --permanent --add-rich-rule 'rule family="ipv6" source address="fd00:10:16::/112" masquerade'

firewall-cmd --reload
```

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2.1.4 □□

2.2 [illegible]

Kube-OVN [illegible] Charts [illegible] Kube-OVN [illegible] Overlay [illegible]

[illegible] Underlay/Vlan [illegible] Underlay [illegible] [illegible]

[illegible] Kube-OVN [illegible]

2.2.1 [illegible]

[illegible]

[illegible] release [illegible]

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/refs/tags/v1.17.0/dist/images/install.sh
```

[illegible] master [illegible]

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/images/install.sh
```

[illegible]

[illegible]

```
REGISTRY="docker.io/kubeovn"            #
VERSION="v1.17.0"                       #     /Tag
POD_CIDR="10.16.0.0/16"                 #      CIDR     SVC/NODE/JOIN CIDR
SVC_CIDR="10.96.0.0/12"                 #     apiserver   service-cluster-ip-range
JOIN_CIDR="100.64.0.0/16"               # Pod         CIDR     SVC/NODE/POD CIDR
LABEL="node-role.kubernetes.io/control-plane" #    OVN DB
IFACE=""                                #                       Kubernetes    Node IP
TUNNEL_TYPE="geneve"                    #           geneve, vxlan   stt stt        ovs
```

[illegible] `IFACE=enp6s0f0,eth.*` [illegible]

[illegible]

> [illegible] root [illegible]

`bash install.sh`

[illegible]

[illegible]

[illegible] Kube-OVN [illegible]

1. [illegible] `[Step 4/6]` [illegible] Pod[illegible]
2. [illegible] Kube-OVN [illegible]

2.2.2 Helm Chart [illegible]

[illegible] Kube-OVN [illegible] Helm [illegible] Kube-OVN [illegible]

[illegible] IP [illegible]

```
# kubectl get node -o wide
NAME                     STATUS     ROLES           AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE           KERNEL-VERSION     CONTAINER-RUNTIME
kube-ovn-control-plane   NotReady   control-plane   20h   v1.26.0   172.18.0.3    <none>        Ubuntu 22.04.1 LTS   5.10.104-linuxkit   containerd://1.6.9
kube-ovn-worker          NotReady   <none>          20h   v1.26.0   172.18.0.2    <none>        Ubuntu 22.04.1 LTS   5.10.104-linuxkit   containerd://1.6.9
```

□□□□□ label

```
# kubectl label node -lbeta.kubernetes.io/os=linux kubernetes.io/os=linux --overwrite
node/kube-ovn-control-plane not labeled
node/kube-ovn-worker not labeled

# kubectl label node -lnode-role.kubernetes.io/control-plane kube-ovn/role=master --overwrite
node/kube-ovn-control-plane labeled

#    label    dpdk        dpdk
# kubectl label node -lovn.kubernetes.io/ovs_dp_type!=userspace ovn.kubernetes.io/ovs_dp_type=kernel --overwrite
node/kube-ovn-control-plane labeled
node/kube-ovn-worker labeled
```

□□ Helm Repo □□

```
# helm repo add kubeovn https://kubeovn.github.io/kube-ovn/
"kubeovn" has been added to your repositories

# helm repo list
NAME            URL
kubeovn         https://kubeovn.github.io/kube-ovn/

# helm repo update kubeovn
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "kubeovn" chart repository
Update Complete. ⎈Happy Helming!⎈

# helm search repo kubeovn
NAME                    CHART VERSION   APP VERSION     DESCRIPTION
kubeovn/kube-ovn        v1.17.0        v1.17.0         Helm chart for Kube-OVN
```

□□ helm install □□ Kube-OVN

Chart □□□□□□□□□□ `values.yaml` □□□□□□□□

```
# helm install kube-ovn kubeovn/kube-ovn --wait -n kube-system --version v1.17.0
NAME: kube-ovn
LAST DEPLOYED: Thu Apr 24 08:30:13 2025
NAMESPACE: kube-system
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

□□

□□□□□ □□□□□□□□□□□□□□ **Helm** □□□□□□□□□□□□□□□□□□ `values.yaml` □□□□□□□□□□□□□□□□□□□

```
helm upgrade -f values.yaml kube-ovn kubeovn/kube-ovn --wait -n kube-system --version v1.17.0
```

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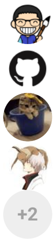
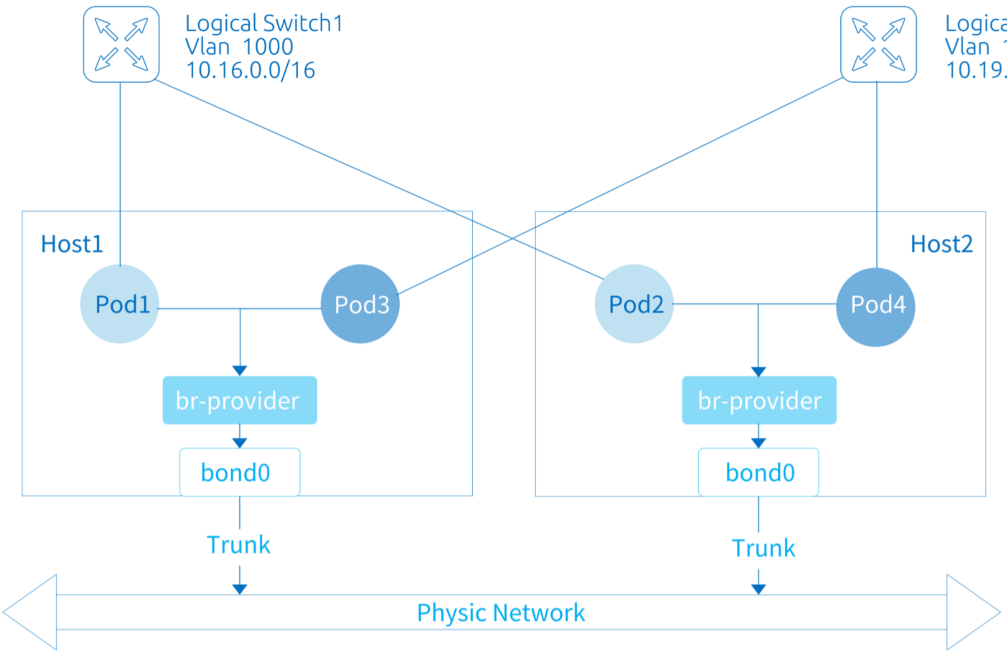


2.2.3 □□

2.3 Underlay [illegible]

[illegible] Kube-OVN [illegible] Geneve [illegible] Overlay [illegible]

[illegible] Kube-OVN [illegible] Underlay [illegible]


Logical Switch1
Vlan 1000
10.16.0.0/16
Logical Switch2
Vlan 1002
10.19.0.0/16
Host1
Host2
Pod1
Pod3
Pod2
Pod4
br-provider
br-provider
bond0
bond0
Trunk
Trunk
Physic Network

2.3.1 [illegible]

[illegible]Overlay [illegible] SNAT/EIP[illegible] [illegible]/[illegible] L3 [illegible]VPC [illegible] Underlay [illegible]

2.3.2 [illegible] Macvlan [illegible]

Kube-OVN [illegible] Underlay [illegible] Macvlan [illegible]

1. [illegible] Macvlan [illegible] OVS [illegible]Macvlan [illegible]
2. Kube-OVN [illegible] arp-proxy [illegible] arp [illegible]
3. [illegible] Macvlan [illegible] netfilter[illegible]Service [illegible] NetworkPolicy [illegible]Kube-OVN [illegible] OVS [illegible] Service [illegible] NetworkPolicy [illegible]
4. Kube-OVN [illegible] Underlay [illegible] Macvlan [illegible] IP [illegible] QoS [illegible]

2.3.3 [illegible]

[illegible] Underlay [illegible]OVS [illegible] OVS [illegible]L2/L3 [illegible] [illegible]Vlan [illegible]

1. [illegible] OpenStack [illegible] VM [illegible] `PortSecurity` [illegible]
2. [illegible] VMware [illegible] vSwitch [illegible] `MAC Address Changes`, `Forged Transmits` [illegible] `Promiscuous Mode Operation` [illegible] `allow`[illegible]
3. [illegible] VMware [illegible] NSX-T [illegible] Underlay [illegible]
4. [illegible] Hyper-V [illegible] `MAC Address Spoofing`[illegible]
5. [illegible] AWS[illegible]GCE[illegible] Mac [illegible] Underlay [illegible] Underlay [illegible] VPC-CNI[illegible]

6. [illegible] Linux Bridge [illegible]

[illegible] Kube-OVN Underlay [illegible] Underlay [illegible]

[illegible] Underlay [illegible]

[illegible]Kube-OVN [illegible] Mac [illegible]IP [illegible] MTU [illegible] OVS Bridge[illegible] Underlay [illegible]OVS Bridge [illegible] `br-PROVIDER_NAME` [illegible] `PROVIDER_NAME` [illegible] `ProviderNetwork` [illegible] provider[illegible]

1. [illegible] OVS Bridge [illegible] NetworkManager [illegible] DHCP [illegible]
2. [illegible] OVS Bridge[illegible] OVS [illegible] Kube-OVN [illegible] OVS [illegible] Underlay [illegible]

2.3.4 [illegible]

[illegible] Underlay [illegible] Pod [illegible] Underlay [illegible]

[illegible]

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

[illegible]

```
ENABLE_ARP_DETECT_IP_CONFLICT #            vlan    arp
NETWORK_TYPE                  #    vlan
VLAN_INTERFACE_NAME           #                    eth1
VLAN_ID                       #        VLAN Tag      0     VLAN
POD_CIDR                      #        CIDR     192.168.1.0/24
POD_GATEWAY                   #             192.168.1.1
EXCLUDE_IPS                   #                   IP       192.168.1.1..192.168.1.100
ENABLE_LB                     #   Underlay        Service       true
EXCHANGE_LINK_NAME            #       provider-network   OVS               false
LS_DNAT_MOD_DL_DST            # DNAT      MAC           Service         true
```

[illegible]

```
bash install.sh
```

2.3.5 [illegible] CRD [illegible] Underlay [illegible]

[illegible] Underlay [illegible] Pod [illegible] `ProviderNetwork` [illegible] `Vlan` [illegible] `Subnet` [illegible]

[illegible] ProviderNetwork

ProviderNetwork [illegible] Underlay [illegible]

[illegible] ProviderNetwork [illegible]:

```yaml
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: net1
spec:
  defaultInterface: eth1
  customInterfaces:
    - interface: eth2
      nodes:
        - node1
  nodeSelector:
    matchLabels:
      kubernetes.io/arch: amd64
      network-type: underlay
    matchExpressions:
      - key: kubernetes.io/hostname
        operator: In
        values:
          - node1
          - node2
```

[illegible]ProviderNetwork [illegible] 12 [illegible]

- `defaultInterface`: [illegible] Bond [illegible] Vlan [illegible]ProviderNetwork [illegible] excludeNodes [illegible] br-net1[illegible] `br-NAME` [illegible] OVS [illegible]
- `customInterfaces`: [illegible]
- `nodeSelector`: [illegible] OVS [illegible] `matchLabels` [illegible] `matchExpressions` [illegible]
- `excludeNodes`: [illegible] `net1.provider-network.ovn.kubernetes.io/exclude=true` [illegible] **`nodeSelector`** [illegible] **`excludeNodes`** [illegible] **`nodeSelector`** [illegible]
- `autoCreateVlanSubinterfaces`: [illegible] `false` [illegible] `true` [illegible] VLAN [illegible] `<interface>.<vlanid>` [illegible]

[illegible]

Key	Value	[illegible]
net1.provider-network.ovn.kubernetes.io/ready	true	[illegible]ProviderNetwork [illegible]
net1.provider-network.ovn.kubernetes.io/interface	eth1	[illegible]
net1.provider-network.ovn.kubernetes.io/mtu	1500	[illegible] MTU

> [illegible] IP[illegible] IP [illegible] OVS [illegible]

[illegible] VLAN [illegible]

[illegible] ProviderNetwork [illegible] `autoCreateVlanSubinterfaces: true` [illegible] VLAN [illegible] VLAN [illegible]

[illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: vlan-provider
spec:
  defaultInterface: eth0.341
  autoCreateVlanSubinterfaces: true
  customInterfaces:
    - interface: eth2.341
      nodes:
        - worker-node-1
        - worker-node-2
```

[illegible]

[illegible] `<interface>.<vlanid>` [illegible] VLAN [illegible] VLAN [illegible] ProviderNetwork [illegible]

[illegible] VLAN

Vlan [illegible] Vlan Tag [illegible] ProviderNetwork [illegible]

[illegible] VLAN [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan1
spec:
  id: 0
  provider: net1
```

- `id`: [illegible] VLAN ID/Tag[illegible]Kube-OVN [illegible] Vlan [illegible] Vlan [illegible] 0 [illegible]vlan [illegible] localnet [illegible]
- `provider`: [illegible] ProviderNetwork [illegible] VLAN [illegible] ProviderNetwork[illegible]

[illegible] Subnet

[illegible] Vlan [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  vlan: vlan1
  disableGatewayCheck: false
```

- `vlan` [illegible] VLAN [illegible] Subnet [illegible] VLAN[illegible]
- `disableGatewayCheck` [illegible] Underlay [illegible] `true` [illegible]

2.3.6 [illegible]

[illegible] IP [illegible]

[illegible] IP [illegible] Pod [illegible] IP [illegible] Mac

2.3.7 [illegible]

[illegible]Kube-OVN [illegible] Underlay [illegible] `spec.logicalGateway` [illegible] `true` [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  vlan: vlan1
  logicalGateway: true
```

[illegible]Pod [illegible] Kube-OVN [illegible]Logical Router[illegible]

2.3.8 Underlay [illegible] Overlay [illegible]

[illegible] Underlay [illegible] Overlay [illegible] Overlay [illegible] Pod [illegible] NAT [illegible] Underlay [illegible] Pod IP[illegible] Underlay [illegible] Pod [illegible] Overlay [illegible] Overlay [illegible] Underlay [illegible] Pod [illegible] Pod IP [illegible] Overlay [illegible] Pod[illegible]

[illegible] Underlay [illegible] Overlay [illegible] `u2oInterconnection` [illegible] `true` [illegible] Kube-OVN [illegible] Underlay IP [illegible] Underlay [illegible] `ovn-cluster` [illegible] Kube-OVN [illegible] Underlay [illegible] Overlay [illegible]

Warning

[illegible] Vlan [illegible] Underlay [illegible] Underlay [illegible] u2o [illegible]

[illegible] IP

[illegible] subnet [illegible] IP [illegible] Underlay Subnet [illegible] `u2oInterconnectionIP` [illegible]

[illegible] Underlay Subnet [illegible] VPC

[illegible] Underlay Subnet [illegible] VPC [illegible] Overlay Subnet [illegible] VPC [illegible] `u2oInterconnection` [illegible] `true` [illegible] `subnet.spec.vpc` [illegible] VPC [illegible]

2.3.9 [illegible]

[illegible] IP [illegible] Netplan [illegible] Ubuntu[illegible] Netplan [illegible] renderer [illegible] NetworkManager[illegible] IP [illegible] DHCP[illegible]

```yaml
network:
  renderer: NetworkManager
  ethernets:
    eth0:
      dhcp4: no
      addresses:
        - 172.16.143.129/24
  version: 2
```

[illegible] IP [illegible] netplan [illegible]

```
netplan generate

nmcli connection reload netplan-eth0
nmcli device set eth0 managed yes
```

[illegible]Kube-OVN [illegible] IP [illegible] OVS [illegible]

[illegible] NetworkManager [illegible] CentOS[illegible]

```
nmcli connection reload eth0
nmcli device set eth0 managed yes
nmcli -t -f GENERAL.STATE device show eth0 | grep -qw unmanaged || nmcli device reapply eth0
```

[illegible] IP [illegible] MAC [illegible]

2.3.10 [illegible]

[illegible] hairpin [illegible] Pod [illegible]

[illegible] hairpin [illegible] Pod [illegible]Pod [illegible] OVS [illegible] MAC [illegible]

[illegible] hairpin[illegible] Kube-OVN [illegible]

Pod [illegible] Pod [illegible]

[illegible] Pod [illegible] 300[illegible] ARP [illegible] OVS [illegible] resubmit [illegible]

```
2022-11-13T08:43:46.782Z|00222|ofproto_dpif_upcall(handler5)|WARN|Flow:
arp,in_port=331,vlan_tci=0x0000,dl_src=00:00:00:25:eb:39,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.213.131.240,arp_tpa=10.213.159.254,arp_op=1,arp_sha=00:00:00:25:e
b:39,arp_tha=ff:ff:ff:ff:ff:ff

bridge("br-int")
----------------
 0. No match.
     >>>> received packet on unknown port 331 <<<<
    drop

Final flow: unchanged
Megaflow: recirc_id=0,eth,arp,in_port=331,dl_src=00:00:00:25:eb:39
Datapath actions: drop
2022-11-13T08:44:34.077Z|00224|ofproto_dpif_xlate(handler5)|WARN|over 4096 resubmit actions on bridge br-int while processing
arp,in_port=13483,vlan_tci=0x0000,dl_src=00:00:00:59:ef:13,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.213.152.3,arp_tpa=10.213.159.254,arp_op=1,arp_sha=00:00:00:59:e
f:13,arp_tha=ff:ff:ff:ff:ff:ff
```

[illegible] OVN NB [illegible] `bcast_arp_req_flood` [illegible] `false` [illegible]

```
kubectl ko nbctl set NB_Global . options:bcast_arp_req_flood=false
```

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2025[illegible]12[illegible]22[illegible]

2022[illegible]5[illegible]20[illegible]

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+2

2.3.11 [illegible]

2.4 [illegible] CNI [illegible]

[illegible] Kube-OVN [illegible] CNI [illegible] CNI [illegible] Cilium[illegible]Calico [illegible] Flannel[illegible]

2.4.1 [illegible]

[illegible] CNI [illegible]Kube-OVN [illegible] CNI [illegible] Pod [illegible] CNI [illegible]eth0[illegible]

- [illegible] CNI [illegible] Kube-OVN [illegible]VPC[illegible]
- [illegible]
- [illegible] Pod [illegible]
- [illegible] Kube-OVN [illegible] VPC NAT [illegible]

2.4.2 [illegible]

[illegible]

1. [illegible] **CNI**[illegible] Kubernetes [illegible] CNI[illegible]Cilium[illegible]Calico[illegible]Flannel [illegible]
2. **Multus CNI**[illegible]
3. **Kube-OVN**[illegible]

[illegible]

1. [illegible] CNI
2. [illegible] Multus CNI
3. [illegible] Kube-OVN

2.4.3 [illegible]

[illegible] Helm Chart v2

```yaml
# values.yaml
cni:
  nonPrimaryCNI: true
```

[illegible] Helm [illegible]

```bash
helm install kube-ovn ./charts/kube-ovn-v2 \
  --namespace kube-system \
  --set cni.nonPrimaryCNI=true
```

[illegible] Helm Chart v1

```yaml
# values.yaml
cni_conf:
  NON_PRIMARY_CNI: true
```

[illegible] Helm [illegible]

```bash
helm install kube-ovn ./charts/kube-ovn \
  --namespace kube-system \
  --set cni_conf.NON_PRIMARY_CNI=true
```

[illegible]

[illegible] `kube-ovn-controller` Deployment [illegible] `kube-ovn-cni` DaemonSet [illegible] Pod [illegible]

```
# kube-ovn-controller Deployment
containers:
- name: kube-ovn-controller
  args:
  - --non-primary-cni-mode=true
```

```
# kube-ovn-cni DaemonSet
containers:
- name: kube-ovn-cni
  args:
  - --non-primary-cni-mode=true
```

2.4.4 [illegible]

[illegible]NADs[illegible]

[illegible] NAD [illegible]

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: kube-ovn-vpc-network
  namespace: default
spec:
  config: |
    {
      "cniVersion": "0.3.1",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "kube-ovn-vpc-network.default.ovn"
    }
```

VPC [illegible]

[illegible] VPC [illegible]

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc-secondary
spec:
  namespaces:
  - default
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet-secondary
spec:
  vpc: vpc-secondary
  cidr: "10.100.0.0/16"
  gateway: "10.100.0.1"
  provider: kube-ovn-vpc-network.default.ovn
```

Pod [illegible]

[illegible] Pod [illegible]

```
apiVersion: v1
kind: Pod
metadata:
  name: multi-network-pod
  annotations:
    k8s.v1.cni.cncf.io/networks: default/kube-ovn-vpc-network
spec:
  containers:
  - name: app
    image: nginx
```

2.4.5 [illegible]

[illegible]eth0[illegible]

- [illegible] CNI[illegible]Cilium[illegible]Calico [illegible]
- [illegible]

- [illegible] Kubernetes [illegible]

[illegible]net1[illegible]net2...[illegible]

- [illegible] Kube-OVN [illegible]
- [illegible]
- [illegible] VPC [illegible]
- [illegible] Kube-OVN [illegible]QoS [illegible] NAT

2.4.6 [illegible]

1. [illegible]

[illegible]

```yaml
#    Pod
apiVersion: v1
kind: Pod
metadata:
  name: frontend
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [
        {"name": "default/public-network"},
        {"name": "default/internal-network"}
      ]
```

2. [illegible]

[illegible]

```yaml
#    A Pod
apiVersion: v1
kind: Pod
metadata:
  name: tenant-a-app
  annotations:
    k8s.v1.cni.cncf.io/networks: default/tenant-a-network

---
#    B Pod
apiVersion: v1
kind: Pod
metadata:
  name: tenant-b-app
  annotations:
    k8s.v1.cni.cncf.io/networks: default/tenant-b-network
```

3. VPC NAT [illegible]

[illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: VpcNatGateway
metadata:
  name: vpc-nat-gw
spec:
  vpc: vpc-secondary
  subnet: subnet-secondary
  lanIp: "10.100.0.254"
```

2.4.7 [illegible]

[illegible]

Pod [illegible]

```yaml
metadata:
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [
        {"name": "default/kube-ovn-network-1", "interface": "net1"},
        {"name": "default/kube-ovn-network-2", "interface": "net2"}
      ]
```

IP [illegible]

[illegible] IP [illegible]

```
#    IP
metadata:
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [{
        "name": "default/kube-ovn-network",
        "ips": ["10.100.0.100/16"]
      }]
```

[illegible]QoS[illegible]

[illegible] QoS [illegible]

```
metadata:
  annotations:
    ovn.kubernetes.io/ingress_rate: "1000"
    ovn.kubernetes.io/egress_rate: "1000"
```

2.4.8 [illegible]

[illegible]

1. **Pod [illegible]** - [illegible] NAD [illegible] - [illegible] kube-ovn-cni [illegible] - [illegible] Multus [illegible]
2. **[illegible]** - [illegible] VPC [illegible] - [illegible] Pod [illegible] - [illegible]
3. **IP [illegible]** - [illegible] CIDR [illegible] - [illegible] IP [illegible] - [illegible] IPAM [illegible]

[illegible]

```
#    Pod
kubectl get pod <pod-name> -o yaml | grep -A 10 "networks-status"

#    Pod
kubectl exec <pod-name> -- ip addr show

#
kubectl exec <pod-name> -- ip route show

#    kube-ovn-cni
kubectl logs -n kube-system daemonset/kube-ovn-cni
```

2.4.9 [illegible]

1. [illegible] Kubernetes [illegible]
2. [illegible]
3. **DNS[illegible]**Pod DNS [illegible]

2.4.10 [illegible]

1. [illegible] IP [illegible]
2. [illegible]
3. [illegible]
4. [illegible]
5. [illegible]

2.4.11 □□

- Multus CNI □□
- □□□□□□□□
- Kube-OVN □□

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2.4.12 □□

2.5 Talos □□

Talos Linux □□ Kubernetes □□□□□ Linux □□□□

2.5.1 □□ Helm Chart □□ Kube-OVN

□□□□□□□□□□□ Talos Linux □□□□□ Kube-OVN□

```
helm install kube-ovn kubeovn/kube-ovn --wait \
    -n kube-system \
    --version v1.17.0 \
    --set OVN_DIR=/var/lib/ovn \
    --set OPENVSWITCH_DIR=/var/lib/openvswitch \
    --set DISABLE_MODULES_MANAGEMENT=true \
    --set cni_conf.MOUNT_LOCAL_BIN_DIR=false
```

□□□□□□□ Underlay □□□□□□□□□□□□ Helm □□□□□□□ Chart □□□□□□□□□

```
helm install kubeovn kubeovn/kube-ovn --wait \
    -n kube-system \
    --version v1.17.0 \
    --set OVN_DIR=/var/lib/ovn \
    --set OPENVSWITCH_DIR=/var/lib/openvswitch \
    --set DISABLE_MODULES_MANAGEMENT=true \
    --set cni_conf.MOUNT_LOCAL_BIN_DIR=false \
    --set networking.NETWORK_TYPE=vlan \
    --set networking.vlan.VLAN_INTERFACE_NAME=enp0s5f1 \
    --set networking.vlan.VLAN_ID=0 \    # VLAN_ID=0      VLAN tag    untagged      chart      100
    --set networking.NET_STACK=ipv4 \
    --set-json networking.EXCLUDE_IPS='"172.99.99.11..172.99.99.99"' \
    --set-json ipv4.POD_CIDR='"172.99.99.0/24"' \
    --set-json ipv4.POD_GATEWAY='"172.99.99.1"'
```

Note

VLAN□Bond□Bridge □□□□□□□□□□□□□ Underlay □□□□□□□□Underlay □□□□□□□□□□ Talos □□□□□□ `ignore=true` □□□□□□

```yaml
machine:
  network:
    interfaces:
      - interface: enp0s5f1
        ignore: true
```

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2.5.2 □□

2.6 [illegible]

Kube-OVN [illegible] pre-v0.0.1 [illegible] Kube-OVN[illegible]

2.6.1 [illegible]

[illegible]

- Linkerd [illegible]
- Elasticsearch [illegible]
- EMQX [illegible]
- KubeSphere [illegible]

[illegible]

2.6.2 OpenVswitch/OVN [illegible]

Kube-OVN [illegible] OpenVswitch [illegible] OVN [illegible]

- OVN [illegible]
- OpenVswitch [illegible]

[illegible] ovn-architecture [illegible]

2.6.3 Kube-OVN [illegible]

[illegible] Kube-OVN [illegible]

[illegible] Kube-OVN [illegible] kubectl [illegible] bash [illegible] Kube-OVN [illegible]

2.6.4 [illegible]

Kube-OVN [illegible]

[illegible] Kube-OVN [illegible] Kube-OVN [illegible] E2E [illegible]

2.6.5 [illegible]

[illegible] Kube-OVN [illegible] OpenTelemetry [illegible] DeepFlow [illegible]

2.6.6 [illegible]

Kube-OVN [illegible] 7*24 [illegible] Github Issue [illegible]

[illegible] Github Issue [illegible] AI [illegible] AI [illegible]

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2.6.7 □□

2.7 [illegible]

[illegible] Kube-OVN [illegible] Kube-OVN [illegible] OVS [illegible] issue [illegible] Kube-OVN [illegible]

2.7.1 [illegible] Kubernetes [illegible]

[illegible]

Script Uninstall **Helm Uninstall**

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/cleanup.sh
bash cleanup.sh
```

```
helm uninstall kube-ovn -n kube-system
```

2.7.2 [illegible]

[illegible] ovsdb [illegible] openvswitch [illegible]

```
rm -rf /var/run/openvswitch
rm -rf /var/run/ovn
rm -rf /etc/origin/openvswitch/
rm -rf /etc/origin/ovn/
rm -rf /etc/cni/net.d/01-kube-ovn.conflist
#        v1.10          00-kube-ovn.conflist
rm -rf /etc/cni/net.d/00-kube-ovn.conflist
rm -rf /var/log/openvswitch
rm -rf /var/log/ovn
rm -fr /var/log/kube-ovn
```

2.7.3 [illegible]

[illegible]iptable/ipset [illegible]

```
reboot
```

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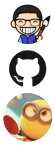
2022[illegible]5[illegible]24[illegible]

GitHub



2.7.4 [illegible]

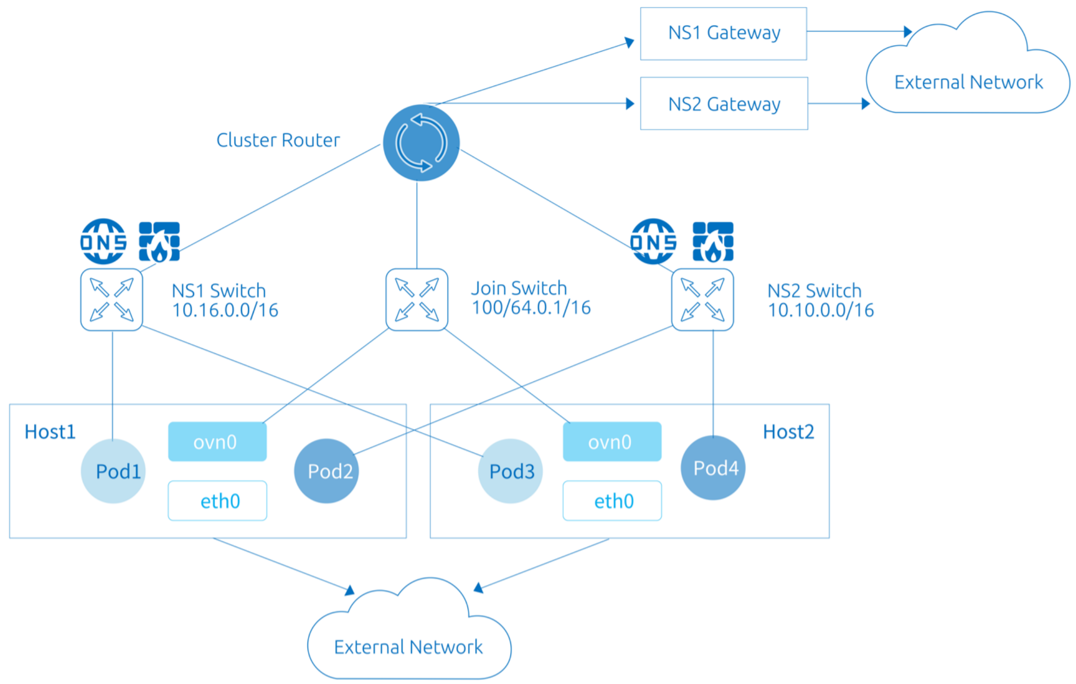
3. ☒☒☒☒

3.1 ☒☒☒☒

☒☒☒ Kube-OVN ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒Kube-OVN ☒☒☒☒☒☒☒ IP ☒☒☒☒☒☒☒☒ Namespace ☒☒☒☒☒☒☒☒☒☒☒ Namespace ☒☒ Pod ☒☒☒☒☒☒☒☒☒☒☒ IP ☒☒☒☒☒☒☒☒☒☒CIDR☒☒☒☒☒☒☒☒☒☒☒☒☒NAT ☒☒☒☒☒

☒☒☒ CNI ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ Kube-OVN ☒☒

☒☒☒☒☒ VPC ☒☒☒☒☒☒☒☒☒☒☒☒☒ IP☒☒☒☒☒ VPC peering ☒☒ VPN ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ IP☒


NS1 Gateway
NS2 Gateway
External Network
Cluster Router
NS1 Switch
10.16.0.0/16
Join Switch
100/64.0.1/16
NS2 Switch
10.10.0.0/16
Host1
ovn0
Pod1
Pod2
eth0
Pod3
ovn0
Pod4
eth0
Host2
External Network

Overlay ☒ Underlay ☒☒

3.1.1 ☒☒☒☒

☒☒☒☒☒☒☒☒☒☒☒☒☒☒Kube-OVN ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ Namespace ☒☒☒☒☒☒☒☒☒☒☒ IP☒ ☒☒☒ ☒☒☒☒☒☒☒☒☒☒☒☒☒☒ CIDR ☒☒☒☒☒☒☒☒☒☒

☒ Overlay ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ NAT ☒☒☒☒☒☒☒ Flannel ☒☒☒☒☒☒☒☒☒☒ ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒

☒ Underlay ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ arping ☒☒☒☒☒☒☒☒

☒☒☒☒☒☒

☒☒☒☒ spec ☒☒ default ☒☒☒ true☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ `ovn-default` ☒

☒☒☒☒☒☒

```
# kubectl get subnet ovn-default -o yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  creationTimestamp: "2019-08-06T09:33:43Z"
  generation: 1
  name: ovn-default
  resourceVersion: "1571334"
  selfLink: /apis/kubeovn.io/v1/subnets/ovn-default
  uid: 7e2451f8-fb44-4f7f-b3e0-cfd27f6fd5d6
spec:
  cidrBlock: 10.16.0.0/16
  default: true
  excludeIps:
  - 10.16.0.1
  gateway: 10.16.0.1
  gatewayType: distributed
  natOutgoing: true
  private: false
  protocol: IPv4
```

3.1.2 Join [illegible]

[illegible] Kubernetes [illegible] Node [illegible] Pod [illegible] Overlay [illegible] Kube-OVN [illegible] `join` [illegible] Node [illegible] ovn0 [illegible] `join` [illegible] Pod [illegible]

[illegible] Pods [illegible] Nodes [illegible] ovn0 [illegible]Node [illegible] Pod [illegible] ovn0 [illegible] ovn0 [illegible]

[illegible] join [illegible] CIDR [illegible] Join [illegible]

[illegible] Join [illegible]

> [illegible] Pod [illegible] `hostport` [illegible] `externalTrafficPolicy: Local` [illegible] NodePort [illegible] Service [illegible]

[illegible] `join` [illegible] CIDR [illegible]

```
# kubectl get subnet join -o yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  creationTimestamp: "2019-08-06T09:33:43Z"
  generation: 1
  name: join
  resourceVersion: "1571333"
  selfLink: /apis/kubeovn.io/v1/subnets/join
  uid: 9c744810-c678-4d50-8a7d-b8ec12ef91b8
spec:
  cidrBlock: 100.64.0.0/16
  default: false
  excludeIps:
  - 100.64.0.1
  gateway: 100.64.0.1
  gatewayNode: ""
  gatewayType: ""
  natOutgoing: false
  private: false
  protocol: IPv4
```

[illegible] Node [illegible] ovn0 [illegible]

```
# ifconfig ovn0
ovn0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1420
        inet 100.64.0.4  netmask 255.255.0.0  broadcast 100.64.255.255
        inet6 fe80::800:ff:fe40:5  prefixlen 64  scopeid 0x20<link>
        ether 0a:00:00:40:00:05  txqueuelen 1000  (Ethernet)
        RX packets 18  bytes 1428 (1.3 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 19  bytes 1810 (1.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

3.1.3 [illegible]

[illegible] Namespace [illegible]

[illegible]

```
cat <<EOF | kubectl create -f -
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 10.66.0.0/16
  excludeIps:
```

```
  - 10.66.0.1..10.66.0.10
  - 10.66.0.101..10.66.0.151
  gateway: 10.66.0.1
  gatewayType: distributed
  natOutgoing: true
  routeTable: ""
  namespaces:
  - ns1
  - ns2
EOF
```

- `cidrBlock`: [illegible] CIDR [illegible] VPC [illegible] Subnet CIDR [illegible]
- `excludeIps`: [illegible] IP [illegible] Underlay [illegible]
- `gateway` [illegible]Overlay [illegible] Kube-OVN [illegible]Underlay [illegible]
- `namespaces`: [illegible] Namespace [illegible] Namespace [illegible] Pod [illegible]
- `routeTable`: [illegible]

[illegible]

```
# kubectl create ns ns1
namespace/ns1 created

# kubectl run nginx --image=docker.io/library/nginx:alpine -n ns1
deployment.apps/nginx created

# kubectl get pod -n ns1 -o wide
NAME                     READY   STATUS    RESTARTS   AGE   IP            NODE    NOMINATED NODE   READINESS GATES
nginx-74d5899f46-n8wtg   1/1     Running   0          10s   10.66.0.11    node1   <none>           <none>
```

Workload [illegible]

[illegible] Pod [illegible] Namespace [illegible] IP [illegible] Namespace [illegible] Workload [illegible] Pod [illegible] Annotation `ovn.kubernetes.io/logical_switch` [illegible]

```
apiVersion: v1
kind: Pod
metadata:
  name: another-subnet
  annotations:
    ovn.kubernetes.io/logical_switch: subnet1
spec:
  containers:
  - name: another-subnet
    image: docker.io/library/nginx:alpine
```

[illegible] Workload [illegible] Deployment[illegible]StatefulSet [illegible] `ovn.kubernetes.io/logical_switch` Annotation [illegible] `spec.template.metadata.annotations` [illegible]

3.1.4 Overlay [illegible]

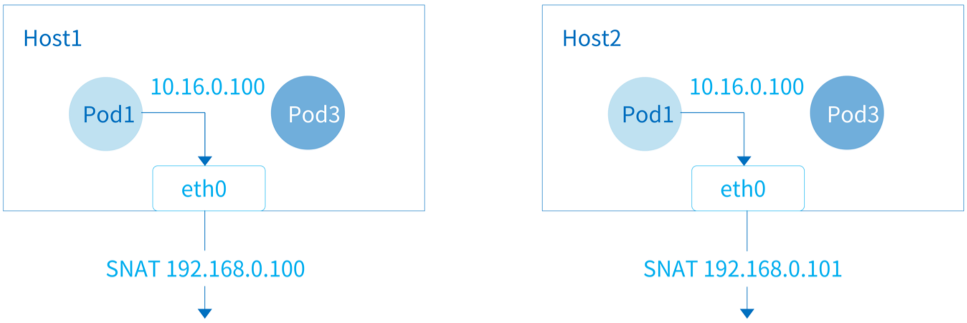
> [illegible] VPC [illegible] Overlay [illegible] Underlay [illegible] VPC [illegible]

Overlay [illegible] Pod [illegible]Kube-OVN [illegible] [illegible]

[illegible] `natOutgoing` [illegible] Pod [illegible] snat[illegible]

[illegible]

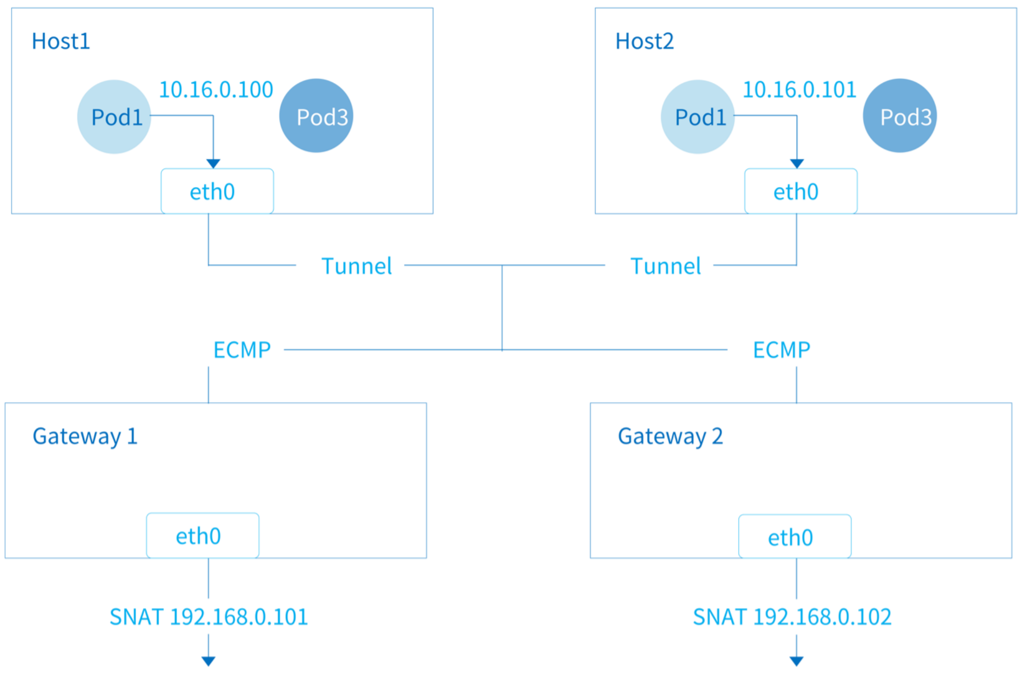
[illegible] Node [illegible] Node [illegible] Pod [illegible] [illegible] `ovn0` [illegible] `natOutgoing` [illegible] `true` [illegible]Pod [illegible] IP[illegible]


Host1
10.16.0.100
Pod1
Pod3
eth0
SNAT 192.168.0.100
Host2
10.16.0.100
Pod1
Pod3
eth0
SNAT 192.168.0.101

[illegible] `gatewayType` [illegible] `distributed` [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: distributed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: distributed
  natOutgoing: true
```

[illegible]


Host1
10.16.0.100
Pod1
Pod3
eth0
Host2
10.16.0.101
Pod1
Pod3
eth0
Tunnel
Tunnel
ECMP
ECMP
Gateway 1
eth0
Gateway 2
eth0
SNAT 192.168.0.101
SNAT 192.168.0.102

[illegible] IP[illegible] [illegible]Pod [illegible] `ovn0` [illegible] `natOutgoing` [illegible] `true` [illegible] Pod [illegible] IP[illegible]

[illegible] `gatewayType` [illegible] `centralized` [illegible] `gatewayNode` [illegible] Kubernetes [illegible] NodeName[illegible] `gatewayNode` [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: centralized
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNode: "node1,node2"
  natOutgoing: true
```

- [illegible] `gatewayNode` [illegible] `kube-ovn-worker:172.18.0.2, kube-ovn-control-plane:172.18.0.3` [illegible]
- [illegible] ECMP [illegible] ECMP [illegible]
- [illegible] Kube-OVN v1.12.0 [illegible] subnet crd [illegible] spec [illegible] `enableEcmp` [illegible] ECMP [illegible] ECMP [illegible] `kube-ovn-controller` Deployment [illegible] `enable-ecmp` [illegible] v1.12.0 [illegible]

> **[illegible]**
>
> [illegible] ECMP [illegible]kube-ovn-controller [illegible] ping [illegible] 5s [illegible] 5s-10s [illegible]
>
> [illegible] Node Ready [illegible]

[illegible]

[illegible] `gatewayNodeSelectors` [illegible]

gatewayNodeSelectors □□□□

- □□ `gatewayNode` □□□□□□□□□□ `gatewayNode` □ `gatewayNodeSelectors` □□□□□
- □□□□□□□□□ OR □□□□□□□□□□□□□□□□□□□□□□□□□□
- □□□□□□□□□□□□□□□□□□□□□□□□□□□

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: centralized-selector
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNodeSelectors:
    - matchLabels:
        role: gateway
    - matchExpressions:
        - key: node-type
          operator: In
          values: ["gateway", "egress"]
  natOutgoing: true
```

3.1.5 □□ ACL □□

Warning

Kube-OVN □□□□ NetworkPolicy□Network Policy API□Subnet ACL □ Security Group □□□□□□□□□□□□□□□□□□□ OVN ACL □□□□□□ NetworkPolicy □ NetworkPolicy API □□□□ □□□

□□□□□□□ ACL □□□□□□□Kube-OVN □ Subnet □□□ ACL □□□□□□□□□□□□□□□□□□□□□□□□

Subnet □□ ACL □□□ OVN □ ACL □□□□□□□□□□□□□□□□□ ovn-nb ACL Table□ `match` □□□□□□□□□□□ ovn-sb Logical Flow Table□

□□ IP □□□ `10.10.0.2` □ Pod □□□□□□□□□□□□□□□□□□□□□□□□□ ACL □□□□□□□□

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: acl
spec:
  allowEWTraffic: false
  acls:
    - action: drop
      direction: to-lport
      match: ip4.dst == 10.10.0.2 && ip
      priority: 1002
    - action: allow-related
      direction: from-lport
      match: ip4.src == 10.10.0.2 && ip
      priority: 1002
  cidrBlock: 10.10.0.0/24
```

□□□□□□□□□□□□□□ ACL □□□□□□□□□□□□□□□□□□□□□□□□□ `allowEWTraffic: true` □□□□

3.1.6 □□□□□□□

> □□ ACL □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ ACL □□□□□□□□□

□□□□□□ Kube-OVN □□□□□□□□□□□□□□□□Pod □□□□□□□□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□□□□ CRD □□ `private` □□□ true□□□□□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□□□ `allowSubnets` □□□ `allowSubnets` □□□□□□□□□□□□□□□□□

□□□□□□□□□□□□□

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: private
spec:
  protocol: IPv4
  default: false
  namespaces:
  - ns1
  - ns2
  cidrBlock: 10.69.0.0/16
  private: true
  allowSubnets:
  - 10.16.0.0/16
  - 10.18.0.0/16
```

3.1.7 Underlay [illegible]

> [illegible] Underlay [illegible]

- `vlan`: [illegible] Underlay [illegible] Subnet [illegible] Vlan CR [illegible] Underlay [illegible]
- `logicalGateway`: [illegible] Underlay [illegible] OVN [illegible] Underlay [illegible] Overlay [illegible] `false` [illegible]

[illegible] MAC [illegible] Underlay [illegible] DHCP[illegible]

[illegible] DHCP [illegible] IP [illegible] CIDR [illegible] Underlay [illegible] `vlan` [illegible] `cidrBlock` [illegible]Kube-OVN [illegible] `Mac` [illegible] `ProtocolMac` [illegible] Pod [illegible] IPAM [illegible] MAC[illegible] IP[illegible] DHCP [illegible] IP[illegible]

MAC-only [illegible] Pod [illegible] IP [illegible]CRI [illegible] Pod [illegible]Pod [illegible] Pending[illegible] Multus [illegible] KubeVirt VM[illegible]

[illegible] `gateway` [illegible] `excludeIps` [illegible] CIDR [illegible] NAT[illegible]CNI [illegible] MAC [illegible] IP[illegible] MAC[illegible] DHCP [illegible]

`vlan` [illegible] `ProviderNetwork` [illegible] `Vlan` [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: dhcp-underlay
spec:
  vlan: vlan-dhcp
  provider: dhcp-net.default.ovn
  cidrBlock: ""
  gateway: ""
  excludeIps: []
```

Pod [illegible] provider [illegible] MAC annotation [illegible] MAC[illegible] `ovn.kubernetes.io/mac_address` [illegible]OVN [illegible] `<nadName>.<nadNamespace>.ovn.kubernetes.io/mac_address` [illegible] NAD [illegible] `<nadName>.<nadNamespace>.kubernetes.io/mac_address.<interfaceName>` [illegible] MAC[illegible] annotation [illegible] provider [illegible] annotation[illegible] IPAM [illegible]

```yaml
metadata:
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name":"dhcp-net","namespace":"default","interface":"net1"}]'
    dhcp-net.default.kubernetes.io/mac_address.net1: "02:00:00:00:10:01"
```

MAC-only [illegible] IP CR [illegible] MAC [illegible] IP[illegible] Pod [illegible]IPAM [illegible] NIC [illegible] MAC [illegible] CIDR [illegible] `cidrBlock` [illegible]MAC-only [illegible] CIDR[illegible] Subnet[illegible]

3.1.8 [illegible]

[illegible] `kube-ovn-cni` [illegible] Pod [illegible] ICMP [illegible] ARP [illegible] Underlay [illegible] ICMP [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: disable-gw-check
spec:
  disableGatewayCheck: true
```

3.1.9 Multicast-Snoop □□

□□□□□ subnet □□ Pod □□□□□□□□□OVN □□□□□□□□□□□□□□□□□□□□ Pod□□□□□ subnet □ multicast snoop □□□OVN □□□ `South Database` □□□□□ `Multicast_Group` □□□□□□□□□□□□

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sample1
spec:
  enableMulticastSnoop: true
```

3.1.10 Subnet MTU □□

□□ Subnet □ Pod □ MTU□□□□□□□□ Subnet □□ Pod □□□

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sample1
spec:
  mtu: 1300
```

3.1.11 □□□□□□□

- IP □□□
- □□ VPC NAT □□□□
- QoS □□
- □□□□□
- DHCP □□
- □□□□□□
- □□□□□□
- □□ IP □□

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3.1.12 □□

3.2 □□□□□□□

□ `Subnet` □□□ `spec.routed: true` □□□□ Pod □□□□□□□□□IPv4 □ `/32` □IPv6 □ `/128` □□ Pod □□□□□□□□□ OVN □□□□□□

3.2.1 □□□□□□□

□□□□□□□□□□ Overlay IPv4 □□□□□

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: routed-subnet
spec:
  cidrBlock: 10.0.0.0/16
  gateway: 10.0.0.1
  protocol: IPv4
  provider: ovn
  routed: true
```

`spec.cidrBlock` □□□□ IPAM □□□□□□□□□□□□□□□ Pod □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ ACL□□□□□□□□□ OVN □□□□□□□□□ `10.0.0.5` □ Pod □□□CNI □□□□

```
ip addr add 10.0.0.5/32 dev eth0
ip route add 10.0.0.1/32 dev eth0
ip route add default via 10.0.0.1
```

IPv6 Pod □□ `/128` □□□□□□ IPv6 □□□□□□□

3.2.2 □□□□□□□

□□□□□□□□□□□□□□ OVN □□□□□ hairpin□□□□□□□□□□□□□□□□□□□□□□□□□□ OVN ACL □□□□□□□□□□□□□□

- □□□□□□□□□□ ARP/ND□
- □□□ ACL □□□□□□□□□□□□□□□□□□ MAC □ IP □□
- □□□□□□□□□□ MAC □□□ IP □□ `from-lport` □□□□□□□□□□□□□□□□□□□□□□□ Pod □□□□ MAC□
- □□ IP□ARP □ ND □□□□□□□

□ `private: true` □□□□□□□□□□□□□□□□□□□□□□□ hairpin□□□ Join CIDR □ `allowSubnets` □□□□□□□

3.2.3 □□□□

□ DPDK □□□□ Provider □□□□□□□ `ovn.kubernetes.io/routes` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□ U2O □□□□□□□□□□□ U2O □□ IP□□□□□□□□□□□□□□ CIDR□□□□□□□□□□□□□ □□□□□□□□□□□ CNI ADD □□□□

□□□□□□□□□ Pod □□□□□□□□□□□□□□□□□ `ovn.kubernetes.io/routed` □□□□□□□ Pod □□□□□□ OVN □□□□□ Pod □□□□□□□□□□□□□□□□□

3.2.4 □□□□

- □□□ OVN Provider □□□Overlay □□□□□□□□□Underlay □□□□□□ `logicalGateway` □ `u2oInterconnection` □□□□ OVN □□□□□□□□
- □□□ `enableDHCP` □ `enableIPv6RA` □ `enableMulticastSnoop` □mac-only/BYO-DHCP □□□□□□ `spec.acls` □
- □□□ DPDK/vhost-user □□□□□□□□□□□□□□□□
- □□□□□ `routed` □□□□□□□□ Pod□□□□□□□□□□ Pod□□□□□□□□□□□□□□□

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2026[illegible]10[illegible]28[illegible]

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3.2.5 [illegible]

3.3 [illegible]

Kube-OVN [illegible] Pod [illegible] Namespace [illegible] IP [illegible] Mac[illegible] [illegible]Kube-OVN [illegible]

- [illegible] Pod [illegible] IP/Mac[illegible]
- [illegible] IP/MAC[illegible]
- Workload [illegible] IP Pool [illegible]
- StatefulSet [illegible]
- KubeVirt VM [illegible]
- [illegible] Multus [illegible]

3.3.1 [illegible] Pod [illegible] IP [illegible] Mac

[illegible] Pod [illegible] annotation [illegible] Pod [illegible] IP/Mac, `kube-ovn-controller` [illegible]

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: static-ip
  annotations:
    ovn.kubernetes.io/ip_address: 10.16.0.15   #           10.16.0.15,fd00:10:16::15
    ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6
spec:
  containers:
  - name: static-ip
    image: docker.io/library/nginx:alpine
```

[illegible] annotation [illegible] Pod IP/Mac [illegible]

1. [illegible] IP/Mac [illegible] IP/Mac [illegible]
2. IP [illegible] CIDR [illegible]
3. [illegible] IP [illegible] Mac[illegible]

3.3.2 [illegible] IP/MAC

[illegible] Pod [illegible] VM [illegible] Multus [illegible] IP/MAC[illegible] Multus [illegible] `k8s.v1.cni.cncf.io/networks` [illegible] `interface` [illegible] annotation[illegible]

- `<nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName>` [illegible] IP
- `<nadName>.<nadNamespace>.kubernetes.io/mac_address.<interfaceName>` [illegible] MAC

[illegible] `<nadName>` [illegible] `<nadNamespace>` [illegible] NetworkAttachmentDefinition [illegible] `<interfaceName>` [illegible] Multus [illegible] `interface` [illegible] annotation [illegible] annotation `ovn.kubernetes.io/ip_address` / `ovn.kubernetes.io/mac_address` [illegible]

[illegible] NAD [illegible] IP/MAC[illegible]

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: multi-if-static
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name": "attachnet", "namespace": "default", "interface": "net1"}, {"name": "attachnet", "namespace": "default",
"interface": "net2"}]'
    attachnet.default.kubernetes.io/ip_address.net1: 172.17.0.100
    attachnet.default.kubernetes.io/mac_address.net1: 00:00:00:53:6B:B1
    attachnet.default.kubernetes.io/ip_address.net2: 172.17.0.101
    attachnet.default.kubernetes.io/mac_address.net2: 00:00:00:53:6B:B2
spec:
  containers:
  - name: multi-if-static
    image: docker.io/library/nginx:alpine
```

3.3.3 Workload □□ IP Pool □□□□

Kube-OVN □□□□ annotation `ovn.kubernetes.io/ip_pool` □ Workload□Deployment/StatefulSet/DaemonSet/Job/CronJob□□□□□ IP□ `kube-ovn-controller` □□□□□ `ovn.kubernetes.io/ip_pool` □□□□ IP □□□□□□□□

IP Pool □ Annotation □□□□ `template` □□ `annotation` □□□□□ Kubernetes □□□ Workload □□□ □□□□□□□□ Workload □□□□□□□□□□□□□□□□□□□

Deployment □□ IP □□

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ippool
  labels:
    app: ippool
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ippool
  template:
    metadata:
      labels:
        app: ippool
      annotations:
        ovn.kubernetes.io/ip_pool: 10.16.0.15,10.16.0.16,10.16.0.17 #
10.16.0.15,fd00:10:16::000E;10.16.0.16,fd00:10:16::000F;10.16.0.17,fd00:10:16::0010
    spec:
      containers:
      - name: ippool
        image: docker.io/library/nginx:alpine
```

□ Workload □□□□ IP □□□□□□□□□

1. `ovn.kubernetes.io/ip_pool` □□ IP □□□□□□□□□ CIDR □□
2. `ovn.kubernetes.io/ip_pool` □□ IP □□□□□□□ IP □□□
3. □ `ovn.kubernetes.io/ip_pool` □□ IP □□□□ replicas □□□□□□□□ Pod □□□□□□□□□□□ Workload □□□□□□□□□□□□□ `ovn.kubernetes.io/ip_pool` □ IP □□□□

3.3.4 StatefulSet □□□□

StatefulSet □□□□□□ IP□□□□□□□ Workload □□□□□□□ `ovn.kubernetes.io/ip_pool` □□□ Pod □□□ IP □□□

□□ StatefulSet □□□□□□□□□□□□□□□□□□□□□□□□□Kube-OVN □□□□□□□□

1. Pod □□□□□□ `ovn.kubernetes.io/ip_pool` □□ IP□□□ StatefulSet □□□□ web□□ web-0 □□□ `ovn.kubernetes.io/ip_pool` □□□□□ IP□ web-1 □□□□□□ IP□□□□□□
2. StatefulSet Pod □□□□□□□□□□ OVN □□ logical_switch_port □□□□□□□□□ Pod □□□□□□ interface □□□□□ Pod □□□□ IP/Mac □□□□□□□□□□□ StatefulSet Volume □□□□□□□□□□
3. □□ 2 □□□□□□□□ `ovn.kubernetes.io/ip_pool` □□□ StatefulSet□Pod □□□□□□□□□□□ IP/Mac□□□□□□ StatefulSet □□□□□□□□□□□□□□□□□□

StatefulSet □□

```yaml
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: web
spec:
  serviceName: "nginx"
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
        ports:
        - containerPort: 80
          name: web
```

[illegible] StatefulSet [illegible] Pod [illegible] Pod IP [illegible]

[illegible] StatefulSet Pod IP

[illegible] StatefulSet [illegible] IP [illegible] Pod Name [illegible] Statefulset [illegible] `ovn.kubernetes.io/ip_pool` Annotation [illegible] Pod [illegible] IP[illegible]

[illegible] StatefulSet Pod IP [illegible] StatefulSet [illegible] scale [illegible] 0[illegible] Annotation [illegible] StatefulSet [illegible]

3.3.5 KubeVirt VM [illegible]

[illegible] KubeVirt [illegible] VM [illegible] `kube-ovn-controller` [illegible] StatefulSet Pod [illegible] IP [illegible] VM [illegible] VM [illegible] IP[illegible]

3.3.6 [illegible] Multus [illegible]

[illegible] Multus [illegible] Pod [illegible] Kube-OVN [illegible] annotation [illegible] Kube-OVN [illegible] CNI[illegible]Kube-OVN [illegible] IPAM [illegible] CNI [illegible]

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3.3.7 [illegible]

3.4 IP [illegible]

IP [illegible]IPPool[illegible]Subnet[illegible] IPAM [illegible] IP [illegible]Namespace[illegible] Workload [illegible]

3.4.1 [illegible]

[illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: IPPool
metadata:
  name: pool-1
spec:
  subnet: ovn-default
  ips:
  - "10.16.0.201"
  - "10.16.0.210/30"
  - "10.16.0.220..10.16.0.230"
  namespaces:
  - ns-1
  enableAddressSet: true
```

[illegible] Workload[illegible]

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ippool
  labels:
    app: ippool
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ippool
  template:
    metadata:
      labels:
        app: ippool
      annotations:
        ovn.kubernetes.io/ip_pool: pool-1
    spec:
      containers:
      - name: ippool
        image: docker.io/library/nginx:alpine
```

3.4.2 [illegible] IP [illegible]

[illegible] `ovn.kubernetes.io/ip_pool` [illegible] IPPool [illegible] `pool-1` [illegible] Pod [illegible] `ovn.kubernetes.io/logical_switch` [illegible] `pool-1.spec.subnet` [illegible] Pod [illegible] Pod [illegible] `logical_switch` [illegible] `cidr` [illegible] `gateway` [illegible]

```yaml
metadata:
  annotations:
    ovn.kubernetes.io/ip_pool: pool-1
```

[illegible] `ovn.kubernetes.io/logical_switch` [illegible] `IPPool.spec.subnet` [illegible]

[illegible] Pod[illegible] `<provider>.kubernetes.io/ip_pool` [illegible] provider [illegible] IP [illegible]provider [illegible] IPPool [illegible] `<provider>.kubernetes.io/logical_switch` [illegible] IP[illegible] IP [illegible] IPv4[illegible]IPv6 [illegible] IP [illegible] IPPool [illegible]

`spec.namespaces` [illegible] IPPool [illegible] IPPool [illegible] Pod[illegible] Pod [illegible]IPPool [illegible] StatefulSet [illegible] VM [illegible] Pod [illegible] Workload [illegible]

[illegible]

[illegible]	[illegible]	[illegible]
subnet	[illegible]	[illegible]
ips	[illegible] IP [illegible]	[illegible] [illegible] [illegible] .. [illegible] IPv6[illegible]
namespaces	[illegible]	[illegible] IPPool [illegible] Pod[illegible]
enableAddressSet	[illegible] AddressSet	[illegible] false[illegible] true [illegible] ACL [illegible] AddressSet [illegible]

3.4.3 [illegible]

1. [illegible] Workload [illegible] IP Pool [illegible] [illegible]IP [illegible] IP [illegible]
2. IP [illegible] `.spec.ips` [illegible] IP [illegible] IP [illegible] `.spec.ips` [illegible] CIDR [illegible]
3. [illegible] IP [illegible]IP [illegible]
4. IP [illegible] `.spec.ips` [illegible]
5. IP [illegible] IP[illegible] IP [illegible] IP [illegible] IP [illegible] IP[illegible]
6. [illegible] IP [illegible] IP [illegible]
7. [illegible] IP [illegible] Namespace[illegible]
8. IP [illegible] `.spec.enableAddressSet` [illegible] `false` [illegible] `true` [illegible] IP [illegible] OVN NB [illegible] AddressSet [illegible] IP [illegible] IP [illegible] AddressSet [illegible] NetworkPolicy [illegible] VPC [illegible] AddressSet [illegible]

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3.4.4 [illegible]

3.5 □□□□□

□□□□□ Pod □□□ Annotations □□□□□□□□□□□□□□□□

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: custom-routes
  annotations:
    ovn.kubernetes.io/routes: |
      [{
        "dst": "192.168.0.101/24",
        "gw": "10.16.0.254"
      }, {
        "gw": "10.16.0.254"
      }]
spec:
  containers:
  - name: nginx
    image: docker.io/library/nginx:alpine
```

> `dst` □□□□□□□□□□□□□□

□□□□□□□ Deployment□DaemonSet □ StatefulSet□□□□ Annotation □□□□□□□□ `.spec.template.metadata.annotations` □□□□□□□

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: custom-routes
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
      annotations:
        ovn.kubernetes.io/routes: |
          [{
            "dst": "192.168.0.101/24",
            "gw": "10.16.0.254"
          }, {
            "gw": "10.16.0.254"
          }]
    spec:
      containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
```

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3.5.1 □□

3.6 EIP [illegible] SNAT [illegible]

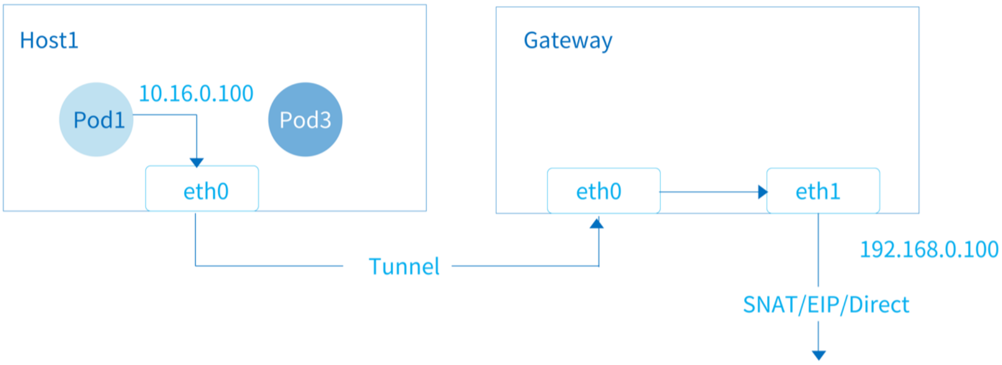
> [illegible] VPC [illegible] VPC [illegible] NAT[illegible]

- VPC Iptables NAT [illegible]
- VPC OVN NAT [illegible]

> [illegible] VPC [illegible], [illegible]

- [illegible] VPC [illegible] OVN NAT [illegible] pod annotation [illegible]
- [illegible] VPC [illegible] OVN NAT [illegible] provider-network[illegible]vlan[illegible]subnet CRD [illegible] ovn-eip[illegible]ovn-dnat[illegible]ovn-fip[illegible]ovn-snat CRD [illegible] EIP [illegible] NAT[illegible]VPC OVN NAT [illegible]
- [illegible] VPC [illegible] Iptables NAT [illegible]VPC Iptables NAT [illegible]

Kube-OVN [illegible] OVN [illegible] L3 Gateway [illegible] Pod [illegible] SNAT [illegible] EIP [illegible] SNAT[illegible] Pod [illegible] IP [illegible] EIP [illegible] Pod [illegible] IP [illegible] EIP [illegible] Pod[illegible]Pod [illegible] EIP [illegible]



3.6.1 [illegible]

- [illegible] OVN [illegible] L3 Gateway [illegible] OVS [illegible] Overlay [illegible] Underlay [illegible]
- [illegible] NAT [illegible] Underlay [illegible]
- [illegible] EIP [illegible] SNAT [illegible]

3.6.2 [illegible]

[illegible] `kube-system` [illegible] ConfigMap `ovn-external-gw-config` [illegible]

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-external-gw-config
  namespace: kube-system
data:
  type: "centralized"
  enable-external-gw: "true"
  external-gw-nodes: "kube-ovn-worker"
  external-gw-nic: "eth1"
  external-gw-addr: "172.56.0.1/16"
  nic-ip: "172.56.0.254/16"
  nic-mac: "16:52:f3:13:6a:25"
```

- `enable-external-gw`: [illegible] SNAT [illegible] EIP [illegible]
- `type`: `centralized` [illegible] `distributed` [illegible] `centralized` [illegible] `distributed` [illegible]
- `external-gw-nodes`: `centralized` [illegible]

- `external-gw-nic`: □□□□□□□□□□□□□□□
- `external-gw-addr`: □□□□□□□ IP □□□□
- `nic-ip`,`nic-mac`: □□□□□□□□□□ IP □ Mac□□□□□□□□□□□ IP □ Mac□

3.6.3 □□ OVN □ OVS □□□□□□□□

□□ OVN-NB □□□□□ `ovn-external` □□□□□□□□□□ `ovn-cluster-ovn-external` □□□□□□□□□□□□□□□□ chassis□

```
# kubectl ko nbctl show
switch 3de4cea7-1a71-43f3-8b62-435a57ef16a6 (external)
    port localnet.external
        type: localnet
        addresses: ["unknown"]
    port external-ovn-cluster
        type: router
        router-port: ovn-cluster-external
router e1eb83ad-34be-4ed5-9a02-fcc8b1d357c4 (ovn-cluster)
    port ovn-cluster-external
        mac: "ac:1f:6b:2d:33:f1"
        networks: ["172.56.0.100/16"]
        gateway chassis: [a5682814-2e2c-46dd-9c1c-6803ef0dab66]
```

□□ OVS □□□□□□□□□□□□□□□□ `br-external` □□□

```
# kubectl ko vsctl ${gateway node name} show
e7d81150-7743-4d6e-9e6f-5c688232e130
    Bridge br-external
        Port br-external
            Interface br-external
                type: internal
        Port eth1
            Interface eth1
        Port patch-localnet.external-to-br-int
            Interface patch-localnet.external-to-br-int
                type: patch
                options: {peer=patch-br-int-to-localnet.external}
```

3.6.4 Pod □□ EIP □ SNAT

□□□□ Pod □□□ `ovn.kubernetes.io/snat` □ `ovn.kubernetes.io/eip` annotation □□□□□ SNAT □ EIP□

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod-snat
  annotations:
    ovn.kubernetes.io/snat: 172.56.0.200
spec:
  containers:
  - name: pod-snat
    image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  name: pod-eip
  annotations:
    ovn.kubernetes.io/eip: 172.56.0.233
spec:
  containers:
  - name: pod-eip
    image: docker.io/library/nginx:alpine
```

□□□ kubectl □□□□□□□□□□ Pod □□□□ EIP □ SNAT □□□□□□□□□□□□□ `ovn.kubernetes.io/routed` annotation □□□□□□□□□

```
kubectl annotate pod pod-eip ovn.kubernetes.io/eip=172.56.0.221 --overwrite
kubectl annotate pod pod-eip ovn.kubernetes.io/routed-
```

□ EIP □ SNAT □□□□□□ `ovn.kubernetes.io/routed` annotation □□□□□□□□

3.6.5 □□□□

`kube-ovn-controller` □□□□□□□□□□□□ SNAT □ EIP □□□□□□□□□□

- `--external-gateway-config-ns`: Configmap `ovn-external-gw-config` □□ Namespace□□□□ `kube-system` □

- `--external-gateway-net`: [illegible] `external` [illegible]
- `--external-gateway-vlanid`: [illegible] Vlan Tag [illegible] 0[illegible] Vlan[illegible]

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3.6.6 [illegible]

3.7 [illegible] QoS [illegible]

Kube-OVN [illegible] Pod [illegible] QoS[illegible]

- [illegible] QoS[illegible]
- `linux-netem` [illegible] QoS[illegible]

> [illegible] Pod [illegible] QoS [illegible] Namespace [illegible] Subnet [illegible] QoS [illegible]

3.7.1 [illegible] QoS

[illegible] QoS [illegible] Pod annotation [illegible] Pod [illegible] `Mbit/s` [illegible]

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: qos
  namespace: ls1
  annotations:
    ovn.kubernetes.io/ingress_rate: "3"
    ovn.kubernetes.io/egress_rate: "1"
spec:
  containers:
  - name: qos
    image: docker.io/library/nginx:alpine
```

[illegible] annotation [illegible] QoS[illegible]

```bash
kubectl annotate --overwrite  pod nginx-74d5899f46-d7qkn ovn.kubernetes.io/ingress_rate=3
```

[illegible] QoS [illegible]

[illegible]

```yaml
kind: DaemonSet
apiVersion: apps/v1
metadata:
  name: perf
  namespace: ls1
  labels:
    app: perf
spec:
  selector:
    matchLabels:
      app: perf
  template:
    metadata:
      labels:
        app: perf
    spec:
      containers:
      - name: nginx
        image: docker.io/kubeovn/perf
```

[illegible] Pod [illegible] iperf3 server[illegible]

```
# kubectl exec -it perf-4n4gt -n ls1 sh
# iperf3 -s
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

[illegible] Pod [illegible] Pod[illegible]

```
# kubectl exec -it perf-d4mqc -n ls1 sh
# iperf3 -c 10.66.0.12
Connecting to host 10.66.0.12, port 5201
[  4] local 10.66.0.14 port 51544 connected to 10.66.0.12 port 5201
[ ID] Interval           Transfer     Bandwidth       Retr  Cwnd
[  4]   0.00-1.00   sec  86.4 MBytes   725 Mbits/sec    3    350 KBytes
[  4]   1.00-2.00   sec  89.9 MBytes   754 Mbits/sec  118    473 KBytes
[  4]   2.00-3.00   sec   101 MBytes   848 Mbits/sec  184    586 KBytes
[  4]   3.00-4.00   sec   104 MBytes   875 Mbits/sec  217    671 KBytes
[  4]   4.00-5.00   sec   111 MBytes   935 Mbits/sec  175    772 KBytes
[  4]   5.00-6.00   sec   100 MBytes   840 Mbits/sec  658    598 KBytes
[  4]   6.00-7.00   sec   106 MBytes   890 Mbits/sec  742    668 KBytes
[  4]   7.00-8.00   sec   102 MBytes   857 Mbits/sec  764    724 KBytes
[  4]   8.00-9.00   sec  97.4 MBytes   817 Mbits/sec  1175    764 KBytes
[  4]   9.00-10.00  sec   111 MBytes   934 Mbits/sec  1083    838 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-10.00  sec  1010 MBytes   848 Mbits/sec  5119             sender
[  4]   0.00-10.00  sec  1008 MBytes   846 Mbits/sec                  receiver

iperf Done.
```

□□□□□ Pod □□□□□ QoS□

```
kubectl annotate --overwrite  pod perf-4n4gt -n ls1 ovn.kubernetes.io/ingress_rate=30
```

□□□□□□ Pod □□□□□ Pod □□□

```
# iperf3 -c 10.66.0.12
Connecting to host 10.66.0.12, port 5201
[  4] local 10.66.0.14 port 52372 connected to 10.66.0.12 port 5201
[ ID] Interval           Transfer     Bandwidth       Retr  Cwnd
[  4]   0.00-1.00   sec  3.66 MBytes  30.7 Mbits/sec    2   76.1 KBytes
[  4]   1.00-2.00   sec  3.43 MBytes  28.8 Mbits/sec    0    104 KBytes
[  4]   2.00-3.00   sec  3.50 MBytes  29.4 Mbits/sec    0    126 KBytes
[  4]   3.00-4.00   sec  3.50 MBytes  29.3 Mbits/sec    0    144 KBytes
[  4]   4.00-5.00   sec  3.43 MBytes  28.8 Mbits/sec    0    160 KBytes
[  4]   5.00-6.00   sec  3.43 MBytes  28.8 Mbits/sec    0    175 KBytes
[  4]   6.00-7.00   sec  3.50 MBytes  29.3 Mbits/sec    0    212 KBytes
[  4]   7.00-8.00   sec  3.68 MBytes  30.9 Mbits/sec    0    294 KBytes
[  4]   8.00-9.00   sec  3.74 MBytes  31.4 Mbits/sec    0    398 KBytes
[  4]   9.00-10.00  sec  3.80 MBytes  31.9 Mbits/sec    0    526 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-10.00  sec  35.7 MBytes  29.9 Mbits/sec    2             sender
[  4]   0.00-10.00  sec  34.5 MBytes  29.0 Mbits/sec                  receiver

iperf Done.
```

3.7.2 linux-netem QoS

> RHEL □□□□□□□□□□ netem □□□□□ yum install -y kernel-modules-extra && modprobe sch_netem

Pod □□□□□□ annotation □□ `linux-netem` □□ QoS□□□ netem □□□□□□□□□□ QoS □□□ Pod □ Ingress □□□□□□□

- `ovn.kubernetes.io/latency` □□□ Pod □□□□□□□□□□□□□□ ms□
- `ovn.kubernetes.io/jitter` □□□ Pod □□□□□□□□□□□□□□□□ ms□
- `ovn.kubernetes.io/limit` □ □ `qdisc` □□□□□□□□□□□□□□□□□□□□□□□□ 1000□
- `ovn.kubernetes.io/loss` □ □□□□□□□□□□□□□□ float □□□□□□□□□ 20□□□□□ 20% □□□□□□

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3.8 [illegible]

Kube-OVN [illegible] IPv4[illegible]IPv6 [illegible] [illegible]

[illegible] Kubernetes [illegible] [illegible] Kubernetes [illegible]

3.8.1 [illegible]

[illegible] CIDR [illegible] `cidr=<IPv4 CIDR>,<IPv6 CIDR>` [illegible] CIDR [illegible] IPv4 [illegible]IPv6 [illegible]

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-test
spec:
  cidrBlock: 10.16.0.0/16,fd00:10:16::/64
  excludeIps:
  - 10.16.0.1
  - fd00:10:16::1
  gateway: 10.16.0.1,fd00:10:16::1
```

[illegible] `install.sh` [illegible] `/112` [illegible] ULA [illegible] `/64` [illegible]/Service CIDR [illegible]

```
POD_CIDR="10.16.0.0/16,fd00:10:16::/112"
JOIN_CIDR="100.64.0.0/16,fd00:100:64::/112"
```

3.8.2 [illegible] Pod [illegible]

[illegible] Pod [illegible] IPv4 [illegible] IPv6 [illegible] Pod [illegible] annotation [illegible]:

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/allocated: "true"
    ovn.kubernetes.io/cidr: 10.16.0.0/16,fd00:10:16::/64
    ovn.kubernetes.io/gateway: 10.16.0.1,fd00:10:16::1
    ovn.kubernetes.io/ip_address: 10.16.0.9,fd00:10:16::9
    ovn.kubernetes.io/logical_switch: ovn-default
    ovn.kubernetes.io/mac_address: 00:00:00:14:88:09
    ovn.kubernetes.io/routed: "true"
...
podIP: 10.16.0.9
  podIPs:
  - ip: 10.16.0.9
  - ip: fd00:10:16::9
```

3.8.3 [illegible] Pod [illegible] IP [illegible]

[illegible] Pod [illegible] IPv4 [illegible] IPv6 [illegible] Pod [illegible] `ip_family` annotation[illegible] [illegible] annotation [illegible]Kube-OVN [illegible] [illegible] Pod [illegible] Subnet [illegible]

[illegible] `ovn.kubernetes.io/ip_family` [illegible]

```
apiVersion: v1
kind: Pod
metadata:
  name: ipv4-only
  annotations:
    ovn.kubernetes.io/ip_family: ipv4
spec:
  containers:
  - name: nginx
    image: docker.io/library/nginx:alpine
```

[illegible] `ipv4` [illegible] `ipv6` [illegible] IPv4 [illegible] `ipv6` [illegible] IPv6 [illegible] `ipv4` [illegible]Pod [illegible]

[illegible] Multus [illegible] `<provider>.kubernetes.io/ip_family` [illegible] `<provider>` [illegible] NetworkAttachmentDefinition [illegible] Subnet [illegible] provider [illegible]

```
apiVersion: v1
kind: Pod
metadata:
  name: attach-ipv6-only
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/ip_family: ipv6
spec:
  containers:
  - name: nginx
    image: docker.io/library/nginx:alpine
```

□□□□ NetworkAttachmentDefinition □□□□□□□□□□□□ `interface` □□□□Kube-OVN □□□□□□□□□ provider □□ annotation□ □□ `net1` □□□ IPv4□ `net2` □□□ IPv6□

```
apiVersion: v1
kind: Pod
metadata:
  name: attach-mixed-family
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name": "attachnet", "namespace": "default", "interface": "net1"}, {"name": "attachnet", "namespace": "default",
"interface": "net2"}]'
    attachnet.default.ovn.net1.kubernetes.io/ip_family: ipv4
    attachnet.default.ovn.net2.kubernetes.io/ip_family: ipv6
spec:
  containers:
  - name: nginx
    image: docker.io/library/nginx:alpine
```

□□□□□□□□□ IP□□□ `ovn.kubernetes.io/ip_address` □ `<provider>.kubernetes.io/ip_address` □□□□ NetworkAttachmentDefinition □□□□□□□ `<nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName>` □□□ IP □□□□□□□ `ip_family` □□□

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3.9 Webhook [illegible]

[illegible] Webhook [illegible] Kube-OVN [illegible] CRD [illegible] Webhook [illegible] IP [illegible] Subnet CIDR [illegible]

[illegible] Webhook [illegible] Subnet [illegible] Pod [illegible] Kube-OVN [illegible] Webhook [illegible] Pod[illegible]

3.9.1 Cert-Manager [illegible]

Webhook [illegible] cert-manager [illegible] Webhook [illegible] cert-manager[illegible]

[illegible] cert-manager:

```
kubectl apply -f https://github.com/cert-manager/cert-manager/releases/download/v1.8.0/cert-manager.yaml
```

[illegible] cert-manager [illegible] cert-manager [illegible]

3.9.2 [illegible] Webhook

[illegible] Webhook [illegible] yaml [illegible]:

```
# kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/yamls/webhook.yaml
deployment.apps/kube-ovn-webhook created
service/kube-ovn-webhook created
validatingwebhookconfiguration.admissionregistration.k8s.io/kube-ovn-webhook created
certificate.cert-manager.io/kube-ovn-webhook-serving-cert created
issuer.cert-manager.io/kube-ovn-webhook-selfsigned-issuer created
```

3.9.3 [illegible] v2 Helm Chart [illegible]

[illegible] `validatingWebhook.certificate.create` [illegible] `kube-ovn-v2` Helm Chart[illegible]

[illegible] `validatingWebhook.enabled` [illegible] `true` [illegible]Chart [illegible] cert-manager Certificate [illegible] Issuer[illegible] `validatingWebhook.certificate.create` [illegible] `true` [illegible] `false` [illegible] Chart [illegible] Certificate[illegible] Webhook [illegible] Deployment[illegible]Service [illegible] ValidatingWebhookConfiguration[illegible]

```yaml
validatingWebhook:
  enabled: true
  certificate:
    create: false
```

[illegible] `validatingWebhook.enabled` [illegible] `false` [illegible] `validatingWebhook.certificate.create` [illegible] Webhook [illegible]

[illegible] Certificate

[illegible] Webhook [illegible] cert-manager [illegible] CA [illegible] cert-manager [illegible] Certificate [illegible] Issuer[illegible] ClusterIssuer[illegible]

[illegible] Chart [illegible] GitOps [illegible] Certificate[illegible] Certificate [illegible] Secret [illegible] Chart [illegible]

[illegible]	[illegible]	[illegible]
Certificate	`kube-ovn-webhook-serving-cert`	[illegible] Webhook [illegible] `cert-manager.io/inject-ca-from` [illegible]
Secret	`kube-ovn-webhook-secret`	[illegible] Webhook Deployment [illegible]

[illegible] Chart [illegible] `namespace` [illegible] `kube-system` [illegible] `clusterDomain` [illegible] `cluster.local` [illegible] `webhook-certificate.yaml` [illegible]

```yaml
apiVersion: cert-manager.io/v1
kind: Certificate
metadata:
  name: kube-ovn-webhook-serving-cert
  namespace: kube-system
spec:
  secretName: kube-ovn-webhook-secret
  dnsNames:
    - kube-ovn-webhook.kube-system
    - kube-ovn-webhook.kube-system.svc
    - kube-ovn-webhook.kube-system.svc.cluster.local
  issuerRef:
    group: cert-manager.io
    kind: Issuer
    name: existing-webhook-issuer
```

[illegible] `existing-webhook-issuer` [illegible] [illegible] ClusterIssuer[illegible] `issuerRef.kind` [illegible] `ClusterIssuer` [illegible] Chart [illegible] DNS [illegible]

[illegible] `<chart-namespace>` [illegible] Chart [illegible] `namespace` [illegible] [illegible] `webhook-certificate.yaml` [illegible] `metadata.namespace` [illegible]

[illegible] Certificate [illegible] `validatingWebhook.certificate.create: false` [illegible] Webhook[illegible]

```
kubectl apply -f webhook-certificate.yaml
kubectl wait --for=condition=Ready certificate/kube-ovn-webhook-serving-cert \
  -n "<chart-namespace>" --timeout=120s
```

Webhook [illegible] `kube-ovn-webhook-secret` [illegible] `tls.crt` [illegible] `tls.key` [illegible] HTTPS [illegible] `ca.crt` [illegible]

cert-manager [illegible] CA [illegible] Webhook [illegible] `cert-manager.io/inject-ca-from` [illegible] `kube-ovn-webhook-serving-cert` [illegible] Certificate[illegible] Certificate [illegible] Secret [illegible] `ca.crt` [illegible] Webhook [illegible] `caBundle` [illegible] API server [illegible] CA [illegible] Webhook [illegible]

[illegible] Webhook Deployment [illegible] CA [illegible]

```
kubectl rollout status deployment/kube-ovn-webhook -n "<chart-namespace>"
kubectl get validatingwebhookconfiguration kube-ovn-webhook \
  -o jsonpath='{.webhooks[0].clientConfig.caBundle}'
```

[illegible]

[illegible] Chart [illegible] Certificate [illegible] Helm [illegible] Certificate [illegible] Issuer[illegible] Certificate [illegible] Secret [illegible] cert-manager [illegible] Certificate [illegible] Helm [illegible] Certificate [illegible]

3.9.4 [illegible] Webhook [illegible]

[illegible] Pod[illegible] Pod IP `10.16.0.15` [illegible]

```
# kubectl get pod -o wide
NAME                      READY   STATUS    RESTARTS   AGE     IP           NODE              NOMINATED NODE   READINESS GATES
static-7584848b74-fw9dm   1/1     Running   0          2d13h   10.16.0.15   kube-ovn-worker   <none>
```

[illegible] yaml [illegible] IP [illegible] Pod[illegible]

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/ip_address: 10.16.0.15
    ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6
  labels:
    app: static
  managedFields:
  name: staticip-pod
  namespace: default
spec:
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: qatest
```

[illegible] yaml [illegible] Pod [illegible] IP [illegible]

```
# kubectl apply -f pod-static.yaml
Error from server (annotation static-ip 10.16.0.15 is conflict with ip crd static-7584848b74-fw9dm.default, ip 10.16.0.15): error when creating "pod-static.yaml": admission webhook "pod-ip-validating.kube-ovn.io" denied the request: annotation static-ip 10.16.0.15 is conflict with ip crd static-7584848b74-fw9dm.default, ip 10.16.0.15
```

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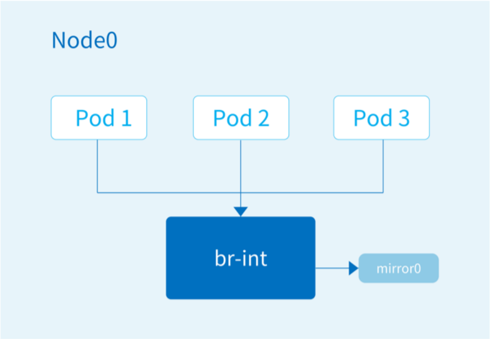
3.10 [illegible]

[illegible] [illegible] [illegible] NPM [illegible]

[illegible] CPU [illegible] 5%~10% [illegible] CPU [illegible]



3.10.1 [illegible]

[illegible] `kube-ovn-cni` DaemonSet [illegible]

- `--enable-mirror=true` [illegible]
- `--mirror-iface=mirror0` : [illegible] [illegible] br-int [illegible]Kube-OVN [illegible] [illegible] `mirror0` [illegible]

[illegible] tcpdump [illegible] `mirror0` [illegible]

```
tcpdump -ni mirror0
```

3.10.2 Pod [illegible]

[illegible] Pod [illegible] Pod [illegible] `ovn.kubernetes.io/mirror` annotation [illegible] Pod [illegible]

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: mirror-pod
  namespace: ls1
  annotations:
    ovn.kubernetes.io/mirror: "true"
spec:
  containers:
  - name: mirror-pod
    image: docker.io/library/nginx:alpine
```

3.10.3 [illegible]

[illegible]

1. Pod to Pod in the same Nodes

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	12.7 us	289 Mbits/sec	12.6 us	(1.8%)	77.9 Mbits/sec
128	15.5 us	517 Mbits/sec	12.7 us	(0%)	155 Mbits/sec
512	12.2 us	1.64 Gbits/sec	12.4 us	(0%)	624 Mbits/sec
1k	13 us	2.96 Gbits/sec	11.4 us	(0.53%)	1.22 Gbits/sec
4k	18 us	7.67 Gbits/sec	25.7 us	(0.41%)	1.50 Gbits/sec

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	11.9 us	324 Mbits/sec	12.2 us	(0.22%)	102 Mbits/sec
128	10.5 us	582 Mbits/sec	9.5 us	(0.21%)	198 Mbits/sec
512	11.6 us	1.84 Gbits/sec	9.32 us	(0.091%)	827 Mbits/sec
1k	10.5 us	3.44 Gbits/sec	10 us	(1.2%)	1.52 Gbits/sec
4k	16.7 us	8.52 Gbits/sec	18.2 us	(1.3%)	2.42 Gbits/sec

2. Pod to Pod in the different Nodes

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	258 us	143 Mbits/sec	237 us	(61%)	28.5 Mbits/sec
128	240 us	252 Mbits/sec	231 us	(64%)	54.9 Mbits/sec
512	236 us	763 Mbits/sec	256 us	(68%)	194 Mbits/sec
1k	242 us	969 Mbits/sec	225 us	(62%)	449 Mbits/sec
4k	352 us	1.12 Gbits/sec	382 us	(0.71%)	21.4 Mbits/sec

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	278 us	140 Mbits/sec	227 us	(24%)	59.6 Mbits/sec
128	249 us	265 Mbits/sec	265 us	(23%)	114 Mbits/sec
512	233 us	914 Mbits/sec	235 us	(21%)	468 Mbits/sec
1k	238 us	1.14 Gbits/sec	240 us	(15%)	891 Mbits/sec
4k	370 us	1.25 Gbits/sec	361 us	(0.43%)	7.54 Mbits/sec

3. Node to Node

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	205 us	162 Mbits/sec	183 us	(11%)	74.2 Mbits/sec
128	222 us	280 Mbits/sec	206 us	(6.3%)	155 Mbits/sec
512	220 us	1.04 Gbits/sec	177 us	(20%)	503 Mbits/sec
1k	213 us	2.06 Gbits/sec	201 us	(8.6%)	1.14 Gbits/sec
4k	280 us	5.01 Gbits/sec	315 us	(37%)	1.20 Gbits/sec

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	204 us	157 Mbits/sec	204 us	(8.8%)	81.9 Mbits/sec
128	213 us	262 Mbits/sec	225 us	(19%)	136 Mbits/sec
512	220 us	1.02 Gbits/sec	227 us	(21%)	486 Mbits/sec
1k	217 us	1.79 Gbits/sec	218 us	(29%)	845 Mbits/sec
4k	275 us	5.27 Gbits/sec	336 us	(34%)	1.21 Gbits/sec

4. Pod to the Node where the Pod is located

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	12.2 us	295 Mbits/sec	12.7 us	(0.27%)	74.1 Mbits/sec
128	14.1 us	549 Mbits/sec	10.6 us	(0.41%)	153 Mbits/sec
512	13.5 us	1.83 Gbits/sec	12.7 us	(0.23%)	586 Mbits/sec
1k	12 us	2.69 Gbits/sec	13 us	(1%)	1.16 Gbits/sec
4k	18.9 us	4.51 Gbits/sec	21.8 us	(0.42%)	1.81 Gbits/sec

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	10.4 us	335 Mbits/sec	12.2 us	(0.75%)	95.4 Mbits/sec
128	12.1 us	561 Mbits/sec	11.3 us	(0.25%)	194 Mbits/sec
512	11.6 us	1.87 Gbits/sec	10.7 us	(0.66%)	745 Mbits/sec
1k	12.7 us	3.12 Gbits/sec	10.9 us	(1.2%)	1.46 Gbits/sec
4k	16.5 us	8.23 Gbits/sec	17.9 us	(1.5%)	2.51 Gbits/sec

5. Pod to the Node where the Pod is not located

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	234 us	153 Mbits/sec	232 us	(63%)	29.4 Mbits/sec
128	237 us	261 Mbits/sec	238 us	(49%)	76.1 Mbits/sec
512	231 us	701 Mbits/sec	238 us	(57%)	279 Mbits/sec
1k	256 us	1.05 Gbits/sec	228 us	(56%)	524 Mbits/sec
4k	330 us	1.08 Gbits/sec	359 us	(1.5%)	35.7 Mbits/sec

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	283 us	141 Mbits/sec	230 us	(26%)	55.8 Mbits/sec
128	234 us	255 Mbits/sec	234 us	(25%)	113 Mbits/sec
512	246 us	760 Mbits/sec	234 us	(22%)	458 Mbits/sec
1k	268 us	1.23 Gbits/sec	242 us	(20%)	879 Mbits/sec
4k	326 us	1.20 Gbits/sec	369 us	(0.5%)	7.87 Mbits/sec

6. Pod to the cluster ip service

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	237 us	133 Mbits/sec	213 us	(65%)	25.5 Mbits/sec
128	232 us	271 Mbits/sec	222 us	(62%)	54.8 Mbits/sec
512	266 us	800 Mbits/sec	234 us	(60%)	232 Mbits/sec
1k	248 us	986 Mbits/sec	239 us	(50%)	511 Mbits/sec
4k	314 us	1.03 Gbits/sec	367 us	(0.6%)	13.2 Mbits/sec

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14305.17	0.87ms	1.48ms	24.46ms
100	29082.07	3.87ms	4.35ms	102.85ms

[illegible]

Size	TCP Latency	TCP Bandwidth	UDP Latency	UDP Lost Rate	UDP Bandwidth
64	241 us	145 Mbits/sec	225 us	(19%)	60.2 Mbits/sec
128	245 us	261 Mbits/sec	212 us	(15%)	123 Mbits/sec
512	252 us	821 Mbits/sec	219 us	(14%)	499 Mbits/sec
1k	253 us	1.08 Gbits/sec	242 us	(16%)	852 Mbits/sec
4k	320 us	1.32 Gbits/sec	360 us	(0.47%)	6.70 Mbits/sec

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	13634.07	0.96ms	1.72ms	30.07ms
100	30215.23	3.59ms	3.20ms	77.56ms

7. Host to the Node port service where the Pod is not located on the target Node

[illegible]

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14802.73	0.88ms	1.66ms	31.49ms
100	29809.58	3.78ms	4.12ms	105.34ms

[illegible]

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14273.33	0.90ms	1.60ms	37.16ms
100	30757.81	3.62ms	3.41ms	59.78ms

8. Host to the Node port service where the Pod is located on the target Node

[illegible]

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	15402.39	802.50us	1.42ms	30.91ms
100	29424.66	4.05ms	4.31ms	90.60ms

[illegible]

TCP-Conn-Number	QPS	Avg-Resp-Time	Stdev-Resp-Time	Max-Resp-Time
10	14649.21	0.91ms	1.72ms	43.92ms
100	32143.61	3.66ms	3.76ms	67.02ms

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3.10.4 □□

3.11 LoadBalancer ▯▯ Service

Kube-OVN ▯▯▯▯▯ VPC ▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯

▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯ `LoadBalancer` ▯▯▯ Service▯▯▯▯▯ LoadBalancerIP ▯▯▯▯▯ VPC ▯▯ Service▯

▯▯▯▯▯▯▯▯▯▯▯▯▯

1. ▯▯▯ `multus-cni` ▯ `macvlan cni` ▯
2. LoadBalancer Service ▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `vpc-nat-gw` ▯▯▯▯▯ macvlan ▯▯▯▯▯▯▯▯▯▯
3. ▯▯▯▯▯ `VPC` ▯▯▯▯▯ VPC ▯▯ LoadBalancer ▯▯▯▯▯ VPC ▯▯▯ VPC ▯▯▯

3.11.1 ▯▯ VPC LoadBalancer Service ▯▯▯▯

▯▯▯▯▯

▯▯ kube-system namespace ▯▯ deployment `kube-ovn-controller` ▯▯ `args` ▯▯▯▯ `--enable-lb-svc=true` ▯▯▯▯▯▯▯▯▯▯▯▯ false▯

```yaml
containers:
- args:
  - /kube-ovn/start-controller.sh
  - --default-cidr=10.16.0.0/16
  - --default-gateway=10.16.0.1
  - --default-gateway-check=true
  - --enable-lb-svc=true                 #       true
```

▯▯ NetworkAttachmentDefinition CRD ▯▯

▯▯▯▯ yaml▯▯▯ `net-attach-def` ▯▯▯

```yaml
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: lb-svc-attachment
  namespace: kube-system
spec:
  # master                   eth0 / ens192 / bond0
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth0",
      "mode": "bridge"
    }'
```

▯▯▯▯▯▯▯▯▯▯ `eth0` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `master` ▯▯▯▯▯▯▯▯▯▯▯▯▯

▯▯ Subnet

▯▯▯ Subnet▯▯▯▯ LoadBalancer Service ▯▯ LoadBalancerIP▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Underlay Subnet ▯▯▯▯▯▯

▯▯▯▯ yaml▯▯▯▯▯▯▯

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attach-subnet
spec:
  protocol: IPv4
  provider: lb-svc-attachment.kube-system    # provider             net-attach-def     Name.Namespace
  cidrBlock: 172.18.0.0/16
  gateway: 172.18.0.1
  excludeIps:
  - 172.18.0.0..172.18.0.10
```

Subnet ▯ `provider` ▯▯▯ `ovn` ▯▯▯ `.ovn` ▯▯▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯▯▯▯▯▯▯▯ `logical switch` ▯▯▯

`provider` ▯ `ovn` ▯▯▯ `.ovn` ▯▯▯▯▯▯ Kube-OVN ▯▯▯ IPAM ▯▯▯▯ IP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

[illegible] LoadBalancer Service

[illegible] yaml[illegible] LoadBalancer Service[illegible]

```yaml
apiVersion: v1
kind: Service
metadata:
  annotations:
    lb-svc-attachment.kube-system.kubernetes.io/logical_switch: attach-subnet   #
    ovn.kubernetes.io/attachmentprovider: lb-svc-attachment.kube-system          #
  labels:
    app: dynamic
  name: test-service
  namespace: default
spec:
  loadBalancerIP: 172.18.0.18                                                   #
  ports:
    - name: test
      protocol: TCP
      port: 80
      targetPort: 80
  selector:
    app: dynamic
  sessionAffinity: None
  type: LoadBalancer
```

[illegible] yaml [illegible]annotation `ovn.kubernetes.io/attachmentprovider` [illegible] `net-attach-def` [illegible] Name.Namespace [illegible] annotation [illegible] Pod [illegible] `net-attach-def` [illegible]

[illegible] annotation [illegible]annotation key [illegible] net-attach-def [illegible] `Name.Namespace.kubernetes.io/logical_switch` [illegible] LoadBalancerIP [illegible] LoadBalancerIP [illegible]

[illegible] LoadBalancerIP [illegible] `spec.loadBalancerIP` [illegible]

[illegible] yaml [illegible] Service [illegible] Service [illegible] Namespace [illegible] Pod [illegible]

```
# kubectl get pod
NAME                                     READY   STATUS    RESTARTS   AGE
lb-svc-test-service-6869d98dd8-cjvll      1/1     Running   0          107m
# kubectl get svc
NAME              TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
test-service      LoadBalancer   10.109.201.193   172.18.0.18   80:30056/TCP   107m
```

[illegible] `service.spec.loadBalancerIP` [illegible] service external-ip [illegible]

[illegible] Pod [illegible] yaml [illegible]

```
# kubectl get pod -o yaml lb-svc-test-service-6869d98dd8-cjvll
apiVersion: v1
kind: Pod
metadata:
  annotations:
    k8s.v1.cni.cncf.io/network-status: |-
      [{
          "name": "kube-ovn",
          "ips": [
              "10.16.0.2"
          ],
          "default": true,
          "dns": {}
      },{
          "name": "default/test-service",
          "interface": "net1",
          "mac": "ba:85:f7:02:9f:42",
          "dns": {}
      }]
    k8s.v1.cni.cncf.io/networks: default/test-service
    k8s.v1.cni.cncf.io/networks-status: |-
      [{
          "name": "kube-ovn",
          "ips": [
              "10.16.0.2"
          ],
          "default": true,
          "dns": {}
      },{
          "name": "default/test-service",
          "interface": "net1",
          "mac": "ba:85:f7:02:9f:42",
          "dns": {}
      }]
    ovn.kubernetes.io/allocated: "true"
    ovn.kubernetes.io/cidr: 10.16.0.0/16
    ovn.kubernetes.io/gateway: 10.16.0.1
    ovn.kubernetes.io/ip_address: 10.16.0.2
    ovn.kubernetes.io/logical_router: ovn-cluster
    ovn.kubernetes.io/logical_switch: ovn-default
    ovn.kubernetes.io/mac_address: 00:00:00:45:F4:29
    ovn.kubernetes.io/pod_nic_type: veth-pair
    ovn.kubernetes.io/routed: "true"
    test-service.default.kubernetes.io/allocated: "true"
    test-service.default.kubernetes.io/cidr: 172.18.0.0/16
    test-service.default.kubernetes.io/gateway: 172.18.0.1
    test-service.default.kubernetes.io/ip_address: 172.18.0.18
    test-service.default.kubernetes.io/logical_switch: attach-subnet
    test-service.default.kubernetes.io/mac_address: 00:00:00:AF:AA:BF
    test-service.default.kubernetes.io/pod_nic_type: veth-pair
```

[illegible] Service [illegible]

```
# kubectl get svc -o yaml test-service
apiVersion: v1
kind: Service
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","kind":"Service","metadata":{"annotations":{"test-service.default.kubernetes.io/logical_switch":"attach-subnet"},"labels":
{"app":"dynamic"},"name":"test-service","namespace":"default"},"spec":{"ports":[{"name":"test","port":80,"protocol":"TCP","targetPort":80}],"selector":
{"app":"dynamic"},"sessionAffinity":"None","type":"LoadBalancer"}}
    ovn.kubernetes.io/vpc: ovn-cluster
    test-service.default.kubernetes.io/logical_switch: attach-subnet
  creationTimestamp: "2022-06-15T09:01:58Z"
  labels:
    app: dynamic
  name: test-service
  namespace: default
  resourceVersion: "38485"
  uid: 161edee1-7f6e-40f5-9e09-5a52c44267d0
spec:
  allocateLoadBalancerNodePorts: true
  clusterIP: 10.109.201.193
  clusterIPs:
  - 10.109.201.193
  externalTrafficPolicy: Cluster
  internalTrafficPolicy: Cluster
  ipFamilies:
  - IPv4
  ipFamilyPolicy: SingleStack
  ports:
  - name: test
    nodePort: 30056
    port: 80
    protocol: TCP
    targetPort: 80
  selector:
    app: dynamic
  sessionAffinity: None
  type: LoadBalancer
status:
  loadBalancer:
    ingress:
    - ip: 172.18.0.18
```

3.11.2 [illegible] LoadBalancerIP [illegible]

[illegible] yaml, [illegible] Pod[illegible] Service [illegible] Endpoints [illegible]

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: dynamic
  name: dynamic
  namespace: default
spec:
  replicas: 2
  selector:
    matchLabels:
      app: dynamic
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: dynamic
    spec:
      containers:
      - image: docker.io/library/nginx:alpine
```

```
        imagePullPolicy: IfNotPresent
        name: nginx
      dnsPolicy: ClusterFirst
      restartPolicy: Always
```

[illegible] Service [illegible] `LoadBalancerIP:Port` [illegible]

```
# curl 172.18.0.18:80
<html>
<head>
        <title>Hello World!</title>
        <link href='//fonts.googleapis.com/css?family=Open+Sans:400,700' rel='stylesheet' type='text/css'>
        <style>
        body {
                background-color: white;
                text-align: center;
                padding: 50px;
                font-family: "Open Sans","Helvetica Neue",Helvetica,Arial,sans-serif;
        }
        #logo {
                margin-bottom: 40px;
        }
        </style>
</head>
<body>
                <h1>Hello World!</h1>
                                <h3>Links found</h3>
        <h3>I am on  dynamic-7d8d7874f5-hsgc4</h3>
        <h3>Cookie                  =</h3>
                                        <b>KUBERNETES</b> listening in 443 available at tcp://10.96.0.1:443<br />
                                                <h3>my name is hanhouchao!</h3>
                        <h3> RequestURI='/'</h3>
</body>
</html>
```

[illegible] Service [illegible] Pod[illegible]

```
# ip a
4: net1@if62: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether ba:85:f7:02:9f:42 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.18.0.18/16 scope global net1
       valid_lft forever preferred_lft forever
    inet6 fe80::b885:f7ff:fe02:9f42/64 scope link
       valid_lft forever preferred_lft forever
36: eth0@if37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP group default
    link/ether 00:00:00:45:f4:29 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.16.0.2/16 brd 10.16.255.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe45:f429/64 scope link
       valid_lft forever preferred_lft forever

# ip rule
0: from all lookup local
32764: from all iif eth0 lookup 100
32765: from all iif net1 lookup 100
32766: from all lookup main
32767: from all lookup default

# ip route show table 100
default via 172.18.0.1 dev net1
10.109.201.193 via 10.16.0.1 dev eth0
172.18.0.0/16 dev net1 scope link

# iptables -t nat -L -n -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 DNAT       tcp  --  *      *       0.0.0.0/0            172.18.0.18          tcp dpt:80 to:10.109.201.193:80

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 MASQUERADE  all  --  *      *       0.0.0.0/0            10.109.201.193
```

[illegible] lb service Pod [illegible] nodeSelector

[illegible] `ovn-vpc-nat-config` ConfigMap [illegible] `nodeSelector` [illegible] LoadBalancer service [illegible] Pod [illegible]

```
apiVersion: v1
data:
  image: docker.io/kubeovn/vpc-nat-gateway:v1.17.0
  nodeSelector: |
    kubernetes.io/hostname: kube-ovn-control-plane
    kubernetes.io/os: linux
kind: ConfigMap
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
```

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3.11.3 □□

3.12 [illegible]

Kube-OVN [illegible] Prometheus [illegible]

[illegible] kube-prometheus [illegible] CRD [illegible] Prometheus [illegible] [illegible] kube-prometheus [illegible] CRD[illegible]Kube-OVN [illegible] Kube-OVN [illegible]

3.12.1 [illegible] Prometheus Monitor

Kube-OVN [illegible] Prometheus Monitor CRD [illegible]

```
#
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/pinger-monitor.yaml
# kube-ovn-controller
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/controller-monitor.yaml
# kube-ovn-cni
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/cni-monitor.yaml
# ovn
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovn-monitor.yaml
```

Prometheus [illegible] 15s[illegible] yaml [illegible] `interval` [illegible]

3.12.2 [illegible] Grafana [illegible]

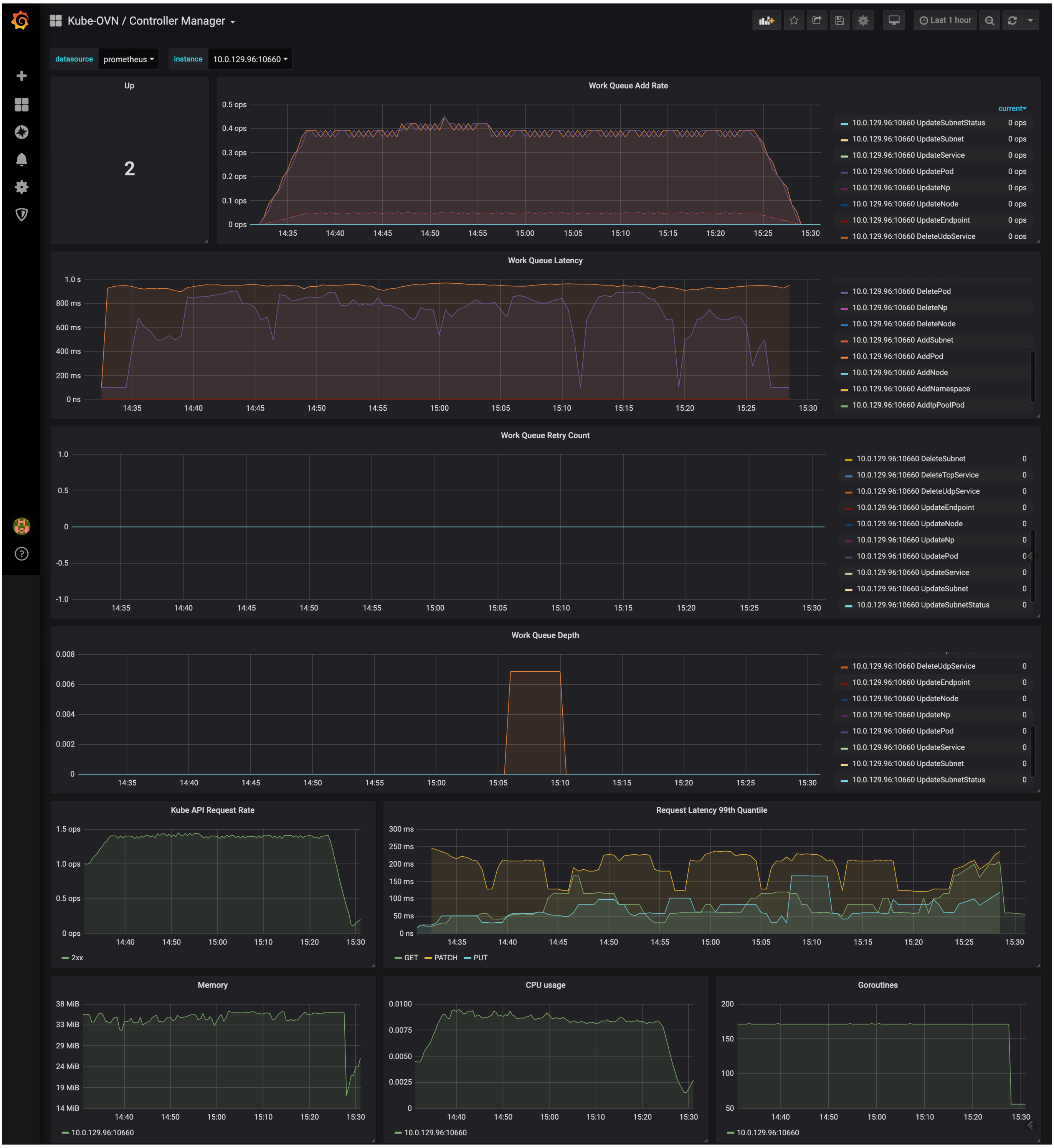
Kube-OVN [illegible] Grafana Dashboard [illegible]

[illegible] Dashboard [illegible]

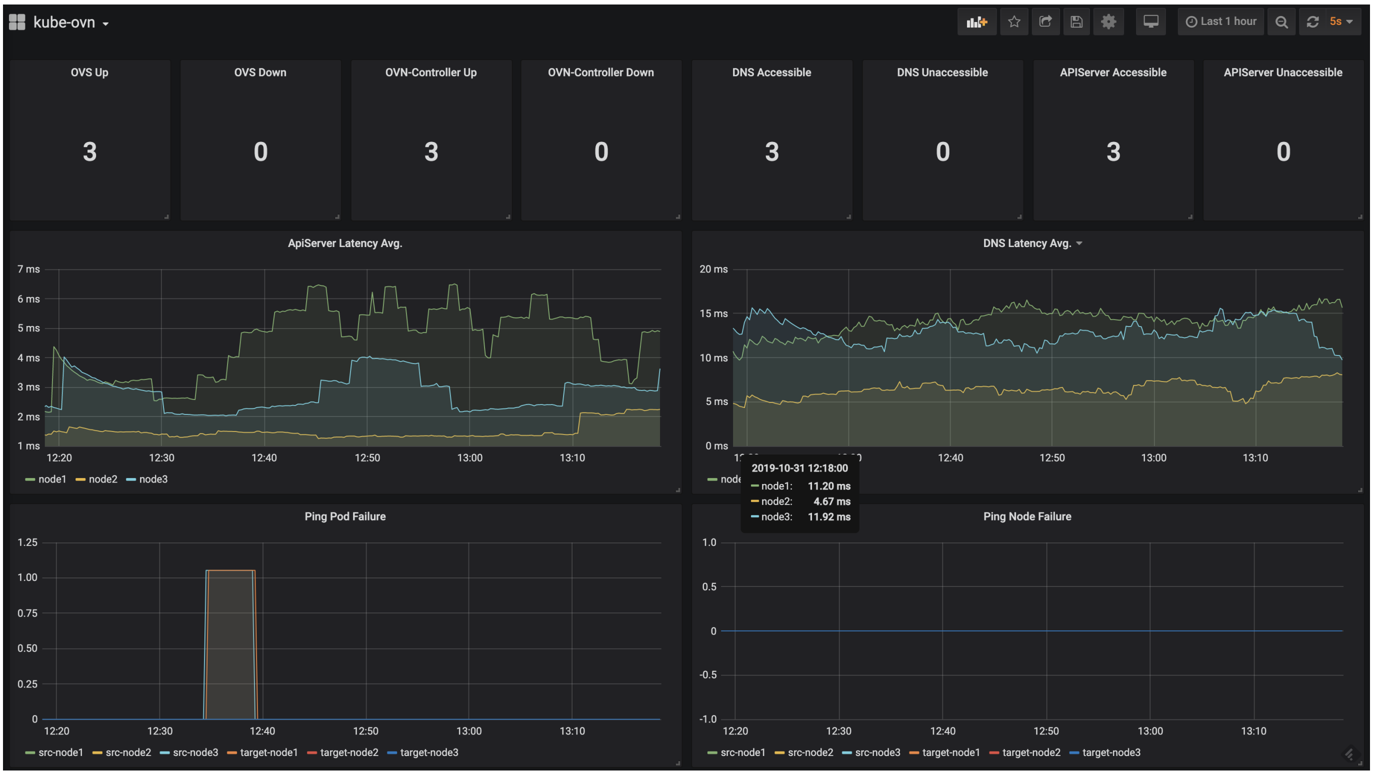
```
#
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/pinger-grafana.json
# kube-ovn-controller
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/controller-grafana.json
# kube-ovn-cni
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/cni-grafana.json
# ovn
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovn-grafana.json
# ovs
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovs-grafana.json
```

[illegible] Grafana [illegible] Prometheus [illegible] Dashboard[illegible]

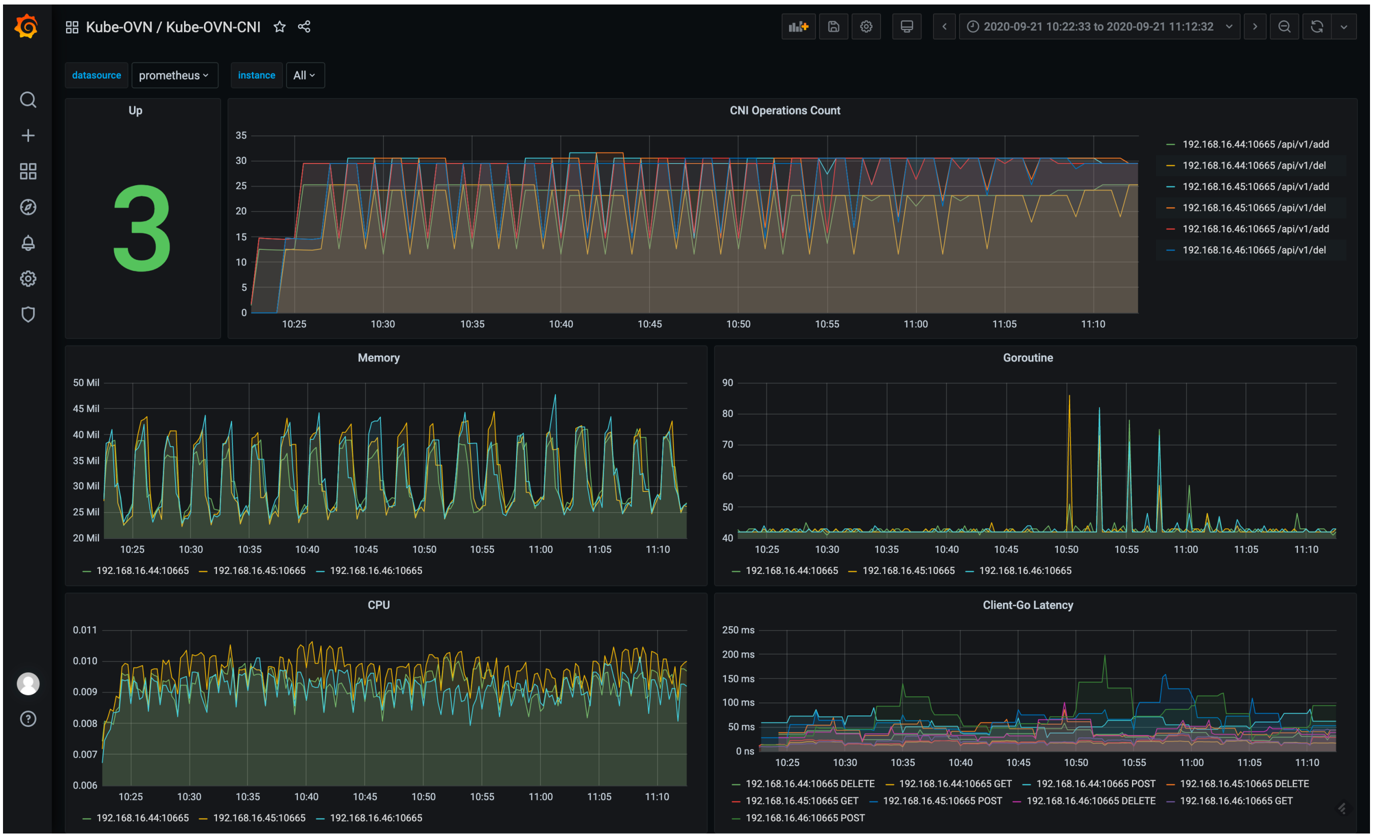
`kube-ovn-controller` [illegible]


Kube-OVN / Controller Manager
datasource prometheus
instance 10.0.129.96:10660
Last 1 hour
Up
2
Work Queue Add Rate
current
10.0.129.96:10660 UpdateSubnetStatus 0 ops
10.0.129.96:10660 UpdateSubnet 0 ops
10.0.129.96:10660 UpdateService 0 ops
10.0.129.96:10660 UpdatePod 0 ops
10.0.129.96:10660 UpdateNp 0 ops
10.0.129.96:10660 UpdateNode 0 ops
10.0.129.96:10660 UpdateEndpoint 0 ops
10.0.129.96:10660 DeleteUdpService 0 ops
Work Queue Latency
10.0.129.96:10660 DeletePod
10.0.129.96:10660 DeleteNp
10.0.129.96:10660 DeleteNode
10.0.129.96:10660 AddSubnet
10.0.129.96:10660 AddPod
10.0.129.96:10660 AddNode
10.0.129.96:10660 AddNamespace
10.0.129.96:10660 AddIpPoolPod
Work Queue Retry Count
10.0.129.96:10660 DeleteSubnet 0
10.0.129.96:10660 DeleteTcpService 0
10.0.129.96:10660 DeleteUdpService 0
10.0.129.96:10660 UpdateEndpoint 0
10.0.129.96:10660 UpdateNode 0
10.0.129.96:10660 UpdateNp 0
10.0.129.96:10660 UpdatePod 0
10.0.129.96:10660 UpdateService 0
10.0.129.96:10660 UpdateSubnet 0
10.0.129.96:10660 UpdateSubnetStatus 0
Work Queue Depth
10.0.129.96:10660 DeleteUdpService 0
10.0.129.96:10660 UpdateEndpoint 0
10.0.129.96:10660 UpdateNode 0
10.0.129.96:10660 UpdateNp 0
10.0.129.96:10660 UpdatePod 0
10.0.129.96:10660 UpdateService 0
10.0.129.96:10660 UpdateSubnet 0
10.0.129.96:10660 UpdateSubnetStatus 0
Kube API Request Rate
2xx
Request Latency 99th Quantile
GET
PATCH
PUT
Memory
10.0.129.96:10660
CPU usage
10.0.129.96:10660
Goroutines
10.0.129.96:10660

`kube-ovn-pinger` □□□□□□□□□


kube-ovn
Last 1 hour
5s
OVS Up
3
OVS Down
0
OVN-Controller Up
3
OVN-Controller Down
0
DNS Accessible
3
DNS Unaccessible
0
APIServer Accessible
3
APIServer Unaccessible
0
ApiServer Latency Avg.
7 ms
6 ms
5 ms
4 ms
3 ms
2 ms
1 ms
12:20
12:30
12:40
12:50
13:00
13:10
node1
node2
node3
DNS Latency Avg.
20 ms
15 ms
10 ms
5 ms
0 ms
2019-10-31 12:18:00
node1: 11.20 ms
node2: 4.67 ms
node3: 11.92 ms
Ping Pod Failure
1.25
1.00
0.75
0.50
0.25
0
src-node1
src-node2
src-node3
target-node1
target-node2
target-node3
Ping Node Failure
1.0
0.5
0
-0.5
-1.0

`kube-ovn-cni` □□□□□□□□□


Kube-OVN / Kube-OVN-CNI
2020-09-21 10:22:33 to 2020-09-21 11:12:32
datasource
prometheus
instance
All
Up
3
CNI Operations Count
35
30
25
20
15
10
5
0
10:25
10:30
10:35
10:40
10:45
10:50
10:55
11:00
11:05
11:10
192.168.16.44:10665 /api/v1/add
192.168.16.44:10665 /api/v1/del
192.168.16.45:10665 /api/v1/add
192.168.16.45:10665 /api/v1/del
192.168.16.46:10665 /api/v1/add
192.168.16.46:10665 /api/v1/del
Memory
50 Mil
45 Mil
40 Mil
35 Mil
30 Mil
25 Mil
20 Mil
192.168.16.44:10665
192.168.16.45:10665
192.168.16.46:10665
Goroutine
90
80
70
60
50
40
CPU
0.011
0.010
0.009
0.008
0.007
0.006
Client-Go Latency
250 ms
200 ms
150 ms
100 ms
50 ms
0 ns
192.168.16.44:10665 DELETE
192.168.16.44:10665 GET
192.168.16.44:10665 POST
192.168.16.45:10665 DELETE
192.168.16.45:10665 GET
192.168.16.45:10665 POST
192.168.16.46:10665 DELETE
192.168.16.46:10665 GET
192.168.16.46:10665 POST

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3.13 ☒☒

3.13.1 NetworkPolicy ☒☒

NetworkPolicy ☒ Kubernetes ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ Pod ☒☒☒☒ Pod ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ Kube-OVN ☒☒ OVN ☒ ACL ☒☒☒☒☒☒ Kubernetes NetworkPolicy ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ NetworkPolicy ☒☒☒☒☒☒☒☒☒☒☒☒

Kube-OVN ☒☒☒☒☒☒

Kube-OVN ☒☒ OVN☒Open Virtual Network☒☒☒☒☒☒☒☒☒ NetworkPolicy☒☒☒☒☒☒☒☒ OVN ☒☒☒

PORT GROUP☒☒☒☒☒☒

☒☒☒☒☒ NetworkPolicy ☒☒Kube-OVN ☒☒☒☒☒☒☒☒☒ Port Group☒☒☒☒☒☒☒☒ `podSelector` ☒ Pod ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ Port Group ☒☒☒☒☒☒ `<   >.<      >` ☒☒☒ `-` ☒☒☒☒☒ `.` ☒

ADDRESS SET☒☒☒☒☒☒

Address Set ☒☒☒☒ NetworkPolicy ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ IP ☒☒☒☒☒☒☒☒☒☒ NetworkPolicy ☒☒☒ Kube-OVN ☒☒☒ `podSelector` ☒ `namespaceSelector` ☒ `ipBlock` ☒☒☒☒☒☒ IP ☒☒☒☒☒☒☒☒ Address Set ☒☒

☒☒☒ NetworkPolicy ☒ Ingress ☒ Egress ☒☒☒☒☒☒☒ Allow ☒ Except ☒☒ Address Set☒

- `<   >.<      >.ingress.allow` ☒☒☒☒☒☒☒ IP ☒☒
- `<   >.<      >.ingress.except` ☒☒☒☒☒☒☒ IP ☒☒
- `<   >.<      >.egress.allow` ☒☒☒☒☒☒☒ IP ☒☒
- `<   >.<      >.egress.except` ☒☒☒☒☒☒☒ IP ☒☒

ACL☒☒☒☒☒☒☒☒☒☒

ACL ☒ OVN ☒☒☒☒☒☒☒☒☒☒☒☒☒☒Kube-OVN ☒ NetworkPolicy ☒☒☒☒☒ OVN ACL ☒☒☒ ☒☒☒☒☒☒☒ Port Group ☒☒ACL ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒

Kube-OVN ☒ NetworkPolicy ☒☒☒ ACL ☒☒☒☒☒☒☒

- Ingress Allow ☒☒☒☒☒☒☒ 2001
- Egress Allow ☒☒☒☒☒☒☒ 2001
- Default Deny ☒☒☒☒☒☒☒ 2000

☒☒☒☒☒☒☒Kube-OVN ☒☒☒☒☒☒☒☒ Kubernetes NetworkPolicy ☒☒☒☒☒☒☒☒☒☒ OVN ☒☒☒☒ ACL ☒☒☒

☒☒☒☒

☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒

Kube-OVN ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒

- **NetworkPolicy**☒Kubernetes ☒☒☒☒☒☒☒☒
- **Network Policy API**☒AdminNetworkPolicy ☒ BaselineAdminNetworkPolicy
- **Subnet ACL**☒☒☒☒☒☒☒☒☒☒☒☒
- **Security Group**☒☒☒☒☒

☒☒☒☒☒☒☒☒☒☒☒☒☒ OVN ACL ☒☒☒☒☒☒☒ NetworkPolicy ☒ Network Policy API ☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒☒ ☒☒☒

NAMED PORT ☒☒☒☒☒☒

NetworkPolicy ☒☒☒☒☒☒☒ Named Port ☒☒☒☒☒☒☒☒☒☒☒

```
ports:
- protocol: TCP
  port: http
```

Kube-OVN [illegible] Named Port [illegible] **Named Port** [illegible]

[illegible] Pod [illegible] Named Port [illegible]NetworkPolicy [illegible] [illegible] Pod A [illegible] `http` [illegible] 8080[illegible] Pod B [illegible] `http` [illegible] 8081[illegible] `port: http` [illegible] NetworkPolicy [illegible]

[illegible] Named Port [illegible] Pod [illegible]

IPBLOCK EXCEPT [illegible]

NetworkPolicy [illegible] `ipBlock` [illegible] `except` [illegible] IP [illegible]

```
egress:
- to:
  - ipBlock:
      cidr: 10.0.0.0/8
      except:
      - 10.0.1.0/24
      - 10.0.2.0/24
```

[illegible] OVN [illegible] `except` [illegible] `except` [illegible] ACL [illegible] [illegible] OVN [illegible]

[illegible] **`except`** [illegible] [illegible] IP [illegible]

- [illegible] CIDR [illegible]
- [illegible] `podSelector` [illegible] `namespaceSelector` [illegible] IP [illegible]

NetworkPolicy [illegible]

Kube-OVN [illegible] NetworkPolicy [illegible]

> ⚠ **Warning**
>
> NetworkPolicy [illegible] [illegible] CPU [illegible]
>
> [illegible]
>
> [illegible] `ovn.kubernetes.io/acl_log_meter_rate` annotation [illegible] ACL [illegible]

[illegible]

[illegible] NetworkPolicy [illegible] annotation `ovn.kubernetes.io/enable_log` [illegible]

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny-ingress
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
spec:
  podSelector: {}
  policyTypes:
  - Ingress
```

[illegible]Drop[illegible]

[illegible]

[illegible] Pod [illegible] `/var/log/ovn/ovn-controller.log` [illegible]

```
# tail -f /var/log/ovn/ovn-controller.log
2022-07-20T05:55:03.229Z|00394|acl_log(ovn_pinctrl0)|INFO|name="np/default-deny-ingress.default/IPv4/0", verdict=drop, severity=warning, direction=to-lport:
udp,vlan_tci=0x0000,dl_src=00:00:00:21:b7:d1,dl_dst=00:00:00:8d:0b:86,nw_src=10.16.0.10,nw_dst=10.16.0.7,nw_tos=0,nw_ecn=0,nw_ttl=63,tp_src=54343,tp_dst=53
```

[illegible] IP[illegible] IP[illegible]

[illegible]

[illegible] Kube-OVN v1.13.0 [illegible] `ovn.kubernetes.io/log_acl_actions` annotation [illegible]Allow[illegible]

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-from-client
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
    ovn.kubernetes.io/log_acl_actions: "allow"
spec:
  podSelector:
    matchLabels:
      app: web
  policyTypes:
  - Ingress
  ingress:
  - from:
    - podSelector:
        matchLabels:
          app: client
```

`ovn.kubernetes.io/log_acl_actions` [illegible]

- `drop` [illegible]
- `allow` [illegible]
- `allow,drop` [illegible]

[illegible]

```
# tail -f /var/log/ovn/ovn-controller.log
2024-08-14T09:27:49.590Z|00004|acl_log(ovn_pinctrl0)|INFO|name="np/allow-from-client.default/ingress/IPv4/0", verdict=allow, severity=info, direction=to-
lport:
icmp,vlan_tci=0x0000,dl_src=96:7b:b0:2f:a0:1a,dl_dst=a6:e5:1b:c2:1b:f8,nw_src=10.16.0.7,nw_dst=10.16.0.12,nw_tos=0,nw_ecn=0,nw_ttl=64,nw_frag=no,icmp_type=8,icm
```

[illegible]

[illegible] `ovn.kubernetes.io/acl_log_meter_rate` annotation [illegible]

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-from-client
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
    ovn.kubernetes.io/log_acl_actions: "allow"
    ovn.kubernetes.io/acl_log_meter_rate: "100"
spec:
  podSelector:
    matchLabels:
      app: web
  policyTypes:
  - Ingress
  ingress:
  - from:
    - podSelector:
        matchLabels:
          app: client
```

`ovn.kubernetes.io/acl_log_meter_rate` [illegible]/[illegible] `100` [illegible] 100 [illegible]

[illegible]

[illegible] annotation `ovn.kubernetes.io/enable_log` [illegible] `false` [illegible]

```
kubectl annotate networkpolicy -n default allow-from-client ovn.kubernetes.io/enable_log=false --overwrite
```

[illegible]

Kube-OVN [illegible]

- **standard**[illegible] NetworkPolicy [illegible] IP [illegible]
- **lax**[illegible] TCP/UDP/SCTP [illegible] ICMP [illegible] L4 [illegible] IP [illegible] DHCP [illegible] UDP [illegible]

[illegible] NetworkPolicy [illegible] annotation [illegible]

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: example-policy
  namespace: default
  annotations:
    ovn.kubernetes.io/network_policy_enforcement: "lax"
spec:
  podSelector: {}
  policyTypes:
  - Ingress
```

[illegible] Kube-OVN [illegible] `--np-enforcement` [illegible]

[illegible]

- [illegible] NetworkPolicy

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2026[illegible]4[illegible]28[illegible]

2025[illegible]10[illegible]23[illegible]

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[illegible]

3.13.2 □□□ NetworkPolicy

□□

□□□□□□Kube-OVN □□ NetworkPolicy □□□ `podSelector` □□□ Pod □□□□ OVN □□□

□□□□□ Pod□□□□□□□□□□□□□□□□□ Provider□

- `ovn.kubernetes.io/network_policy_for`

□□□□□□□□□□□□□□□□□□□□□ Pod □□□□□□

□□□□

□□□□□□□□□

```yaml
metadata:
  annotations:
    ovn.kubernetes.io/network_policy_for: "ovn,default/net-a,default/net-b"
```

□□□□□□□□

- `ovn` □□□ OVN Provider□
- `<namespace>/<net-attach-def>`

□□□

- `ovn`
- `default/net-a`
- `ovn,default/net-a`

Provider □□□□

- □□□□□□□□□□□□□□□□□□□□ OVN Provider□
- □□□□□□□□□□□□□□
- □□□□□□□□□□□□□□□□□ Provider□□□□□□□□□□
- □□□□□□□□□□

`<namespace>/<net-attach-def>` □ Kube-OVN □□□□□□ Provider □□□□□

- `<nad-name>.<nad-namespace>.ovn`

□□

□□ Pod □□□□□□

- □□ OVN Provider□ `ovn` □
- `default/net-a`
- `default/net-b`

□□□

- □□□ `network_policy_for` □
- □□□□□ `ovn` □ `net-a` □ `net-b`
- `network_policy_for: default/net-a` □
- □□□□□□ `net-a`
- `network_policy_for: ovn,default/net-b` □

- [illegible] `ovn` [illegible] `net-b`

[illegible]

- [illegible] Provider/[illegible] Kubernetes NetworkPolicy [illegible]
- [illegible]

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2026[illegible]4[illegible]28[illegible]

2026[illegible]2[illegible]24[illegible]

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[illegible]

3.13.3 [illegible]

Kubernetes [illegible] NetworkPolicy [illegible] L3 [illegible] L4 [illegible] AdminNetworkPolicy (ANP) [illegible] Pod [illegible] DNSNameResolver CoreDNS [illegible]

[illegible]

[illegible] NetworkPolicy [illegible] OVN [illegible] AddressSet [illegible] IP [illegible] IP [illegible] OVN [illegible] AddressSet [illegible] DNS [illegible]

[illegible]

1. kube-ovn-controller [illegible] AdminNetworkPolicy [illegible] DNSNameResolver CR [illegible]
2. CoreDNS [illegible] DNSNameResolver CR [illegible] IP [illegible] DNSNameResolver status [illegible]
3. kube-ovn-controller [illegible] DNSNameResolver CR [illegible] status [illegible] AddressSet[illegible]

[illegible]

[illegible] IP [illegible] Deny [illegible] Allow [illegible] Allow [illegible] Deny [illegible]

[illegible]

[illegible] ANP [illegible] BANP CRD

[illegible] AdminNetworkPolicy [illegible] CRD[illegible]

```
kubectl apply -f https://raw.githubusercontent.com/kubernetes-sigs/network-policy-api/refs/heads/main/config/crd/experimental/
policy.networking.k8s.io_adminnetworkpolicies.yaml
kubectl apply -f https://raw.githubusercontent.com/kubernetes-sigs/network-policy-api/refs/heads/main/config/crd/experimental/
policy.networking.k8s.io_baselineadminnetworkpolicies.yaml
```

[illegible] DNSNAMERESOLVER [illegible]

[illegible] DNSNameResolver [illegible]

```
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsnameresolver/refs/heads/main/manifest/crd.yaml
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsnameresolver/refs/heads/main/manifest/rbac.yaml
kubectl apply -f https://raw.githubusercontent.com/kubeovn/dnsnameresolver/refs/heads/main/manifest/cm.yaml
```

[illegible] COREDNS [illegible]

[illegible] DNSNameResolver [illegible] CoreDNS[illegible] tag[illegible] `dev` [illegible] kubeovn/dnsnameresolver releases[illegible]

```
kubectl set image deployment/coredns coredns=kubeovn/dnsnameresolver:<release-tag> -n kube-system
```

[illegible] CoreDNS [illegible]

```
kubectl get pod -n kube-system -l k8s-app=kube-dns
```

[illegible] ANP [illegible]

[illegible] kube-ovn-controller [illegible]

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: kube-ovn-controller
spec:
  template:
    spec:
      containers:
      - name: kube-ovn-controller
        args:
        - --enable-anp=true
        - --enable-dns-name-resolver=true
        # ...
```

[illegible]

[illegible]

```yaml
apiVersion: policy.networking.k8s.io/v1alpha1
kind: AdminNetworkPolicy
metadata:
  name: deny-external-domains
spec:
  priority: 55
  subject:
    namespaces:
      matchLabels:
        kubernetes.io/metadata.name: kube-system
  egress:
  - action: Deny
    name: deny-baidu-google
    to:
    - domainNames:
      - '*.baidu.com.'
      - '*.google.com.'
```

[illegible]

[illegible]	[illegible]
`priority`	[illegible]
`subject`	[illegible]Pod [illegible]
`egress`	[illegible]
`action`	[illegible] `Allow` [illegible] `Deny` [illegible] `Pass`
`domainNames`	[illegible] `.` [illegible]

[illegible]

[illegible] kube-ovn-pinger [illegible]

```
#
kubectl exec -it -n kube-system kube-ovn-pinger-xxxxx -- ping baidu.com
```

> [illegible] DNS [illegible] ACL [illegible]

[illegible] DNSNameResolver [illegible]

```
# kubectl get dnsnameresolver
NAME                              DNS NAME        RESOLVED IPS
anp-deny-external-domains-88dc32ab   *.google.com.
anp-deny-external-domains-fb3029ce   *.baidu.com.    220.181.7.203
```

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2025[illegible]9[illegible]17[illegible]

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[illegible]

3.13.4 [illegible]

[illegible] Pod [illegible]Pod [illegible] Kube-OVN IPAM [illegible] MAC [illegible] IP [illegible]

[illegible]

- [illegible] Pod [illegible] IP [illegible]
- [illegible] Pod [illegible] MAC [illegible] ARP [illegible]
- [illegible]

[illegible]

[illegible] OVN [illegible] Port Security [illegible] Pod [illegible]Kube-OVN [illegible] OVN [illegible]

- [illegible] MAC [illegible] IP [illegible]
- OVN [illegible] MAC [illegible] IP [illegible]
- [illegible] IPAM [illegible]
- [illegible] OVN [illegible]

[illegible] OVN [illegible]

[illegible]

[illegible] Pod [illegible] `ovn.kubernetes.io/port_security` annotation [illegible]

[illegible]

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: secure-pod
  annotations:
    ovn.kubernetes.io/port_security: "true"
spec:
  containers:
  - name: nginx
    image: docker.io/library/nginx:alpine
```

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2025[illegible]10[illegible]24[illegible]

2025[illegible]10[illegible]24[illegible]

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[illegible]

4. KubeVirt

4.1 VM [illegible] IP

[illegible] IP [illegible] VM [illegible] VM [illegible] IP [illegible]

[illegible] CNI [illegible]

- [illegible] IP [illegible] VM [illegible]VM [illegible] IP [illegible]
- IP [illegible] Node [illegible]VM [illegible] IP[illegible]
- [illegible] IP [illegible] VM [illegible] IP [illegible]

[illegible] KubeVirt [illegible] `masquerade` [illegible] iptables [illegible] VM [illegible] VM [illegible] IP [illegible] `masquerade` [illegible] `bridge` [illegible]

- Pod IP [illegible] VM IP [illegible] Pod IP [illegible]
- `masquerade` [illegible] iptables [illegible] `bridge` [illegible]
- `masquerade` [illegible]
- `masquerade` [illegible] conntrack [illegible]

Kube-OVN [illegible] KubeVirt [illegible] `bridge` [illegible] `managedTap` [illegible] IP [illegible] VM [illegible] IP [illegible] VM [illegible] annotation [illegible] VM [illegible] IP [illegible] Kube-OVN [illegible] Multus [illegible]

4.1.1 IP [illegible] VM [illegible]

[illegible] VM [illegible] IP [illegible] IP [illegible] VM [illegible]Kube-OVN [illegible] IPAM [illegible] VM [illegible] VM [illegible] IP [illegible]

[illegible] `bridge` [illegible] VM[illegible] IP [illegible]

1. [illegible] VM

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
      networks:
      - name: default
        pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
```

```
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
```

1. [illegible] VM [illegible]

```
kubectl get vmi testvm
```

1. [illegible] VM

```
virtctl restart testvm
```

1. [illegible] VM

```
virtctl migrate testvm
```

[illegible] bridge [illegible] VM [illegible]IP [illegible]

4.1.2 [illegible] IP/Mac [illegible]

[illegible] VM [illegible] IP/Mac [illegible] VM [illegible] annotation [illegible] VM [illegible] IP [illegible] KubeVirt [illegible] VM [illegible] IP/MAC[illegible] `<nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName>` [illegible] `<nadName>.<nadNamespace>.kubernetes.io/mac_address.<interfaceName>` [illegible] - [illegible] IP/MAC[illegible]

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        ovn.kubernetes.io/ip_address: 10.16.0.15 #(1)
        ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6 #(2)
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
      networks:
      - name: default
        pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
```

1. 🙋 [illegible] IP [illegible]
2. 🙋 [illegible] Mac [illegible]

> ⚠️ **Warning**
>
> KubeVirt VM API [illegible] `spec.template.spec.domain.devices.interfaces.macAddress` [illegible] Mac [illegible] Mac [illegible] Kube-OVN [illegible] annotation [illegible] Mac [illegible] Mac [illegible]

4.1.3 VM □□ IP

Kube-OVN □□□□□□□□□ VM IP□□□□□ IP □□□□□ VM □□□□□□

□□□□□□□□□ VM IP□

1. □□ VM Annotation□□□□□□□□ IP □□□
2. □□ `virtctl restart <vm name>` □□ VM□□□□ IP □□□□□

4.1.4 □□□□

□□□□□ VM □□□□□□□ IP□□□□□□□□ VM □□□□□□□□□□□ `ovn.kubernetes.io/logical_switch` □□□□

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        ovn.kubernetes.io/logical_switch: subnet1 #(1)
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
      networks:
      - name: default
        pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
```

1. 🙋 □□□□□ VM □□□□□

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4.1.5 □□

4.2 [illegible]

KubeVirt [illegible] Clone [illegible] Label [illegible] Annotation[illegible] Kube-OVN [illegible] Annotation [illegible] IP[illegible]MAC [illegible]

4.2.1 [illegible] Annotation

[illegible] `VirtualMachineClone` [illegible] Kube-OVN [illegible] Annotation[illegible]

```yaml
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  template:
    annotationFilters:
      - "ovn.kubernetes.io/*"
```

4.2.2 [illegible]

> **Note**
>
> `patches` [illegible] KubeVirt 1.6 [illegible]

[illegible] IP [illegible]

```yaml
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  patches:
  - '{"op": "replace", "path": "/spec/template/metadata/annotations/ovn.kubernetes.io~1ip_address", "value": "10.16.0.15"}'
```

[illegible]

```yaml
kind: VirtualMachineClone
apiVersion: "clone.kubevirt.io/v1beta1"
metadata:
  name: testclone
spec:
  source:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-source
  target:
    apiGroup: kubevirt.io
    kind: VirtualMachine
    name: vm-target
  patches:
  - '{"op": "remove", "path": "/spec/template/metadata/annotations/ovn.kubernetes.io~1ip_address"}'
```

[illegible] Annotation [illegible] KubeVirt Clone API[illegible]

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4.2.3 □□

4.3 □□□□

□ KubeVirt □ Bridge □□□□□□DHCP □□□ virt-launcher □□□□□□□□ KubeVirt □□□□ IPv4 □□□ DHCP□□□ Bridge □□□ KubeVirt VM □□□□ RA □□□□□□ IPv6 □□□□□ Kube-OVN □□□ DHCP □ RA □□□□□□ KubeVirt □□□□□ DHCP/RA □□□□□□□□□□□□

□ KubeVirt 1.4.0 □□□□□□□□ Network Binding Plugin □□□□□ Bridge □□□□□ `managedTap` □□□□□ KubeVirt □□□ DHCP □□□□□□□□□ `managedTap` □□□ Kube-OVN □ DHCP/RA □□□□□□ VM □□□□□□□□□□□□

4.3.1 □□□□ DHCP

□ Kube-OVN □ Subnet □□□ DHCP □ IPv6 RA □□□□□□□ YAML □□□□□

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: dual-stack-subnet
spec:
  cidrBlock: "10.244.0.0/16,fd00:10:244::/64"
  enableDHCP: true
  enableIPv6RA: true
```

4.3.2 □□ `managedTap` □□□□

□ KubeVirt □□□ `managedTap` □ Network Binding Plugin:

```
# kubectl patch kubevirts -n kubevirt kubevirt --type=json -p=\
'[{"op": "add", "path": "/spec/configuration/network",   "value": {
    "binding": {
        "managedtap": {
            "domainAttachmentType": "managedTap"
        }
    }
}}]'
```

4.3.3 □□□□□□□□□□ `managedTap` □□□□

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: dual-stack-vm
  namespace: default
spec:
  running: false
  template:
    spec:
      domain:
        devices:
          interfaces:
            - name: default
              binding:
                name: managedtap
      networks:
      - name: default
        pod: {}
```

□□□□□□□□□□□□ VM □□ DHCP □ IPv6 RA □□□□□□□□□□ IPv4/IPv6 □□□

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4.3.4 [illegible]

4.4 [illegible]

[illegible]

KubeVirt [illegible]

- KubeVirt [illegible] Bridge [illegible]
- KubeVirt [illegible]
- [illegible] IP [illegible]
- [illegible]

Kube-OVN [illegible] 0.5 [illegible] TCP [illegible]

4.4.1 [illegible]

[illegible] VM Spec [illegible] `kubevirt.io/allow-pod-bridge-network-live-migration: "true"` [illegible] annotation[illegible]Kube-OVN [illegible]

1. [illegible] VM

```
kubectl apply -f - <<EOF
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
      networks:
      - name: default
        pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
EOF
```

1. SSH [illegible]

```
# password: gocubsgo
virtctl ssh cirros@testvm
ping 8.8.8.8
```

1. [illegible]

```
virtctl migrate testvm
```

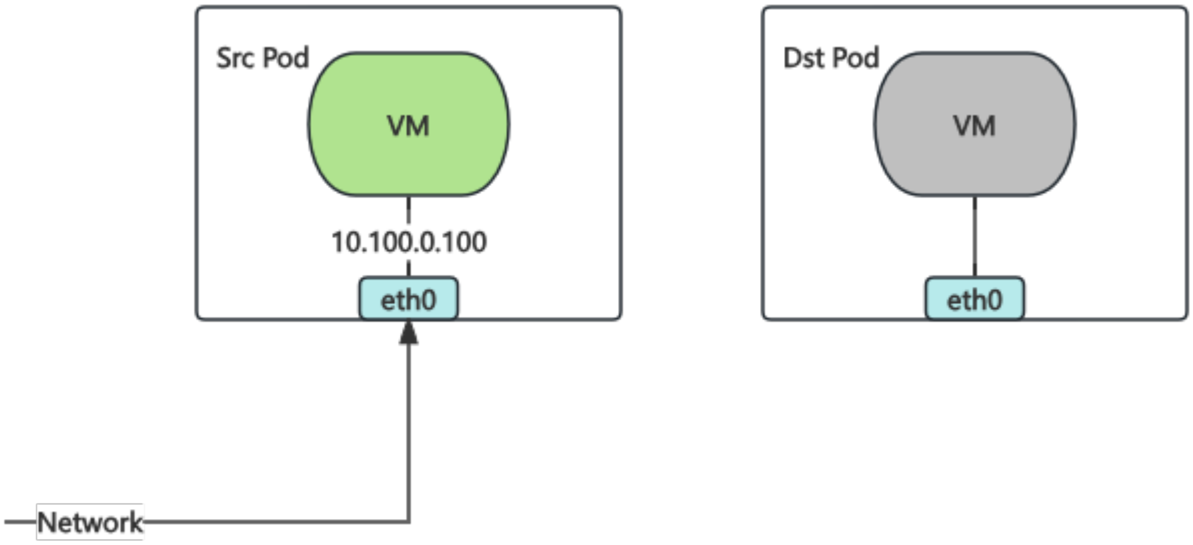
[illegible] VM [illegible] SSH [illegible] ping [illegible]

4.4.2 [illegible]

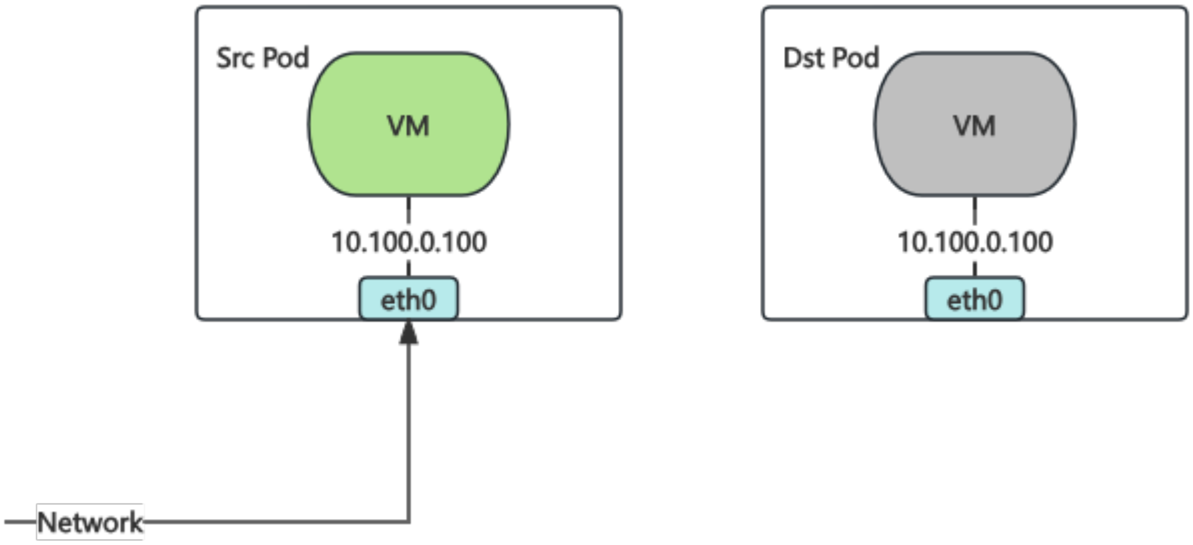
Kube-OVN [illegible]Live migration - Reducing downtime with multchassis port bindings[illegible]

[illegible] IP [illegible]

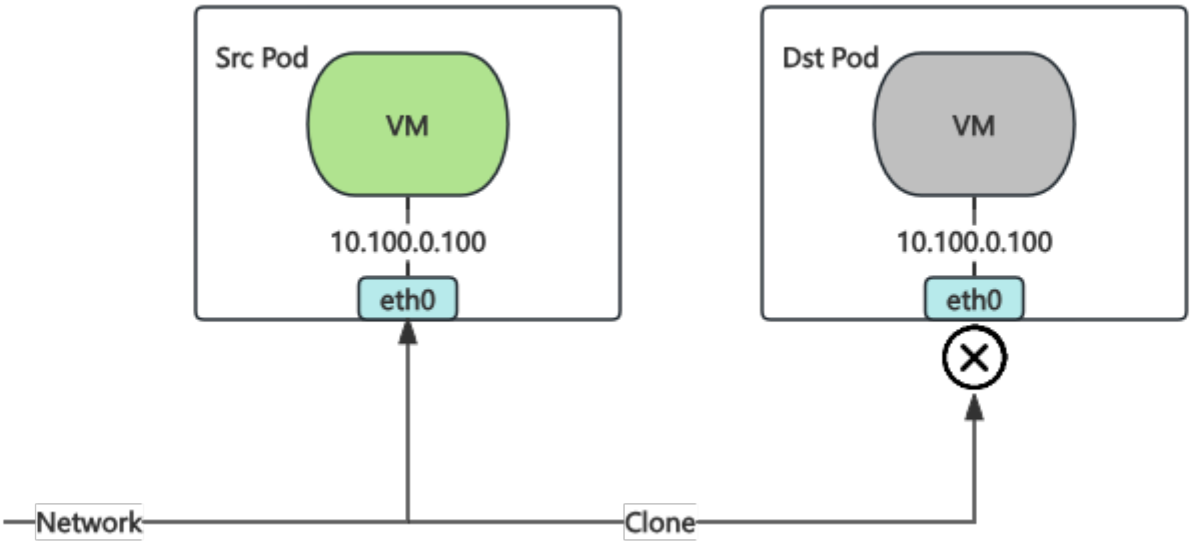
1. KubeVirt [illegible] Pod[illegible]


Src Pod
VM
10.100.0.100
eth0
Dst Pod
VM
eth0
Network

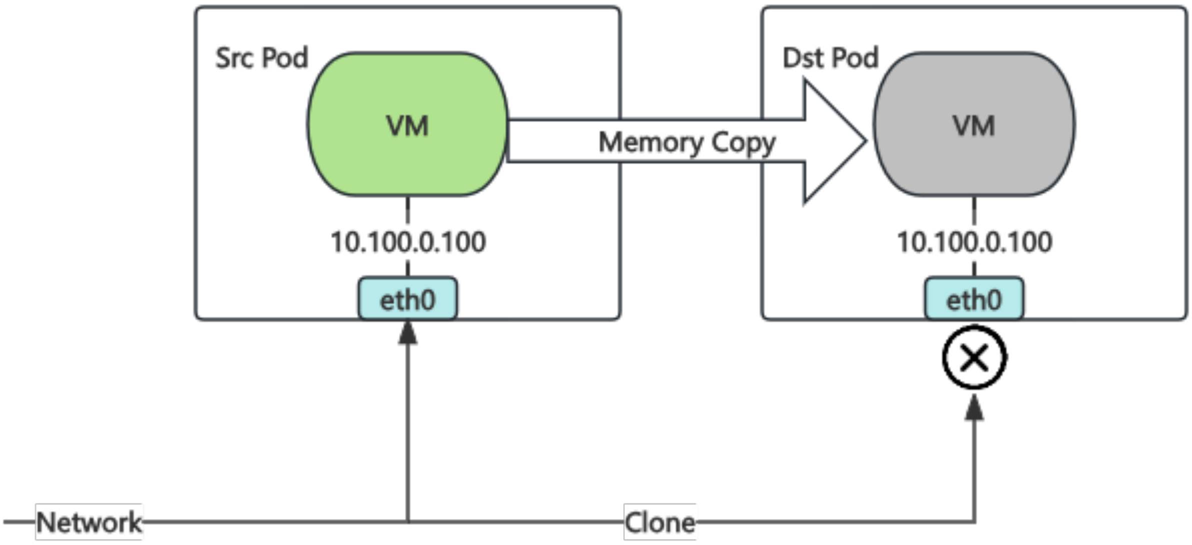
1. Kube-OVN [illegible] Pod [illegible] Pod[illegible] Pod [illegible]


Src Pod
VM
10.100.0.100
eth0
Dst Pod
VM
10.100.0.100
eth0
Network

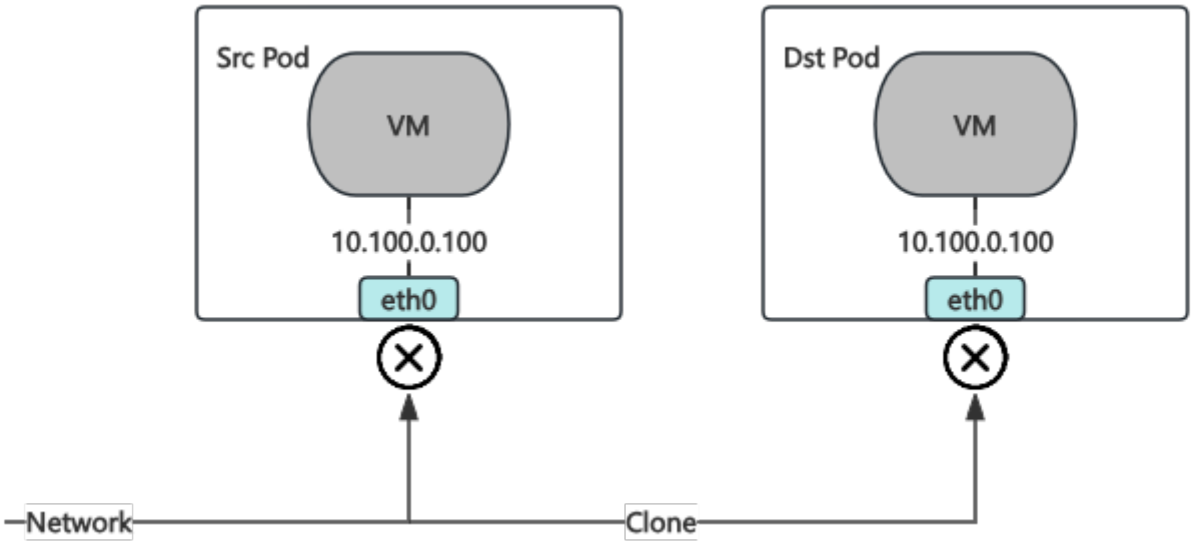
1. Kube-OVN [illegible] Pod [illegible] Pod[illegible] Pod [illegible] Pod [illegible]


Src Pod
VM
10.100.0.100
eth0
Dst Pod
VM
10.100.0.100
eth0
Network
Clone

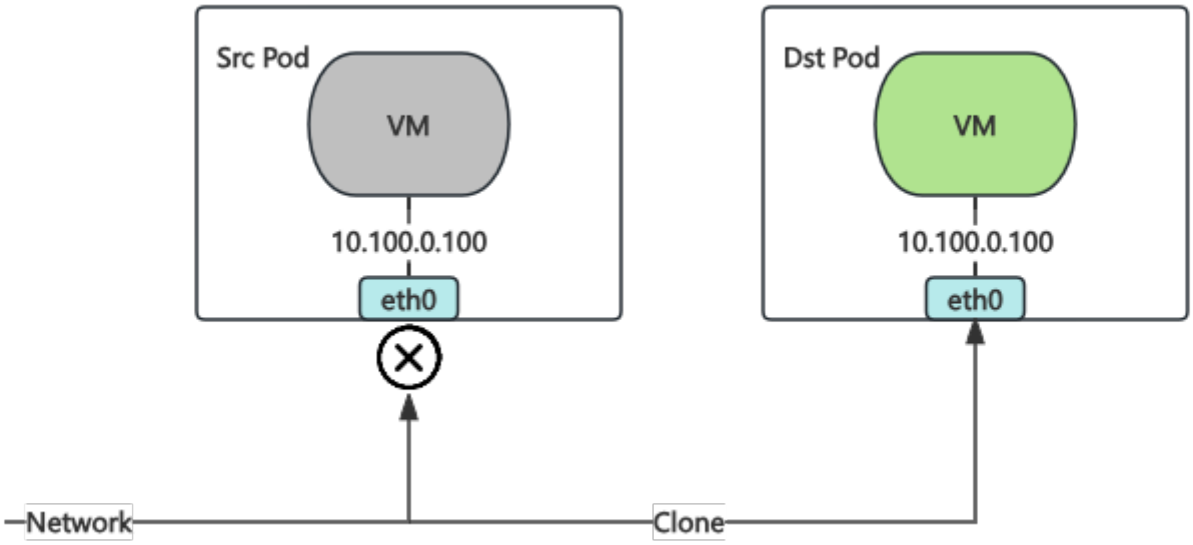
1. KubeVirt [illegible] VM [illegible]


Src Pod
VM
10.100.0.100
eth0
Memory Copy
Dst Pod
VM
10.100.0.100
eth0
Network
Clone

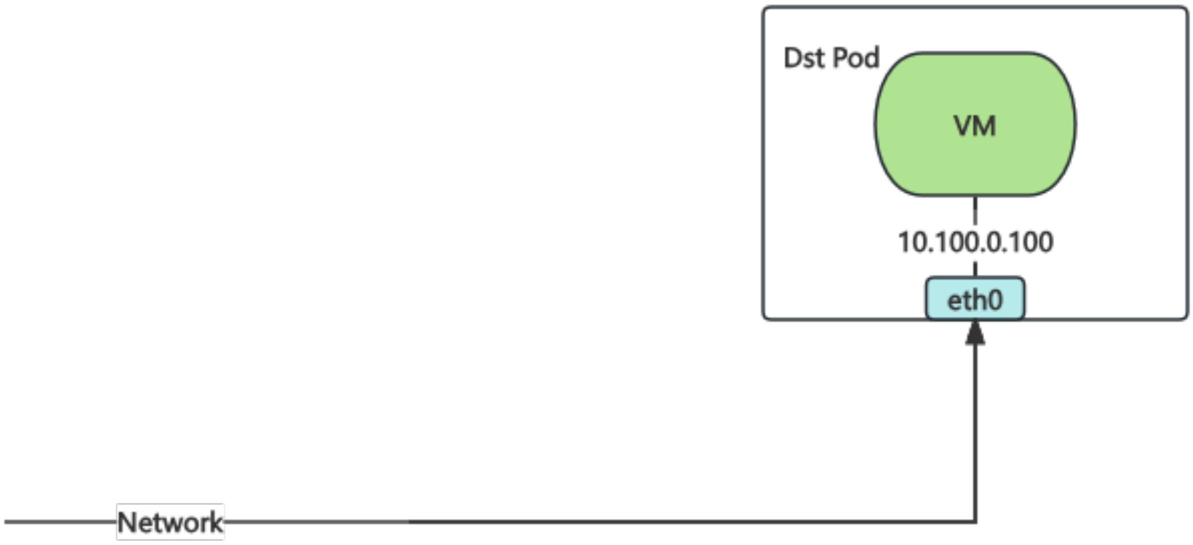
1. KubeVirt [illegible] Pod[illegible] Pod [illegible]


Src Pod
VM
10.100.0.100
eth0
Dst Pod
VM
10.100.0.100
eth0
Network
Clone

1. KubeVirt [illegible] Pod[illegible] libvirt [illegible] RARP [illegible] Pod [illegible] Pod [illegible]


Src Pod
VM
10.100.0.100
eth0
Dst Pod
VM
10.100.0.100
eth0
Network
Clone

1. KubeVirt [illegible] Pod[illegible]Kube-OVN [illegible] Watch Migration CR [illegible]


Dst Pod
VM
10.100.0.100
eth0
Network

[illegible] 5 [illegible] 6 [illegible] libvirt [illegible] RARP [illegible] 0.5 [illegible] TCP [illegible]

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2025[illegible]3[illegible]3[illegible]

GitHub

4.4.3 [illegible]

4.5 [illegible]

Kube-OVN [illegible] Multus Dynamic Networks Controller [illegible] KubeVirt [illegible] v1.4.0 [illegible] VM [illegible]

4.5.1 [illegible]

[illegible] Thick [illegible] Multus[illegible]

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/refs/heads/master/deployments/multus-daemonset-thick.yml
```

[illegible] Multus Dynamic Networks Controller[illegible]

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-dynamic-networks-controller/refs/heads/main/manifests/dynamic-networks-controller.yaml
```

4.5.2 [illegible]

[illegible] NetworkAttachmentDefinition

[illegible] `provider` [illegible] `ovn` [illegible]

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "attachnet.default.ovn"
    }'
```

- `spec.config.type` : [illegible] `kube-ovn` [illegible] CNI [illegible] Kube-OVN [illegible]
- `server_socket` : Kube-OVN [illegible] socket [illegible] `/run/openvswitch/kube-ovn-daemon.sock` [illegible]
- `provider` : [illegible] NetworkAttachmentDefinition [illegible] `<name>.<namespace>.ovn` , Kube-OVN [illegible] Subnet [illegible] ovn[illegible]

[illegible] Kube-OVN Subnet

[illegible] Kube-OVN [illegible] `provider` [illegible] NetworkAttachmentDefinition [illegible] `<name>.<namespace>.ovn` [illegible] `ovn` [illegible] Kube-OVN [illegible] Subnet [illegible]

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: attachnet.default.ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

4.5.3 [illegible] VM [illegible]

[illegible] yaml [illegible] VM[illegible]

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
spec:
  runStrategy: Always
  template:
    spec:
      domain:
        devices:
          disks:
          - disk:
              bus: virtio
            name: containerdisk
          interfaces:
          - masquerade: {}
            name: defaultnetwork
          rng: {}
        resources:
          requests:
            memory: 1024M
      networks:
      - name: defaultnetwork
        pod: {}
      terminationGracePeriodSeconds: 0
      volumes:
      - containerDisk:
          image: quay.io/kubevirt/fedora-with-test-tooling-container-disk:devel
        name: containerdisk
```

[illegible]

[illegible] VM [illegible] Spec [illegible]

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
template:
  spec:
    domain:
      devices:
        interfaces:
        - name: defaultnetwork
          masquerade: {}
          # new interface
        - name: dyniface1
          bridge: {}
    networks:
    - name: defaultnetwork
      pod: {}
      # new network
    - name: dyniface1
      multus:
        networkName: attachnet
```

[illegible]

[illegible] `interface` [illegible] `state` [illegible] `absent` [illegible]

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
template:
  spec:
    domain:
      devices:
        interfaces:
          - name: defaultnetwork
            masquerade: {}
          # set the interface state to absent
          - name: dyniface1
            state: absent
            bridge: {}
    networks:
      - name: defaultnetwork
        pod: {}
      - name: dyniface1
        multus:
          networkName: attachnet
```

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4.5.4 [illegible]

4.6 DHCP

⊠⊠⊠ `managedTap` ⊠ `SR-IOV` ⊠ `DPDK` ⊠⊠⊠⊠⊠⊠KubeVirt ⊠⊠⊠ DHCP ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠Kube-OVN ⊠⊠⊠⊠ OVN ⊠ DHCP ⊠⊠⊠⊠⊠⊠⊠⊠ Pod ⊠⊠⊠⊠ DHCP ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ KubeVirt ⊠⊠⊠⊠⊠⊠ DHCP ⊠⊠⊠⊠⊠ IP ⊠⊠⊠⊠⊠ Kube-OVN ⊠ DHCP ⊠⊠⊠⊠ DHCPv6, IPv6RA, DNS⊠TFTP ⊠ DHCP ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ DHCP ⊠⊠⊠⊠⊠⊠⊠⊠⊠

> **Warning**
>
> ⊠⊠ `bridge` ⊠⊠⊠⊠⊠KubeVirt ⊠ DHCP ⊠⊠⊠ Kube-OVN ⊠⊠⊠⊠⊠ DHCP ⊠⊠⊠⊠⊠ Kube-OVN ⊠⊠⊠ DHCP ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ Kube-OVN ⊠⊠⊠⊠⊠ DHCP ⊠⊠⊠ ⊠⊠⊠ `managedTap` ⊠⊠⊠⊠⊠⊠ `bridge` ⊠⊠⊠⊠ `managedTap` ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ managedTap ⊠⊠⊠⊠⊠

4.6.1 ⊠⊠⊠⊠ DHCP ⊠⊠

⊠⊠ DHCP ⊠⊠⊠⊠⊠⊠

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sn-dual
spec:
  cidrBlock: "10.0.0.0/24,240e::a00/120"
  default: false
  disableGatewayCheck: true
  disableInterConnection: false
  excludeIps:
    - 10.0.0.1
    - 240e::a01
  gateway: 10.0.0.1,240e::a01
  gatewayNode: ''
  gatewayType: distributed
  natOutgoing: false
  private: false
  protocol: Dual
  provider: ovn
  vpc: vpc-test
  enableDHCP: true
  dhcpV4Options: "lease_time=3600,router=10.0.0.1,server_id=169.254.0.254,server_mac=00:00:00:2E:2F:B8"
  dhcpV6Options: "server_id=00:00:00:2E:2F:C5"
  enableIPv6RA: true
  ipv6RAConfigs: "address_mode=dhcpv6_stateful,max_interval=30,min_interval=5,send_periodic=true"
```

- `enableDHCP` : ⊠⊠⊠⊠⊠⊠⊠ DHCP ⊠⊠⊠
- `dhcpV4Options` , `dhcpV6Options` : ⊠⊠⊠⊠⊠⊠⊠ ovn-nb ⊠ DHCP ⊠⊠⊠⊠⊠⊠⊠⊠ DHCP Options⊠ ⊠⊠⊠⊠⊠⊠ `"lease_time=3600, router=$ipv4_gateway, server_id=169.254.0.254, server_mac=$random_mac"` ⊠ `server_id=$random_mac` ⊠
- `enableIPv6RA` : ⊠⊠⊠⊠ DHCPv6 ⊠⊠⊠⊠⊠⊠⊠⊠⊠
- `ipv6RAConfigs` ⊠⊠⊠⊠⊠⊠⊠⊠ ovn-nb ⊠ Logical_Router_Port ⊠⊠⊠⊠⊠⊠⊠⊠ Logical Router Port⊠⊠⊠⊠⊠ `address_mode=dhcpv6_stateful, max_interval=30, min_interval=5, send_periodic=true` ⊠

4.6.2 Pod ⊠⊠ DHCP ⊠⊠

⊠⊠⊠⊠⊠⊠⊠ DHCP ⊠⊠⊠⊠Kube-OVN ⊠⊠⊠⊠⊠ Pod ⊠⊠⊠⊠⊠ Pod ⊠⊠⊠⊠⊠⊠⊠⊠⊠ DHCP ⊠⊠⊠Pod ⊠⊠⊠ DHCP ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ DHCP ⊠⊠⊠⊠⊠⊠⊠⊠⊠ `enableDHCP` ⊠⊠⊠⊠⊠⊠

⊠⊠⊠⊠

```
#     provider   "ovn"
ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1,dns_server=8.8.8.8"
ovn.kubernetes.io/dhcp-v6-options: "server_id=00:00:00:00:00:01"

#      provider   "net1.ns1.ovn"
net1.ns1.ovn.kubernetes.io/dhcp-v4-options: "lease_time=7200"
```

⊠⊠⊠ key ⊠⊠⊠ `<provider>.kubernetes.io/dhcp-v4-options` ⊠ `<provider>.kubernetes.io/dhcp-v6-options` ⊠⊠⊠ `<provider>` ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ provider ⊠ `ovn` ⊠⊠⊠⊠⊠ Multus ⊠⊠⊠⊠⊠⊠⊠⊠provider ⊠⊠⊠ `<net-attach-def-name>.<namespace>.ovn` ⊠

[illegible]

[illegible] Pod [illegible] DHCPv4 [illegible]

```
apiVersion: v1
kind: Pod
metadata:
  name: dhcp-pod
  annotations:
    ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1,dns_server=8.8.8.8"
    ovn.kubernetes.io/dhcp-v6-options: "server_id=00:00:00:00:00:01"
spec:
  containers:
    - name: test
      image: docker.io/library/nginx:alpine
```

[illegible] Multus [illegible] provider [illegible] DHCP [illegible]

```
apiVersion: v1
kind: Pod
metadata:
  name: multi-nic-dhcp-pod
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name": "net1", "namespace": "ns1"}]'
    ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1"
    net1.ns1.ovn.kubernetes.io/dhcp-v4-options: "lease_time=7200,router=10.0.1.1"
spec:
  containers:
    - name: test
      image: docker.io/library/nginx:alpine
```

[illegible]

DHCP [illegible]

1. **Pod** [illegible] `<provider>.kubernetes.io/dhcp-v4-options` [illegible] `<provider>.kubernetes.io/dhcp-v6-options` [illegible] Pod [illegible] DHCP [illegible]
2. [illegible] Subnet CRD [illegible] `dhcpV4Options` [illegible] `dhcpV6Options` [illegible] DHCP [illegible]

[illegible] Pod [illegible] DHCP [illegible] Pod [illegible] DHCP [illegible]

> **Note**
>
> [illegible] Pod [illegible] DHCP [illegible] DHCP [illegible] Pod [illegible]

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GitHub

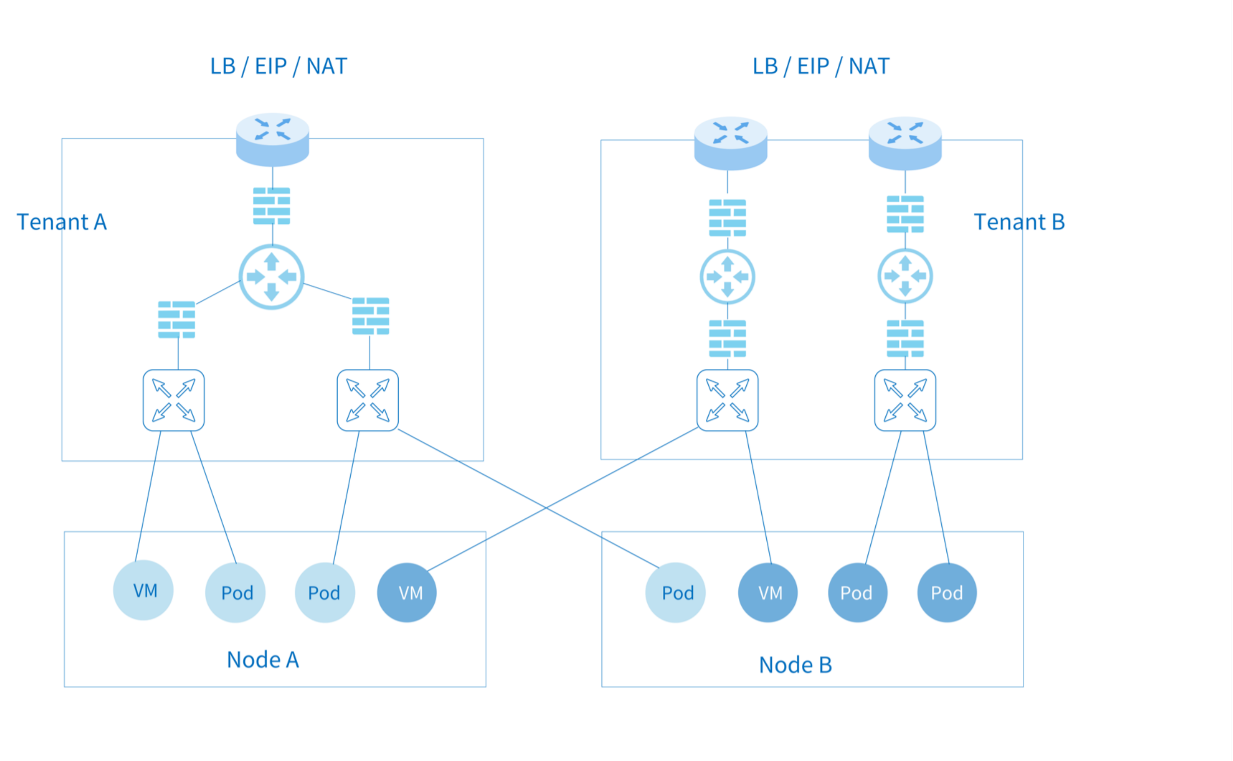
4.6.3 [illegible]

5. VPC [illegible]

5.1 VPC [illegible]

Kube-OVN [illegible] VPC [illegible] VPC [illegible] Subnet [illegible] [illegible]EIP [illegible]

> VPC [illegible] Kubernetes [illegible] Pod [illegible]NodePort [illegible] DNS [illegible] Kubernetes [illegible] Kube-OVN [illegible] VPC [illegible] VPC [illegible] Subnet [illegible] Kubernetes [illegible] VPC [illegible]EIP [illegible] NAT [illegible] VPC [illegible] ACL [illegible] VPC [illegible] VPC [illegible] VPC [illegible] Underlay [illegible]VPC [illegible] Underlay [illegible]


LB / EIP / NAT
LB / EIP / NAT
Tenant A
Tenant B
VM
Pod
Pod
VM
Node A
Pod
VM
Pod
Pod
Node B

5.1.1 [illegible]

[illegible] Kube-OVN [illegible] VPC [illegible] OVN [illegible] IP [illegible] VPC [illegible] OVN [illegible] Datapath ID [illegible] Datapath ID [illegible] IP [illegible] VPC [illegible] IP [illegible]OVN Architecture Design Decisions[illegible]

5.1.2 [illegible] VPC

[illegible] VPC[illegible]

```yaml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  namespaces:
  - ns1
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-2
spec:
  namespaces:
  - ns2
```

- `namespaces` [illegible] Namespace [illegible] VPC[illegible]

[illegible] VPC [illegible] CIDR:

```yaml
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net1
spec:
  vpc: test-vpc-1
  cidrBlock: 10.0.1.0/24
  protocol: IPv4
  namespaces:
    - ns1
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: test-vpc-2
  cidrBlock: 10.0.1.0/24
  protocol: IPv4
  namespaces:
    - ns2
```

[illegible] Namespace [illegible] Pod:

```yaml
apiVersion: v1
kind: Pod
metadata:
  namespace: ns1
  name: vpc1-pod
spec:
  containers:
    - name: vpc1-pod
      image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  namespace: ns2
  name: vpc2-pod
spec:
  containers:
    - name: vpc2-pod
      image: docker.io/library/nginx:alpine
```

[illegible] Pod [illegible] CIDR[illegible] VPC[illegible] Pod [illegible]

5.1.3 [illegible] VPC [illegible]

VPC [illegible] VPC [illegible]VPC [illegible] IP[illegible]SNAT [illegible] DNAT [illegible]

VPC [illegible] Multus-CNI [illegible] multus-cni[illegible]

Note

[illegible] VPC [illegible] VPC [illegible]

[illegible] VPC [illegible]VPC NAT [illegible]OVN [illegible] Egress Gateway[illegible] VPC NAT [illegible] Kube-OVN [illegible] VPC NAT [illegible] Pod[illegible] VPC [illegible] Macvlan [illegible] Pod [illegible] iptables [illegible]

OVN [illegible] OVN [illegible] NAT [illegible] OVN [illegible] BFD [illegible] OVN [illegible] OVN [illegible]

Egress Gateway [illegible] VPC NAT [illegible]

[illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-vpc-external-network
spec:
  protocol: IPv4
  provider: ovn-vpc-external-network.kube-system
  cidrBlock: 192.168.0.0/24
  gateway: 192.168.0.1  # IP address of the physical gateway
  excludeIps:
```

```
  - 192.168.0.1..192.168.0.10
---
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-vpc-external-network
  namespace: kube-system
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth1",
      "mode": "bridge",
      "ipam": {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "ovn-vpc-external-network.kube-system"
      }
    }'

         `ipam`        VPC     net1     Kube-OVN    IP
---
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-vpc-external-network
  namespace: kube-system
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth1",
      "mode": "bridge"
    }'
```

- □ Subnet □□□□□□□□□□□□□□□□□□□□□□ Macvlan □□□ VPC □□□□□□□□□□□□□□□□□□□□ IP□
- VPC □□□□ Macvlan □□□□□□□□ `NetworkAttachmentDefinition` □ `master` □□□□□□□□□□□□□□□
- `name` □□□□□□□

□ Macvlan □□□□□□□□□□□□□□□□□□□□□□□□□□□L2/L3 □□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□Vlan □□□□□□□□□

1. □□ OpenStack □ VM □□□□□□□□□□□□□ `PortSecurity` □□□
2. □□ VMware □ vSwitch □□□□□□ `MAC Address Changes`, `Forged Transmits` □ `Promiscuous Mode Operation` □□□ `allow` □
3. □□ Hyper-V □□□□□□□□□□□□□□□□□□□ `MAC Address Spoofing` □
4. □□□□□□ AWS□GCE□□□□□□□□□□□□□□□ Mac □□□□ Macvlan □□□□□
5. □□ Macvlan □□□□□□Macvlan □□□□□□□□□□□□□□□□□□□□□ VpcNATGateway Pod □□□□□□□□□□□□□□ Pod□
6. □□□□□□□□□□□□□□ Trunk □□□□□□□□□□□□□□□□□□□ Macvlan □□□

□□ VPC □□□□

VPC □□□□□□□□ `kube-system` □□□□□□ `ovn-vpc-nat-gw-config` □□□□□□□□ `ovn-vpc-nat-config` ConfigMap □□□□□ `image` □□□ Pod □□□□□□□□□ `nodeSelector` □□□□□ `nodeSelector` □□□ `LoadBalancer` □□ Service □ lb-svc □□□VpcNATGateway Pod □□□□ `VpcNatGateway.spec.selector` □□□□□□

```
---
kind: ConfigMap
apiVersion: v1
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
data:
  image: 'docker.io/kubeovn/vpc-nat-gateway:v1.17.0'
  nodeSelector: |
    kubernetes.io/hostname: kube-ovn-control-plane
---
kind: ConfigMap
apiVersion: v1
metadata:
  name: ovn-vpc-nat-gw-config
  namespace: kube-system
data:
  enable-vpc-nat-gw: 'true'
```

- `image`: □□ Pod □□□□□□□
- `enable-vpc-nat-gw` □□□□□□□ VPC □□□□□

[illegible] VPC [illegible]

```yaml
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
  externalSubnets:
    - ovn-vpc-external-network
  noDefaultEIP: false
```

- `vpc` [illegible] VpcNatGateway [illegible] VPC[illegible]
- `subnet` [illegible] VPC [illegible] Subnet [illegible]VPC [illegible] Pod [illegible] `lanIp` [illegible]
- `lanIp` [illegible] `subnet` [illegible] IP[illegible]VPC [illegible] Pod [illegible] IP[illegible] VPC [illegible] VpcNatGateway [illegible] `nextHopIP` [illegible] `lanIp` [illegible]
- `selector` [illegible]VpcNatGateway Pod [illegible] Kubernetes [illegible] NodeSelector [illegible]
- `externalSubnets` [illegible]VPC [illegible] `ovn-vpc-external-network` [illegible]
- `noDefaultEIP` [illegible]VPC [illegible] EIP[illegible] `false` [illegible] v1.15 [illegible] BGP [illegible] `true` [illegible] Underlay [illegible]

[illegible]

- `tolerations` [illegible] VPC [illegible] [illegible][illegible]
- `affinity` [illegible] VPC [illegible] Pod [illegible] [illegible][illegible]

[illegible] VPC-NAT-GW [illegible]

1. nat gw pod [illegible]net1 arp [illegible]ping [illegible] eip [illegible] arp[illegible] ping [illegible]

[illegible] EIP

EIP [illegible] IP [illegible] VPC [illegible] DNAT[illegible]SNAT [illegible] IP [illegible]

[illegible] EIP[illegible]

```yaml
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-random
spec:
  natGwDp: gw1
```

[illegible] EIP [illegible]

```yaml
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  natGwDp: gw1
  v4ip: 192.168.0.100
```

[illegible] EIP [illegible]

```yaml
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-random
spec:
  natGwDp: gw1
  externalSubnet: ovn-vpc-external-network
```

- `externalSubnet` [illegible]EIP [illegible] `ovn-vpc-external-network` [illegible]IptablesEIP [illegible] VpcNatGateway [illegible] `externalSubnets` [illegible]EIP [illegible]

[illegible] DNAT [illegible]

[illegible] DNAT [illegible] EIP [illegible] VPC [illegible] IP [illegible]

```yaml
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eipd01
spec:
  natGwDp: gw1

---
kind: IptablesDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: dnat01
spec:
  eip: eipd01
  externalPort: '8888'
  internalIp: 10.0.1.10
  internalPort: '80'
  protocol: tcp
```

[illegible] SNAT [illegible]

[illegible] SNAT [illegible]VPC [illegible] Pod [illegible] EIP [illegible] SNAT[illegible]

```yaml
---
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eips01
spec:
  natGwDp: gw1
---
kind: IptablesSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat01
spec:
  eip: eips01
  internalCIDR: 10.0.1.0/24
```

[illegible] IP

[illegible] IP [illegible]VPC [illegible] IP [illegible] EIP [illegible] EIP [illegible] VPC [illegible] IP[illegible]VPC [illegible] IP [illegible] SNAT [illegible] EIP[illegible]

```yaml
---
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eipf01
spec:
  natGwDp: gw1

---
kind: IptablesFIPRule
apiVersion: kubeovn.io/v1
metadata:
  name: fip01
spec:
  eip: eipf01
  internalIp: 10.0.1.5
```

5.1.4 [illegible]

[illegible] VPC [illegible] Kube-OVN [illegible]

[illegible]

```yaml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  staticRoutes:
    - cidr: 0.0.0.0/0
      nextHopIP: 10.0.1.254
      policy: policyDst
    - cidr: 172.31.0.0/24
      nextHopIP: 10.0.1.253
      policy: policySrc
      routeTable: "rtb1"
```

- `policy`: [illegible] `policyDst` [illegible] `policySrc` [illegible]
- [illegible]CIDR [illegible]
- `routeTable`: [illegible]

[illegible]

[illegible] OVN [illegible] Logical Router Policy[illegible]

[illegible]

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  policyRoutes:
    - action: drop
      match: ip4.src==10.0.1.0/24 && ip4.dst==10.0.1.250
      priority: 11
    - action: reroute
      match: ip4.src==10.0.1.0/24
      nextHopIP: 10.0.1.252
      priority: 10
```

5.1.5 [illegible]

Kubernetes [illegible] Service [illegible] Kubernetes [illegible] Service [illegible] IP [illegible] VPC [illegible] VPC [illegible] Kubernetes [illegible] Service [illegible]

[illegible]Kube-OVN [illegible] `SwitchLBRule` [illegible]

[illegible] `SwitchLBRule` [illegible]

```
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
```

- `vip` [illegible]
- `namespace` [illegible] Pod [illegible] Namespace[illegible]
- `sessionAffinity` [illegible] Service [illegible] `sessionAffinity` [illegible]
- `selector` [illegible] Service [illegible] `selector` [illegible]
- `ports` [illegible] Service [illegible] `port` [illegible]

[illegible]

```
# kubectl get slr
NAME               VIP         PORT(S)                  SERVICE                             AGE
vpc-dns-test-cjh2  10.96.0.3   53/UDP,53/TCP,9153/TCP   kube-system/slr-vpc-dns-test-cjh2   88m
```

5.1.6 [illegible] vpc-dns

[illegible] VPC [illegible] VPC [illegible]VPC [illegible] Pod [illegible] coredns [illegible] VPC [illegible] CoreDNS [illegible] Service [illegible] Kube-OVN [illegible] vpc-dns [illegible]

□□□□□

```yaml
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-nad
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "ovn-nad.default.ovn"
    }'
```

□□ vpc-dns □ ConfigMap

□ kube-system □□□□□□□ configmap□□□ vpc-dns □□□□□□□□□□□ vpc-dns □□□

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-config
  namespace: kube-system
data:
  coredns-vip: 10.96.0.3
  enable-vpc-dns: "true"
  nad-name: ovn-nad
  nad-provider: ovn-nad.default.ovn
```

- `enable-vpc-dns` □□□□□□ `true` □□□□□ `false` □□□□□□□ `true` □
- `coredns-image` □□□□□□□dns □□□□□□□□□□ coredns □□□□□
- `coredns-vip` □□ coredns □□ lb □□□ vip□
- `nad-name` □□□□ `network-attachment-definitions` □□□□□
- `nad-provider` □□□□ provider □□□
- `k8s-service-host` □□□□□□□□ coredns □□ k8s apiserver □□□ ip□
- `k8s-service-port` □□□□□□□□ coredns □□ k8s apiserver □□□ port□

□□ vpc-dns □□□□

```yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: system:vpc-dns
rules:
  - apiGroups:
    - ""
    resources:
    - endpoints
    - services
    - pods
    - namespaces
    verbs:
    - list
    - watch
  - apiGroups:
    - discovery.k8s.io
    resources:
    - endpointslices
    verbs:
    - list
    - watch
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  annotations:
    rbac.authorization.kubernetes.io/autoupdate: "true"
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: vpc-dns
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: system:vpc-dns
subjects:
```

```
- kind: ServiceAccount
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-corefile
  namespace: kube-system
data:
  Corefile: |
    .:53 {
        errors
        health {
          lameduck 5s
        }
        ready
        kubernetes cluster.local in-addr.arpa ip6.arpa {
          pods insecure
          fallthrough in-addr.arpa ip6.arpa
        }
        prometheus :9153
        forward . /etc/resolv.conf {
          prefer_udp
        }
        cache 30
        loop
        reload
        loadbalance
    }
```

□□ vpc-dns

```
kind: VpcDns
apiVersion: kubeovn.io/v1
metadata:
  name: test-cjh1
spec:
  vpc: cjh-vpc-1
  subnet: cjh-subnet-1
```

- `vpc` □□□□□ dns □□□ vpc □□□
- `subnet` □□□□□ dns □□□□□□□□

□□□□□□□

```
[root@hci-dev-mst-1 kubeovn]# kubectl get vpc-dns
NAME        ACTIVE   VPC         SUBNET
test-cjh1   false    cjh-vpc-1   cjh-subnet-1
test-cjh2   true     cjh-vpc-1   cjh-subnet-2
```

- `ACTIVE` : `true` □□□□□□□□ dns □□□ `false` □□□

□□

- □□ VPC □□□□□□□□□□ DNS □□□
- □□□ VPC □□□□□ vpc-dns □□□□□□□ VPC □□□ subnet□□□□□□ vpc-dns □□□□□ `true` □□□□ `false` □
- □ `true` □ vpc-dns □□□□□□□□□□ `false` □ vpc-dns □□□□□

5.1.7 □□□□

□□□□□ VPC □□□□□□□□□□□□□□□□□□□□ VPC □□□□□□□

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  namespaces:
```

```
  - ns1
  defaultSubnet: test
```

- `defaultSubnet` [illegible]VPC [illegible]

[illegible] Namespace [illegible] `ovn.kubernetes.io/logical_switch` [illegible] `ovn.kubernetes.io/logical_switch` [illegible] Pod [illegible]

[illegible] VPC Pod [illegible] livenessProbe [illegible] readinessProbe

[illegible] VPC [illegible] Pod [illegible] kubelet [illegible] VPC [illegible] Pod[illegible]Kube-OVN [illegible] TProxy [illegible] kubelet [illegible] VPC [illegible] Pod[illegible]

[illegible] DaemonSet `kube-ovn-cni` [illegible] `--enable-tproxy=true` [illegible]

```
spec:
  template:
    spec:
      containers:
      - args:
        - --enable-tproxy=true
```

[illegible]

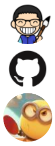
1. [illegible] VPC [illegible] Pod [illegible] IP[illegible]
2. [illegible] `tcpSocket` [illegible] `httpGet` [illegible]

PDF **Slack** **Support**

2026[illegible]4[illegible]28[illegible]

2022[illegible]5[illegible]24[illegible]

GitHub



5.1.8 [illegible]

5.2 VPC Egress Gateway

Note

[illegible] VPC [illegible] VPC [illegible]

[illegible] VPC [illegible]VPC NAT [illegible]OVN [illegible] Egress Gateway[illegible] VPC NAT [illegible] Kube-OVN [illegible] VPC NAT [illegible] Pod[illegible] VPC [illegible] Macvlan [illegible] Pod [illegible] iptables [illegible]

OVN [illegible] OVN [illegible] NAT [illegible] OVN [illegible] BFD [illegible] OVN [illegible] OVN [illegible]

Egress Gateway [illegible] VPC NAT [illegible]

VPC Egress Gateway [illegible] VPC[illegible] VPC[illegible] Pod [illegible]

- [illegible] ECMP [illegible] Active-Active [illegible]
- [illegible] BFD [illegible] <1s [illegible]
- [illegible] IPv6 [illegible]
- [illegible] Namespace [illegible] Pod [illegible]
- [illegible] Node [illegible] Egress Gateway [illegible]

VPC Egress Gateway [illegible] BGP [illegible] EVPN/VXLAN [illegible] Egress Gateway BGP/EVPN [illegible]

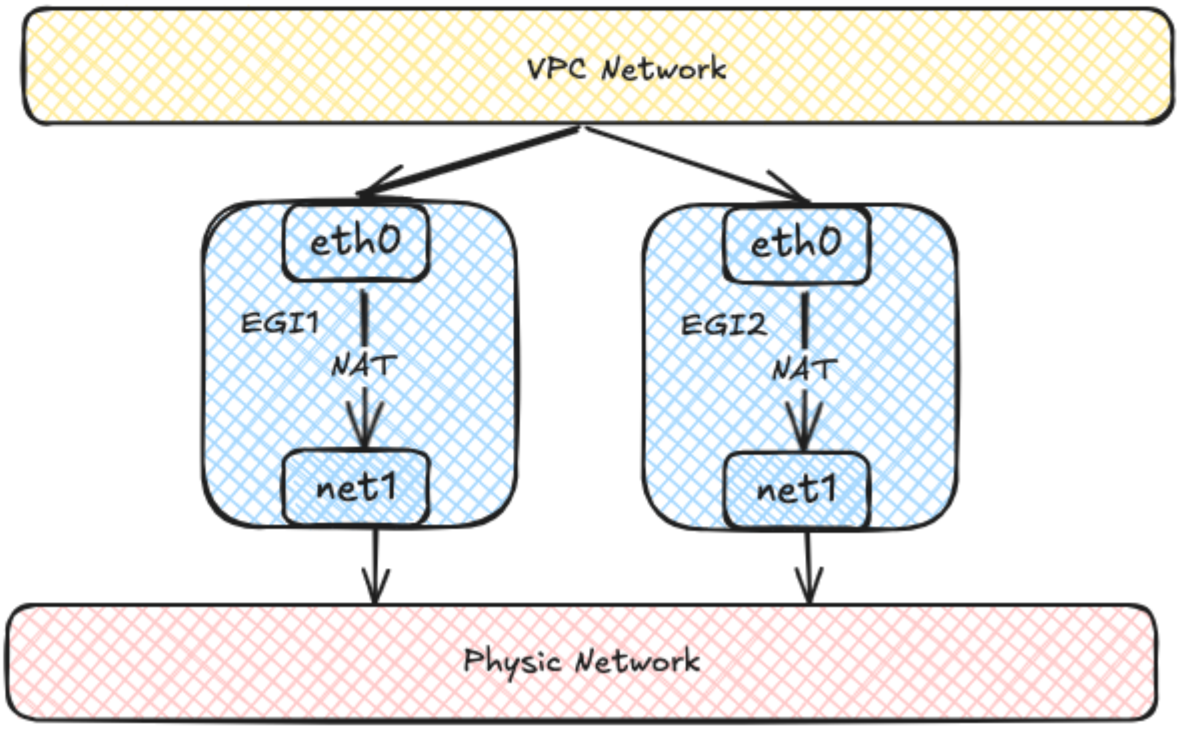
[illegible]conntrack NAT [illegible] JSON [illegible] VPC Egress Gateway [illegible]

[illegible] VPC Egress Gateway [illegible]

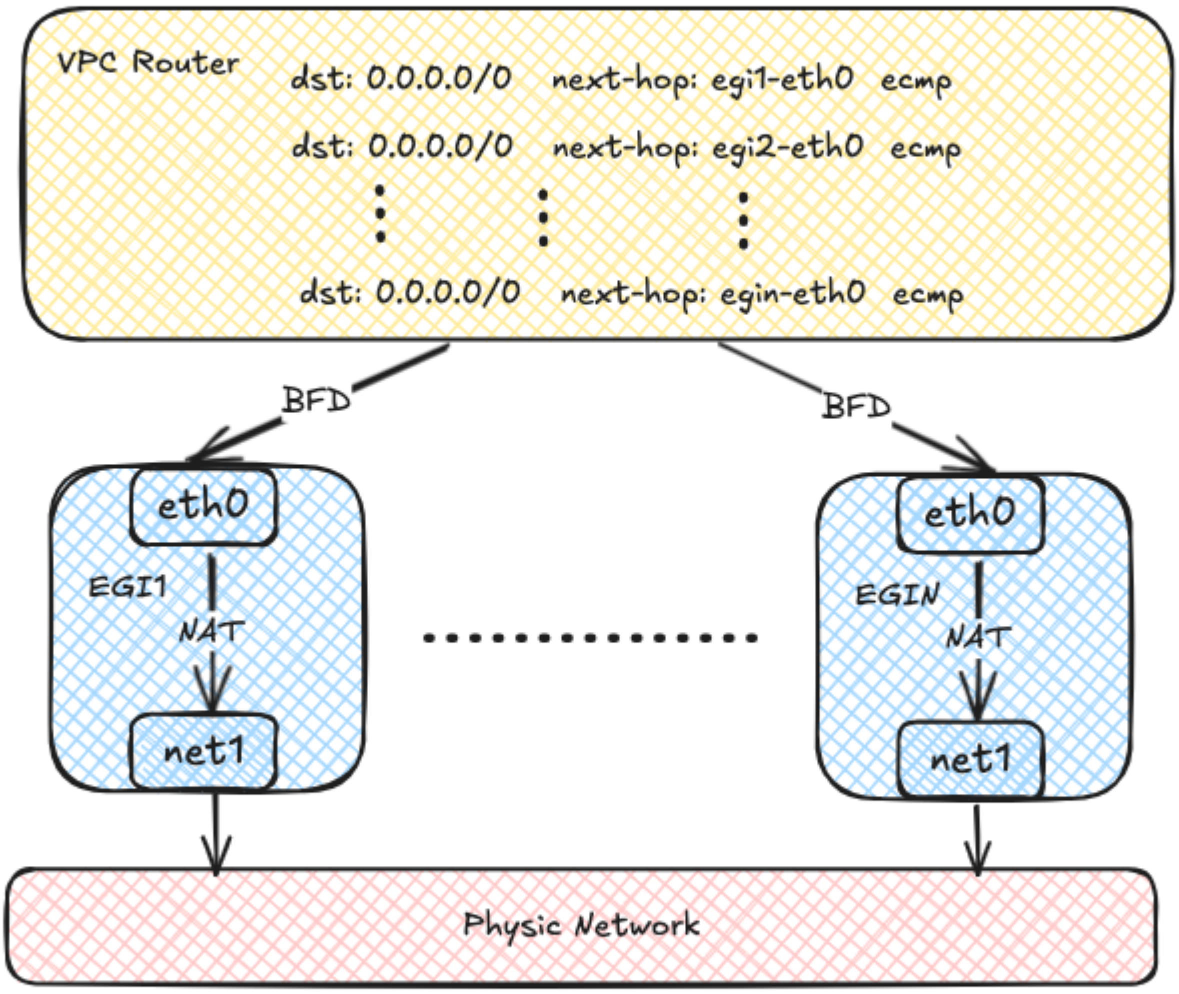
- [illegible] Macvlan [illegible] Underlay
- Gateway [illegible] Egress IP
- [illegible] SNAT[illegible] EIP [illegible] DNAT
- [illegible]

5.2.1 [illegible]

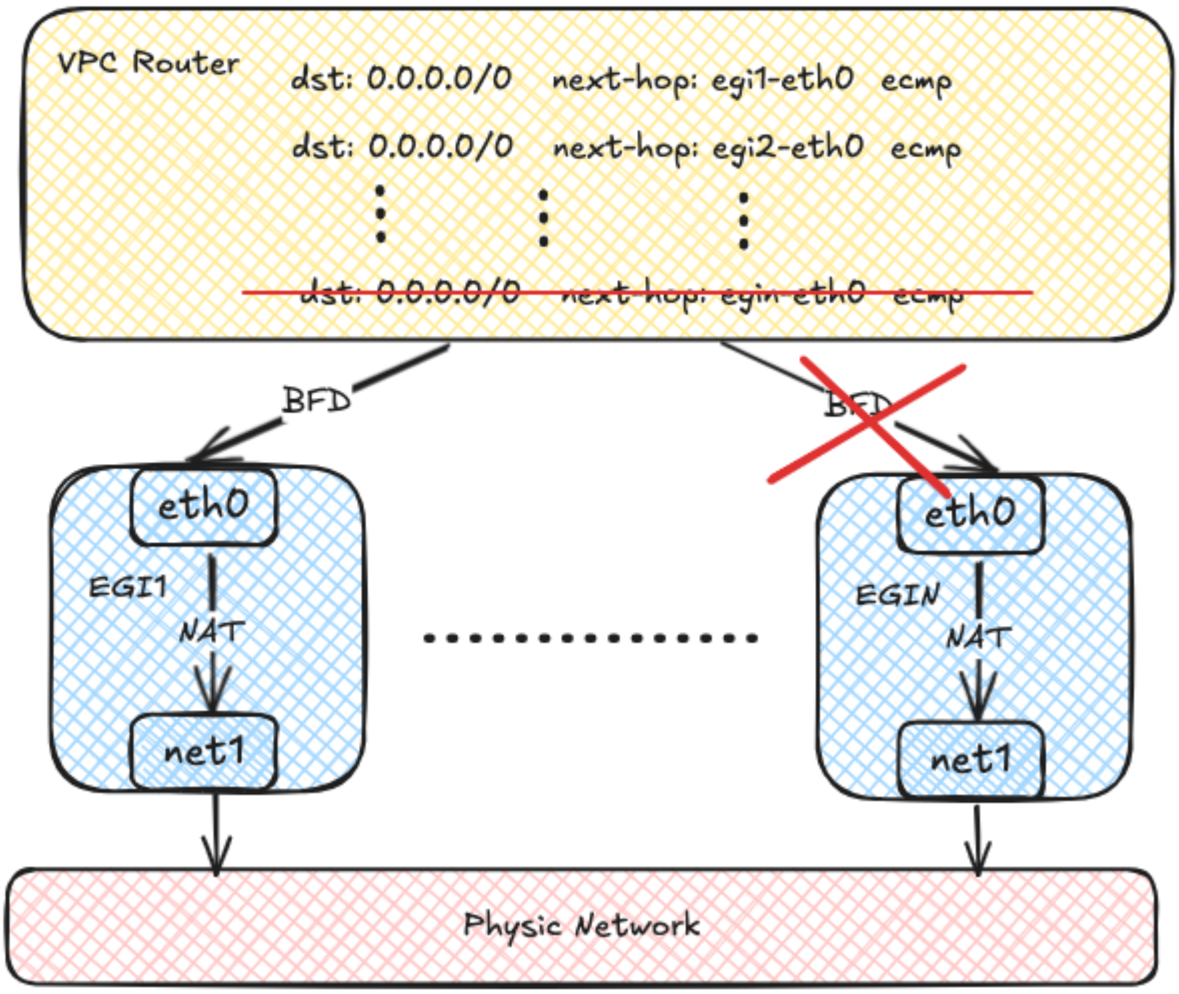
[illegible] Egress Gateway [illegible] Pod [illegible] Pod [illegible] VPC [illegible] Macvlan [illegible] Egress Gateway [illegible] NAT [illegible]


VPC Network
eth0
EGI1
NAT
net1
eth0
EGI2
NAT
net1
Physic Network

[illegible] Egress Gateway [illegible] OVN [illegible] VPC [illegible] Pod [illegible]OVN [illegible] Egress Gateway [illegible] Egress Gateway [illegible]


VPC Router
dst: 0.0.0.0/0 next-hop: egi1-eth0 ecmp
dst: 0.0.0.0/0 next-hop: egi2-eth0 ecmp
dst: 0.0.0.0/0 next-hop: egin-eth0 ecmp
BFD
BFD
eth0
EGI1
NAT
net1
eth0
EGIN
NAT
net1
Physic Network

OVN [illegible] BFD [illegible] Egress Gateway [illegible] Egress Gateway [illegible]OVN [illegible]


VPC Router
dst: 0.0.0.0/0 next-hop: egi1-eth0 ecmp
dst: 0.0.0.0/0 next-hop: egi2-eth0 ecmp
~~dst: 0.0.0.0/0 next-hop: egin-eth0 ecmp~~
BFD
BFD
eth0
EGI1
NAT
net1
eth0
EGIN
NAT
net1
Physic Network

5.2.2 ⊠⊠⊠⊠

VPC Egress Gateway ⊠ VPC NAT Gateway ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠ Multus-CNI⊠

> ⊠⊠ VPC Egress Gateway ⊠⊠⊠⊠⊠⊠ ConfigMap⊠

5.2.3 ⊠⊠⊠⊠

⊠⊠ NetworkAttachmentDefinition

VPC Egress Gateway ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ VPC ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ NetworkAttachmentDefinition ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ `macvlan` ⊠⊠⊠⊠ Kube-OVN ⊠⊠ IPAM ⊠⊠⊠⊠⊠⊠

```yaml
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: eth1
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth1",
      "mode": "bridge",
      "ipam": {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "eth1.default"
      }
    }'
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan1
```

```
spec:
  protocol: IPv4
  provider: eth1.default
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
    - 172.17.0.0..172.17.0.10
```

> □□□□□□ CNI □□□□ NetworkAttachmentDefinition □ VPC Egress Gateway □□□□□□□□□

□□□□□□□□□□□□□□□□ □□□□□□

□□ VPC Egress Gateway

□□ VPC Egress Gateway □□□□□□□□

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalSubnet: macvlan1
  policies:
    - snat: true
      subnets:
        - ovn-default
```

□□□□□ default □□□□□ VPC `ovn-cluster` □□□□□ gateway1 □□□ VPC Egress Gateway□ `ovn-cluster` □ `ovn-default` □□□10.16.0.0/16□□□□ Pod □□□ `macvlan1` □□□ SNAT □□□□□□□□

□□□□□□□ VPC Egress Gateway□

```
$ kubectl get veg gateway1
NAME       VPC           REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   AGE
gateway1   ovn-cluster   1          false         macvlan1          Completed   true    13s
```

□□□□□□

```
kubectl get veg gateway1 -o wide
NAME       VPC           REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   INTERNAL IPS     EXTERNAL IPS     WORKING NODES         AGE
gateway1   ovn-cluster   1          false         macvlan1          Completed   true    ["10.16.0.12"]   ["172.17.0.11"]   ["kube-ovn-worker"]   82s
```

□□□□□□

```
$ kubectl get deployment -l ovn.kubernetes.io/vpc-egress-gateway=gateway1
NAME       READY   UP-TO-DATE   AVAILABLE   AGE
gateway1   1/1     1            1           4m40s

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway1 -o wide
NAME                       READY   STATUS    RESTARTS   AGE     IP           NODE              NOMINATED NODE   READINESS GATES
gateway1-b9f8b4448-76lhm   1/1     Running   0          4m48s   10.16.0.12   kube-ovn-worker   <none>           <none>
```

□□ Pod □□ IP□□□□ iptables □□□

```
$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: net1@if13: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 62:d8:71:90:7b:86 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.17.0.11/16 brd 172.17.255.255 scope global net1
       valid_lft forever preferred_lft forever
    inet6 fe80::60d8:71ff:fe90:7b86/64 scope link
       valid_lft forever preferred_lft forever
17: eth0@if18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP group default
    link/ether 36:7c:6b:c7:82:6b brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.16.0.12/16 brd 10.16.255.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::347c:6bff:fec7:826b/64 scope link
       valid_lft forever preferred_lft forever

$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- ip route show
default via 172.17.0.1 dev net1
10.16.0.0/16 dev eth0 proto kernel scope link src 10.16.0.12
```

```
172.17.0.0/16 dev net1 proto kernel scope link src 172.17.0.11

$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- iptables -t nat -S
-P PREROUTING ACCEPT
-P INPUT ACCEPT
-P OUTPUT ACCEPT
-P POSTROUTING ACCEPT
-A POSTROUTING -s 10.16.0.0/16 -j MASQUERADE --random-fully
```

⊠ Gateway Pod ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

```
$ kubectl exec -ti gateway1-b9f8b4448-76lhm -c gateway -- bash
nobody@gateway1-b9f8b4448-76lhm:/kube-ovn$ tcpdump -i any -nnve icmp and host 172.17.0.1
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
06:50:58.936528 eth0  In  ifindex 17 92:26:b8:9e:f2:1c ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 63, id 30481, offset 0, flags [DF], proto ICMP (1),
length 84)
    10.16.0.9 > 172.17.0.1: ICMP echo request, id 37989, seq 0, length 64
06:50:58.936574 net1  Out ifindex 2 62:d8:71:90:7b:86 ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 62, id 30481, offset 0, flags [DF], proto ICMP (1),
length 84)
    172.17.0.11 > 172.17.0.1: ICMP echo request, id 39449, seq 0, length 64
06:50:58.936613 net1  In  ifindex 2 02:42:39:79:7f:08 ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 64, id 26701, offset 0, flags [none], proto ICMP
(1), length 84)
    172.17.0.1 > 172.17.0.11: ICMP echo reply, id 39449, seq 0, length 64
06:50:58.936621 eth0  Out ifindex 17 36:7c:6b:c7:82:6b ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 63, id 26701, offset 0, flags [none], proto ICMP
(1), length 84)
    172.17.0.1 > 10.16.0.9: ICMP echo reply, id 37989, seq 0, length 64
```

OVN Logical Router ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

```
$ kubectl ko nbctl lr-policy-list ovn-cluster
Routing Policies
     31000                            ip4.dst == 10.16.0.0/16           allow
     31000                           ip4.dst == 100.64.0.0/16           allow
     30000                              ip4.dst == 172.18.0.2         reroute                100.64.0.3
     30000                              ip4.dst == 172.18.0.3         reroute                100.64.0.2
     30000                              ip4.dst == 172.18.0.4         reroute                100.64.0.4
     29100                            ip4.src == 10.16.0.0/16         reroute                10.16.0.12
     29000 ip4.src == $ovn.default.kube.ovn.control.plane_ip4         reroute                100.64.0.2
     29000       ip4.src == $ovn.default.kube.ovn.worker2_ip4         reroute                100.64.0.4
     29000        ip4.src == $ovn.default.kube.ovn.worker_ip4         reroute                100.64.0.3
```

⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ `.spec.replicas` ⊠⊠⊠⊠⊠⊠⊠⊠

```
$ kubectl scale veg gateway1 --replicas=2
vpcegressgateway.kubeovn.io/gateway1 scaled

$ kubectl get veg gateway1
NAME       VPC           REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   AGE
gateway1   ovn-cluster   2          false         macvlan           Completed   true    39m

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway1 -o wide
NAME                       READY   STATUS    RESTARTS   AGE   IP           NODE               NOMINATED NODE   READINESS GATES
gateway1-b9f8b4448-76lhm   1/1     Running   0          40m   10.16.0.12   kube-ovn-worker    <none>           <none>
gateway1-b9f8b4448-zd4dl   1/1     Running   0          64s   10.16.0.13   kube-ovn-worker2   <none>           <none>

$ kubectl ko nbctl lr-policy-list ovn-cluster
Routing Policies
     31000                            ip4.dst == 10.16.0.0/16           allow
     31000                           ip4.dst == 100.64.0.0/16           allow
     30000                              ip4.dst == 172.18.0.2         reroute                100.64.0.3
     30000                              ip4.dst == 172.18.0.3         reroute                100.64.0.2
     30000                              ip4.dst == 172.18.0.4         reroute                100.64.0.4
     29100                            ip4.src == 10.16.0.0/16         reroute                10.16.0.12, 10.16.0.13
     29000 ip4.src == $ovn.default.kube.ovn.control.plane_ip4         reroute                100.64.0.2
     29000       ip4.src == $ovn.default.kube.ovn.worker2_ip4         reroute                100.64.0.4
     29000        ip4.src == $ovn.default.kube.ovn.worker_ip4         reroute                100.64.0.3
```

⊠⊠ Egress Gateway IP ⊠⊠⊠⊠⊠

⊠⊠⊠⊠ `externalIPs` ⊠ `nodeSelector` ⊠⊠⊠⊠ Egress Gateway Pods ⊠⊠⊠⊠ Egress IP ⊠⊠⊠⊠⊠⊠⊠⊠⊠

```yaml
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 2
  externalSubnet: macvlan1
  policies:
    - snat: true
      subnets:
        - ovn-default
  externalIPs:
    - 172.17.0.10
    - 172.17.0.11
  nodeSelector:
    - matchLabels:
        kubernetes.io/hostname: kube-ovn-worker
```

⊠⊠⊠⊠⊠⊠ IP ⊠⊠⊠⊠⊠⊠⊠⊠⊠ `internalIPPool` ⊠ `externalIPPool` ⊠⊠⊠⊠⊠ `IPPool` ⊠⊠⊠IPPool ⊠⊠⊠⊠⊠⊠⊠⊠ `internalSubnet` ⊠ `externalSubnet` ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ `internalIPs` ⊠ `externalIPs` ⊠⊠⊠⊠⊠

```
spec:
  internalIPPool: veg-internal
  externalIPPool: veg-external
```

⊠⊠ **BFD** ⊠⊠⊠

BFD ⊠⊠⊠⊠⊠ VPC ⊠ BFD LRP ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ VPC ⊠⊠⊠⊠⊠ BFD Port⊠⊠⊠⊠⊠

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc1
spec:
  bfdPort:
    enabled: true
    ip: 10.255.255.255
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  vpc: vpc1
  protocol: IPv4
  cidrBlock: 192.168.0.0/24
```

⊠⊠ BFD Port ⊠⊠⊠⊠⊠ OVN LR ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ BFD ⊠ LRP⊠

```
$ kubectl ko nbctl show vpc1
router 0c1d1e8f-4c86-4d96-88b2-c4171c7ff824 (vpc1)
    port bfd@vpc1
        mac: "8e:51:4b:16:3c:90"
        networks: ["10.255.255.255"]
    port vpc1-subnet1
        mac: "de:c9:5c:38:7a:61"
        networks: ["192.168.0.1/24"]
```

⊠⊠⊠⊠⊠ VPC Egress Gateway ⊠ `.spec.bfd.enabled` ⊠⊠⊠ `true` ⊠⊠⊠⊠⊠⊠⊠

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway2
  namespace: default
spec:
  vpc: vpc1
  replicas: 2
  internalSubnet: subnet1
  externalSubnet: macvlan
  bfd:
    enabled: true
  policies:
    - snat: true
      ipBlocks:
        - 192.168.0.0/24
```

⊠⊠ VPC Egress Gateway ⊠⊠⊠

```
$ kubectl get veg gateway2 -o wide
NAME       VPC    REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   INTERNAL IPS                    EXTERNAL IPS                    WORKING
NODES                            AGE
gateway2   vpc1   2          true          macvlan           Completed   true    ["192.168.0.2","192.168.0.3"]   ["172.17.0.13","172.17.0.14"]   ["kube-ovn-
worker","kube-ovn-worker2"]   58s

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway2 -o wide
NAME                       READY   STATUS    RESTARTS   AGE     IP            NODE               NOMINATED NODE   READINESS GATES
gateway2-fcc6b8b87-8lgvx   1/1     Running   0          2m18s   192.168.0.3   kube-ovn-worker2   <none>           <none>
gateway2-fcc6b8b87-wmww6   1/1     Running   0          2m18s   192.168.0.2   kube-ovn-worker    <none>           <none>

$ kubectl ko nbctl lr-policy-list vpc1
Routing Policies
     29100                       ip4.src == 192.168.0.0/24         reroute                192.168.0.2, 192.168.0.3
```

```
       29090                       ip4.src == 192.168.0.0/24            drop

$ kubectl ko nbctl list bfd
_uuid               : 223ede10-9169-4c7d-9524-a546e24bfab5
detect_mult         : 3
dst_ip              : "192.168.0.2"
external_ids        : {af="4", vendor=kube-ovn, vpc-egress-gateway="default/gateway2"}
logical_port        : "bfd@vpc1"
min_rx              : 1000
min_tx              : 1000
options             : {}
status              : up

_uuid               : b050c75e-2462-470b-b89c-7bd38889b758
detect_mult         : 3
dst_ip              : "192.168.0.3"
external_ids        : {af="4", vendor=kube-ovn, vpc-egress-gateway="default/gateway2"}
logical_port        : "bfd@vpc1"
min_rx              : 1000
min_tx              : 1000
options             : {}
status              : up
```

[illegible] Pod [illegible] BFD [illegible]

```
$ kubectl exec gateway2-fcc6b8b87-8lgvx -c bfdd -- bfdd-control status
There are 1 sessions:
Session 1
 id=1 local=192.168.0.3 (p) remote=10.255.255.255 state=Up

$ kubectl exec gateway2-fcc6b8b87-wmww6 -c bfdd -- bfdd-control status
There are 1 sessions:
Session 1
 id=1 local=192.168.0.2 (p) remote=10.255.255.255 state=Up
```

[illegible]

VPC BFD PORT

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
`enabled`	`boolean`	[illegible]	`false`	[illegible] BFD Port[illegible]	`true`
`ip`	`string`	[illegible]	-	BFD Port [illegible] IP [illegible] IPv6 [illegible]	`169.255.255.2` `fdff::1` / `169.255.255.2`
`nodeSelector`	`object`	[illegible]	-	[illegible] BFD Port [illegible]BFD Port [illegible] OVN HA Chassis Group[illegible] Active/ Backup [illegible] Active [illegible] nodeSelector[illegible] Kube-OVN [illegible] `kubectl ko nbctl list ha_chassis_group` [illegible] OVN HA Chassis Group [illegible]	-
`nodeSelector.matchLabels`	`dict/map`	[illegible]	-	[illegible]	-
`nodeSelector.matchExpressions`	`object array`	[illegible]	-	[illegible]	-

VPC EGRESS GATEWAY

Spec⌧

`vpc`	`string`		VPC ovn-cluster	VPC	`vpc1`
`replicas`	`integer/int32`		`1`		`2`
`prefix`	`string`		-	Deployment	`veg-`
`image`	`string`		-	Deployment	`docker.io/kubeovn/kube-ovn:v1.14.0-debug`
`internalSubnet`	`string`		VPC	VPC	`subnet1`
`externalSubnet`	`string`		-		`ext1`
`internalIPs`	`string array`		-	VPC IP IPv6 IP `<replicas> + 1` Pod	`10.16.0.101` / `fd00::11` / `10.16.0.101,fd00::11`
`externalIPs`	`string array`		-	IP IPv6 IP `<replicas> + 1` Pod	`10.16.0.101` / `fd00::11` / `10.16.0.101,fd00::11`
`internalIPPool`	`string`		-	`IPPool` IPPool `internalSubnet` `internalIPs`	`veg-internal`
`externalIPPool`	`string`		-	`IPPool` IPPool `externalSubnet` `externalIPs`	`veg-external`
`bfd`	`object`		-	BFD	-
`policies`	`object array`		-	Egress `selectors`	-
`selectors`	`object array`		-	Namespace Selector Pod Selector Egress Pod SNAT/ MASQUERADE `policies`	-
`nodeSelector`	`object array`		-	Deployment/ Pod	-

`trafficPolicy`	`string`		`Cluster`	`Cluster` / `Local` **BFD** `Local` Egress VPC Egress Gateway VPC Egress Gateway Egress	`Local`
`bandwidth`	`object`		-		-
`bandwidth.ingress`	`integer` `string`		-	VPC	`1024` / `1Gi`
`bandwidth.egress`	`integer` `string`		-	VPC	`1024` / `1Gi`

 Mbps `M` `Mi` `G` `Gi` quantity bit/s Mbps `100M` `100Mi` `1G` `1Gi` 100105 1000 1074 Mbps

 9223372036854 Mbps `ingress` `egress`

```yaml
spec:
  bandwidth:
    ingress: 1024
    egress: 1Gi
```

0 9223372036854 Mbps Kube-OVN `vpc-egress-gateways.kubeovn.io` CRD `x-kubernetes-int-or-string: true`

Helm Chart `crds/` CRD `kube-ovn-v2` Chart `KUBE_OVN_VERSION` Kube-OVN CRD bundle

```
kubectl apply -f "https://raw.githubusercontent.com/kubeovn/kube-ovn/${KUBE_OVN_VERSION}/charts/kube-ovn-v2/crds/kube-ovn-crd.yaml"
```

 quantity Mbps

BFD

`enabled`	`boolean`		`false`	BFD	`true`
`minRX`	`integer/int32`		`1000`	BFD minRX ms	`500`
`minTX`	`integer/int32`		`1000`	BFD minTX ms	`500`
`multiplier`	`integer/int32`		`3`	BFD multiplier	`1`

Egress [illegible]

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
`snat`	`boolean`	[illegible]	`false`	[illegible] SNAT/ MASQUERADE[illegible]	`true`
`ipBlocks`	`string array`	[illegible]	-	[illegible] Gateway [illegible] IP [illegible] IPv6[illegible]	`192.168.0.1` / `192.168.0.0/24`
`subnets`	`string array`	[illegible]	-	[illegible] Gateway [illegible] VPC [illegible] IPv6 [illegible]	`subnet1`

Selectors[illegible]

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
`namespaceSelector`	`object`	[illegible]	-	Namespace [illegible] Namespace[illegible]	-
`namespaceSelector.matchLabels`	`dict/map`	[illegible]	-	[illegible]	-
`namespaceSelector.matchExpressions`	`object array`	[illegible]	-	[illegible]	-
`podSelector`	`object`	[illegible]	-	Pod [illegible] Pod[illegible]	-
`podSelector.matchLabels`	`dict/map`	[illegible]	-	[illegible]	-
`podSelector.matchExpressions`	`object array`	[illegible]	-	[illegible]	-

[illegible]

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
`matchLabels`	`dict/map`	[illegible]	-	[illegible]	-
`matchExpressions`	`object array`	[illegible]	-	[illegible]	-
`matchFields`	`object array`	[illegible]	-	[illegible]	-

Status[illegible]

[illegible]	[illegible]	[illegible]	[illegible]
`ready`	`boolean`	Gateway [illegible]	`true`
`phase`	`string`	Gateway [illegible]	`Pending` / `Processing` / `Completed`
`internalIPs`	`string array`	[illegible] VPC [illegible] IP [illegible]	-
`externalIPs`	`string array`	[illegible] IP [illegible]	-
`workload`	`object`	[illegible]	-
`workload.apiVersion`	`string`	[illegible] API [illegible]	`apps/v1`
`workload.kind`	`string`	[illegible]	`Deployment`
`workload.name`	`string`	[illegible]	`gateway1`
`workload.nodes`	`string array`	[illegible]	-
`conditions`	`object array`	-	-

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2026[illegible]8[illegible]21[illegible]

2024[illegible]12[illegible]12[illegible]

GitHub

5.2.4 [illegible]

5.3 VPC Egress Gateway [illegible]

VPC Egress Gateway [illegible] Pod [illegible] sidecar[illegible] sidecar [illegible] conntrack NAT [illegible] NAT [illegible]

5.3.1 [illegible]

- observer [illegible] restartable init container[illegible] Kubernetes 1.29 [illegible] `SidecarContainers` [illegible] Deployment [illegible] dry-run[illegible] API Server [illegible] restartable init [illegible] observer[illegible] `ObservabilityConfigured=False` [illegible] observer [illegible]
- [illegible] `spec.observability` [illegible] `vpc-egress-gateways.kubeovn.io` CRD[illegible]Helm [illegible] CRD[illegible] CRD [illegible] Kube-OVN [illegible] CRD[illegible]
- conntrack [illegible] Pod [illegible] `NET_ADMIN` [illegible] observer binary [illegible] `CAP_NET_ADMIN` file capability[illegible] security context [illegible] `NET_ADMIN` [illegible] capability bounding set[illegible] `allowPrivilegeEscalation` [illegible] file capability [illegible] root launcher [illegible] `exec` [illegible]observer [illegible] UID [illegible] GID 65534 [illegible] capability[illegible]
- Prometheus Operator [illegible] ServiceMonitor CRD [illegible]

5.3.2 [illegible]

[illegible] JSON [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 2
  externalSubnet: macvlan1
  policies:
    - snat: true
      subnets:
        - ovn-default
  observability:
    resources:
      requests:
        cpu: 20m
        memory: 64Mi
      limits:
        cpu: 200m
        memory: 256Mi
    interfaceMetrics:
      enabled: true
    conntrack:
      metrics:
        enabled: true
      log:
        enabled: true
        events: [start, end]
        rateLimit:
          recordsPerSecond: 100
          burst: 1000
        filters:
          include: []
          exclude: []
    serviceMonitor:
      labels:
        monitoring: platform
      annotations: {}
```

`resources` [illegible]observer [illegible] request [illegible] `20m` CPU [illegible] `64Mi` [illegible]limit [illegible] `200m` CPU [illegible] `256Mi` [illegible] interface-only [illegible] `20MiB` [illegible] request [illegible] conntrack cache [illegible]

[illegible] ConfigMap [illegible]filter [illegible] rate limit [illegible] Pod [illegible] Deployment [illegible] Pod[illegible]

- [illegible]
- [illegible]
- [illegible] `observability.resources` [illegible]
- [illegible]

□□□□□□□ConfigMap □ Service □□□□□□□□□□□□□□□□□ condition□□□□□□ observer □□□volume □□ Pod□

□□□□□□□□□□□□□□□□□ `/kube-ovn/vpc-egress-gateway-observer` □launcher □□□□□ `sleep infinity` □□□□□ binary □□□□□□□□□□□ observer □□□□□□□□□□□ `CAP_NET_ADMIN` file capability□□□□□□□□□□□□ conntrack □□□□□□□□□□□□□□□

5.3.3 □□

observer □ TCP □□ `10666` □□□ `/metrics` □ `/healthz` □observer □□□□ Prometheus registry□□□□ `go_*` □ `process_*` □ `promhttp_*` □□□□□□□□□□ `namespace` □ `name` □ `pod` □ `node` □□□□□

□□□□□□□liveness probe □□□□□□□ observer binary□□□□ loopback □□ `/healthz` □□□ observer □□□□□□□□□ Pod □□□□□□□□□□□□□□□□□□□ observer binary □□□ `sleep infinity` □□□□□□□□□□

□□□□

□□ scrape □□□ `/proc/net/dev` □□□□□□□observer □□□□□ scrape □□□ Pod `network-status` annotation □□□□□□□□ Multus □□□□□□□□□□□□□□□□□□□

□□	□□□□	□□
`kube_ovn_vpc_egress_gateway_interface_rx_bytes_total`	`interface` □ `type`	□□□□□□
`kube_ovn_vpc_egress_gateway_interface_tx_bytes_total`	`interface` □ `type`	□□□□□□
`kube_ovn_vpc_egress_gateway_interface_rx_packets_total`	`interface` □ `type`	□□□□□□
`kube_ovn_vpc_egress_gateway_interface_tx_packets_total`	`interface` □ `type`	□□□□□□
`kube_ovn_vpc_egress_gateway_interface_drops_total`	`interface` □ `type` □ `direction`	□□□□□□
`kube_ovn_vpc_egress_gateway_interface_errors_total`	`interface` □ `type` □ `direction`	□□□□□□

`type` □□□ `internal` □ `external` □ `direction` □ `rx` □ `tx` □

Conntrack □□

Conntrack □□□□□ IP □□□□□□□□□□□□□□□□□□□

□□	□□□□	□□
`kube_ovn_vpc_egress_gateway_conntrack_nat_flows_active`	`address_family` □ `protocol` □ `nat_type`	□□ NAT □□□□ conntrack table dump □□□□□□□□□□
`kube_ovn_vpc_egress_gateway_conntrack_nat_flows_started_total`	`address_family` □ `protocol` □ `nat_type`	observer □□□□□□□□ NAT □□□□□□
`kube_ovn_vpc_egress_gateway_conntrack_nat_flows_ended_total`	`address_family` □ `protocol` □ `nat_type`	□□□□ NAT □□□□□□□□
`kube_ovn_vpc_egress_gateway_conntrack_nat_packets_total`	`address_family` □ `protocol` □ `nat_type` □ `direction`	conntrack accounting □□□□□□□□□□□
`kube_ovn_vpc_egress_gateway_conntrack_nat_bytes_total`	`address_family` □ `protocol` □ `nat_type` □ `direction`	conntrack accounting □□□□□□□□□□□

`address_family` □ `ipv4` □ `ipv6` □□□ `protocol` □□ `tcp` □ `udp` □ `sctp` □ `icmp` □ `icmpv6` □□□□□□□□□□ `other` □ `nat_type` □ `snat` □ `dnat` □ `snat_dnat` □ `direction` □ `original` □ `reply` □

□□□□□□ `kube_ovn_vpc_egress_gateway_observability_*` □□□□□□□□□□□□□□□□□□□□□□□conntrack □□□□□□cache □□□□□□□□□□□□□□□□□□□□□□□ conntrack accounting □□□□□□□□□□□□□□□ `/metrics` □□□□□□□□□□□□□□□□□□

5.3.4 ServiceMonitor □□

□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ `publishNotReadyAddresses: true` □ headless Service□□□□ `targetPort` □ `10666` □□□□□□□□□□□□□□□□□□ ServiceMonitor□ `observability.serviceMonitor` □□□□ label □ annotation □□□□ ServiceMonitor □□□□□□□□□□□□□□□□ selector label□

□□ ServiceMonitor CRD □□□□□□□□□□□ `ServiceMonitorReady=False` □□□□□□CRD □□□□□□□□□□□□□ ServiceMonitor□□ condition □□□□□ `Ready` condition □□□□□

5.3.5 JSON □□□

□□□□ JSON Lines □□□□□ `observability` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

```
kubectl logs -n default <gateway-pod> -c observability
```

`v1` schema □□ `start` □ `end` □□□□□□□□□□□□ conntrack update □□□□□□□□□□□

```
{"schemaVersion":"v1","timestamp":"2026-08-05T03:40:00Z","event":"end","conntrackID":2471802592,"zone":0,"namespace":"default","name":"gateway1","pod":"gateway1
r9x8m","node":"worker-1","addressFamily":"ipv4","protocol":"tcp","protocolNumber":6,"natType":["snat"],"original":
{"sourceIP":"10.16.0.25","sourcePort":42136,"destinationIP":"203.0.113.10","destinationPort":443},"translated":
{"sourceIP":"172.17.0.11","sourcePort":42136,"destinationIP":"203.0.113.10","destinationPort":443},"counters":
{"originalPackets":8,"originalBytes":624,"replyPackets":7,"replyBytes":591}}
```

□□□□□ `other` □□□□ `protocolNumber` □□□□ IP □□□□ `natType` □□□□□□□□□□□□□□ `dnat` □ `snat` □conntrack accounting □□□□□□ `counters` □□□observer □□□□ `nf_conntrack_acct` □

observer □ dump □□ conntrack table □□□ conntrack □□□□□ NAT □□□□□ active-flow □□□ cache□□□□□□□□□ `start` □□□cache □□□ 65,536 □□□□netlink □□ channel □□□□□□□□□□□ 4,096 □□□□□□□□□ consumer □□□□□□□□□□□□□□□□□□□□□□□□□□□□ conntrack □□□

□□□□□□□□□□□□ NAT □□□□□□□□□□□□□ Pod □□□□□ 100 □□□□□□□□□□□□ 1,000□□□□□□□□□□ best-effort □□□□□□□□□□□□□□□□□□

5.3.6 □□□ Filter

□□ include □ exclude □□□□□□□□□□□□□NAT □□□□□□□□□□ tuple □□□□□□□□□□□□ OR □□□□□□□□□□□□□□□□ AND □□□□□□□□□□□□□ OR □□□exclude □□□□□ include □□□□ include □□□□□□ NAT □□

□□□□□□□□□□□□□□□□□□□□□□□□□ `10.16.0.0/16` □□□□□ 80 □ 443 □ TCP SNAT □□□□□□□□□□□□□□□ `192.0.2.0/24` □□□

```
spec:
  observability:
    conntrack:
      log:
        enabled: true
        filters:
          include:
            - protocols: [tcp]
              natTypes: [snat]
              original:
                sourceCIDRs: [10.16.0.0/16]
                destinationPorts:
                  - start: 80
                    end: 443
          exclude:
            - translated:
                destinationCIDRs: [192.0.2.0/24]
```

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5.3.7 □□

5.4 Egress Gateway BGP [illegible] EVPN [illegible]

[illegible]

BGP [illegible] EVPN [illegible]

[illegible]

- [illegible] L3VPN[illegible] L2VPN
- [illegible] FRR [illegible] Pod[illegible]
- [illegible] BGP BFD

5.4.1 [illegible]

VPC Egress Gateway [illegible] BGP [illegible] EVPN/VXLAN [illegible] Egress Gateway Pod [illegible] FRR[illegible]Free Range Routing[illegible] BGP [illegible] EVPN [illegible] Gateway [illegible] init [illegible] BgpConf [illegible] EvpnConf [illegible] FRR [illegible]

[illegible] BGP [illegible]Egress Gateway [illegible] BGP [illegible] EVPN [illegible]init [illegible] Pod [illegible] Linux VRF[illegible]bridge [illegible] VXLAN [illegible] L2VPN EVPN [illegible] VXLAN [illegible] Egress Gateway Pod [illegible]Kubernetes [illegible] Geneve[illegible]

> [illegible] BGP [illegible] kube-ovn-speaker [illegible] Pod/[illegible]/Service [illegible]

5.4.2 [illegible]

[illegible] BGP/EVPN [illegible] VPC Egress Gateway [illegible] Multus-CNI [illegible]

[illegible] BGP[illegible] EVPN[illegible] BGP [illegible]

5.4.3 [illegible]

[illegible] **BgpConf** [illegible]

BgpConf [illegible] BGP [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: BgpConf
metadata:
  name: bgp-conf-6502
spec:
  localASN: 65002
  peerASN: 65001
  neighbours:
    - 10.0.1.1
  holdTime: 90s
  keepaliveTime: 30s
  connectTime: 10s
  ebgpMultiHop: true
```

[illegible] AS [illegible] 65002[illegible] AS [illegible] 65001 [illegible] BGP [illegible] 10.0.1.1[illegible] EBGP Multi-Hop[illegible]

[illegible] **EvpnConf** [illegible]

EvpnConf [illegible] EVPN [illegible]EvpnConf [illegible] BgpConf [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: EvpnConf
metadata:
  name: evpn-conf-1016
spec:
  vni: 1016
  routeTargets:
    - "65000:1016"
```

[illegible] VNI [illegible] 1016 [illegible] EVPN [illegible]Route Target [illegible] `65000:1016` [illegible]

[illegible] BGP/EVPN [illegible] VPC Egress Gateway

[illegible] VPC Egress Gateway [illegible] NetworkAttachmentDefinition [illegible]

[illegible] VpcEgressGateway [illegible] `.spec` [illegible] `bgpConf` [illegible] `evpnConf` [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalIPs:
    - 10.0.1.13
  internalIPs:
    - 10.16.0.13
  externalSubnet: macvlan1
  bgpConf: bgp-conf-6502
  evpnConf: evpn-conf-1016
  policies:
    - snat: false
      subnets:
        - ovn-default
    - snat: false
      ipBlocks:
        - 10.17.0.0/16
```

[illegible] BGP [illegible] EVPN [illegible] VPC Egress Gateway[illegible]Gateway Pod [illegible] FRR [illegible] BGP EVPN [illegible] VXLAN [illegible] BGP/EVPN [illegible] `snat` [illegible] `false` [illegible]

[illegible] BGP[illegible] EVPN[illegible]

[illegible] BGP [illegible] EVPN/VXLAN[illegible] `bgpConf` [illegible] `evpnConf` [illegible] FRR [illegible] BGP IPv4 Unicast [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway-bgp-only
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalSubnet: macvlan1
  bgpConf: bgp-conf-6502
  policies:
    - snat: true
      subnets:
        - ovn-default
```

5.4.4

BgpConf

`localASN`	`uint32`		-	AS	`65002`
`peerASN`	`uint32`		-	AS	`65001`
`routerId`	`string`		Pod IP	BGP Router ID	`10.0.1.13`
`neighbours`	`string array`		-	BGP IP	`10.0.1.1`
`password`	`string`		-	BGP	`secret`
`holdTime`	`string (duration)`		-	BGP Hold	`90s`
`keepaliveTime`	`string (duration)`		-	BGP Keepalive	`30s`
`connectTime`	`string (duration)`		-	BGP	`10s`
`ebgpMultiHop`	`boolean`		`false`	EBGP Multi-Hop	`true`

EvpnConf

`vni`	`uint32`		-	VXLAN Network Identifier	`1016`
`routeTargets`	`string array`		-	Route Target	`65000:1016`

VpcEgressGateway

 VpcEgressGateway `.spec` BGP/EVPN VpcEgressGateway Egress Gateway

`bgpConf`	`string`		-	BgpConf Egress Gateway BGP	`bgp-conf-6502`
`evpnConf`	`string`		-	EvpnConf `bgpConf` Egress Gateway EVPN/VXLAN	`evpn-conf-1016`

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5.4.5 □□

5.5 VPC QoS

Kube-OVN [illegible] QoSPolicy CRD [illegible] VPC [illegible]

5.5.1 [illegible]

QoSPolicy [illegible] `priority` [illegible] QoS [illegible]

[illegible]

[illegible]	[illegible]	[illegible]
EIP QoS	1	[illegible] EIP [illegible]
NATGW [illegible] QoS	2	[illegible] NATGW [illegible] IP [illegible]
NATGW net1 [illegible] QoS	3	[illegible]

[illegible]

- `shared=false` [illegible] QoS [illegible] `shared=true` [illegible]
- EIP [illegible] QoS [illegible] NATGW [illegible] QoS

5.5.2 EIP QoS

[illegible] EIP [illegible] 1Mbps[illegible] 1[illegible] `shared=false` [illegible] QoSPolicy [illegible] EIP [illegible] QoSPolicy [illegible] QoS [illegible]

QoSPolicy [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-eip-example
spec:
  shared: false
  bindingType: EIP
  bandwidthLimitRules:
  - name: eip-ingress
    rateMax: "1" # Mbps
    burstMax: "1" # Mbps
    priority: 1
    direction: ingress
  - name: eip-egress
    rateMax: "1" # Mbps
    burstMax: "1" # Mbps
    priority: 1
    direction: egress
```

IptablesEIP [illegible]

```yaml
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-1
spec:
  natGwDp: gw1
  qosPolicy: qos-eip-example
```

`.spec.qosPolicy` [illegible]

5.5.3 [illegible] QoS [illegible] EIP

[illegible] `label` [illegible] qos [illegible] eip[illegible]

```
# kubectl get eip  -l ovn.kubernetes.io/qos=qos-eip-example
NAME    IP             MAC                 NAT   NATGWDP   READY
eip-1   172.18.11.24   00:00:00:34:41:0B   fip   gw1       true
```

5.5.4 VPC NATGW net1 [illegible] QoS

[illegible] VPC NATGW [illegible] net1 [illegible] 10Mbps[illegible] 3[illegible] `shared=true` [illegible] QoSPolicy [illegible] QoSPolicy [illegible]

QoSPolicy [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-natgw-example
spec:
  shared: true
  bindingType: NATGW
  bandwidthLimitRules:
  - name: net1-ingress
    interface: net1
    rateMax: "10" # Mbps
    burstMax: "10" # Mbps
    priority: 3
    direction: ingress
  - name: net1-egress
    interface: net1
    rateMax: "10" # Mbps
    burstMax: "10" # Mbps
    priority: 3
    direction: egress
```

VpcNatGateway [illegible]

```yaml
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  qosPolicy: qos-natgw-example
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
```

`.spec.qosPolicy` [illegible]

5.5.5 net1 [illegible] QoS

[illegible] net1 [illegible] 5Mbps[illegible] 2[illegible] `shared=true` [illegible] QoSPolicy [illegible] QoSPolicy [illegible]

QoSPolicy [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-natgw-example
spec:
  shared: true
  bindingType: NATGW
  bandwidthLimitRules:
  - name: net1-extip-ingress
    interface: net1
    rateMax: "5" # Mbps
    burstMax: "5" # Mbps
    priority: 2
    direction: ingress
    matchType: ip
    matchValue: src 172.18.11.22/32
  - name: net1-extip-egress
    interface: net1
    rateMax: "5" # Mbps
    burstMax: "5" # Mbps
    priority: 2
    direction: egress
    matchType: ip
    matchValue: dst 172.18.11.23/32
```

VpcNatGateway [illegible]

```yaml
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  qosPolicy: qos-natgw-example
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
```

5.5.6 □□□□□□ QoS □ NATGW

□□ `label` □□□□□□□□ qos □ eip□

```
# kubectl get vpc-nat-gw  -l ovn.kubernetes.io/qos=qos-natgw-example
NAME   VPC          SUBNET   LANIP
gw1    test-vpc-1   net1     10.0.1.254
```

5.5.7 □□ qos □□

```
# kubectl get qos -A
NAME                SHARED   BINDINGTYPE
qos-eip-example     false    EIP
qos-natgw-example   true     NATGW
```

5.5.8 □□

- □□□□□□□□□□□ QoS □□□□□□□□□ QoS □□□□□□□□□□□□ QoS □ EIP □ NATGW□□□□□□ `spec.qosPolicy` □□□

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5.5.9 □□

5.6 [illegible] VPC [illegible]

Kubernetes [illegible] Service [illegible] VPC [illegible] Service [illegible]

1. Service IP [illegible] VPC [illegible]
2. [illegible] IP [illegible]

[illegible]Kube-OVN [illegible] 1.11 [illegible] `SwitchLBRule` CRD[illegible] VPC [illegible]

`SwitchLBRule` [illegible] VPC [illegible]

5.6.1 `Selector` [illegible]

[illegible] `selector` [illegible] `label` [illegible] `pod` [illegible]

`SwitchLBRule` [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
```

- `selector`, `sessionAffinity` [illegible] `port` [illegible] Kubernetes Service[illegible]
- `vip` [illegible] IP [illegible]
- `namespace` [illegible] `selector` [illegible] Pod [illegible]

Kube-OVN [illegible] `SwitchLBRule` [illegible] Pod [illegible] Pod [illegible] VPC [illegible] L2 LB[illegible]

5.6.2 `Endpoints` [illegible]

[illegible] `endpoints` [illegible] `selector` [illegible] `kubevirt` [illegible] `vm` [illegible]

`SwitchLBRule` [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  endpoints:
    - 192.168.0.101
    - 192.168.0.102
    - 192.168.0.103
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
```

- `sessionAffinity` [illegible] `port` [illegible] Kubernetes Service[illegible]
- `vip` [illegible] IP [illegible]
- `namespace` [illegible]
- `endpoints` [illegible] IP [illegible]

□□□□□□□□ `selector` □ `endpoints`，□□□□□ `selector` □□□

5.6.3 □□□□

`OVN` □□ `IPv4` □□□□□□□□□□□□□□□□□□□ □□□

[Health Checks](https://www.ovn.org/support/dist-docs/ovn-nb.5.html)

□□ `ovn` □□□□□□□□□□□□□□ `SwitchLBRule` □□□□□□□□□□ `SwitchLBRule` □□□□□□□□ `VPC` □ `subnet` □□□□□□□□□ `vip` □□□□□□□□□□□□□ `ip_port_mappings` □ `load_balancer_health_check` □□□□□□□□□□□

- □□□□ `vip` □□□□□□□ `subnet` □□□□□□□□□□□ `subnet` □□□□□□□□□□□□□□□□□□□□□□□ `SwitchLBRule` □□□□□□□□□□
- □□□□□□□ `Selector` □□□□□□□□□□□

□□□□□□□□□

```
root@server:~# kubectl get po -o wide -n vulpecula
NAME                     READY   STATUS    RESTARTS   AGE     IP          NODE     NOMINATED NODE   READINESS GATES
nginx-78d9578975-f4qn4   1/1     Running   3          4d16h   10.16.0.4   worker   <none>           <none>
nginx-78d9578975-t8tm5   1/1     Running   3          4d16h   10.16.0.6   worker   <none>           <none>

#    slr
root@server:~# cat << END > slr.yaml
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  nginx
  namespace:  vulpecula
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
END
root@server:~# kubectl apply -f slr.yaml
root@server:~# kubectl get slr
NAME              VIP       PORT(S)    SERVICE                        AGE
vulpecula-nginx   1.1.1.1   8888/TCP   default/slr-vulpecula-nginx    3d21h
```

□□□□□ `subnet` □□□ `vip` □□□□□□

```
#       vip

root@server:~# kubectl get vip
NAME          NS    V4IP        MAC                 V6IP    PMAC   SUBNET           READY   TYPE
vulpecula-subnet    10.16.0.2   00:00:00:39:95:C1   <nil>          vulpecula-subnet    true
```

□□□□□□□□□□□ `Load_Balancer_Health_Check` □ `Service_Monitor` □

```
root@server:~# kubectl ko nbctl list Load_Balancer
_uuid               : 3cbb6d43-44aa-4028-962f-30d2dba9f0b8
external_ids        : {}
health_check        : [5bee3f12-6b54-411c-9cc8-c9def8f67356]
ip_port_mappings    : {"10.16.0.4"="nginx-78d9578975-f4qn4.default:10.16.0.2", "10.16.0.6"="nginx-78d9578975-t8tm5.default:10.16.0.2"}
name                : cluster-tcp-session-loadbalancer
options             : {affinity_timeout="10800"}
protocol            : tcp
selection_fields    : [ip_src]
vips                : {"1.1.1.1:8888"="10.16.0.4:80,10.16.0.6:80"}

root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
_uuid               : 5bee3f12-6b54-411c-9cc8-c9def8f67356
external_ids        : {switch_lb_subnet=vulpecula-subnet}
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
vip                 : "1.1.1.1:8888"

root@server:~# kubectl ko sbctl list Service_Monitor
_uuid               : 1bddc541-cc49-44ea-9935-a4208f627a91
external_ids        : {}
ip                  : "10.16.0.4"
logical_port        : nginx-78d9578975-f4qn4.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
```

```
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online

_uuid               : 84dd24c5-e1b4-4e97-9daa-13687ed59785
external_ids        : {}
ip                  : "10.16.0.6"
logical_port        : nginx-78d9578975-t8tm5.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online
```

[illegible] `vip` [illegible]

```
root@server:~# kubectl exec -it -n vulpecula nginx-78d9578975-t8tm5 -- curl 1.1.1.1:8888
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

[illegible]

[illegible] `pod` [illegible]

```
kubectl delete po nginx-78d9578975-f4qn4
kubectl get po -o wide -n vulpecula
NAME                     READY   STATUS    RESTARTS   AGE     IP           NODE     NOMINATED NODE   READINESS GATES
nginx-78d9578975-lxmvh   1/1     Running   0          31s     10.16.0.8    worker   <none>           <none>
nginx-78d9578975-t8tm5   1/1     Running   3          4d16h   10.16.0.6    worker   <none>           <none>
```

[illegible] `Load_Balancer_Health_Check` [illegible] `Service_Monitor` [illegible]

```
root@server:~# kubectl ko nbctl list Load_Balancer
_uuid               : 3cbb6d43-44aa-4028-962f-30d2dba9f0b8
external_ids        : {}
health_check        : [5bee3f12-6b54-411c-9cc8-c9def8f67356]
ip_port_mappings    : {"10.16.0.4"="nginx-78d9578975-f4qn4.default:10.16.0.2", "10.16.0.6"="nginx-78d9578975-t8tm5.default:10.16.0.2",
"10.16.0.8"="nginx-78d9578975-lxmvh.default:10.16.0.2"}
name                : cluster-tcp-session-loadbalancer
options             : {affinity_timeout="10800"}
protocol            : tcp
selection_fields    : [ip_src]
vips                : {"1.1.1.1:8888"="10.16.0.6:80,10.16.0.8:80"}

root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
_uuid               : 5bee3f12-6b54-411c-9cc8-c9def8f67356
external_ids        : {switch_lb_subnet=vulpecula-subnet}
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
vip                 : "1.1.1.1:8888"

root@server:~# kubectl ko sbctl list Service_Monitor
_uuid               : 84dd24c5-e1b4-4e97-9daa-13687ed59785
external_ids        : {}
ip                  : "10.16.0.6"
logical_port        : nginx-78d9578975-t8tm5.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online

_uuid               : 5917b7b7-a999-49f2-a42d-da81f1eeb28f
external_ids        : {}
ip                  : "10.16.0.8"
logical_port        : nginx-78d9578975-lxmvh.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online
```

[illegible] `SwitchLBRule` [illegible] `Load_Balancer_Health_Check` [illegible] `Service_Monitor` [illegible] `vip` [illegible]

```
root@server:~# kubectl delete -f slr.yaml
switchlbrule.kubeovn.io "vulpecula-nginx" deleted
root@server:~# kubectl get vip
No resources found
```

```
root@server:~# kubectl ko sbctl list Service_Monitor
root@server:~#
root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
root@server:~#
```

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5.6.4 □□

5.7 ▯▯▯ VPC ▯▯ DNS

▯▯▯▯▯▯▯ VPC ▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯ coredns▯ ▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯ Kubernetes ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `vpc-dns` CRD ▯▯▯▯▯▯▯

▯ CRD ▯▯▯▯▯▯▯ coredns▯▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯

Note

▯ DNS ▯▯▯▯▯▯▯▯ Pod ▯ VM ▯▯▯▯▯▯▯▯▯▯ Webhook ▯▯▯▯▯▯▯▯▯▯ `/etc/resolv.conf` ▯▯▯▯

5.7.1 ▯▯ vpc-dns ▯▯▯▯▯▯▯

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: system:vpc-dns
rules:
  - apiGroups:
    - ""
    resources:
    - endpoints
    - services
    - pods
    - namespaces
    verbs:
    - list
    - watch
  - apiGroups:
    - discovery.k8s.io
    resources:
    - endpointslices
    verbs:
    - list
    - watch
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  annotations:
    rbac.authorization.kubernetes.io/autoupdate: "true"
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: vpc-dns
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: system:vpc-dns
subjects:
- kind: ServiceAccount
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-corefile
  namespace: kube-system
data:
  Corefile: |
    .:53 {
        errors
        health {
          lameduck 5s
        }
        ready
        kubernetes cluster.local in-addr.arpa ip6.arpa {
          pods insecure
          fallthrough in-addr.arpa ip6.arpa
        }
        prometheus :9153
        forward . /etc/resolv.conf {
          prefer_udp
        }
        cache 30
        loop
        reload
        loadbalance
    }
```

 nat-gw-pod

5.7.2

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-nad
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "ovn-nad.default.ovn"
    }'
```

5.7.3 vpc-dns Configmap

 kube-system configmap vpc-dns vpc-dns

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-config
  namespace: kube-system
data:
  coredns-vip: 10.96.0.3
  enable-vpc-dns: "true"
  nad-name: ovn-nad
  nad-provider: ovn-nad.default.ovn
```

- `enable-vpc-dns` `true`
- `coredns-image` dns coredns
- `coredns-vip` coredns lb vip
- `nad-name` `network-attachment-definitions`
- `nad-provider` provider
- `k8s-service-host` coredns k8s apiserver ip apiserver
- `k8s-service-port` coredns k8s apiserver port apiserver

5.7.4 vpc-dns

 vpc-dns yaml

```
kind: VpcDns
apiVersion: kubeovn.io/v1
metadata:
  name: test-cjh1
spec:
  vpc: cjh-vpc-1
  subnet: cjh-subnet-1
  replicas: 2
```

- `vpc`   dns vpc
- `subnet` dns
- `replicas`: vpc dns deployment replicas

```
# kubectl get vpc-dns
NAME        ACTIVE   VPC         SUBNET
test-cjh1   false    cjh-vpc-1   cjh-subnet-1
test-cjh2   true     cjh-vpc-1   cjh-subnet-2
```

`ACTIVE` : `true` [illegible] dns [illegible] `false` [illegible]

[illegible] VPC [illegible] DNS [illegible]

- [illegible] VPC [illegible] vpc-dns [illegible] VPC [illegible] subnet[illegible] vpc-dns [illegible] `true` [illegible] `false` [illegible]
- [illegible] `true` [illegible] vpc-dns [illegible] `false` [illegible] vpc-dns [illegible]

5.7.5 [illegible]

[illegible] vpc-dns Pod [illegible] label `app=vpc-dns` [illegible] vpc-dns Pod [illegible]

```
# kubectl -n kube-system get pods -l app=vpc-dns
NAME                                 READY   STATUS    RESTARTS   AGE
vpc-dns-test-cjh1-7b878d96b4-g5979   1/1     Running   0          28s
vpc-dns-test-cjh1-7b878d96b4-ltmf9   1/1     Running   0          28s
```

[illegible] slr [illegible]

```
# kubectl -n kube-system get slr
NAME                VIP         PORT(S)                  SERVICE                             AGE
vpc-dns-test-cjh1   10.96.0.3   53/UDP,53/TCP,9153/TCP   kube-system/slr-vpc-dns-test-cjh1   113s
```

[illegible] VPC [illegible] Pod[illegible] dns [illegible]:

```
nslookup kubernetes.default.svc.cluster.local 10.96.0.3
```

[illegible] VPC [illegible] switch lb rule [illegible] VPC [illegible] Pod [illegible]

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5.7.6 [illegible]

5.8 SecurityGroup [illegible]

Kube-OVN [illegible] Pod [illegible]

Warning

Kube-OVN [illegible] NetworkPolicy[illegible]Network Policy API[illegible]Subnet ACL [illegible] Security Group [illegible] OVN ACL [illegible] NetworkPolicy [illegible] NetworkPolicy API [illegible]

5.8.1 [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-example
spec:
  allowSameGroupTraffic: true
  egressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: all
    remoteAddress: 10.16.0.13 # 10.16.0.0/16
    remoteType: address
  ingressRules:
  - ipVersion: ipv4
    policy: deny
    priority: 1
    protocol: icmp
    remoteAddress: 10.16.0.14
    remoteType: address
```

[illegible] Kube-OVN [illegible]

Pod [illegible] `ovn.kubernetes.io/security_groups` annotation [illegible]

```yaml
    ovn.kubernetes.io/security_groups: sg-example
```

[illegible]Port Security[illegible]

5.8.2 [illegible]

[illegible] `tier` [illegible] ACL [illegible] ACL [illegible]

- **`tier`** [illegible] `0` [illegible] `1` [illegible] `0` [illegible]tier `0` [illegible] tier `0` [illegible] `policy: pass` [illegible] ACL [illegible] tier `1` [illegible]
- **`policy: pass`** [illegible] `allow` [illegible] `deny` [illegible] tier [illegible] 1 [illegible] `pass` [illegible]

[illegible]

[illegible] tier-0 [illegible] tier-1 [illegible]

[illegible]

[illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-tier0
spec:
  tier: 0
  allowSameGroupTraffic: true
  ingressRules:
  - ipVersion: ipv4
    policy: pass
    priority: 1
    protocol: tcp
    remoteAddress: 10.16.0.0/16
    remoteType: address
  - ipVersion: ipv4
    policy: deny
    priority: 2
    protocol: all
    remoteAddress: 0.0.0.0/0
    remoteType: address
---
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-tier1
spec:
  tier: 1
  allowSameGroupTraffic: true
  ingressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: tcp
    remoteAddress: 10.16.0.0/16
    remoteType: address
    portRangeMin: 80
    portRangeMax: 443
```

⌧⌧⌧⌧⌧⌧ `sg-tier0` ⌧⌧⌧ `10.16.0.0/16` ⌧⌧⌧ TCP ⌧⌧⌧⌧⌧⌧ tier 1⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ `sg-tier1` ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ 80-443 ⌧⌧⌧ TCP ⌧⌧⌧

⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ Pod⌧⌧⌧ annotation ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧

```yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: web
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-tier0,sg-tier1'
  name: multi-tier-pod
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: nginx
```

5.8.3 ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧

⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ `localAddress` ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧

- **`localAddress`** ⌧⌧⌧⌧⌧⌧⌧ IP ⌧⌧⌧ CIDR⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧
- **`sourcePortRangeMin`** / **`sourcePortRangeMax`** ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧1-65535⌧⌧⌧⌧⌧⌧ TCP ⌧ UDP ⌧⌧⌧

⌧⌧⌧⌧⌧⌧⌧⌧

```yaml
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-local-filter
spec:
  allowSameGroupTraffic: true
  ingressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: tcp
    remoteAddress: 10.16.0.0/16
    remoteType: address
    portRangeMin: 8080
    portRangeMax: 8080
    localAddress: 10.16.0.100
    sourcePortRangeMin: 1024
    sourcePortRangeMax: 65535
```

⌧⌧⌧⌧⌧⌧ `10.16.0.0/16` ⌧ `10.16.0.100` ⌧ 8080 ⌧⌧⌧⌧⌧ TCP ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ 1024-65535⌧

5.8.4 ⌧⌧⌧⌧

- ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ACL ⌧⌧⌧⌧⌧⌧⌧⌧⌧OVN ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ACL ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ACL ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧

- [illegible] `priority` [illegible] 1-16384[illegible] ACL [illegible] ACL [illegible] `ACL [illegible] = 18484 - [illegible]` [illegible] ACL [illegible]
- `tier` [illegible] `0` [illegible] `1` [illegible] `policy: pass` [illegible] tier `0` [illegible] tier `1` [illegible]
- [illegible]Kube-OVN [illegible] CNI[illegible] Pod [illegible] Pod [illegible] Pod [illegible] ContainerCreating [illegible] Running [illegible]

5.8.5 [illegible]

[illegible] YAML [illegible] Pod[illegible] annotation [illegible]

```yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: static
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-example'
  name: sg-test-pod
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: qatest
```

[illegible]

```
# kubectl get pod -o wide
NAME                  READY   STATUS              RESTARTS   AGE     IP           NODE                     NOMINATED NODE   READINESS GATES
sg-test-pod           0/1     ContainerCreating   0          5h32m   <none>       kube-ovn-worker          <none>           <none>
test-99fff7f86-52h9r  1/1     Running             0          5h41m   10.16.0.14   kube-ovn-control-plane   <none>           <none>
test-99fff7f86-qcgjw  1/1     Running             0          5h43m   10.16.0.13   kube-ovn-worker          <none>           <none>
```

[illegible] kubectl describe pod [illegible] Pod [illegible]

```
# kubectl describe pod sg-test-pod
Name:         sg-test-pod
Namespace:    default
Priority:     0
Node:         kube-ovn-worker/172.18.0.2
Start Time:   Tue, 28 Feb 2023 10:29:36 +0800
Labels:       app=static
Annotations:  ovn.kubernetes.io/allocated: true
              ovn.kubernetes.io/cidr: 10.16.0.0/16
              ovn.kubernetes.io/gateway: 10.16.0.1
              ovn.kubernetes.io/ip_address: 10.16.0.15
              ovn.kubernetes.io/logical_router: ovn-cluster
              ovn.kubernetes.io/logical_switch: ovn-default
              ovn.kubernetes.io/mac_address: 00:00:00:FA:17:97
              ovn.kubernetes.io/pod_nic_type: veth-pair
              ovn.kubernetes.io/port_security: true
              ovn.kubernetes.io/routed: true
              ovn.kubernetes.io/security_groups: sg-allow-reject
Status:       Pending
IP:
IPs:          <none>
.
.
.
Events:
  Type     Reason                  Age                   From     Message
  ----     ------                  ----                  ----     -------
  Warning  FailedCreatePodSandBox  5m3s (x70 over 4h59m)  kubelet  (combined from similar events): Failed to create pod sandbox: rpc error: code = Unknown
desc = failed to setup network for sandbox "40636e0c7f1ade5500fa958486163d74f2e2300051a71522a9afd7ba0538afb6": plugin type="kube-ovn" failed (add): RPC
failed; request ip return 500 configure nic failed 10.16.0.15 network not ready after 200 ping 10.16.0.1
```

[illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-gw-both
spec:
  allowSameGroupTraffic: true
  egressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 2
    protocol: all
    remoteAddress: 10.16.0.13
    remoteType: address
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: all
    remoteAddress: 10.16.0.1
    remoteType: address
  ingressRules:
  - ipVersion: ipv4
    policy: deny
    priority: 2
    protocol: icmp
    remoteAddress: 10.16.0.14
    remoteType: address
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: icmp
    remoteAddress: 10.16.0.1
    remoteType: address
```

[illegible]

[illegible] yaml [illegible] Pod [illegible] Pod [illegible]

```yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: static
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-gw-both'
  name: sg-gw-both
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: qatest
```

[illegible] Pod [illegible]

```
# kubectl get pod -o wide
NAME                   READY   STATUS              RESTARTS   AGE     IP            NODE                     NOMINATED NODE   READINESS GATES
sg-test-pod            0/1     ContainerCreating   0          5h41m   <none>        kube-ovn-worker          <none>           <none>
sg-gw-both             1/1     Running             0          5h37m   10.16.0.19    kube-ovn-worker          <none>           <none>
```

[illegible]

PDF Slack Support

2026[illegible]3[illegible]17[illegible]

2023[illegible]2[illegible]28[illegible]

GitHub

5.8.6 [illegible]

5.9 OVN EIP FIP SNAT DNAT [illegible]

Note

[illegible] VPC [illegible] VPC [illegible]

[illegible] VPC [illegible]VPC NAT [illegible]OVN [illegible] Egress Gateway[illegible] VPC NAT [illegible] Kube-OVN [illegible] VPC NAT [illegible] Pod[illegible] VPC [illegible] Macvlan [illegible] Pod [illegible] iptables [illegible]

OVN [illegible] OVN [illegible] NAT [illegible] OVN [illegible] BFD [illegible] OVN [illegible] OVN [illegible]

Egress Gateway [illegible] VPC NAT [illegible]

[illegible] VPC [illegible] OVN NAT [illegible] `provider-network vlan (external) subnet` [illegible] VPC EIP/SNAT[illegible]

5.9.1 [illegible]

- [illegible] `kube-ovn-controller` [illegible] `kube-ovn-cni` [illegible] `ovn-external-gw-config` [illegible] `VPC spec enableExternal` [illegible]
- `CRD` [illegible] `provider-network` `vlan` `subnet` [illegible] `VPC spec extraExternalSubnets` [illegible] `ovn-eip` [illegible] `ovn-dnat` [illegible] `ovn-fip` [illegible] `ovn-snat` [illegible] CRD [illegible]

```
graph LR

pod-->subnet-->vpc-->lrp--bind-->gw-chassis-->snat-->lsp-->external-subnet
lrp-.-peer-.-lsp
```

Pod [illegible] SNAT [illegible] Pod [illegible] Fip [illegible]

```
graph LR


pod-->subnet-->vpc-->lrp--bind-->local-chassis-->snat-->lsp-->external-subnet


lrp-.-peer-.-lsp
```

Pod [illegible] FIP (dnat_and_snat) [illegible]

[illegible] CRD [illegible] iptables nat gw [illegible]

- ovn eip: [illegible] ip [illegible] underlay provider network vlan subnet [illegible]
- ovn fip[illegible] dnat snat[illegible] VPC [illegible] ip [illegible] vip [illegible]
- ovn snat[illegible] VPC [illegible] ip [illegible] snat [illegible]
- ovn dnat[illegible] router lb [illegible], [illegible] ip + [illegible] VPC [illegible] endpoints

5.9.2 1. [illegible]

[illegible] [illegible]

[illegible] OpenStack Neutron ovn[illegible] provider network [illegible] VPC EIP/SNAT [illegible]

[illegible] vlan[illegible] vlan 0[illegible] vlan id[illegible]

```
#
# 1. kube-ovn-controller
          - --external-gateway-vlanid=204
          - --external-gateway-switch=external204

# 2. kube-ovn-cni          :
          - --external-gateway-switch=external204
```

```
###                vlan id                   underlay
```

[illegible]

- [illegible] provider network[illegible]vlan[illegible]subnet [illegible]
- [illegible] VPC enable_eip_snat [illegible] vlan[illegible]subnet [illegible] ip [illegible] ipam[illegible]
- [illegible] VPC [illegible] enable_eip_snat [illegible], [illegible] pod annotation [illegible] fip snat[illegible]
- [illegible] VPC [illegible] enable_eip_snat [illegible] vlan[illegible]subnet [illegible] VPC [illegible] eip snat [illegible]

1.1 [illegible] underlay [illegible]

```
#    provider-network  vlan  subnet
# cat 01-provider-network.yaml

apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: external204
spec:
  defaultInterface: vlan

# cat 02-vlan.yaml

apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan204
spec:
  id: 204
  provider: external204

# cat 03-vlan-subnet.yaml

apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: external204
spec:
  protocol: IPv4
  cidrBlock: 10.5.204.0/24
  gateway: 10.5.204.254
  vlan: vlan204
  excludeIps:
  - 10.5.204.1..10.5.204.100
```

1.2 [illegible] VPC [illegible] eip_snat

```
#      VPC     underlay    provider subnet
# cat 00-centralized-external-gw-no-ip.yaml

apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-external-gw-config
  namespace: kube-system
data:
  enable-external-gw: "true"
  external-gw-nodes: "pc-node-1,pc-node-2,pc-node-3"
  type: "centralized"
  external-gw-nic: "vlan" #      ovs
  external-gw-addr: "10.5.204.254/24" # underlay       ip
```

[illegible] logical router port (lrp) ip [illegible] mac[illegible] underlay [illegible] lrp [illegible] ovn eip [illegible]

[illegible] ip [illegible] lrp [illegible] ovn-eip[illegible] lrp [illegible] ovn eip[illegible]

1.3 [illegible] VPC [illegible] eip snat fip [illegible]

```
# cat 00-ns.yml

apiVersion: v1
kind: Namespace
metadata:
  name: vpc1

# cat 01-vpc-external.yml
```

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc1
spec:
  namespaces:
  - vpc1
  enableExternal: true
  staticRoutes:
    - cidr: 0.0.0.0/0
      nextHopIP: 10.5.204.254
      policy: policyDst

# VPC    enableExternal       lrp

# cat 02-subnet.yml

apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vpc1-subnet1
spec:
  cidrBlock: 192.168.0.0/24
  default: false
  disableGatewayCheck: false
  disableInterConnection: true
  gatewayNode: ""
  gatewayType: distributed
  #gatewayType: centralized
  natOutgoing: false
  private: false
  protocol: IPv4
  provider: ovn
  vpc: vpc1
  namespaces:
  - vpc1

#                   subnet
```

[illegible]

```
# kubectl ko nbctl show vpc1

router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    nat 21d853b0-f7b4-40bd-9a53-31d2e2745739
        external ip: "10.5.204.115"
        logical ip: "192.168.0.0/24"
        type: "snat"
```

```
# kubectl ko nbctl lr-route-list vpc1

IPv4 Routes
Route Table <main>:
                0.0.0.0/0              10.5.204.254 dst-ip
#    VPC CRD
```

> [illegible] enableExternal [illegible] VPC CRD [illegible]

1.4 [illegible]

1.4.1 [illegible] UNDERLAY [illegible]

[illegible] eip snat fip [illegible] 1 [illegible] eip snat fip [illegible]

```
#    provider-network  vlan  subnet
# cat 01-extra-provider-network.yaml
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: extra
spec:
  defaultInterface: vlan
# cat 02-extra-vlan.yaml
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan0
spec:
  id: 0
  provider: extra
```

```
# cat 03-extra-vlan-subnet.yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: extra
spec:
  protocol: IPv4
  cidrBlock: 10.10.204.0/24
  gateway: 10.10.204.254
  vlan: vlan0
  excludeIps:
  - 10.10.204.1..10.10.204.100
```

1.4.2 [illegible] VPC [illegible]

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc1
spec:
  namespaces:
  - vpc1
  enableExternal: true  #    enableExternal   VPC                external   ls
  extraExternalSubnets: #    extraExternalSubnets
  - extra
```

[illegible]

```
# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    port vpc1-extra
        mac: "00:00:00:EF:6A:C7"
        networks: ["10.10.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
```

5.9.3 2. ovn-eip

[illegible] iptables-eip [illegible]ovn-eip [illegible] type

- nat: [illegible] ovn dnat[illegible]fip, snat [illegible] nat [illegible]
- lrp: [illegible] underlay [illegible] lrp [illegible] ip [illegible] dnat snat [illegible]
- lsp: [illegible]LSP[illegible] EIP

```
---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  externalSubnet: external204
  type: nat

#       eip           fip
```

externalSubnet [illegible] external204[illegible]

[illegible] externalSubnet [illegible] extra[illegible]

2.1 ovn-fip [illegible] pod [illegible] fip

```
# kubectl get po -o wide -n vpc1 vpc-1-busybox01
NAME               READY   STATUS    RESTARTS   AGE     IP            NODE
vpc-1-busybox01    1/1     Running   0          3d15h   192.168.0.2   pc-node-2

# kubectl get ip vpc-1-busybox01.vpc1
NAME                    V4IP          V6IP   MAC                 NODE        SUBNET
vpc-1-busybox01.vpc1    192.168.0.2          00:00:00:0A:DD:27   pc-node-2   vpc1-subnet1

---

kind: OvnEip
apiVersion: kubeovn.io/v1
```

```
metadata:
  name: eip-static
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  ovnEip: eip-static
  ipName: vpc-1-busybox01.vpc1  #       ip crd
  type: "centralized"           # centralized    distributed

--
#          VPC       ip

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  ovnEip: eip-static
  vpc: vpc1
  v4Ip: 192.168.0.2
  type: "centralized"           # centralized    distributed
```

```
# kubectl get ofip
NAME          VPC    V4EIP          V4IP          READY   IPTYPE   IPNAME
eip-for-vip   vpc1   10.5.204.106   192.168.0.3   true    vip      test-fip-vip
eip-static    vpc1   10.5.204.101   192.168.0.2   true             vpc-1-busybox01.vpc1
# kubectl get ofip eip-static
NAME         VPC    V4EIP          V4IP          READY   IPTYPE   IPNAME
eip-static   vpc1   10.5.204.101   192.168.0.2   true             vpc-1-busybox01.vpc1

[root@pc-node-1 03-cust-vpc]# ping 10.5.204.101
PING 10.5.204.101 (10.5.204.101) 56(84) bytes of data.
64 bytes from 10.5.204.101: icmp_seq=2 ttl=62 time=1.21 ms
64 bytes from 10.5.204.101: icmp_seq=3 ttl=62 time=0.624 ms
64 bytes from 10.5.204.101: icmp_seq=4 ttl=62 time=0.368 ms
^C
--- 10.5.204.101 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3049ms
rtt min/avg/max/mdev = 0.368/0.734/1.210/0.352 ms
[root@pc-node-1 03-cust-vpc]#

#       node ping    VPC    pod     ip
```

```
#     ip
# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    nat 813523e7-c68c-408f-bd8c-cba30cb2e4f4
        external ip: "10.5.204.101"
        logical ip: "192.168.0.2"
        type: "dnat_and_snat"
```

2.2 ovn-fip ⌧ vip ⌧⌧⌧⌧ fip

⌧⌧⌧⌧⌧⌧⌧ vip ⌧⌧⌧⌧⌧⌧⌧⌧⌧ kubevirt ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ vip ⌧⌧⌧ keepalived⌧kube-vip ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧

⌧⌧⌧⌧⌧⌧⌧ fip ⌧⌧ VPC ⌧⌧⌧ vip ⌧⌧⌧⌧⌧⌧⌧ vip ⌧⌧⌧⌧⌧⌧⌧

```
#     vip eip    eip     vip
# cat vip.yaml

apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: test-fip-vip
spec:
  subnet: vpc1-subnet1

# cat 04-fip.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
```

```
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  ovnEip: eip-for-vip
  ipType: vip         #       fip     pod ip              vip
  ipName: test-fip-vip

---
#          VPC       ip

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  ovnEip: eip-for-vip
  ipType: vip         #       fip     pod ip              vip
  vpc: vpc1
  v4Ip: 192.168.0.3
```

```
# kubectl get ofip
NAME          VPC    V4EIP          V4IP          READY   IPTYPE   IPNAME
eip-for-vip   vpc1   10.5.204.106   192.168.0.3   true    vip      test-fip-vip


[root@pc-node-1 fip-vip]# ping  10.5.204.106
PING 10.5.204.106 (10.5.204.106) 56(84) bytes of data.
64 bytes from 10.5.204.106: icmp_seq=1 ttl=62 time=0.694 ms
64 bytes from 10.5.204.106: icmp_seq=2 ttl=62 time=0.436 ms

#   node    ping


# pod     ip

[root@pc-node-1 fip-vip]# kubectl -n vpc1 exec -it vpc-1-busybox03 -- bash
[root@vpc-1-busybox03 /]#
[root@vpc-1-busybox03 /]#
[root@vpc-1-busybox03 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
1568: eth0@if1569: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:56:40:e5 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.5/24 brd 192.168.0.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.0.3/24 scope global secondary eth0  #      vip
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe56:40e5/64 scope link
       valid_lft forever preferred_lft forever

[root@vpc-1-busybox03 /]# tcpdump -i eth0 host  192.168.0.3 -netvv
tcpdump: listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
00:00:00:ed:8e:c7 > 00:00:00:56:40:e5, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 62, id 44830, offset 0, flags [DF], proto ICMP (1), length 84)
    10.5.32.51 > 192.168.0.3: ICMP echo request, id 177, seq 1, length 64
00:00:00:56:40:e5 > 00:00:00:ed:8e:c7, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 64, id 43962, offset 0, flags [none], proto ICMP (1), length 84)
    192.168.0.3 > 10.5.32.51: ICMP echo reply, id 177, seq 1, length 64

# pod        fip     icmp
```

5.9.4 3. ovn-snat

3.1 ovn-snat □□□□ subnet □ cidr

□□□□ iptables-snat □□□□□□□□□□□□

```
# cat 03-subnet-snat.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnSnatRule
```

```
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpcSubnet: vpc1-subnet1 # eip

---
#          VPC       subnet cidr

kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpc: vpc1
  v4IpCidr: 192.168.0.0/24 #        cidr      ip
```

□□□□□□□□□□□□□□□□□□□□□ externalSubnet □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ extra□

3.2 ovn-snat □□□□□□ pod ip

□□□□□ iptables-snat □□□□□□□□□□□□□□

```
# cat 03-pod-snat.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-pod-vpc-ip
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat01
spec:
  ovnEip: snat-for-pod-vpc-ip
  ipName: vpc-1-busybox02.vpc1 # eip      pod ip

---
#          VPC       ip

kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpc: vpc1
  v4IpCidr: 192.168.0.4
```

□□□□□□□□□□□□□□□□□□□□□ externalSubnet □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ extra□

□□□□□□□□□□□□□□□ snat □□□□□□□□□□□□□□□

```
# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    nat 21d853b0-f7b4-40bd-9a53-31d2e2745739
        external ip: "10.5.204.115"
        logical ip: "192.168.0.0/24"
        type: "snat"
    nat da77a11f-c523-439c-b1d1-72c664196a0f
        external ip: "10.5.204.116"
        logical ip: "192.168.0.4"
        type: "snat"
```

```
[root@pc-node-1 03-cust-vpc]# kubectl get po -A -o wide  | grep busy
vpc1          vpc-1-busybox01                                 1/1     Running   0                3d15h   192.168.0.2   pc-node-2   <none>           <none>
vpc1          vpc-1-busybox02                                 1/1     Running   0                17h     192.168.0.4   pc-node-1   <none>           <none>
vpc1          vpc-1-busybox03                                 1/1     Running   0                17h     192.168.0.5   pc-node-1   <none>           <none>
vpc1          vpc-1-busybox04                                 1/1     Running   0                17h     192.168.0.6   pc-node-3   <none>           <none>
vpc1          vpc-1-busybox05                                 1/1     Running   0                17h     192.168.0.7   pc-node-1   <none>           <none>
```

```
# kubectl exec -it -n vpc1            vpc-1-busybox04   bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
[root@vpc-1-busybox04 /]#
[root@vpc-1-busybox04 /]#
[root@vpc-1-busybox04 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
17095: eth0@if17096: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:76:94:55 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.6/24 brd 192.168.0.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe76:9455/64 scope link
       valid_lft forever preferred_lft forever
[root@vpc-1-busybox04 /]# ping 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=1 ttl=114 time=22.2 ms
64 bytes from 223.5.5.5: icmp_seq=2 ttl=114 time=21.8 ms

[root@pc-node-1 03-cust-vpc]# kubectl exec -it -n vpc1            vpc-1-busybox02   bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
1566: eth0@if1567: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:0b:e9:d0 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.4/24 brd 192.168.0.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe0b:e9d0/64 scope link
       valid_lft forever preferred_lft forever
[root@vpc-1-busybox02 /]# ping 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=2 ttl=114 time=22.7 ms
64 bytes from 223.5.5.5: icmp_seq=3 ttl=114 time=22.6 ms
64 bytes from 223.5.5.5: icmp_seq=4 ttl=114 time=22.1 ms
^C
--- 223.5.5.5 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3064ms
rtt min/avg/max/mdev = 22.126/22.518/22.741/0.278 ms

#        pod           snat
```

5.9.5 4. ovn-dnat

4.1 ovn-dnat □ pod □□□□ dnat

```yaml
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  externalSubnet: underlay

---
kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ovnEip: eip-dnat
  ipName: vpc-1-busybox01.vpc1 #       pod ip crd
  protocol: tcp
  internalPort: "22"
  externalPort: "22"


---
#          VPC       ip

kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ovnEip: eip-dnat
  protocol: tcp
  internalPort: "22"
  externalPort: "22"
  vpc: vpc1
  v4Ip: 192.168.0.3
```

□□□□□□□□□□□□□□□□□ externalSubnet □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ extra□

OvnDnatRule □□□□ IptablesDnatRule □□

```
# kubectl get oeip eip-dnat
NAME       V4IP        V6IP   MAC                 TYPE   READY
eip-dnat   10.5.49.4          00:00:00:4D:CE:49   dnat   true

# kubectl get odnat
NAME                   EIP                    PROTOCOL   V4EIP        V4IP           INTERNALPORT   EXTERNALPORT   IPNAME                                READY
eip-dnat               eip-dnat               tcp        10.5.49.4    192.168.0.3    22             22             vpc-1-busybox01.vpc1                  true
```

4.2 ovn-dnat □ vip □□□□ dnat

```yaml
kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ipType: vip  #       dnat     pod ip              vip
  ovnEip: eip-dnat
  ipName: test-dnat-vip
  protocol: tcp
  internalPort: "22"
  externalPort: "22"

---
#          VPC       ip

kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ipType: vip  #       dnat     pod ip              vip
  ovnEip: eip-dnat
  ipName: test-dnat-vip
  protocol: tcp
  internalPort: "22"
  externalPort: "22"
  vpc: vpc1
  v4Ip: 192.168.0.4
```

OvnDnatRule □□□□ IptablesDnatRule □□

```
# kubectl get vip test-dnat-vip
NAME            V4IP          PV4IP   MAC                 PMAC   V6IP   PV6IP   SUBNET         READY
test-dnat-vip   192.168.0.4           00:00:00:D0:C0:B5                         vpc1-subnet1   true

# kubectl get oeip eip-dnat
NAME       V4IP        V6IP   MAC                 TYPE   READY
eip-dnat   10.5.49.4          00:00:00:4D:CE:49   dnat   true

# kubectl get odnat eip-dnat
NAME       EIP        PROTOCOL   V4EIP       V4IP          INTERNALPORT   EXTERNALPORT   IPNAME          READY
eip-dnat   eip-dnat   tcp        10.5.49.4   192.168.0.4   22             22             test-dnat-vip   true
```

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5.9.6 □□

5.10 VPC [illegible]

VPC [illegible] VPC [illegible] VPC [illegible] [illegible] NAT [illegible]

5.10.1 [illegible]

1. [illegible] VPC[illegible]
2. [illegible] VPC [illegible] CIDR [illegible]
3. [illegible] VPC [illegible] VPC [illegible]

5.10.2 [illegible]

[illegible] VPC[illegible] VPC [illegible] Subnet[illegible]Subnet [illegible] CIDR [illegible]

```yaml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-1
spec: {}
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net1
spec:
  vpc: vpc-1
  cidrBlock: 10.0.0.0/16
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-2
spec: {}
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: vpc-2
  cidrBlock: 172.31.0.0/16
```

[illegible] VPC [illegible] `vpcPeerings` [illegible]

```yaml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-1
spec:
  vpcPeerings:
    - remoteVpc: vpc-2
      localConnectIP: 169.254.0.1/30
  staticRoutes:
    - cidr: 172.31.0.0/16
      nextHopIP: 169.254.0.2
      policy: policyDst
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-2
spec:
  vpcPeerings:
    - remoteVpc: vpc-1
      localConnectIP: 169.254.0.2/30
  staticRoutes:
    - cidr: 10.0.0.0/16
      nextHopIP: 169.254.0.1
      policy: policyDst
```

- `remoteVpc` : [illegible] VPC [illegible]
- `localConnectIP` : [illegible] IP [illegible] CIDR[illegible] IP [illegible] CIDR[illegible]
- `cidr` [illegible] Subnet [illegible] CIDR[illegible]
- `nextHopIP` [illegible] VPC [illegible] `localConnectIP` [illegible]

□□□□□ Subnet □□□ Pod

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/logical_switch: net1
  name: vpc-1-pod
spec:
  containers:
    - name: vpc-1-pod
      image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/logical_switch: net2
  name: vpc-2-pod
spec:
  containers:
    - name: vpc-2-pod
      image: docker.io/library/nginx:alpine
```

□□□□□□□

```
# kubectl exec -it vpc-1-pod -- ping $(kubectl get pod vpc-2-pod -o jsonpath='{.status.podIP}')
PING 172.31.0.2 (172.31.0.2): 56 data bytes
64 bytes from 172.31.0.2: seq=0 ttl=62 time=0.655 ms
64 bytes from 172.31.0.2: seq=1 ttl=62 time=0.086 ms
64 bytes from 172.31.0.2: seq=2 ttl=62 time=0.098 ms
^C
--- 172.31.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.086/0.279/0.655 ms
# kubectl exec -it vpc-2-pod -- ping $(kubectl get pod vpc-1-pod -o jsonpath='{.status.podIP}')
PING 10.0.0.2 (10.0.0.2): 56 data bytes
64 bytes from 10.0.0.2: seq=0 ttl=62 time=0.594 ms
64 bytes from 10.0.0.2: seq=1 ttl=62 time=0.093 ms
64 bytes from 10.0.0.2: seq=2 ttl=62 time=0.088 ms
^C
--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.088/0.258/0.594 ms
```

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5.10.3 □□

6. □□□□

6.1 kubectl □□□□

□□□□□□□□□□□□Kube-OVN □□□ kubectl □□□□□□□□□□ □□□□□□□□□□□□□□□□□□□ OVN □□□□□□□□□OVN □□□ □□□□□□OVS □□□□□□□tcpdump □□□□□□□□□□□□□□□ □□□□□□□□□□□□

6.1.1 □□□□

Kube-OVN □□□□□□□□□□□□□□□□□□□□ kubectl □□□□□□□□□ □□□□□□□□□□□□□□□□□□

□□ `kubectl-ko` □□□

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/kubectl-ko
```

□□□□□□□ `$PATH` □□□□

```
mv kubectl-ko /usr/local/bin/kubectl-ko
```

□□□□□□□□

```
chmod +x /usr/local/bin/kubectl-ko
```

□□□□□□□□□□□□□□

```
# kubectl plugin list
The following compatible plugins are available:

/usr/local/bin/kubectl-ko
```

6.1.2 □□□□

□□ `kubectl ko` □□□□□□□□□□□□□□□□□□□□□□□□□□

```
# kubectl ko
kubectl ko {subcommand} [option...]
Available Subcommands:
  [nb|sb] [status|kick|backup|dbstatus|restore]     ovn-db operations show cluster status, kick stale server, backup database, get db consistency status or
restore ovn nb db when met 'inconsistent data' error
  nbctl [ovn-nbctl options ...]    invoke ovn-nbctl
  sbctl [ovn-sbctl options ...]    invoke ovn-sbctl
  vsctl {nodeName} [ovs-vsctl options ...]   invoke ovs-vsctl on the specified node
  ofctl {nodeName} [ovs-ofctl options ...]   invoke ovs-ofctl on the specified node
  dpctl {nodeName} [ovs-dpctl options ...]   invoke ovs-dpctl on the specified node
  appctl {nodeName} [ovs-appctl options ...]   invoke ovs-appctl on the specified node
  tcpdump {namespace/podname} [tcpdump options ...]     capture pod traffic
  {trace|ovn-trace} ...    trace ovn microflow of specific packet
    {trace|ovn-trace} {namespace/podname} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]    trace ICMP/TCP/UDP
    {trace|ovn-trace} {namespace/podname} {target ip address} [target mac address] arp {request|reply}                     trace ARP request/reply
    {trace|ovn-trace} {node//nodename} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]       trace ICMP/TCP/UDP
    {trace|ovn-trace} {node//nodename} {target ip address} [target mac address] arp {request|reply}                        trace ARP request/reply
  diagnose {all|node|subnet|IPPorts} [nodename|subnetName|{proto1}-{IP1}-{Port1},{proto2}-{IP2}-{Port2}]    diagnose connectivity of all nodes or a specific
node or specify subnet's ds pod or IPPorts like 'tcp-172.18.0.2-53,udp-172.18.0.3-53'
  env-check    check the environment configuration
  reload    restart all kube-ovn components
  log {kube-ovn|ovn|ovs|linux|all}    save log to ./kubectl-ko-log/
  perf [image] performance test default image is docker.io/kubeovn/test:v1.13.0
  icnbctl [ovn-nbctl options ...]    invoke ovn-ic-nbctl
  icsbctl [ovn-sbctl options ...]    invoke ovn-ic-sbctl
```

□□□□□□□□□□□□□□□□□□□□

[nb | sb] [status | kick | backup | dbstatus | restore]

□□□□□□□□ OVN □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□

[illegible]

[illegible] OVN [illegible] leader [illegible] `ovs-appctl cluster/status` [illegible]:

```
# kubectl ko nb status
306b
Name: OVN_Northbound
Cluster ID: 9a87 (9a872522-3e7d-47ca-83a3-d74333e1a7ca)
Server ID: 306b (306b256b-b5e1-4eb0-be91-4ca96adf6bad)
Address: tcp:[172.18.0.2]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 280309 ms ago, reason: timeout
Last Election won: 280309 ms ago
Election timer: 5000
Log: [139, 139]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-8723 ->8723 <-85d6 ->85d6
Disconnections: 0
Servers:
    85d6 (85d6 at tcp:[172.18.0.4]:6643) next_index=139 match_index=138 last msg 763 ms ago
    8723 (8723 at tcp:[172.18.0.3]:6643) next_index=139 match_index=138 last msg 763 ms ago
    306b (306b at tcp:[172.18.0.2]:6643) (self) next_index=2 match_index=138
status: ok
```

[illegible] `Server` [illegible] `match_index` [illegible] `last msg` [illegible] Server [illegible] [illegible]

[illegible]

[illegible] OVN [illegible] [illegible] `172.18.0.3` [illegible]:

```
# kubectl ko nb kick 8723
started removal
```

[illegible]

```
# kubectl ko nb status
306b
Name: OVN_Northbound
Cluster ID: 9a87 (9a872522-3e7d-47ca-83a3-d74333e1a7ca)
Server ID: 306b (306b256b-b5e1-4eb0-be91-4ca96adf6bad)
Address: tcp:[172.18.0.2]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 324356 ms ago, reason: timeout
Last Election won: 324356 ms ago
Election timer: 5000
Log: [140, 140]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-85d6 ->85d6
Disconnections: 2
Servers:
    85d6 (85d6 at tcp:[172.18.0.4]:6643) next_index=140 match_index=139 last msg 848 ms ago
    306b (306b at tcp:[172.18.0.2]:6643) (self) next_index=2 match_index=139
status: ok
```

[illegible]

[illegible] OVN [illegible]

```
# kubectl ko nb backup
tar: Removing leading `/' from member names
backup ovn-nb db to /root/ovnnb_db.060223191654183154.backup
```

[illegible]

[illegible]

```
# kubectl ko nb dbstatus
status: ok
```

[illegible] `inconsistent data` [illegible]

[illegible]

[illegible] `inconsistent data` [illegible]

```
# kubectl ko nb restore
deployment.apps/ovn-central scaled
ovn-central original replicas is 3
first nodeIP is 172.18.0.5
ovs-ovn pod on node 172.18.0.5 is ovs-ovn-8jxv9
ovs-ovn pod on node 172.18.0.3 is ovs-ovn-sjzb6
ovs-ovn pod on node 172.18.0.4 is ovs-ovn-t87zk
backup nb db file
restore nb db file, operate in pod ovs-ovn-8jxv9
deployment.apps/ovn-central scaled
finish restore nb db file and ovn-central replicas
recreate ovs-ovn pods
pod "ovs-ovn-8jxv9" deleted
pod "ovs-ovn-sjzb6" deleted
pod "ovs-ovn-t87zk" deleted
```

[nbctl | sbctl] [options ...]

[illegible] OVN [illegible] leader [illegible] `ovn-nbctl` [illegible] `ovn-sbctl` [illegible] OVN [illegible] ovn-nbctl(8) [illegible] ovn-sbctl(8)[illegible]

```
# kubectl ko nbctl show
switch c7cd17e8-ceee-4a91-9bb3-e5a313fe1ece (snat)
    port snat-ovn-cluster
        type: router
        router-port: ovn-cluster-snat
switch 20e0c6d0-023a-4756-aec5-200e0c60f95d (join)
    port node-liumengxin-ovn3-192.168.137.178
        addresses: ["00:00:00:64:FF:A8 100.64.0.4"]
    port node-liumengxin-ovn1-192.168.137.176
        addresses: ["00:00:00:AF:98:62 100.64.0.2"]
    port node-liumengxin-ovn2-192.168.137.177
        addresses: ["00:00:00:D9:58:B8 100.64.0.3"]
    port join-ovn-cluster
        type: router
        router-port: ovn-cluster-join
switch 0191705c-f827-427b-9de3-3c3b7d971ba5 (central)
    port central-ovn-cluster
        type: router
        router-port: ovn-cluster-central
switch 2a45ff05-388d-4f85-9daf-e6fccd5833dc (ovn-default)
    port alertmanager-main-0.monitoring
        addresses: ["00:00:00:6C:DF:A3 10.16.0.19"]
    port kube-state-metrics-5d6885d89-4nf8h.monitoring
        addresses: ["00:00:00:6F:02:1C 10.16.0.15"]
    port fake-kubelet-67c55dfd89-pv86k.kube-system
        addresses: ["00:00:00:5C:12:E8 10.16.19.177"]
    port ovn-default-ovn-cluster
        type: router
        router-port: ovn-cluster-ovn-default
router 212f73dd-d63d-4d72-864b-a537e9afbee1 (ovn-cluster)
    port ovn-cluster-snat
        mac: "00:00:00:7A:82:8F"
        networks: ["172.22.0.1/16"]
    port ovn-cluster-join
        mac: "00:00:00:F8:18:5A"
        networks: ["100.64.0.1/16"]
    port ovn-cluster-central
        mac: "00:00:00:4D:8C:F5"
        networks: ["192.101.0.1/16"]
    port ovn-cluster-ovn-default
        mac: "00:00:00:A3:F8:18"
        networks: ["10.16.0.1/16"]
```

vsctl {nodeName} [options ...]

[illegible] `nodeName` [illegible] `ovs-ovn` [illegible] `ovs-vsctl` [illegible] `vswitchd` [illegible] OVS [illegible] ovs-vsctl(8)[illegible]

```
# kubectl ko vsctl kube-ovn-01 show
0d4c4675-c9cc-440a-8c1a-878e17f81b88
    Bridge br-int
        fail_mode: secure
        datapath_type: system
        Port a2c1a8a8b83a_h
            Interface a2c1a8a8b83a_h
        Port "4fa5c4cbb1a5_h"
            Interface "4fa5c4cbb1a5_h"
        Port ovn-eef07d-0
            Interface ovn-eef07d-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.178"}
        Port ovn0
```

```
            Interface ovn0
                type: internal
        Port mirror0
            Interface mirror0
                type: internal
        Port ovn-efa253-0
            Interface ovn-efa253-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.177"}
        Port br-int
            Interface br-int
                type: internal
    ovs_version: "2.17.2"
```

ofctl {nodeName} [options ...]

□□□□□□□□ `nodeName` □□ `ovs-ovn` □□□□□□□□□ `ovs-ofctl` □□□□□□□□ OpenFlow□ □□□□□□□□□□□□□□□ OVS □□□□□ ovs-ofctl(8)□

```
# kubectl ko ofctl kube-ovn-01 dump-flows br-int
NXST_FLOW reply (xid=0x4): flags=[more]
 cookie=0xcf3429e6, duration=671791.432s, table=0, n_packets=0, n_bytes=0, idle_age=65534, hard_age=65534, priority=100,in_port=2 actions=load:0x4-
>NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x1->NXM_NX_REG14[],resubmit(,8)
 cookie=0xc91413c6, duration=671791.431s, table=0, n_packets=907489, n_bytes=99978275, idle_age=0, hard_age=65534, priority=100,in_port=7 actions=load:0x1-
>NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x4->NXM_NX_REG14[],resubmit(,8)
 cookie=0xf180459, duration=671791.431s, table=0, n_packets=17348582, n_bytes=2667811214, idle_age=0, hard_age=65534, priority=100,in_port=6317
actions=load:0xa->NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x9->NXM_NX_REG14[],resubmit(,8)
 cookie=0x7806dd90, duration=671791.431s, table=0, n_packets=3235428, n_bytes=833821312, idle_age=0, hard_age=65534, priority=100,in_port=1 actions=load:0xd-
>NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x3->NXM_NX_REG14[],resubmit(,8)
...
```

dpctl {nodeName} [options ...]

□□□□□□□□ `nodeName` □□ `ovs-ovn` □□□□□□□□□ `ovs-dpctl` □□□□□□□□ OVS datapath□ □□□□□□□□□□□□□□□ OVS □□□□□ ovs-dpctl(8)□

```
# kubectl ko dpctl kube-ovn-01 show
system@ovs-system:
  lookups: hit:350805055 missed:21983648 lost:73
  flows: 105
  masks: hit:1970748791 total:22 hit/pkt:5.29
  port 0: ovs-system (internal)
  port 1: ovn0 (internal)
  port 2: mirror0 (internal)
  port 3: br-int (internal)
  port 4: stt_sys_7471 (stt: packet_type=ptap)
  port 5: eeb4d9e51b5d_h
  port 6: a2c1a8a8b83a_h
  port 7: 4fa5c4cbb1a5_h
```

appctl {nodeName} [options ...]

□□□□□□□□ `nodeName` □□ `ovs-ovn` □□□□□□□□□ `ovs-appctl` □□□□□□□□ daemon □□□ □□□□□□□□□□□□□□□ OVS □□□□□ ovs-appctl(8)□

```
# kubectl ko appctl kube-ovn-01 vlog/list
                 console    syslog    file
                 -------    ------    ------
backtrace          OFF        ERR       INFO
bfd                OFF        ERR       INFO
bond               OFF        ERR       INFO
bridge             OFF        ERR       INFO
bundle             OFF        ERR       INFO
bundles            OFF        ERR       INFO
...
```

tcpdump {namespace/podname} [tcpdump options ...]

□□□□□□ `namespace/podname` □□□□ `kube-ovn-cni` □□□□□□ `tcpdump` □□□□□□ veth □□ □□□□□□□□□□□□□□□□□□□□□□□□

```
# kubectl ko tcpdump default/ds1-l6n7p icmp
+ kubectl exec -it kube-ovn-cni-wlg4s -n kube-ovn -- tcpdump -nn -i d7176fe7b4e0_h icmp
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on d7176fe7b4e0_h, link-type EN10MB (Ethernet), capture size 262144 bytes
06:52:36.619688 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 1, length 64
06:52:36.619746 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 1, length 64
06:52:37.619588 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 2, length 64
06:52:37.619630 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 2, length 64
06:52:38.619933 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 3, length 64
06:52:38.619973 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 3, length 64
```

trace [arguments ...]

[illegible] Pod [illegible] OVN [illegible] Openflow [illegible]

[illegible]

```
kubectl ko trace {namespace/podname} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]
kubectl ko trace {namespace/podname} {target ip address} [target mac address] arp {request|reply}
kubectl ko trace {node//nodename} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]
kubectl ko trace {node//nodename} {target ip address} [target mac address] arp {request|reply}
```

[illegible]

```
# kubectl ko trace default/ds1-l6n7p 8.8.8.8 icmp
+ kubectl exec ovn-central-5bc494cb5-np9hm -n kube-ovn -- ovn-trace --ct=new ovn-default 'inport == "ds1-l6n7p.default" && ip.ttl == 64 && icmp && eth.src == 
0a:00:00:10:00:05 && ip4.src == 10.16.0.4 && eth.dst == 00:00:00:B8:CA:43 && ip4.dst == 8.8.8.8'
#
icmp,reg14=0xf,vlan_tci=0x0000,dl_src=0a:00:00:10:00:05,dl_dst=00:00:00:b8:ca:43,nw_src=10.16.0.4,nw_dst=8.8.8.8,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=0,icmp_
code=0

ingress(dp="ovn-default", inport="ds1-l6n7p.default")
-----------------------------------------------------
 0. ls_in_port_sec_l2 (ovn-northd.c:4143): inport == "ds1-l6n7p.default" && eth.src == {0a:00:00:10:00:05}, priority 50, uuid 39453393
    next;
 1. ls_in_port_sec_ip (ovn-northd.c:2898): inport == "ds1-l6n7p.default" && eth.src == 0a:00:00:10:00:05 && ip4.src == {10.16.0.4}, priority 90, uuid 81bcd485
    next;
 3. ls_in_pre_acl (ovn-northd.c:3269): ip, priority 100, uuid 7b4f4971
    reg0[0] = 1;
    next;
 5. ls_in_pre_stateful (ovn-northd.c:3396): reg0[0] == 1, priority 100, uuid 36cdd577
    ct_next;

ct_next(ct_state=new|trk)
-------------------------
 6. ls_in_acl (ovn-northd.c:3759): ip && (!ct.est || (ct.est && ct_label.blocked == 1)), priority 1, uuid 7608af5b
    reg0[1] = 1;
    next;
10. ls_in_stateful (ovn-northd.c:3995): reg0[1] == 1, priority 100, uuid 2aba1b90
    ct_commit(ct_label=0/0x1);
    next;
16. ls_in_l2_lkup (ovn-northd.c:4470): eth.dst == 00:00:00:b8:ca:43, priority 50, uuid 5c9c3c9f
    outport = "ovn-default-ovn-cluster";
    output;

...
```

[illegible] trace [illegible] Underlay [illegible] Mac [illegible]

```
kubectl ko trace default/virt-handler-7lvml 8.8.8.8 82:7c:9f:83:8c:01 icmp
```

diagnose {all|node|subnet|IPPorts} [nodename|subnetName|{proto1}-{IP1}-{Port1},{proto2}-{IP2}-{Port2}]

[illegible] `kube-ovn-pinger` [illegible]

```
# kubectl ko diagnose all
switch c7cd17e8-ceee-4a91-9bb3-e5a313fe1ece (snat)
    port snat-ovn-cluster
        type: router
        router-port: ovn-cluster-snat
switch 20e0c6d0-023a-4756-aec5-200e0c60f95d (join)
    port node-liumengxin-ovn3-192.168.137.178
        addresses: ["00:00:00:64:FF:A8 100.64.0.4"]
    port node-liumengxin-ovn1-192.168.137.176
        addresses: ["00:00:00:AF:98:62 100.64.0.2"]
    port join-ovn-cluster
        type: router
        router-port: ovn-cluster-join
switch 0191705c-f827-427b-9de3-3c3b7d971ba5 (central)
    port central-ovn-cluster
        type: router
        router-port: ovn-cluster-central
switch 2a45ff05-388d-4f85-9daf-e6fccd5833dc (ovn-default)
    port ovn-default-ovn-cluster
        type: router
        router-port: ovn-cluster-ovn-default
    port prometheus-k8s-1.monitoring
        addresses: ["00:00:00:AA:37:DF 10.16.0.23"]
router 212f73dd-d63d-4d72-864b-a537e9afbee1 (ovn-cluster)
    port ovn-cluster-snat
        mac: "00:00:00:7A:82:8F"
        networks: ["172.22.0.1/16"]
    port ovn-cluster-join
        mac: "00:00:00:F8:18:5A"
        networks: ["100.64.0.1/16"]
    port ovn-cluster-central
        mac: "00:00:00:4D:8C:F5"
        networks: ["192.101.0.1/16"]
    port ovn-cluster-ovn-default
        mac: "00:00:00:A3:F8:18"
        networks: ["10.16.0.1/16"]
Routing Policies
     31000                            ip4.dst == 10.16.0.0/16           allow
     31000                           ip4.dst == 100.64.0.0/16           allow
     30000                         ip4.dst == 192.168.137.177         reroute                100.64.0.3
     30000                         ip4.dst == 192.168.137.178         reroute                100.64.0.4
     29000                 ip4.src == $ovn.default.fake.6_ip4         reroute               100.64.0.22
     29000                 ip4.src == $ovn.default.fake.7_ip4         reroute               100.64.0.21
     29000                 ip4.src == $ovn.default.fake.8_ip4         reroute               100.64.0.23
     29000 ip4.src == $ovn.default.liumengxin.ovn3.192.168.137.178_ip4         reroute                100.64.0.4
     20000 ip4.src == $ovn.default.liumengxin.ovn1.192.168.137.176_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPv4         reroute
100.64.0.2
     20000 ip4.src == $ovn.default.liumengxin.ovn2.192.168.137.177_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPv4         reroute
100.64.0.3
     20000 ip4.src == $ovn.default.liumengxin.ovn3.192.168.137.178_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPv4         reroute
100.64.0.4
IPv4 Routes
Route Table <main>:
                0.0.0.0/0                100.64.0.1 dst-ip
UUID                                    LB                  PROTO      VIP                     IPs
e9bcfd9d-793e-4431-9073-6dec96b75d71    cluster-tcp-load    tcp        10.100.209.132:10660    192.168.137.176:10660
                                                            tcp        10.101.239.192:6641     192.168.137.177:6641
                                                            tcp        10.101.240.101:3000     10.16.0.7:3000
                                                            tcp        10.103.184.186:6642     192.168.137.177:6642
35d2b7a5-e3a7-485a-a4b7-b4970eb0e63b    cluster-tcp-sess    tcp        10.100.158.128:8080     10.16.0.10:8080,10.16.0.5:8080,10.16.63.30:8080
                                                            tcp        10.107.26.215:8080      10.16.0.19:8080,10.16.0.20:8080,10.16.0.21:8080
                                                            tcp        10.107.26.215:9093      10.16.0.19:9093,10.16.0.20:9093,10.16.0.21:9093
                                                            tcp        10.98.187.99:8080       10.16.0.22:8080,10.16.0.23:8080
                                                            tcp        10.98.187.99:9090       10.16.0.22:9090,10.16.0.23:9090
f43303e4-89aa-4d3e-a3dc-278a552fe27b    cluster-udp-load    udp        10.96.0.10:53           10.16.0.4:53,10.16.0.9:53
_uuid               : 06776304-5a96-43ed-90c4-c4854c251699
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : node_liumengxin_ovn2_192.168.137.177_underlay_v6

_uuid               : 62690625-87d5-491c-8675-9fd83b1f433c
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : node_liumengxin_ovn1_192.168.137.176_underlay_v6

_uuid               : b03a9bae-94d5-4562-b34c-b5f6198e180b
addresses           : ["10.16.0.0/16", "100.64.0.0/16", "172.22.0.0/16", "192.101.0.0/16"]
external_ids        : {vendor=kube-ovn}
name                : ovn.cluster.overlay.subnets.IPv4

_uuid               : e1056f3a-24cc-4666-8a91-75ee6c3c2426
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : ovn.cluster.overlay.subnets.IPv6

_uuid               : 3e5d5fff-e670-47b2-a2f5-a39f4698a8c5
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : node_liumengxin_ovn3_192.168.137.178_underlay_v6
_uuid               : 2d85dbdc-d0db-4abe-b19e-cc806d32b492
action              : drop
direction           : from-lport
external_ids        : {}
label               : 0
log                 : false
match               : "inport==@ovn.sg.kubeovn_deny_all && ip"
meter               : []
name                : []
options             : {}
priority            : 2003
severity            : []

_uuid               : de790cc8-f155-405f-bb32-5a51f30c545f
action              : drop
direction           : to-lport
external_ids        : {}
label               : 0
log                 : false
match               : "outport==@ovn.sg.kubeovn_deny_all && ip"
meter               : []
name                : []
options             : {}
priority            : 2003
severity            : []
Chassis "e15ed4d4-1780-4d50-b09e-ea8372ed48b8"
    hostname: liumengxin-ovn1-192.168.137.176
    Encap stt
        ip: "192.168.137.176"
        options: {csum="true"}
    Port_Binding node-liumengxin-ovn1-192.168.137.176
    Port_Binding perf-6vxkn.default
    Port_Binding kube-state-metrics-5d6885d89-4nf8h.monitoring
    Port_Binding alertmanager-main-0.monitoring
    Port_Binding kube-ovn-pinger-6ftdf.kube-system
    Port_Binding fake-kubelet-67c55dfd89-pv86k.kube-system
    Port_Binding prometheus-k8s-0.monitoring
Chassis "eef07da1-f8ad-4775-b14d-bd6a3b4eb0d5"
    hostname: liumengxin-ovn3-192.168.137.178
    Encap stt
        ip: "192.168.137.178"
        options: {csum="true"}
    Port_Binding kube-ovn-pinger-7twb4.kube-system
    Port_Binding prometheus-adapter-86df476d87-rl88g.monitoring
    Port_Binding prometheus-k8s-1.monitoring
    Port_Binding node-liumengxin-ovn3-192.168.137.178
    Port_Binding perf-ff475.default
    Port_Binding alertmanager-main-1.monitoring
    Port_Binding blackbox-exporter-676d976865-tvsjd.monitoring
Chassis "efa253c9-494d-4719-83ae-b48ab0f11c03"
    hostname: liumengxin-ovn2-192.168.137.177
    Encap stt
        ip: "192.168.137.177"
        options: {csum="true"}
    Port_Binding grafana-6c4c6b8fb7-pzd2c.monitoring
    Port_Binding node-liumengxin-ovn2-192.168.137.177
    Port_Binding alertmanager-main-2.monitoring
    Port_Binding coredns-6789c94dd8-9jqsz.kube-system
    Port_Binding coredns-6789c94dd8-25d4r.kube-system
    Port_Binding prometheus-operator-7bbc99fc8b-wgjm4.monitoring
    Port_Binding prometheus-adapter-86df476d87-gdxmc.monitoring
    Port_Binding perf-fjnws.default
    Port_Binding kube-ovn-pinger-vh2xg.kube-system
ds kube-proxy ready
kube-proxy ready
deployment ovn-central ready
deployment kube-ovn-controller ready
ds kube-ovn-cni ready
ds ovs-ovn ready
deployment coredns ready
ovn-nb leader check ok
ovn-sb leader check ok
ovn-northd leader check ok
### kube-ovn-controller recent log

### start to diagnose node liumengxin-ovn1-192.168.137.176
#### ovn-controller log:
2022-06-03T00:56:44.897Z|16722|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:06:44.912Z|16723|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:16:44.925Z|16724|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:26:44.936Z|16725|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:36:44.959Z|16726|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:46:44.974Z|16727|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:56:44.988Z|16728|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:06:45.001Z|16729|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:16:45.025Z|16730|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:26:45.040Z|16731|inc_proc_eng|INFO|User triggered force recompute.

#### ovs-vswitchd log:
2022-06-02T23:03:00.137Z|00079|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:f9d1
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-02T23:23:31.840Z|00080|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:15b2
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T00:09:15.659Z|00081|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:dc:e3:63,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.63.30,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:e5a5
 with metadata skb_priority(0),tunnel(tun_id=0x150017000004,src=192.168.137.178,dst=192.168.137.176,ttl=64,tp_src=9239,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.63.30,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T00:30:13.409Z|00064|dpif(handler2)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:6b4a
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T02:02:33.832Z|00082|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:a819
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0

#### ovs-vsctl show results:
0d4c4675-c9cc-440a-8c1a-878e17f81b88
    Bridge br-int
        fail_mode: secure
        datapath_type: system
        Port a2c1a8a8b83a_h
            Interface a2c1a8a8b83a_h
        Port "4fa5c4cbb1a5_h"
            Interface "4fa5c4cbb1a5_h"
        Port ovn-eef07d-0
            Interface ovn-eef07d-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.178"}
        Port ovn0
```

```
            Interface ovn0
                type: internal
        Port "04d03360e9a0_h"
            Interface "04d03360e9a0_h"
        Port eeb4d9e51b5d_h
            Interface eeb4d9e51b5d_h
        Port mirror0
            Interface mirror0
                type: internal
        Port "8e5d887ccd80_h"
            Interface "8e5d887ccd80_h"
        Port ovn-efa253-0
            Interface ovn-efa253-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.177"}
        Port "17512d5be1f1_h"
            Interface "17512d5be1f1_h"
        Port br-int
            Interface br-int
                type: internal
    ovs_version: "2.17.2"

#### pinger diagnose results:
I0603 10:35:04.349404   17619 pinger.go:19]
-------------------------------------------------------------------------------
Kube-OVN:
  Version:       v1.17.0
  Build:         2022-04-24_08:02:50
  Commit:        git-73f9d15
  Go Version:    go1.17.8
  Arch:          amd64
-------------------------------------------------------------------------------
I0603 10:35:04.376797   17619 config.go:166] pinger config is &{KubeConfigFile: KubeClient:0xc000493380 Port:8080 DaemonSetNamespace:kube-system DaemonSetName:kube-ovn-pinger Interval:5 Mode:job ExitCode:0 InternalDNS:kubernetes.default ExternalDNS: NodeName:liumengxin-ovn1-192.168.137.176 HostIP:192.168.137.176 PodName:kube-ovn-pinger-6ftdf PodIP:10.16.0.10 PodProtocols:[IPv4] ExternalAddress: NetworkMode:kube-ovn PollTimeout:2 PollInterval:15 SystemRunDir:/var/run/openvswitch DatabaseVswitchName:Open_vSwitch DatabaseVswitchSocketRemote:unix:/var/run/openvswitch/db.sock DatabaseVswitchFileDataPath:/etc/openvswitch/conf.db DatabaseVswitchFileLogPath:/var/log/openvswitch/ovsdb-server.log DatabaseVswitchFilePidPath:/var/run/openvswitch/ovsdb-server.pid DatabaseVswitchFileSystemIDPath:/etc/openvswitch/system-id.conf ServiceVswitchdFileLogPath:/var/log/openvswitch/ovs-vswitchd.log ServiceVswitchdFilePidPath:/var/run/openvswitch/ovs-vswitchd.pid ServiceOvnControllerFileLogPath:/var/log/ovn/ovn-controller.log ServiceOvnControllerFilePidPath:/var/run/ovn/ovn-controller.pid}
I0603 10:35:04.449166   17619 exporter.go:75] liumengxin-ovn1-192.168.137.176: exporter connect successfully
I0603 10:35:04.554011   17619 ovn.go:21] ovs-vswitchd and ovsdb are up
I0603 10:35:04.651293   17619 ovn.go:33] ovn_controller is up
I0603 10:35:04.651342   17619 ovn.go:39] start to check port binding
I0603 10:35:04.749613   17619 ovn.go:135] chassis id is 1d7f3d6c-eec5-4b3c-adca-2969d9cdfd80
I0603 10:35:04.763487   17619 ovn.go:49] port in sb is [node-liumengxin-ovn1-192.168.137.176 perf-6vxkn.default kube-state-metrics-5d6885d89-4nf8h.monitoring alertmanager-main-0.monitoring kube-ovn-pinger-6ftdf.kube-system fake-kubelet-67c55dfd89-pv86k.kube-system prometheus-k8s-0.monitoring]
I0603 10:35:04.763583   17619 ovn.go:61] ovs and ovn-sb binding check passed
I0603 10:35:05.049309   17619 ping.go:259] start to check apiserver connectivity
I0603 10:35:05.053666   17619 ping.go:268] connect to apiserver success in 4.27ms
I0603 10:35:05.053786   17619 ping.go:129] start to check pod connectivity
I0603 10:35:05.249590   17619 ping.go:159] ping pod: kube-ovn-pinger-6ftdf 10.16.0.10, count: 3, loss count 0, average rtt 16.30ms
I0603 10:35:05.354135   17619 ping.go:159] ping pod: kube-ovn-pinger-7twb4 10.16.63.30, count: 3, loss count 0, average rtt 1.81ms
I0603 10:35:05.458460   17619 ping.go:159] ping pod: kube-ovn-pinger-vh2xg 10.16.0.5, count: 3, loss count 0, average rtt 1.92ms
I0603 10:35:05.458523   17619 ping.go:83] start to check node connectivity
```

⊠⊠ diagnose ⊠⊠⊠⊠⊠⊠ subnet ⊠⊠⊠⊠⊠ subnet ⊠⊠⊠ daemonset⊠⊠ `kube-ovn-pinger` ⊠⊠⊠⊠⊠ daemonset ⊠⊠⊠ pod ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ daemonset⊠

⊠⊠ diagnose ⊠⊠⊠⊠⊠⊠ IPPorts ⊠⊠⊠⊠⊠⊠⊠ `kube-ovn-pinger` pod ⊠⊠⊠⊠⊠⊠⊠IP⊠Port ⊠⊠⊠⊠⊠

reload

⊠⊠⊠⊠⊠⊠⊠ Kube-OVN ⊠⊠⊠⊠⊠

```
# kubectl ko reload
pod "ovn-central-8684dd94bd-vzgcr" deleted
Waiting for deployment "ovn-central" rollout to finish: 0 of 1 updated replicas are available...
deployment "ovn-central" successfully rolled out
pod "ovs-ovn-bsnvz" deleted
pod "ovs-ovn-m9b98" deleted
pod "kube-ovn-controller-8459db5ff4-64c62" deleted
Waiting for deployment "kube-ovn-controller" rollout to finish: 0 of 1 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
pod "kube-ovn-cni-2klnh" deleted
pod "kube-ovn-cni-t2jz4" deleted
Waiting for daemon set "kube-ovn-cni" rollout to finish: 0 of 2 updated pods are available...
Waiting for daemon set "kube-ovn-cni" rollout to finish: 1 of 2 updated pods are available...
daemon set "kube-ovn-cni" successfully rolled out
pod "kube-ovn-pinger-ln72z" deleted
pod "kube-ovn-pinger-w8lrk" deleted
Waiting for daemon set "kube-ovn-pinger" rollout to finish: 0 of 2 updated pods are available...
Waiting for daemon set "kube-ovn-pinger" rollout to finish: 1 of 2 updated pods are available...
daemon set "kube-ovn-pinger" successfully rolled out
pod "kube-ovn-monitor-7fb67d5488-7q6zb" deleted
Waiting for deployment "kube-ovn-monitor" rollout to finish: 0 of 1 updated replicas are available...
deployment "kube-ovn-monitor" successfully rolled out
```

log

[illegible] kube-ovn [illegible] Kube-OVN[illegible]OVN[illegible]Open vSwitch [illegible] log [illegible] linux [illegible] debug [illegible]

```
# kubectl ko log all
Collecting kube-ovn logging files
Collecting ovn logging files
Collecting openvswitch logging files
Collecting linux dmesg files
Collecting linux iptables-legacy files
Collecting linux iptables-nft files
Collecting linux route files
Collecting linux link files
Collecting linux neigh files
Collecting linux memory files
Collecting linux top files
Collecting linux sysctl files
Collecting linux netstat files
Collecting linux addr files
Collecting linux ipset files
Collecting linux tcp files
Collected files have been saved in the directory /root/kubectl-ko-log
```

[illegible]

```
# tree kubectl-ko-log/
kubectl-ko-log/
|-- kube-ovn-control-plane
|   |-- kube-ovn
|   |   |-- kube-ovn-cni.log
|   |   |-- kube-ovn-monitor.log
|   |   `-- kube-ovn-pinger.log
|   |-- linux
|   |   |-- addr.log
|   |   |-- dmesg.log
|   |   |-- ipset.log
|   |   |-- iptables-legacy.log
|   |   |-- iptables-nft.log
|   |   |-- link.log
|   |   |-- memory.log
|   |   |-- neigh.log
|   |   |-- netstat.log
|   |   |-- route.log
|   |   |-- sysctl.log
|   |   |-- tcp.log
|   |   `-- top.log
|   |-- openvswitch
|   |   |-- ovs-vswitchd.log
|   |   `-- ovsdb-server.log
|   `-- ovn
|       |-- ovn-controller.log
|       |-- ovn-northd.log
|       |-- ovsdb-server-nb.log
|       `-- ovsdb-server-sb.log
```

perf [image]

[illegible] Kube-OVN [illegible]

1. [illegible]
2. Hostnetwork [illegible]
3. [illegible]
4. OVN-NB, OVN-SB, OVN-Northd leader [illegible]

[illegible] image [illegible] Pod [illegible] `docker.io/kubeovn/test:v1.13.0` [illegible]

```
# kubectl ko perf
============================== Prepareing Performance Test Resources ===============================
pod/test-client created
pod/test-host-client created
pod/test-server created
pod/test-host-server created
service/test-server created
pod/test-client condition met
pod/test-host-client condition met
pod/test-host-server condition met
pod/test-server condition met
====================================================================================================
============================ Start Pod Network Unicast Performance Test ============================
Size            TCP Latency     TCP Bandwidth   UDP Latency     UDP Lost Rate   UDP Bandwidth
64              82.8 us         97.7 Mbits/sec  67.6 us         (0%)            8.42 Mbits/sec
128             85.4 us         167 Mbits/sec   67.2 us         (0%)            17.2 Mbits/sec
```

```
512             85.8 us         440 Mbits/sec   68.7 us         (0%)            68.4 Mbits/sec
1k              85.1 us         567 Mbits/sec   68.7 us         (0%)            134 Mbits/sec
4k              138 us          826 Mbits/sec   78.1 us         (1.4%)          503 Mbits/sec
====================================================================================================
=============================== Start Host Network Performance Test ================================
Size            TCP Latency     TCP Bandwidth   UDP Latency     UDP Lost Rate   UDP Bandwidth
64              49.7 us         120 Mbits/sec   37.9 us         (0%)            18.6 Mbits/sec
128             49.7 us         200 Mbits/sec   38.1 us         (0%)            35.5 Mbits/sec
512             51.9 us         588 Mbits/sec   38.9 us         (0%)            142 Mbits/sec
1k              51.7 us         944 Mbits/sec   37.2 us         (0%)            279 Mbits/sec
4k              74.9 us         1.66 Gbits/sec  39.9 us         (0%)            1.20 Gbits/sec
====================================================================================================
============================== Start Service Network Performance Test ==============================
Size            TCP Latency     TCP Bandwidth   UDP Latency     UDP Lost Rate   UDP Bandwidth
64              111 us          96.3 Mbits/sec  88.4 us         (0%)            7.59 Mbits/sec
128             83.7 us         150 Mbits/sec   69.2 us         (0%)            16.9 Mbits/sec
512             87.4 us         374 Mbits/sec   75.8 us         (0%)            60.9 Mbits/sec
1k              88.2 us         521 Mbits/sec   73.1 us         (0%)            123 Mbits/sec
4k              148 us          813 Mbits/sec   77.6 us         (0.0044%)       451 Mbits/sec
====================================================================================================
=========================== Start Pod Multicast Network Performance Test ===========================
Size            UDP Latency     UDP Lost Rate   UDP Bandwidth
64              0.014 ms        (0.17%)         5.80 Mbits/sec
128             0.012 ms        (0%)            11.4 Mbits/sec
512             0.016 ms        (0%)            46.1 Mbits/sec
1k              0.023 ms        (0.073%)        89.8 Mbits/sec
4k              0.035 ms        (1.3%)          126 Mbits/sec
====================================================================================================
============================= Start Host Multicast Network Performance ==============================
Size            UDP Latency     UDP Lost Rate   UDP Bandwidth
64              0.007 ms        (0%)            9.95 Mbits/sec
128             0.005 ms        (0%)            21.8 Mbits/sec
512             0.008 ms        (0%)            86.8 Mbits/sec
1k              0.013 ms        (0.045%)        168 Mbits/sec
4k              0.010 ms        (0.31%)         242 Mbits/sec
====================================================================================================
================================= Start Leader Recover Time Test ===================================
Delete ovn central nb pod
pod "ovn-central-5cb9c67d75-tlz9w" deleted
Waiting for ovn central nb pod running
=============================== OVN nb Recovery takes 3.305236803 s ================================
Delete ovn central sb pod
pod "ovn-central-5cb9c67d75-szx4c" deleted
Waiting for ovn central sb pod running
=============================== OVN sb Recovery takes 3.462698535 s ================================
Delete ovn central northd pod
pod "ovn-central-5cb9c67d75-zqmqv" deleted
Waiting for ovn central northd pod running
============================= OVN northd Recovery takes 2.691291403 s ==============================
====================================================================================================
================================= Remove Performance Test Resource =================================
rm -f unicast-test-client.log
rm -f unicast-test-host-client.log
rm -f unicast-test-client.log
kubectl ko nbctl lb-del test-server
rm -f multicast-test-server.log
kubectl exec ovs-ovn-gxdrf -n kube-system -- ip maddr del 01:00:5e:00:00:64 dev eth0
kubectl exec ovs-ovn-h57bf -n kube-system -- ip maddr del 01:00:5e:00:00:64 dev eth0
rm -f multicast-test-host-server.log
pod "test-client" deleted
pod "test-host-client" deleted
pod "test-host-server" deleted
pod "test-server" deleted
service "test-server" deleted
====================================================================================================
```

PDF **Slack** **Support**

2026□4□28□

2022□5□24□

GitHub

6.1.3 □□

6.2 [illegible]

[illegible] Kubernetes [illegible] `ovs-ovn` [illegible] `ovn-controller` [illegible] `ovn-central` [illegible] Kubernetes [illegible]

[illegible] `ovn-central` [illegible] ovn-central [illegible]

6.2.1 [illegible]

```
# kubectl drain kube-ovn-worker --ignore-daemonsets --force
node/kube-ovn-worker cordoned
WARNING: ignoring DaemonSet-managed Pods: kube-system/kube-ovn-cni-zt74b, kube-system/kube-ovn-pinger-5rxfs, kube-system/kube-proxy-jpmnm, kube-system/ovs-
ovn-v2kll
evicting pod kube-system/coredns-64897985d-qsgpt
evicting pod local-path-storage/local-path-provisioner-5ddd94ff66-llss6
evicting pod kube-system/kube-ovn-controller-8459db5ff4-94lxb
pod/kube-ovn-controller-8459db5ff4-94lxb evicted
pod/coredns-64897985d-qsgpt evicted
pod/local-path-provisioner-5ddd94ff66-llss6 evicted
node/kube-ovn-worker drained
```

6.2.2 [illegible] kubelet [illegible] docker

[illegible] `ovs-ovn` [illegible] `ovn-central` [illegible]

```
systemctl stop kubelet
systemctl stop docker
```

Containerd

> [illegible] containerd [illegible] CRI [illegible] Pod/[illegible] `ovs-ovn` [illegible]

[illegible] containerd [illegible] `ovs-ovn` [illegible]

1. [illegible] kubelet

   ```
   systemctl stop kubelet
   ```

2. [illegible] ovs-ovn Pod

   ```
   crictl rm -f $(crictl ps | grep openvswitch | awk '{print $1}')
   ```

3. [illegible] containerd

   ```
   systemctl stop containerd
   ```

6.2.3 [illegible] Node [illegible]

```
rm -rf /var/run/openvswitch
rm -rf /var/run/ovn
rm -rf /etc/origin/openvswitch/
rm -rf /etc/origin/ovn/
rm -rf /etc/cni/net.d/00-kube-ovn.conflist
rm -rf /etc/cni/net.d/01-kube-ovn.conflist
rm -rf /var/log/openvswitch
rm -rf /var/log/ovn
```

6.2.4 [illegible] kubectl [illegible]

```
kubectl delete no kube-ovn-01
```

6.2.5 □□□□□□□□□ ovn-sb □□□

□□□□□ `kube-ovn-worker` □□□□□□

```
# kubectl ko sbctl show
Chassis "b0564934-5a0d-4804-a4c0-476c93596a17"
  hostname: kube-ovn-worker
  Encap geneve
      ip: "172.18.0.2"
      options: {csum="true"}
  Port_Binding kube-ovn-pinger-5rxfs.kube-system
Chassis "6a29de7e-d731-4eaf-bacd-2f239ee52b28"
  hostname: kube-ovn-control-plane
  Encap geneve
      ip: "172.18.0.3"
      options: {csum="true"}
  Port_Binding coredns-64897985d-nbfln.kube-system
  Port_Binding node-kube-ovn-control-plane
  Port_Binding local-path-provisioner-5ddd94ff66-h4tn9.local-path-storage
  Port_Binding kube-ovn-pinger-hf2p6.kube-system
  Port_Binding coredns-64897985d-fhwlw.kube-system
```

6.2.6 □□□□□□ chassis □□□□□□□□□□□

uuid □□□□□□□□□ Chassis □□ id□

```
# kubectl ko sbctl chassis-del b0564934-5a0d-4804-a4c0-476c93596a17
# kubectl ko sbctl show
Chassis "6a29de7e-d731-4eaf-bacd-2f239ee52b28"
  hostname: kube-ovn-control-plane
  Encap geneve
      ip: "172.18.0.3"
      options: {csum="true"}
  Port_Binding coredns-64897985d-nbfln.kube-system
  Port_Binding node-kube-ovn-control-plane
  Port_Binding local-path-provisioner-5ddd94ff66-h4tn9.local-path-storage
  Port_Binding kube-ovn-pinger-hf2p6.kube-system
  Port_Binding coredns-64897985d-fhwlw.kube-system
```

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6.2.7 □□

6.3 □□ ovn-central □□

□□ `ovn-central` □□ `ovn-nb` □ `ovn-sb` □□□□□□□ etcd □ Raft □□□□□□□ `ovn-central` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□

6.3.1 ovn-central □□□□

□□□□□□□□□□□□□□□□ `kube-ovn-control-plane2` □□□□□□□□□□□ `ovn-central` □□□□□□

```
# kubectl -n kube-system get pod -o wide | grep central
ovn-central-6bf58cbc97-2cdhg                      1/1     Running   0             21m   172.18.0.3   kube-ovn-control-plane    <none>           <none>
ovn-central-6bf58cbc97-crmfp                      1/1     Running   0             21m   172.18.0.5   kube-ovn-control-plane2   <none>           <none>
ovn-central-6bf58cbc97-lxmpl                      1/1     Running   0             21m   172.18.0.4   kube-ovn-control-plane3   <none>           <none>
```

□□ ovn-nb □□□□□□□□

□□□□□□□□□□□ ID□□□□□□□□□

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2135194 ms ago, reason: timeout
Last Election won: 2135188 ms ago
Election timer: 5000
Log: [135, 135]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-d64b ->d64b <-4984 ->4984
Disconnections: 0
Servers:
    4984 (4984 at tcp:[172.18.0.4]:6643) next_index=135 match_index=134 last msg 1084 ms ago
    1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=2 match_index=134
    d64b (d64b at tcp:[172.18.0.5]:6643) next_index=135 match_index=134 last msg 1084 ms ago
status: ok
```

`kube-ovn-control-plane2` □□□□ IP □ `172.18.0.5` □□□□□□□ ID □ `d64b` □□□□□ ovn-nb □□□□□□□□□

```
# kubectl ko nb kick d64b
started removal
```

□□□□□□□□□

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2297649 ms ago, reason: timeout
Last Election won: 2297643 ms ago
Election timer: 5000
Log: [136, 136]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-4984 ->4984
Disconnections: 2
Servers:
    4984 (4984 at tcp:[172.18.0.4]:6643) next_index=136 match_index=135 last msg 1270 ms ago
    1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=2 match_index=135
status: ok
```

[illegible] ovn-sb [illegible]

[illegible] ovn-sb [illegible] ID[illegible]

```
kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2395317 ms ago, reason: timeout
Last Election won: 2395316 ms ago
Election timer: 5000
Log: [130, 130]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-e9f7 ->e9f7 <-6e84 ->6e84
Disconnections: 0
Servers:
    e9f7 (e9f7 at tcp:[172.18.0.5]:6644) next_index=130 match_index=129 last msg 1006 ms ago
    6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=130 match_index=129 last msg 1004 ms ago
    3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=2 match_index=129
status: ok
```

`kube-ovn-control-plane2` [illegible] IP [illegible] `172.18.0.5` [illegible] ID [illegible] `e9f7` [illegible] ovn-sb [illegible]

```
# kubectl ko sb kick e9f7
started removal
```

[illegible]

```
# kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2481636 ms ago, reason: timeout
Last Election won: 2481635 ms ago
Election timer: 5000
Log: [131, 131]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-6e84 ->6e84
Disconnections: 2
Servers:
    6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=131 match_index=130 last msg 642 ms ago
    3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=2 match_index=130
status: ok
```

[illegible] ovn-central

[illegible] ovn-central [illegible] `NODE_IPS` [illegible]

```
kubectl label node kube-ovn-control-plane2 kube-ovn/role-
kubectl scale deployment -n kube-system ovn-central --replicas=2
kubectl set env deployment/ovn-central -n kube-system NODE_IPS="172.18.0.3,172.18.0.4"
kubectl rollout status deployment/ovn-central -n kube-system
```

[illegible] ovn-central [illegible]

[illegible] `ovs-ovn` [illegible]

```
# kubectl set env daemonset/ovs-ovn -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4"
daemonset.apps/ovs-ovn env updated
# kubectl delete pod -n kube-system -lapp=ovs
pod "ovs-ovn-4f6jc" deleted
pod "ovs-ovn-csn2w" deleted
pod "ovs-ovn-mpbmb" deleted
```

[illegible] `kube-ovn-controller` [illegible]

```
# kubectl set env deployment/kube-ovn-controller -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4"
deployment.apps/kube-ovn-controller env updated

# kubectl rollout status deployment/kube-ovn-controller -n kube-system
Waiting for deployment "kube-ovn-controller" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "kube-ovn-controller" rollout to finish: 2 of 3 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
```

[illegible]

[illegible] `kube-ovn-control-plane2` [illegible]

```
rm -rf /etc/origin/ovn
```

[illegible] Kubernetes [illegible]

6.3.2 ovn-central [illegible]

[illegible] Kubernetes [illegible] `ovn-central` [illegible]

[illegible]

[illegible] `/etc/origin/ovn` [illegible] `ovnnb_db.db` [illegible] `ovnsb_db.db` [illegible]

```
rm -rf /etc/origin/ovn
```

[illegible] **ovn-central** [illegible]

[illegible] `ovn-central` [illegible]

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 44
Leader: self
Vote: self

Last Election started 1855739 ms ago, reason: timeout
Last Election won: 1855729 ms ago
Election timer: 5000
Log: [147, 147]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: ->4984 <-4984
Disconnections: 0
Servers:
    4984 (4984 at tcp:[172.18.0.4]:6643) next_index=147 match_index=146 last msg 367 ms ago
    1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=140 match_index=146
status: ok

# kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 33
Leader: self
Vote: self

Last Election started 1868589 ms ago, reason: timeout
Last Election won: 1868579 ms ago
Election timer: 5000
Log: [142, 142]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: ->6e84 <-6e84
Disconnections: 0
Servers:
    6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=142 match_index=141 last msg 728 ms ago
    3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=134 match_index=141
status: ok
```

□□□□□□□□□□

□□□□ ovn-central □□□□ `NODE_IPS` □□□□□□□□□□□□□□

```
kubectl label node kube-ovn-control-plane2 kube-ovn/role=master
kubectl scale deployment -n kube-system ovn-central --replicas=3
kubectl set env deployment/ovn-central -n kube-system NODE_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
kubectl rollout status deployment/ovn-central -n kube-system
```

□□□□□□□□ ovn-central □□

□□ `ovs-ovn` □□□□□□□□□□□□□□

```
# kubectl set env daemonset/ovs-ovn -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
daemonset.apps/ovs-ovn env updated
# kubectl delete pod -n kube-system -lapp=ovs
pod "ovs-ovn-4f6jc" deleted
pod "ovs-ovn-csn2w" deleted
pod "ovs-ovn-mpbmb" deleted
```

□□ `kube-ovn-controller` □□□□□□□□□□□□□□

```
# kubectl set env deployment/kube-ovn-controller -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
deployment.apps/kube-ovn-controller env updated

# kubectl rollout status deployment/kube-ovn-controller -n kube-system
Waiting for deployment "kube-ovn-controller" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "kube-ovn-controller" rollout to finish: 2 of 3 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
```

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6.3.3 □□

6.4 OVN [illegible]

[illegible]

6.4.1 [illegible]

[illegible] kubectl [illegible] backup [illegible]

```
# kubectl ko nb backup
tar: Removing leading `/' from member names
backup ovn-nb db to /root/ovnnb_db.060223191654183154.backup

# kubectl ko sb backup
tar: Removing leading `/' from member names
backup ovn-sb db to /root/ovnsb_db.060223191654183154.backup
```

6.4.2 [illegible]

[illegible] [illegible]

[illegible]

[illegible] `/var/log/ovn/ovn-northd.log` [illegible]

```
 * ovn-northd is not running
ovsdb-server: ovsdb error: error reading record 2739 from OVN_Northbound log: record 2739 advances commit index to 6308 but last log index is 6307
 * Starting ovsdb-nb
```

[illegible]

[illegible] `OVN_Northbound` [illegible] `OVN_Southbound` [illegible] [illegible] `OVN_Northbound` [illegible] ovn-nb [illegible]

```
# kubectl ko nb status
9182
Name: OVN_Northbound
Cluster ID: e75f (e75fa340-49ed-45ab-990e-26cb865ebc85)
Server ID: 9182 (9182e8dd-b5b0-4dd8-8518-598cc1e374f3)
Address: tcp:[10.0.128.61]:6643
Status: cluster member
Role: leader
Term: 1454
Leader: self
Vote: self

Last Election started 1732603 ms ago, reason: timeout
Last Election won: 1732587 ms ago
Election timer: 1000
Log: [7332, 12512]
Entries not yet committed: 1
Entries not yet applied: 1
Connections: ->f080 <-f080 <-e631 ->e631
Disconnections: 1
Servers:
    f080 (f080 at tcp:[10.0.129.139]:6643) next_index=12512 match_index=12510 last msg 63 ms ago
    9182 (9182 at tcp:[10.0.128.61]:6643) (self) next_index=10394 match_index=12510
    e631 (e631 at tcp:[10.0.131.173]:6643) next_index=12512 match_index=0
```

[illegible]

```
kubectl ko nb kick e631
```

[illegible]

```
mv /etc/origin/ovn/ovnnb_db.db /tmp
```

[illegible] `ovn-central` Pod[illegible]

```
kubectl delete pod -n kube-system ovn-central-xxxx
```

6.4.3 □□□□□□□□□□□□

□□□□□□□□□□□□□□ leader□□□□□□□□□□□□□□

□□ ovn-central

□□□□ `ovn-central` □□□□□□□□ `ovn-central` □□□□□□□□□□□□□□

```
kubectl scale deployment -n kube-system ovn-central --replicas=0
```

□□□□

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> □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□ `ovsdb-tool cluster-to-standalone` □□□□□□□

□□ `ovn-central` □□□□ `NODE_IPS` □□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□ `/etc/origin/ovn` □□□□□□□□□□ □ □□□□□□□□□□□□□□□

□□□□□□□□□ `docker` □

```
docker run -it -v /etc/origin/ovn:/etc/ovn kubeovn/kube-ovn:v1.17.0 bash
cd /etc/ovn/
ovsdb-tool cluster-to-standalone ovnnb_db_standalone.db ovnnb_db.db
ovsdb-tool cluster-to-standalone ovnsb_db_standalone.db ovnsb_db.db
```

□□□□□ containerd□□ docker□□□□□□ `ctr` □□□□□□□□

```
ctr -n k8s.io image pull docker.io/kubeovn/kube-ovn:v1.17.0
ctr -n k8s.io run --rm -t \
  --mount type=bind,src=/etc/origin/ovn,dst=/etc/ovn,options=rbind:rw \
  docker.io/kubeovn/kube-ovn:v1.17.0 ovn-recover bash
cd /etc/ovn/
ovsdb-tool cluster-to-standalone ovnnb_db_standalone.db ovnnb_db.db
ovsdb-tool cluster-to-standalone ovnsb_db_standalone.db ovnsb_db.db
exit
```

□□□□ `ovn-central` Pod □□□ `ovsdb-tool cluster-to-standalone` □□ `kubectl cp` □□□□□□□

□□□□ ovn-central □□□□□□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

```
mv /etc/origin/ovn/ovnnb_db.db /tmp
mv /etc/origin/ovn/ovnsb_db.db /tmp
```

□□□□□□□

□□□□□□□□□□□□ `ovnnb_db.db` □ `ovnsb_db.db` □□□□□ `ovn-central` □□□□ `NODE_IPS` □□□□□□□ `/etc/origin/ovn/` □□□□

```
mv /etc/origin/ovn/ovnnb_db_standalone.db /etc/origin/ovn/ovnnb_db.db
mv /etc/origin/ovn/ovnsb_db_standalone.db /etc/origin/ovn/ovnsb_db.db
```

□□ `ovn-central` □□□□□

```
kubectl scale deployment -n kube-system ovn-central --replicas=3
kubectl rollout status deployment/ovn-central -n kube-system
```

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6.5 [illegible] CIDR

[illegible] CIDR [illegible]

> [illegible] CIDR [illegible] Pod [illegible] [illegible] CIDR [illegible] Join [illegible] CIDR [illegible][illegible] Join [illegible] CIDR[illegible]

6.5.1 [illegible]

[illegible] `kubectl edit` [illegible] `cidrBlock`[illegible]`gateway` [illegible] `excludeIps`[illegible]

```
kubectl edit subnet test-subnet
```

6.5.2 [illegible] Namespace [illegible] Pod

[illegible] `test` Namespace [illegible]

```
for pod in $(kubectl get pod --no-headers -n test --field-selector spec.restartPolicy=Always -o custom-columns=NAME:.metadata.name,HOST:spec.hostNetwork |
awk '{if ($2!="true") print $1}'); do
  kubectl delete pod "$pod" -n test --ignore-not-found
done
```

[illegible] host [illegible] Pod[illegible]

```
for ns in $(kubectl get ns --no-headers -o custom-columns=NAME:.metadata.name); do
  for pod in $(kubectl get pod --no-headers -n "$ns" --field-selector spec.restartPolicy=Always -o custom-columns=NAME:.metadata.name,HOST:spec.hostNetwork |
awk '{if ($2!="true") print $1}'); do
    kubectl delete pod "$pod" -n "$ns" --ignore-not-found
  done
done
```

6.5.3 [illegible]

[illegible] CIDR [illegible] `kube-ovn-controller` Deployment [illegible]

```
args:
- --default-cidr=10.17.0.0/16
- --default-gateway=10.17.0.1
- --default-exclude-ips=10.17.0.1
```

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2026[illegible]4[illegible]28[illegible]

2022[illegible]5[illegible]24[illegible]

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6.5.4 [illegible]

6.6 □□ Join □□ CIDR

□□□□□□□ Join □□ CIDR □□□□□□□□□□□□□□□□□□□□□□□□□

> □□ Join □□ CIDR □□□□□□□□ Pod □□□□□□□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□

6.6.1 □□ Join □□

```
kubectl patch subnet join --type='json' -p '[{"op": "replace", "path": "/metadata/finalizers", "value": []}]'
kubectl delete subnet join
```

6.6.2 □□□□□□□□

```
kubectl annotate node ovn.kubernetes.io/allocated=false --all --overwrite
```

6.6.3 □□ Join □□□□□□

□□ `kube-ovn-controller` □ Join □□□□□□□□

```
kubectl edit deployment -n kube-system kube-ovn-controller
```

□□□□□□□□

```
args:
- --node-switch-cidr=100.51.0.0/16
```

□□ `kube-ovn-controller` □□ `join` □□□

```
kubectl delete pod -n kube-system -lapp=kube-ovn-controller
```

□□□□ Join □□□□□□

```
# kubectl get subnet
NAME          PROVIDER   VPC           PROTOCOL   CIDR            PRIVATE   NAT     DEFAULT   GATEWAYTYPE   V4USED   V4AVAILABLE   V6USED   V6AVAILABLE
EXCLUDEIPS
join          ovn        ovn-cluster   IPv4       100.51.0.0/16   false     false   false     distributed   2        65531         0        0
["100.51.0.1"]
ovn-default   ovn        ovn-cluster   IPv4       10.17.0.0/16    false     true    true      distributed   5        65528         0        0
["10.17.0.1"]
```

6.6.4 □□□□ ovn0 □□□□

□□□□□ `ovn0` □□□□□□□□□□□□□□□□□□ `kube-ovn-cni` □□□□

```
kubectl delete pod -n kube-system -l app=kube-ovn-cni
```

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6.6.5 □□

6.7 [illegible]

[illegible] `kube-ovn.yaml` [illegible]

```
vi kube-ovn.yaml
# ...
        - name: kube-ovn-controller
          image: "docker.io/kubeovn/kube-ovn:v1.17.0"
          imagePullPolicy: IfNotPresent
          args:
          - /kube-ovn/start-controller.sh
          - --v=3
# ...
#
```

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2025[illegible]9[illegible]10[illegible]

2023[illegible]4[illegible]4[illegible]

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6.7.1 [illegible]

6.8 [illegible]

[illegible] [illegible] [illegible] Kube-OVN [illegible]

6.8.1 [illegible]

1. [illegible]
    - [illegible] Pod [illegible] Pod [illegible]
    - [illegible] Pod [illegible]
    - [illegible] Pod [illegible] Service [illegible]
    - [illegible] Pod [illegible]
    - [illegible] Pod [illegible]
    - [illegible] `kube-ovn-pinger` [illegible]
2. [illegible]
    - [illegible] Pod [illegible]
    - [illegible] kube-ovn-cni [illegible] CNI [illegible]
    - [illegible] ovs-ovn [illegible] OVS [illegible]
    - [illegible] kube-ovn-controller [illegible]
    - [illegible] ovn-central [illegible] OVN [illegible]
    - [illegible] dmesg [illegible]
    - [illegible] `netstat -s` [illegible]
3. [illegible] CPU, [illegible] IO [illegible]

[illegible] `kubectl ko logs` [illegible]

6.8.2 Pod [illegible] IP [illegible]

[illegible]

Pod [illegible] Running [illegible]kubectl describe Pod [illegible] `duplicate IPv4 address <ip> found on logical switch port <port>`

[illegible]

1. [illegible] Pod [illegible] IP Pod[illegible] IP [illegible] IP [illegible] IP [illegible] IP Pod
2. [illegible] kube-ovn-controller [illegible] IP [illegible]
3. [illegible] kube-ovn-controller [illegible] IP [illegible]
4. `kubectl ko nbctl show` [illegible] OVN [illegible] IP [illegible]
5. [illegible] OVN [illegible] Kubernetes [illegible] IP [illegible] `kubectl ko nbctl lsp-del <port>` [illegible]

6.8.3 Pod [illegible] ping gateway failed

[illegible]

Pod [illegible] Running [illegible]kubectl describe Pod [illegible] `network <ip> with gateway <gw ip> is not ready for interface eth0 after 30 checks`

[illegible]

1. `kubectl ko sbctl show` [illegible] Pod [illegible]
2. [illegible] ovn-central [illegible] ovs-ovn [illegible] ovn-central [illegible] ovs-ovn [illegible]

3. [illegible] Pod [illegible]
4. [illegible] Underlay [illegible] Underlay [illegible] [illegible]

6.8.4 Pod [illegible]

[illegible]

[illegible] VPC [illegible] Pod [illegible]

[illegible]

1. [illegible] `kubectl ko trace` [illegible] OVN [illegible] ACL [illegible]
2. [illegible] ACL [illegible] ACL [illegible]
3. [illegible] ACL [illegible] Subnet stats [illegible]
4. [illegible] Subnet Spec [illegible]
5. [illegible] `kube-ovn-controller` [illegible]

6.8.5 Pod IP [illegible] CIDR [illegible]

[illegible]

Pod [illegible] IP [illegible] CIDR [illegible]

[illegible]

1. [illegible] `/etc/cni/net.d/` [illegible] Kube-OVN [illegible] CNI [illegible]
2. [illegible]
3. [illegible] kubelet [illegible] Pod

6.8.6 Debug Pod [illegible]

[illegible]

kubectl debug [illegible] Pod [illegible] ContainerCreating [illegible]Pod Event [illegible] network not ready [illegible] no address allocated

[illegible]

[illegible] debug Pod [illegible] yaml [illegible] yaml [illegible] Annotation[illegible]

```
ovn.kubernetes.io/ip_address
ovn.kubernetes.io/mac_address
ovn.kubernetes.io/allocated
ovn.kubernetes.io/routed
```

[illegible] debug Pod[illegible] yaml [illegible] debug Pod[illegible]

6.8.7 ARM [illegible]

[illegible]

ARM [illegible] Offload [illegible]

[illegible] `netstat` [illegible]

```
# netstat -us
IcmpMsg:
    InType0: 22
    InType3: 24
    InType8: 117852
    OutType0: 117852
    OutType3: 29
    OutType8: 22
Udp:
    3040636 packets received
    0 packets to unknown port received.
    4 packet receive errors
    602 packets sent
    0 receive buffer errors
    0 send buffer errors
    InCsumErrors: 4
UdpLite:
IpExt:
    InBcastPkts: 10244
    InOctets: 4446320361
    OutOctets: 1496815600
    InBcastOctets: 3095950
    InNoECTPkts: 7683903
```

[illegible] `InCsumErrors` [illegible]

[illegible]

[illegible] `tx offload` [illegible] TCP [illegible]

```
ethtool -K eth0 tx off
```

[illegible]CentOS 7 [illegible] `4.19.90-25.16.v2101` [illegible]

6.8.8 Pod [illegible] Service [illegible]

[illegible]

Pod [illegible] Service [illegible] `dmesg` [illegible]

```
netlink: Unknown conntrack attr (type=6, max=5)
openvswitch: netlink: Flow actions may not be safe on all matching packets.
```

[illegible] OVS [illegible] NAT [illegible]

[illegible]

1. [illegible] OVS [illegible]
2. [illegible] Overlay [illegible] `kube-ovn-controller` [illegible] `--enable-lb=false` [illegible] OVN LB [illegible] kube-proxy [illegible] Service [illegible]

6.8.9 ovn-central [illegible]

[illegible]

[illegible] v1.11.x [illegible]1 [illegible] Pod [illegible] OVN NB [illegible] SB [illegible] Kube-OVN [illegible] ovsdb-server/compact [illegible]

[illegible]

[illegible] v1.13 [illegible] `ENABLE_COMPACT` [illegible] `false` [illegible] `dist/images/install.sh` [illegible] `start-db.sh` [illegible] ovn-central [illegible] `true` [illegible]

```
- name: ENABLE_COMPACT
  value: "false"
```

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2022年5月24日

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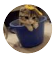


6.8.10 [illegible]

7. [illegible]

7.1 [illegible]

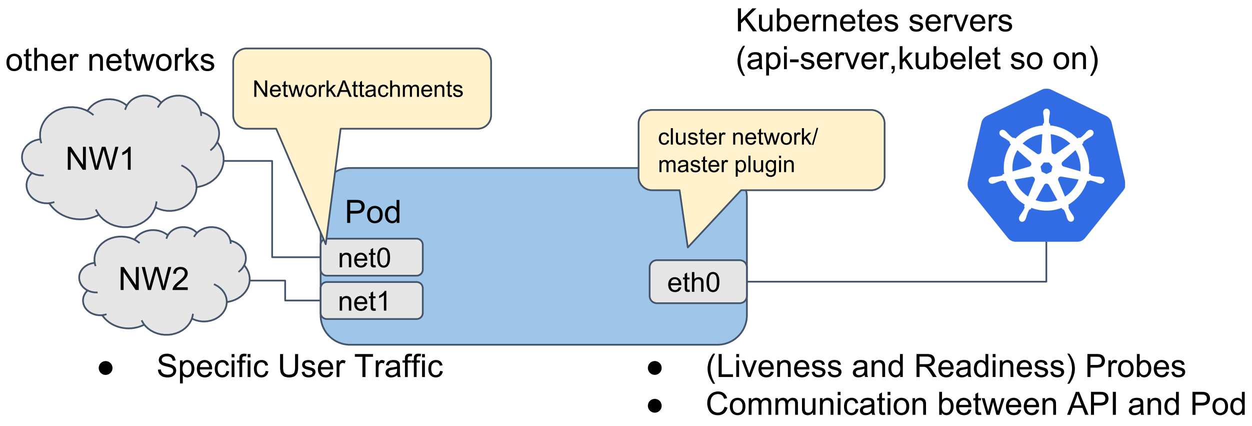
Kube-OVN [illegible] CNI [illegible] macvlan[illegible]vlan[illegible]host-device [illegible] IPAM [illegible] Kube-OVN [illegible] IP [illegible]

Kube-OVN [illegible] Kube-OVN [illegible]

7.1.1 [illegible]

[illegible]

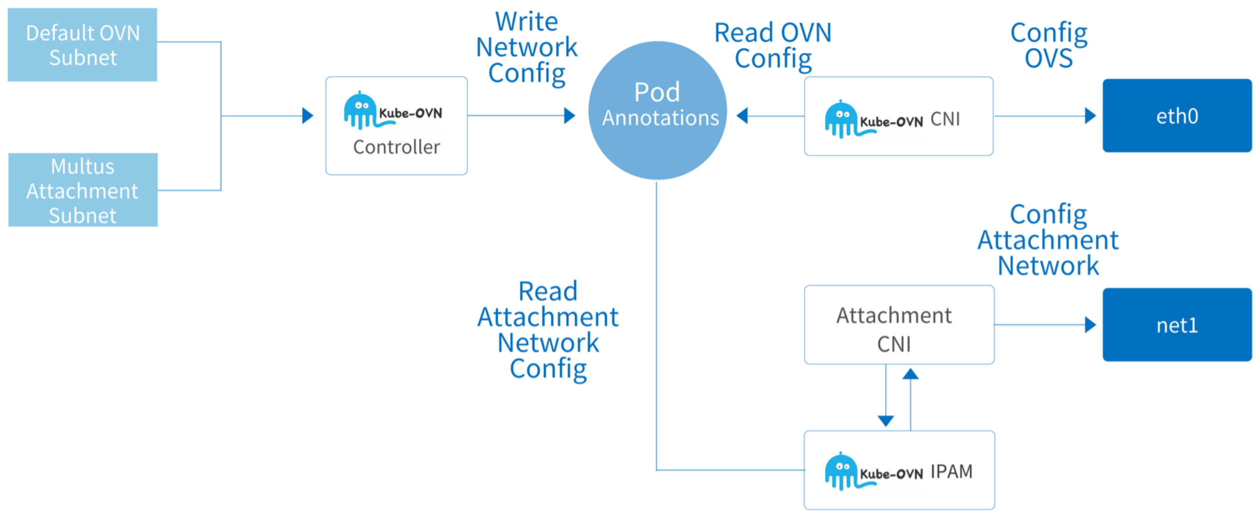
[illegible] Multus CNI [illegible] pod [illegible] pod [illegible]eth0[illegible]net0 [illegible] net1[illegible] eth0 [illegible] kubernetes [illegible] kubernetes [illegible] kubernetes api-server[illegible]kubelet [illegible] net0 [illegible] net1 [illegible] CNI [illegible] vlan/vxlan/ptp[illegible]


other networks
NetworkAttachments
Kubernetes servers
(api-server,kubelet so on)
cluster network/
master plugin
NW1
NW2
Pod
net0
net1
eth0
Specific User Traffic
(Liveness and Readiness) Probes
Communication between API and Pod

IPAM[illegible]

[illegible] Multus CNI[illegible] Pod [illegible] IP [illegible] Kube-OVN [illegible] Subnet [illegible] IP [illegible] CRD [illegible] IP [illegible]IP [illegible] Kube-OVN [illegible] IPAM [illegible]

□□□□


Default OVN Subnet
Multus Attachment Subnet
Kube-OVN Controller
Write Network Config
Pod Annotations
Read OVN Config
Kube-OVN CNI
Config OVS
eth0
Config Attachment Network
Read Attachment Network Config
Attachment CNI
net1
Kube-OVN IPAM

□□□□□□□□□ Kube-OVN □□□□□□□□□□ IP □□□□□□□□ eth0 □□□□ OVN □□□net1 □□□□□□ CNI □□□ net1 □□□□□□□□□□ multus-cni □□ NetworkAttachmentDefinition □□□□□

□ Pod □□□□ `kube-ovn-controller` □□□□ Pod □□□□□□□□ Pod □□ annotation □□□□□□□ Subnet □□□□□ IP □□□□□□□ □□ Pod □□□□□□□□□□ Pod annotation □□

□□□□□□□□CNI □□□□□□□□ `kube-ovn-cni` □□□ ipam □□□ `kube-ovn-cni` □□□□ Pod annotation □□□□□□□□ CNI □□□□□□□□□□□□ CNI □□□

7.1.2 □□□□□

`NetworkAttachmentDefinition` □□ `spec` □□□multus □□□□ `defaultConfDir` □□□□□ CNI □□□□□□□□□□

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: macvlan-conf-2
```

□□ `kube-ovn-controller` □□□□□□□□ `NetworkAttachmentDefinition` □□ `provider` □□□□□□□□□□□□□□□□□□□□□□ `spec` □□□□□□□□□□□□□□□□ `kube-ovn-controller` □□□□□□□ `provider` □□□□□ `spec` □□□□□□ Kube-OVN □ IPAM □□□□□□□□□□

7.1.3 □□□□

□□ Kube-OVN □ Multus

□□□ Kube-OVN □□□□ □ Multus how to use □□□ Kube-OVN □ Multus-CNI□

□□□ CNI □□ IPAM

□□□□□□ Kube-OVN □□□□□□□□□□□□□ CNI□

□□ NETWORKATTACHMENTDEFINITION

□□□□□□ macvlan □□□□□□□□□□□□□□□□ ipam □□□ `kube-ovn` □

```
#    macvlan
sudo modprobe macvlan
```

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: macvlan
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth0",
      "mode": "bridge",
      "ipam": {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "macvlan.default"
      }
    }'
```

- `spec.config.ipam.type`: □□□□□ `kube-ovn` □□□ kube-ovn □□□□□□□□□□□
- `server_socket`: Kube-OVN □□□□□ socket □□□□□□□□□ `/run/openvswitch/kube-ovn-daemon.sock` □
- `provider`: □□ NetworkAttachmentDefinition □ `<name>.<namespace>` □Kube-OVN □□□□□□□□□□□□□□ Subnet □□□
- `master`: □□□□□□□□□

Info

□□□ `provider` □ Underlay □□ `ProviderNetwork` □□□□□□□□□□□□□□□□□□□□□□□□□□

□□□□ KUBE-OVN SUBNET

□□□□ Kube-OVN Subnet□□□□□□ `cidrBlock` □ `exclude_ips` □ `provider` □□□□□□□□ NetworkAttachmentDefinition □ `<name>.<namespace>` □ □□□ macvlan □□ □□□□□□□ Subnet □□□

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan
spec:
  protocol: IPv4
  provider: macvlan.default
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

> □□□ `gateway` □ `private` □ `nat` □□□□ `provider` □□□ ovn □□□□□□□□□□ attachment network□□□□□□□

□□□□□□□□ Pod

□□□□□□□□□ Pod□□□□□□□□ annotation `k8s.v1.cni.cncf.io/networks` □□□□□□□ NetworkAttachmentDefinition □ `<namespace>/<name>` □

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

□□□□ IP □ Pod

□□□□ IP □ Pod□□□ `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/ip_address` annotation□

```
apiVersion: v1
kind: Pod
metadata:
  name: static-ip
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
    ovn.kubernetes.io/ip_address: 10.16.0.15
    ovn.kubernetes.io/mac_address: 00:00:00:53:6B:B6
    macvlan.default.kubernetes.io/ip_address: 172.17.0.100
    macvlan.default.kubernetes.io/mac_address: 00:00:00:53:6B:BB
spec:
  containers:
  - name: static-ip
    image: docker.io/library/nginx:alpine
```

□□□□□□ IP □□□□□

□□□□ ippool □□□□□□□□ `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/ip_pool` annotation□

```
apiVersion: apps/v1
kind: Deployment
metadata:
  namespace: default
  name: static-workload
  labels:
    app: static-workload
spec:
  replicas: 2
  selector:
    matchLabels:
      app: static-workload
  template:
    metadata:
      labels:
        app: static-workload
      annotations:
        k8s.v1.cni.cncf.io/networks: default/macvlan
        ovn.kubernetes.io/ip_pool: 10.16.0.15,10.16.0.16,10.16.0.17
        macvlan.default.kubernetes.io/ip_pool: 172.17.0.200,172.17.0.201,172.17.0.202
    spec:
      containers:
      - name: static-workload
        image: docker.io/library/nginx:alpine
```

□□□□□□□□ macvlan □ Pod

□□□□ macvlan □□□□□□□□ Pod□□□□□□□□□□□□ Pod □□□□□□□□□□□□□□ annotation□□□ `default-route` □□□□□□

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod-route
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{
      "name": "macvlan",
      "namespace": "default",
      "default-route": ["172.17.0.1"]
    }]'
spec:
  containers:
  - name: samplepod-route
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

□□□□□□□ macvlan □ Pod

□□□□ macvlan □□□□□□ Pod□□□□□□□□ annotation `v1.multus-cni.io/default-network` □□□□□□□ NetworkAttachmentDefinition □ `<namespace>/<name>` □

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod-macvlan
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: default/macvlan
spec:
  containers:
  - name: samplepod-macvlan
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

□□□□ KUBE-OVN SUBNET□PROVIDER OVN□

□□□□ `provider` □□□ ovn □ subnet □□□ IP □□□□□□□□ `provider` □ ovn □ Kube-OVN Subnet□□□□□□ `cidrBlock` □ `exclude_ips` □□□ Subnet □□□

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan
spec:
  protocol: IPv4
  provider: ovn
```

```yaml
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

□□□□□□□□ Pod

□□□□□ `provider` □□□ ovn □ subnet □□□ IP □ Pod□□□ annotation `k8s.v1.cni.cncf.io/networks` □ `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/logical_switch` □□□□□

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
    macvlan.default.kubernetes.io/logical_switch: macvlan
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

- `k8s.v1.cni.cncf.io/networks`: □□□□□□ NetworkAttachmentDefinition □ `<namespace>/<name>`
- `macvlan.default.kubernetes.io/logical_switch`: □□□□□□

> □□□
>
> - □□ `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/logical_switch` □□□□□□□□□□□□ provider □□□□□
> - □□ ovn □□□ subnet □□ ipam □□□□□□□□ IP □ Pod□□□□□□□ IP □□□□□□□□□□□□ macvlan □ Pod□
> - □□□□□□□□□□ macvlan □ Pod□

□□□□□ Kube-OVN □□□□

□□□□□□□□ Kube-OVN □□□□□

□□ NETWORKATTACHMENTDEFINITION

□ `provider` □□□□□□ `ovn` □

```yaml
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "attachnet.default.ovn"
    }'
```

- `spec.config.type`: □□□ `kube-ovn` □□□ CNI □□□□ Kube-OVN □□□
- `server_socket`: Kube-OVN □□□□□ socket □□□□□□□□ `/run/openvswitch/kube-ovn-daemon.sock` □
- `provider`: □□ NetworkAttachmentDefinition □ `<name>.<namespace>.ovn` □Kube-OVN □□□□□□□□□□□□ Subnet □□□□□□□□□□□ ovn□

□□□□ KUBE-OVN SUBNET

□□□ Kube-OVN □□□□□□□□ `provider` □□□□□□□□ NetworkAttachmentDefinition □ `<name>.<namespace>.ovn` □□□□ `ovn` □□□□□□□ □ Kube-OVN □□□□□□□□□ Subnet □□□□□

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: attachnet.default.ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

□□□□□□□□ Pod

□□□□□□□□□ Pod□□□□□□□□ annotation `k8s.v1.cni.cncf.io/networks` □□□□□□□ NetworkAttachmentDefinition □ `<namespace>/<name>` □

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

□□□□□□□□□□□□□ IP/MAC

□□□ NetworkAttachmentDefinition □□□ IP/MAC□□ `k8s.v1.cni.cncf.io/networks` □□□□□□□□□□□ `interface` □□□□□□□□□□ annotation□ `<nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName>` □ `<nadName>.<nadNamespace>.kubernetes.io/mac_address.<interfaceName>` □□□□□□□ - □□□□□□□□□□□□□ IP/MAC□

□□□□□□□□□□□□

□□□□□□□ Kube-OVN □□□ Pod□□□□□ `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/routes` annotation □□□□□□□□□

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: custom-routes-attach
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/routes: |
      [{
        "dst": "192.168.0.101/24",
        "gw": "172.17.0.254"
      }, {
        "gw": "172.17.0.254"
      }]
spec:
  containers:
  - name: custom-routes-attach
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

> `dst` □□□□□□□□□□□□□□

□□□□□□□ Deployment□DaemonSet □ StatefulSet□□□□ Annotation □□□□□□□□ `.spec.template.metadata.annotations` □□

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: custom-routes-attach
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
      annotations:
        k8s.v1.cni.cncf.io/networks: default/attachnet
        attachnet.default.ovn.kubernetes.io/routes: |
          [{
            "dst": "192.168.0.101/24",
            "gw": "172.17.0.254"
          }, {
            "gw": "172.17.0.254"
          }]
    spec:
      containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
```

□□□□ KUBE-OVN SUBNET□PROVIDER OVN□

□□□□ `provider` □□□ ovn □ subnet □□□ IP □□□□□□□□ `provider` □ ovn □ Kube-OVN Subnet□□□□□□ `cidrBlock` □ `exclude_ips` □□□ Subnet □□□

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

[illegible] Pod

[illegible] `provider` [illegible] ovn [illegible] subnet [illegible] IP [illegible] Pod[illegible] annotation `k8s.v1.cni.cncf.io/networks` [illegible] `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/logical_switch` [illegible]

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/logical_switch: attachnet
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

- `k8s.v1.cni.cncf.io/networks` : [illegible] NetworkAttachmentDefinition [illegible] `<namespace>/<name>`
- `attachnet.default.ovn.kubernetes.io/logical_switch` : [illegible]

> [illegible]
>
> - [illegible] `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/logical_switch` [illegible] provider [illegible]
> - [illegible] Kube-OVN [illegible] Pod[illegible] IP [illegible] Pod[illegible] IP [illegible] macvlan [illegible] Pod[illegible] Kube-OVN [illegible] Pod[illegible]
> - [illegible] Kube-OVN [illegible] Pod[illegible] `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/routes` annotation [illegible]

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7.1.4 [illegible]

7.2 [illegible]

[illegible]Kube-OVN [illegible] [illegible]

[illegible]

[illegible]Kube-OVN [illegible]

7.2.1 [illegible]

[illegible]

1. CNI [illegible] CNI [illegible]
2. [illegible] 10ns [illegible] 20ns[illegible]
3. [illegible] CPU[illegible] CPU [illegible]
4. [illegible] Macvlan [illegible] SR-IOV [illegible]

7.2.2 [illegible]

> [illegible] [illegible]

Overlay [illegible]

[illegible]

- Kubernetes: 1.22.0
- OS: CentOS 7
- Kube-OVN: 1.8.0 *Overlay* [illegible]
- CPU: Intel(R) Xeon(R) E-2278G
- Network: 2*10Gbps, xmit_hash_policy=layer3+4

[illegible] `qperf -t 60 <server ip> -ub -oo msg_size:1 -vu tcp_lat tcp_bw udp_lat udp_bw` [illegible] 1 [illegible] tcp/udp [illegible]

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Mb/s)	udp_bw(Mb/s)
Kube-OVN Default	25.7	22.9	27.1	1.59
Kube-OVN Optimized	13.9	12.9	27.6	5.57
HOST Network	13.1	12.4	28.2	6.02

Overlay[illegible] Underlay [illegible]

[illegible] Kube-OVN [illegible] Overlay [illegible] Underlay [illegible]

Environment:

- Kubernetes: 1.22.0
- OS: CentOS 7
- Kube-OVN: 1.8.0
- CPU: AMD EPYC 7402P 24-Core Processor
- Network: Intel Corporation Ethernet Controller XXV710 for 25GbE SFP28

`qperf -t 60 <server ip> -ub -oo msg_size:1 -vu tcp_lat tcp_bw udp_lat udp_bw`

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Mb/s)	udp_bw(Mb/s)
Kube-OVN Overlay	15.2	14.6	23.6	2.65
Kube-OVN Underlay	14.3	13.8	24.2	3.46
HOST Network	16.6	15.4	24.8	2.64

`qperf -t 60 <server ip> -ub -oo msg_size:1K -vu tcp_lat tcp_bw udp_lat udp_bw`

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Gb/s)	udp_bw(Gb/s)
Kube-OVN Overlay	16.5	15.8	10.2	2.77
Kube-OVN Underlay	15.9	14.5	9.6	3.22
HOST Network	18.1	16.6	9.32	2.66

`qperf -t 60 <server ip> -ub -oo msg_size:4K -vu tcp_lat tcp_bw udp_lat udp_bw`

Type	tcp_lat (us)	udp_lat (us)	tcp_bw (Gb/s)	udp_bw(Gb/s)
Kube-OVN Overlay	34.7	41.6	16.0	9.23
Kube-OVN Underlay	32.6	44	15.1	6.71
HOST Network	35.9	45.9	14.6	5.59

> [illegible] netfilter[illegible] `kube-proxy` [illegible] netfilter[illegible]

7.2.3 [illegible]

[illegible]

CPU [illegible]

[illegible] CPU [illegible] CPU [illegible]

```
cpupower frequency-set -g performance
```

[illegible]

[illegible]

[illegible]

```
# ethtool -g eno1
 Ring parameters for eno1:
 Pre-set maximums:
 RX:             4096
 RX Mini:        0
 RX Jumbo:       0
 TX:             4096
 Current hardware settings:
 RX:             255
 RX Mini:        0
 RX Jumbo:       0
 TX:             255
```

[illegible]

```
ethtool -G eno1 rx 4096
ethtool -G eno1 tx 4096
```

[illegible] tuned [illegible]

tuned [illegible] profile [illegible]

[illegible]

```
tuned-adm profile network-latency
```

[illegible]

```
tuned-adm profile network-throughput
```

[illegible]

[illegible] `irqbalance` [illegible] CPU [illegible] CPU [illegible]

[illegible] OVN LB

OVN [illegible] L2 LB [illegible] `conntrack` [illegible] recirculate [illegible] CPU [illegible] 20% [illegible] CPU [illegible] Overlay [illegible] `kube-proxy` [illegible] Service [illegible] Pod-to-Pod [illegible] `kube-ovn-controller` [illegible]

```
command:
- /kube-ovn/start-controller.sh
args:
...
- --enable-lb=false
...
```

> Underlay [illegible] `kube-proxy` [illegible] iptables [illegible] ipvs [illegible] LB [illegible] Service [illegible]

[illegible] conntrack [illegible]

[illegible] OVN LB [illegible] Service [illegible] Service [illegible] NetworkPolicy [illegible] Subnet A [illegible] Pod [illegible] Subnet B [illegible] Service[illegible] kube-ovn-controller [illegible] `--skip-conntrack-dst-cidrs` [illegible] conntrack [illegible]

```
    --skip-conntrack-dst-cidrs="10.17.0.0/16,169.254.169.245/32"
```

[illegible] FastPath [illegible]

[illegible] network ns[illegible] netfilter [illegible] 20% [illegible] CPU [illegible] netfilter [illegible] `FastPath` [illegible] netfilter [illegible] CPU [illegible]

> [illegible] netfilter [illegible] iptables[illegible]ipvs[illegible]nftables [illegible]

[illegible]

[illegible] FastPath [illegible]

[illegible] `insmod kube_ovn_fastpath.ko` [illegible] `FastPath` [illegible] `dmesg` [illegible]

```
# dmesg
...
[619631.323788] init_module,kube_ovn_fastpath_local_out
[619631.323798] init_module,kube_ovn_fastpath_post_routing
[619631.323800] init_module,kube_ovn_fastpath_pre_routing
[619631.323801] init_module,kube_ovn_fastpath_local_in
...
```

OVS [illegible]

OVS [illegible] flow [illegible] 10% [illegible] CPU [illegible] x86 CPU [illegible] `popcnt` [illegible] `sse4.2` [illegible]flow [illegible] CPU [illegible] 5% [illegible]

[illegible] `FastPath` [illegible]

[illegible] CPU [illegible]

```
cat /proc/cpuinfo  | grep popcnt
cat /proc/cpuinfo  | grep sse4_2
```

CENTOS [illegible]

[illegible]

```
yum install -y gcc kernel-devel-$(uname -r) python3 autoconf automake libtool rpm-build openssl-devel
```

[illegible] OVS [illegible] RPM [illegible]:

```
git clone -b branch-3.5 --depth=1 https://github.com/openvswitch/ovs.git
cd ovs
curl -s  https://github.com/kubeovn/ovs/commit/2d2c83c26d4217446918f39d5cd5838e9ac27b32.patch |  git apply
./boot.sh
./configure --with-linux=/lib/modules/$(uname -r)/build CFLAGS="-g -O2 -mpopcnt -msse4.2"
make rpm-fedora-kmod
cd rpm/rpmbuild/RPMS/x86_64/
```

[illegible] RPM [illegible]

```
rpm -i openvswitch-kmod-3.5.1-1.el7.x86_64.rpm
```

[illegible] Kube-OVN[illegible] OVS [illegible]

UBUNTU [illegible]

[illegible]

```
apt install -y autoconf automake libtool gcc build-essential libssl-dev
```

[illegible] OVS [illegible]

```
apt install -y autoconf automake libtool gcc build-essential libssl-dev

git clone -b branch-3.5 --depth=1 https://github.com/openvswitch/ovs.git
cd ovs
curl -s  https://github.com/kubeovn/ovs/commit/2d2c83c26d4217446918f39d5cd5838e9ac27b32.patch |  git apply
./boot.sh
./configure --prefix=/usr/ --localstatedir=/var --enable-ssl --with-linux=/lib/modules/$(uname -r)/build
make -j `nproc`
make install
make modules_install

cat > /etc/depmod.d/openvswitch.conf << EOF
override openvswitch * extra
override vport-* * extra
EOF

depmod -a
cp debian/openvswitch-switch.init /etc/init.d/openvswitch-switch
/etc/init.d/openvswitch-switch force-reload-kmod
```

[illegible] Kube-OVN[illegible] OVS [illegible]

[illegible] STT [illegible]

Warning

OpenVswitch [illegible] 3.6 [illegible] STT [illegible] Tunnel [illegible]

[illegible] Geneve [illegible] Vxlan [illegible] UDP [illegible] UDP [illegible] TCP [illegible] TCP [illegible] offload [illegible] TCP [illegible] CPU [illegible] TCP [illegible]

STT [illegible] TCP [illegible] TCP [illegible] TCP [illegible] TCP [illegible] TCP [illegible]

STT [illegible] OVS [illegible]OVS [illegible]

STT □□□□□

```
kubectl set env daemonset/ovs-ovn -n kube-system TUNNEL_TYPE=stt

kubectl delete pod -n kube-system -lapp=ovs
```

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7.2.4 □□

7.3 □□□□ FastPath □□

□□□□□□□□□ Profile□ `Netfilter` □□□□□□□□□□□□□□□□□ 20% □□□ CPU □□□FastPath □□□□□ `Netfilter` □□ □□ CPU □□□□□□□□□□□□□□□□□□□□□□□□□□□□□ FastPath □□□

7.3.1 □□□□□□□□□□□□

```
git clone --depth=1 https://github.com/kubeovn/kube-ovn.git
```

7.3.2 □□□□

□□ RPM □□□□ CentOS□RHEL□□

```
yum install -y kernel-devel-$(uname -r) gcc elfutils-libelf-devel
```

□□ DEB □□□□ Ubuntu□Debian□□

```
apt install -y linux-headers-$(uname -r) build-essential
```

7.3.3 □□□□□□

□□ 3.x □□□□

```
cd kube-ovn/fastpath/3.x
make all
```

□□ 4.x ~ 6.x □□□□

```
cd kube-ovn/fastpath/4.x-6.x
make all
```

7.3.4 □□□□□□

□ `kube_ovn_fastpath.ko` □□□□□□□□□□□□□□□□□□□□

```
insmod kube_ovn_fastpath.ko
```

□□ `dmesg` □□□□□□

```
# dmesg
[619631.323788] init_module,kube_ovn_fastpath_local_out
[619631.323798] init_module,kube_ovn_fastpath_post_routing
[619631.323800] init_module,kube_ovn_fastpath_pre_routing
[619631.323801] init_module,kube_ovn_fastpath_local_in
```

□□□□□□□□□□□□

```
rmmod kube_ovn_fastpath.ko
```

> □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

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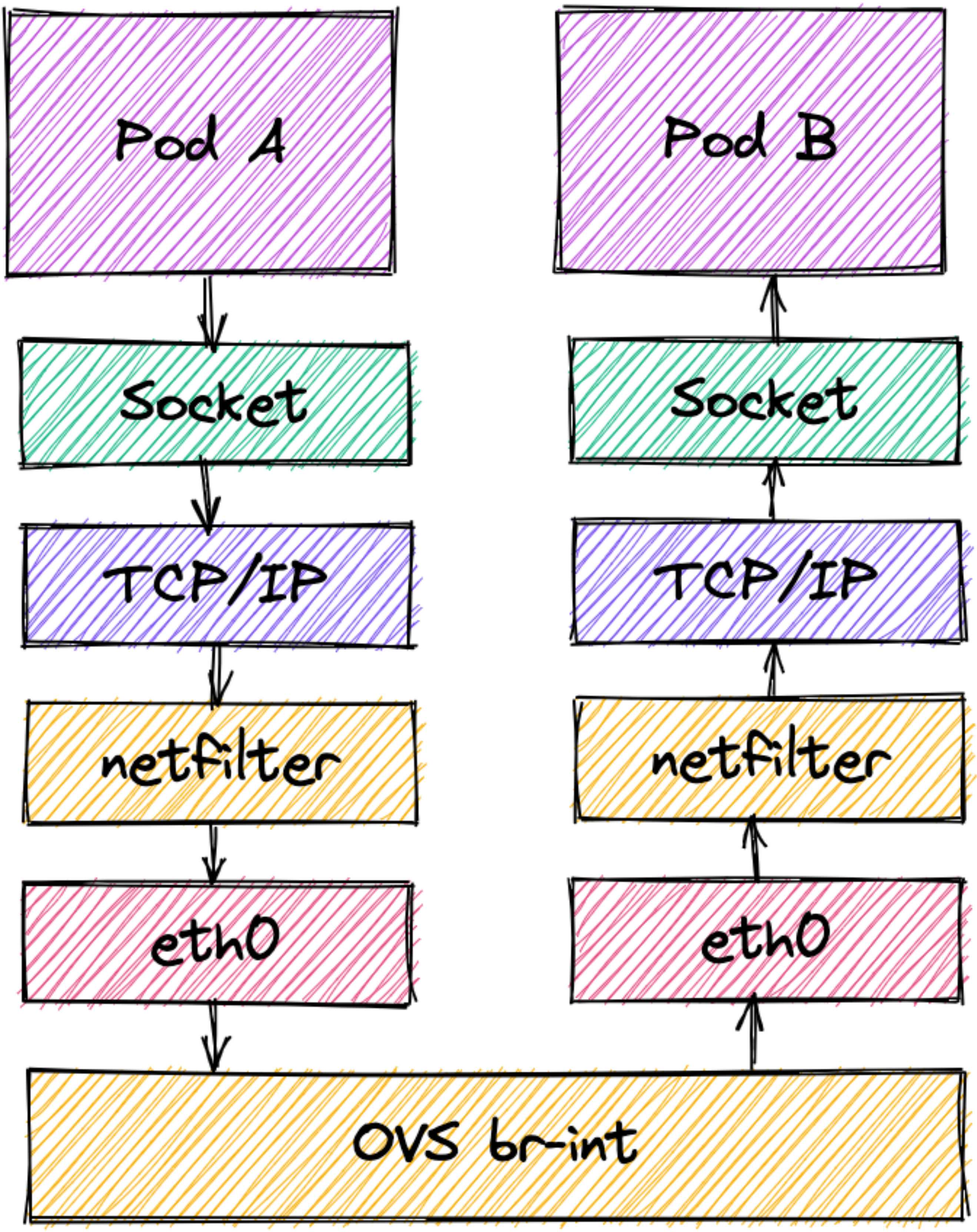
7.3.5 □□

7.4 [illegible] eBPF [illegible] TCP [illegible]

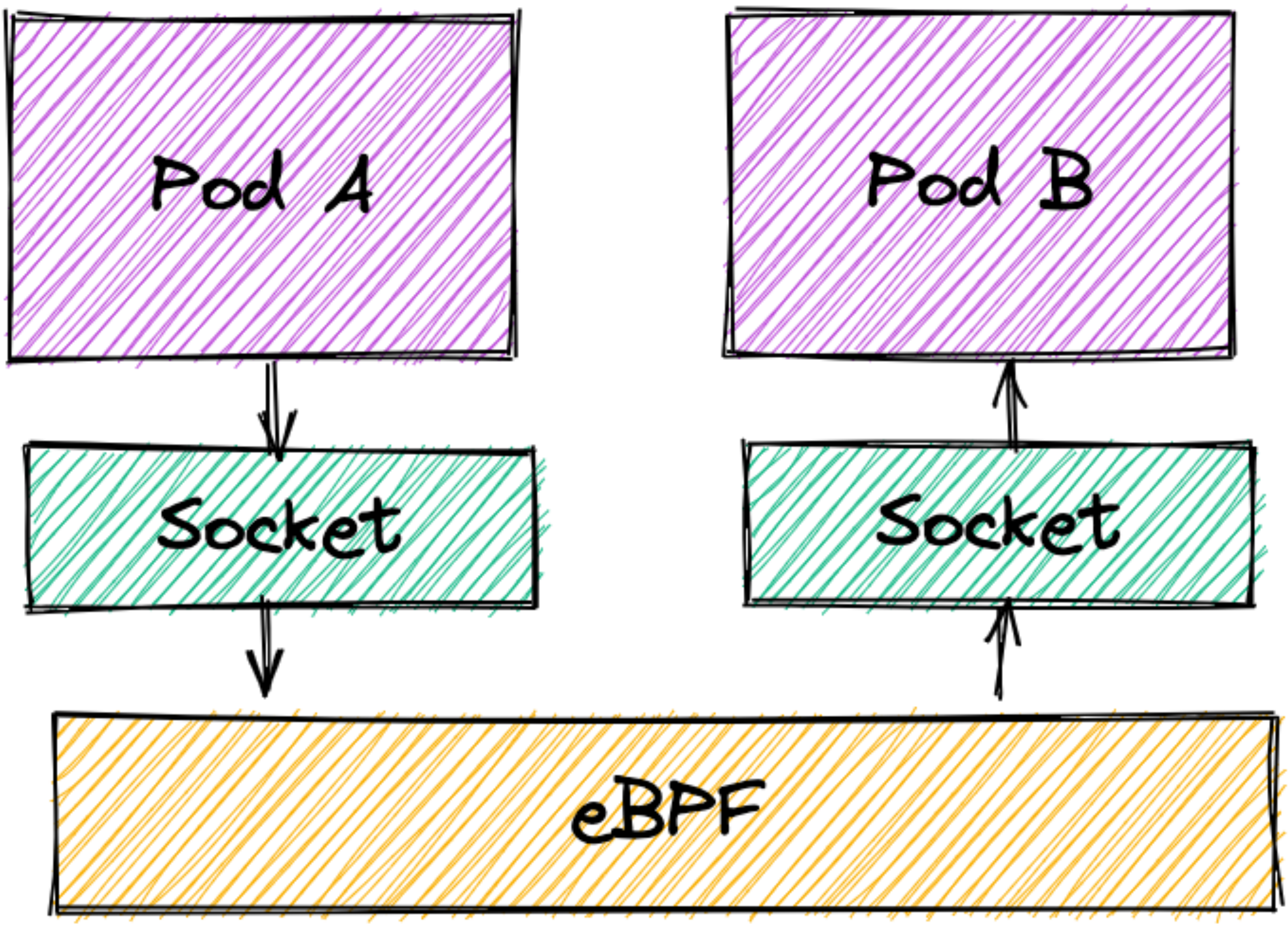
[illegible] 5G [illegible] Pod [illegible] TCP [illegible] Intel [illegible] istio-tcpip-bypass [illegible]Pod [illegible] eBPF [illegible] TCP/IP [illegible] socket [illegible]

7.4.1 [illegible]

[illegible] Pod [illegible] TCP [illegible] TCP/IP, netfilter[illegible]OVS [illegible]


Pod A
Pod B
Socket
Socket
TCP/IP
TCP/IP
netfilter
netfilter
eth0
eth0
OVS br-int

istio-tcpip-bypass [illegible] TCP [illegible] socket [illegible]



[illegible] TCP [illegible] Service Mesh [illegible] Service Mesh [illegible]

[illegible] Tanzu Service Mesh Acceleration using eBPF[illegible]

7.4.2 [illegible]

eBPF [illegible] Ubuntu 20.04 [illegible] Linux 5.4.0-74-generic [illegible]

7.4.3 [illegible]

[illegible] Pod[illegible] `nodeSelector` [illegible]

```
# kubectl create deployment perf --image=kubeovn/perf:dev --replicas=2
deployment.apps/perf created
# kubectl get pod -o wide
NAME                        READY   STATUS    RESTARTS   AGE   IP            NODE     NOMINATED NODE   READINESS GATES
perf-7697bc6ddf-b2cpv   1/1     Running   0          28s   100.64.0.3    sealos   <none>           <none>
perf-7697bc6ddf-p2xpt   1/1     Running   0          28s   100.64.0.2    sealos   <none>           <none>
```

[illegible] Pod [illegible] qperf server[illegible] Pod [illegible] qperf client [illegible]

```
# kubectl exec -it perf-7697bc6ddf-b2cpv sh
/ # qperf

# kubectl exec -it perf-7697bc6ddf-p2xpt sh
/ # qperf -t 60 100.64.0.3 -ub -oo msg_size:1:16K:*4 -vu tcp_lat tcp_bw
```

[illegible] istio-tcpip-bypass [illegible]

```
kubectl apply -f https://raw.githubusercontent.com/intel/istio-tcpip-bypass/main/bypass-tcpip-daemonset.yaml
```

[illegible] perf client [illegible]

```
# kubectl exec -it perf-7697bc6ddf-p2xpt sh
/ # qperf -t 60 100.64.0.3 -ub -oo msg_size:1:16K:*4 -vu tcp_lat tcp_bw
```

7.4.4 [illegible]

[illegible] TCP [illegible] 40% ~ 60% [illegible] 1024 [illegible] 40% ~ 80% [illegible]

Packet Size (byte)	eBPF tcp_lat (us)	Default tcp_lat (us)	eBPF tcp_bw (Mb/s)	Default tcp_bw(Mb/s)
1	20.2	44.5	1.36	4.27
4	20.2	48.7	5.48	16.7
16	19.6	41.6	21.7	63.5
64	18.8	41.3	96.8	201
256	19.2	36	395	539
1024	18.3	42.4	1360	846
4096	16.5	62.6	4460	2430
16384	20.2	58.8	9600	6900

> [illegible] 512 [illegible] eBPF [illegible] TCP [illegible] eBPF [illegible] eBPF TCP [illegible]

7.4.5 [illegible]

1. istio-tcpip-bypass
2. Deep Dive TCP/IP Bypass with eBPF in Service Mesh
3. Tanzu Service Mesh Acceleration using eBPF

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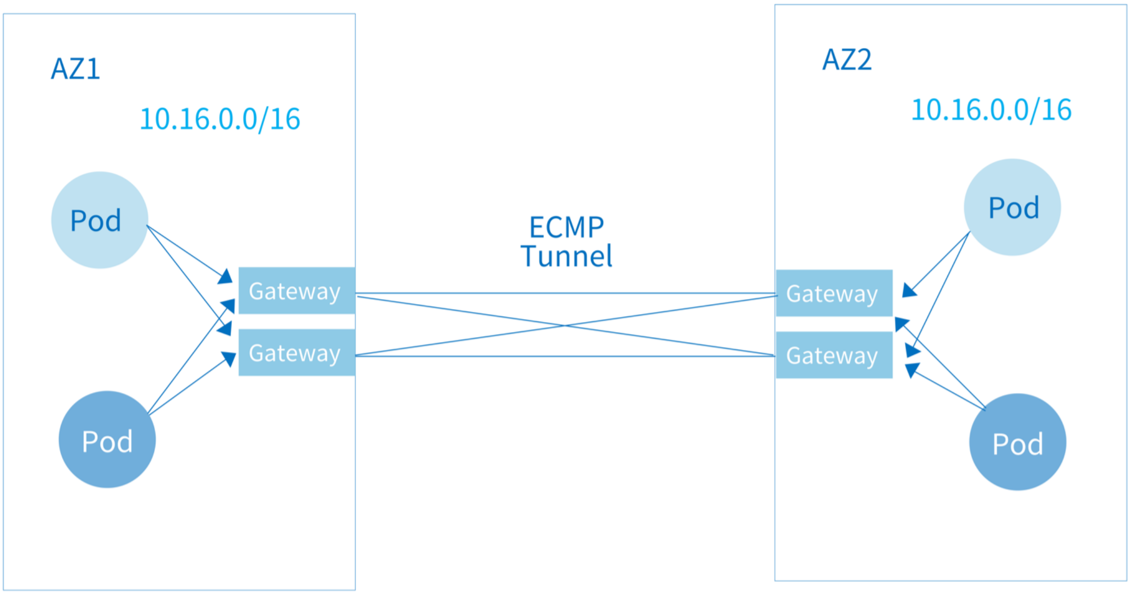
7.4.6 [illegible]

7.5 ⊠⊠ OVN-IC ⊠⊠⊠⊠⊠⊠⊠

Kube-OVN ⊠⊠⊠⊠ OVN-IC ⊠⊠⊠ Kubernetes ⊠⊠ Pod ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ Pod ⊠⊠⊠⊠ Pod IP ⊠⊠⊠⊠⊠⊠⊠ Kube-OVN ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ IP ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

> ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ Overlay ⊠⊠⊠⊠⊠Underlay ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠


AZ1
10.16.0.0/16
Pod
Gateway
Gateway
Pod
ECMP
Tunnel
AZ2
10.16.0.0/16
Pod
Gateway
Gateway
Pod

⚠ **Limitation**

OVN-IC ⊠⊠⊠⊠⊠⊠⊠ Pod IP ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ Service⊠DNS ⊠ NetworkPolicy ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ Istio ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

7.5.1 ⊠⊠⊠⊠

1. 1.11.16 ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ `install.sh` ⊠⊠⊠⊠⊠⊠⊠⊠

   ```
   ENABLE_IC=true
   ```

   ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ deployment ovn-ic-controller⊠

2. ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ CIDR ⊠⊠
3. ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ `kube-ovn-controller` ⊠⊠ IP ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠
4. ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ IP ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠
5. ⊠⊠⊠⊠⊠⊠⊠ VPC ⊠⊠⊠⊠⊠⊠⊠⊠ VPC ⊠⊠⊠⊠⊠⊠⊠⊠⊠

7.5.2 ⊠⊠⊠⊠⊠ OVN-IC ⊠⊠⊠

⊠⊠⊠⊠⊠⊠⊠ 1

⊠⊠⊠⊠⊠⊠ 1⊠Kube-OVN v1.11.16 ⊠⊠⊠⊠⊠

⊠⊠⊠⊠⊠⊠ "⊠⊠⊠" ⊠⊠ "⊠⊠⊠⊠⊠⊠" ⊠⊠⊠⊠⊠⊠⊠⊠ Deployment ⊠⊠⊠⊠⊠⊠ master ⊠⊠⊠⊠⊠⊠ master ⊠⊠⊠ 1⊠⊠⊠⊠⊠⊠⊠⊠master ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

[illegible] `install-ovn-ic.sh` [illegible]

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install-ic-server.sh
```

[illegible] `TS_NUM` [illegible] ECMP Path [illegible]

```
sed 's/VERSION=.*/VERSION=v1.17.0/' dist/images/install-ic-server.sh | TS_NUM=3 bash
```

[illegible]

```
deployment.apps/ovn-ic-server created
Waiting for deployment spec update to be observed...
Waiting for deployment "ovn-ic-server" rollout to finish: 0 out of 3 new replicas have been updated...
Waiting for deployment "ovn-ic-server" rollout to finish: 0 of 3 updated replicas are available...
Waiting for deployment "ovn-ic-server" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "ovn-ic-server" rollout to finish: 2 of 3 updated replicas are available...
deployment "ovn-ic-server" successfully rolled out
OVN IC Server installed Successfully
```

[illegible] `kubectl ko icsbctl show` [illegible]

```
kubectl ko icsbctl show
availability-zone az0
    gateway 059b5c54-c540-4d77-b009-02d65f181a02
        hostname: kube-ovn-worker
        type: geneve
            ip: 172.18.0.3
        port ts-az0
            transit switch: ts
            address: ["00:00:00:B4:8E:BE 169.254.100.97/24"]
    gateway 74ee4b9a-ba48-4a07-861e-1a8e4b9f905f
        hostname: kube-ovn-worker2
        type: geneve
            ip: 172.18.0.2
        port ts1-az0
            transit switch: ts1
            address: ["00:00:00:19:2E:F7 169.254.101.90/24"]
    gateway 7e2428b6-344c-4dd5-a0d5-972c1ccec581
        hostname: kube-ovn-control-plane
        type: geneve
            ip: 172.18.0.4
        port ts2-az0
            transit switch: ts2
            address: ["00:00:00:EA:32:BA 169.254.102.103/24"]
availability-zone az1
    gateway 034da7cb-3826-4318-81ce-6a877a9bf285
        hostname: kube-ovn1-worker
        type: geneve
            ip: 172.18.0.6
        port ts-az1
            transit switch: ts
            address: ["00:00:00:25:3A:B9 169.254.100.51/24"]
    gateway 2531a683-283e-4fb8-a619-bdbcb33539b8
        hostname: kube-ovn1-worker2
        type: geneve
            ip: 172.18.0.5
        port ts1-az1
            transit switch: ts1
            address: ["00:00:00:52:87:F4 169.254.101.118/24"]
    gateway b0efb0be-e5a7-4323-ad4b-317637a757c4
        hostname: kube-ovn1-control-plane
        type: geneve
            ip: 172.18.0.8
        port ts2-az1
            transit switch: ts2
            address: ["00:00:00:F6:93:1A 169.254.102.17/24"]
```

[illegible] 2

[illegible] `kube-ovn-controller` [illegible] IP [illegible] `OVN-IC` [illegible]

[illegible] `docker` [illegible] `OVN-IC` [illegible]

```
docker run --name=ovn-ic-db -d --env "ENABLE_OVN_LEADER_CHECK="false"" --network=host --privileged  -v /etc/ovn/:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /
var/log/ovn:/var/log/ovn kubeovn/kube-ovn:v1.17.0 bash start-ic-db.sh
```

[illegible] `containerd` [illegible] `docker` [illegible]

```
ctr -n k8s.io run -d --env "ENABLE_OVN_LEADER_CHECK="false"" --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --
mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw" docker.io/kubeovn/
kube-ovn:v1.17.0 ovn-ic-db bash start-ic-db.sh
```

7.5.3 [illegible]

[illegible] VPC [illegible] Subnet [illegible] CIDR [illegible] `OVN-IC` [illegible] Subnet CIDR [illegible]

[illegible] `kube-system` Namespace [illegible] `ovn-ic-config` ConfigMap[illegible]

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

- `enable-ic`: [illegible]
- `az-name`: [illegible]
- `ic-db-host`: [illegible] `OVN-IC` [illegible]
- `ic-nb-port`: `OVN-IC` [illegible] 6645[illegible]
- `ic-sb-port`: `OVN-IC` [illegible] 6646[illegible]
- `gw-nodes`: [illegible]
- `auto-route`: [illegible]

[illegible] `ovn-ic-config` [illegible] ConfigMap [illegible] ConfigMap[illegible] ConfigMap[illegible]

[illegible] `ovn-ic` [illegible] `ts` [illegible]

```
# ovn-ic-sbctl show
availability-zone az1
    gateway deee03e0-af16-4f45-91e9-b50c3960f809
        hostname: az1-gw
        type: geneve
            ip: 192.168.42.145
        port ts-az1
            transit switch: ts
            address: ["00:00:00:50:AC:8C 169.254.100.45/24"]
availability-zone az2
    gateway e94cc831-8143-40e3-a478-90352773327b
        hostname: az2-gw
        type: geneve
            ip: 192.168.42.149
        port ts-az2
            transit switch: ts
            address: ["00:00:00:07:4A:59 169.254.100.63/24"]
```

[illegible]

```
# kubectl ko nbctl lr-route-list ovn-cluster
IPv4 Routes
                10.42.1.1            169.254.100.45 dst-ip (learned)
                10.42.1.3                100.64.0.2 dst-ip
                10.16.0.2                100.64.0.2 src-ip
                10.16.0.3                100.64.0.2 src-ip
                10.16.0.4                100.64.0.2 src-ip
                10.16.0.6                100.64.0.2 src-ip
             10.17.0.0/16            169.254.100.45 dst-ip (learned)
            100.65.0.0/16            169.254.100.45 dst-ip (learned)
```

[illegible] 1 [illegible] Pod [illegible] `ping` [illegible] 2 [illegible] Pod IP [illegible]

[illegible] Subnet [illegible] `disableInterConnection` [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: no-advertise
spec:
  cidrBlock: 10.199.0.0/16
  disableInterConnection: true
```

7.5.4 ⊠⊠⊠⊠⊠⊠

⊠⊠⊠⊠⊠⊠⊠⊠⊠ CIDR ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

⊠ `kube-system` Namespace ⊠⊠⊠ `ovn-ic-config` ConfigMap⊠⊠⊠ `auto-route` ⊠⊠⊠ `false` ⊠

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "false"
```

⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

```
[root@az1 ~]# kubectl ko nbctl show
switch a391d3a1-14a0-4841-9836-4bd930c447fb (ts)
    port ts-az1
        type: router
        router-port: az1-ts
    port ts-az2
        type: remote
        addresses: ["00:00:00:4B:E2:9F 169.254.100.31/24"]

[root@az2 ~]# kubectl ko nbctl show
switch da6138b8-de81-4908-abf9-b2224ec4edf3 (ts)
    port ts-az2
        type: router
        router-port: az2-ts
    port ts-az1
        type: remote
        addresses: ["00:00:00:FB:2A:F7 169.254.100.79/24"]
```

⊠⊠⊠⊠⊠⊠⊠⊠⊠ `az1` ⊠ ⊠⊠ `az2` ⊠⊠⊠⊠⊠⊠ `169.254.100.31` ⊠ `az2` ⊠ `az1` ⊠⊠⊠⊠⊠⊠ `169.254.100.79` ⊠

⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ `az1` ⊠⊠⊠⊠ CIDR ⊠ `10.16.0.0/24` ⊠⊠⊠ `az2` ⊠⊠⊠⊠ CIDR ⊠ `10.17.0.0/24` ⊠

⊠⊠⊠ `az1` ⊠⊠⊠⊠⊠ `az2` ⊠⊠⊠⊠

```
kubectl ko nbctl lr-route-add ovn-cluster 10.17.0.0/24 169.254.100.31
```

⊠⊠⊠ `az2` ⊠⊠⊠⊠⊠ `az1` ⊠⊠⊠⊠

```
kubectl ko nbctl lr-route-add ovn-cluster 10.16.0.0/24 169.254.100.79
```

7.5.5 ⊠⊠⊠ OVN-IC ⊠⊠⊠⊠⊠

⊠⊠⊠⊠⊠⊠⊠⊠ 1

⊠⊠⊠⊠⊠⊠ 1⊠Kube-OVN v1.11.16 ⊠⊠⊠⊠⊠

⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ 1

⊠⊠⊠⊠⊠⊠⊠⊠ 2

`OVN-IC` ⊠⊠⊠⊠⊠⊠⊠⊠⊠ Raft ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ 3 ⊠⊠⊠⊠

⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ `OVN-IC` ⊠⊠⊠⊠ leader⊠

⊠⊠ `docker` ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

```
docker run --name=ovn-ic-db -d --env "ENABLE_OVN_LEADER_CHECK="false"" --network=host --privileged -v /etc/ovn/:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/
log/ovn:/var/log/ovn -e LOCAL_IP="192.168.65.3"  -e NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1"   kubeovn/kube-ovn:v1.17.0 bash start-ic-db.sh
```

[illegible] `containerd` [illegible]

```
ctr -n k8s.io run -d --env "ENABLE_OVN_LEADER_CHECK="false"" --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --
mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw"  --
env="NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1"" --env="LOCAL_IP="192.168.65.3"" docker.io/kubeovn/kube-ovn:v1.17.0 ovn-ic-db bash start-ic-db.sh
```

- `LOCAL_IP` [illegible] IP [illegible]
- `NODE_IPS` [illegible] `OVN-IC` [illegible] IP [illegible]

[illegible] `OVN-IC` [illegible] follower[illegible]

[illegible] `docker` [illegible]

```
docker run --name=ovn-ic-db -d --network=host --privileged -v /etc/ovn/:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn -e
LOCAL_IP="192.168.65.2"  -e NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1" -e LEADER_IP="192.168.65.3"  kubeovn/kube-ovn:v1.17.0 bash start-ic-db.sh
```

[illegible] `containerd` [illegible]

```
ctr -n k8s.io run -d --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --mount="type=bind,src=/var/run/ovn,dst=/var/run/
ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw"  --env="NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1"" --
env="LOCAL_IP="192.168.65.2"" --env="LEADER_IP="192.168.65.3"" docker.io/kubeovn/kube-ovn:v1.17.0 ovn-ic-db bash start-ic-db.sh
```

- `LOCAL_IP` [illegible] IP [illegible]
- `NODE_IPS` [illegible] `OVN-IC` [illegible] IP [illegible]
- `LEADER_IP`: [illegible] `OVN-IC` [illegible] leader [illegible] IP [illegible]

[illegible] `ovn-ic-config` [illegible] `OVN-IC` [illegible]

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3,192.168.65.2,192.168.65.1"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

7.5.6 [illegible] ECMP

[illegible] 1[illegible]

[illegible] ECMP[illegible]ECMP path [illegible] 3[illegible] ECMP path [illegible]

```
kubectl edit deployment ovn-ic-server -n kube-system
```

[illegible] 'TS_NUM' [illegible] `TS_NUM` [illegible] ECMP Path [illegible]

7.5.7 [illegible]

[illegible]

[illegible] `ovn-ic-config` Configmap[illegible]

```
kubectl -n kube-system delete cm ovn-ic-config
```

[illegible] `ts` [illegible]

```
kubectl ko nbctl ls-del ts
```

[illegible]

7.5.8 [illegible] az-name

[illegible] `kubectl edit` [illegible] `ovn-ic-config` [illegible] configmap [illegible] `az-name` [illegible] [illegible] ovn-cni pod [illegible] 10 [illegible]

```
ovn-appctl -t ovn-controller inc-engine/recompute
```

7.5.9 [illegible]

[illegible] `ovn-ic-config` Configmap[illegible]

```
kubectl -n kube-system delete cm ovn-ic-config
```

[illegible] `ts` [illegible]

```
kubectl ko nbctl ls-del ts
```

[illegible] OVN-IC [illegible]

[illegible] `docker` [illegible]

```
docker stop ovn-ic-db
docker rm ovn-ic-db
```

[illegible] `containerd` [illegible]

```
ctr -n k8s.io task kill ovn-ic-db
ctr -n k8s.io containers rm ovn-ic-db
```

[illegible] deployment `ovn-ic-server` [illegible]

```
kubectl delete deployment ovn-ic-server -n kube-system
```

[illegible] master [illegible] DB[illegible]

```
rm -f /etc/origin/ovn/ovn_ic_nb_db.db
rm -f /etc/origin/ovn/ovn_ic_sb_db.db
```

PDF Slack Support

2025[illegible]6[illegible]19[illegible]

2022[illegible]5[illegible]24[illegible]

GitHub +1

7.5.10 [illegible]

7.6 □□ Submariner □□□□□□□

Submariner □□□□□□□□ Kubernetes □□ Pod □ Service □□□□□□□□□□□□□□ Kube-OVN □□□□□□□□

□□□□ OVN-IC □□□□□□□□□□□Submariner □□□□ Kube-OVN □□ Kube-OVN □□□□□□□ □□□ Service □□□□□□□□□ Submariner □□□□□□□□□□□□□□□□□□□□□□□□□□□□

7.6.1 □□□□

- □□□□□ Service CIDR □□□□□□ CIDR □□□□□

7.6.2 □□ Submariner

Note

Submariner □ v0.22.0 □□□ nftables □□□□□□□□□□□□ Kube-OVN □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ Submariner□

```
kubectl create namespace submariner-operator
kubectl create configmap submariner-global --namespace=submariner-operator --from-literal=use-nftables=false
```

□□ `subctl` □□□□□□□□□□□□□□□□

```
curl -Ls https://get.submariner.io | bash
export PATH=$PATH:~/.local/bin
echo export PATH=\$PATH:~/.local/bin >> ~/.profile
```

□□ `kubeconfig` □□□□□ `submariner-broker` □□□□□□□□

```
subctl deploy-broker
```

□□□□□ `cluster0` □□□□□ CIDR □ `10.16.0.0/16` □join □□ CIDR □ `100.64.0.0/16` □ `cluster1` □□□□□ CIDR □ `11.16.0.0/16` □join □□ CIDR □ `100.68.0.0/16` □

□□ `kubeconfig` □ `cluster0` □□□□□ broker□□□□□□□□:

```
subctl  join broker-info.subm --clusterid  cluster0 --clustercidr 100.64.0.0/16,10.16.0.0/16  --natt=false --cable-driver vxlan --health-check=false
kubectl label nodes cluster0 submariner.io/gateway=true
```

□□ `kubeconfig` □ `cluster1` □□□□□ broker□□□□□□□□:

```
subctl  join broker-info.subm --clusterid  cluster1 --clustercidr 100.68.0.0/16,11.16.0.0/16  --natt=false --cable-driver vxlan --health-check=false
kubectl label nodes cluster1 submariner.io/gateway=true
```

□□□□ `join` □□□□□□□□ `gateway, routeagent` pod □□□□, □□ `submariner-operator` □□ `clusterrole` □□□□□□:

```
- apiGroups:
  - "apps"
  resources:
  - daemonsets
  verbs:
  - create
  - get
  - list
  - watch
  - update
```

□□□□□□□□□□□□□□□ `subnet` `ovn-default` □□□□□□□ `centralized` □□□□□□ `gatewayNode` □□□□□□□□□□□□□□□□□□□

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-default
spec:
  gatewayType: centralized
  gatewayNode: "node1,node2"   #   submariner     gateway
```

□ submariner □□□ `gateway` □□□□□ `gatewayNode` □□□□□□□□□□□□□

[illegible] Pod [illegible] IP [illegible]

[illegible] `subctl` [illegible]

```
subctl show all
subctl diagnose all
```

[illegible] Submariner [illegible] Submariner [illegible]

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2026[illegible]4[illegible]28[illegible]

2022[illegible]5[illegible]24[illegible]

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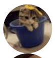

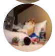

7.6.3 [illegible]

7.7 Overlay [illegible]

[illegible] Underlay [illegible] Pod [illegible] IP [illegible] [illegible]

> [illegible] VPC [illegible] Overlay [illegible]Pod IP [illegible] IP [illegible]

7.7.1 [illegible]

- [illegible] `ip_forward`[illegible]
- [illegible] iptables [illegible] forward [illegible] Drop [illegible]
- [illegible] ct [illegible] `INVALID` [illegible]

7.7.2 [illegible]

[illegible] `natOutgoing` [illegible] `false` [illegible] nat [illegible] Pod IP [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: routed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: distributed
  natOutgoing: false
```

[illegible]Pod [illegible]

[illegible] Kubernetes [illegible]

```
ip route add 10.166.0.0/16 via 192.168.2.10 dev eth0
```

`10.166.0.0/16` [illegible] `192.168.2.10` [illegible] Kubernetes [illegible]

[illegible]

[illegible] [illegible] IP [illegible] Keepalived [illegible] VIP[illegible] VIP[illegible]

[illegible] [illegible] Subnet [illegible] `gatewayType` [illegible] `centralized` [illegible] `gatewayNode` [illegible] IP[illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: routed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNode: "node1"
  natOutgoing: false
```

[illegible] nat [illegible] VPC NAT [illegible]

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2023年12月25日

2022年7月6日

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7.7.3 ▯▯

7.8 Overlay [illegible]

[illegible] Overlay [illegible] tunnel [illegible]

- [illegible]
- [illegible]
- [illegible]

7.8.1 [illegible]

[illegible]Kube-OVN [illegible] Kubernetes Node IP [illegible]

[illegible]

```
IFACE=eth1
```

[illegible] `ens[a-z0-9]*,eth[a-z0-9]*` [illegible]

[illegible] `kube-ovn-cni` DaemonSet [illegible]

```
args:
- --iface=eth1
```

[illegible] annotation `ovn.kubernetes.io/tunnel_interface` [illegible] annotation [illegible] `iface` [illegible] annotation[illegible]

```
kubectl annotate node no1 ovn.kubernetes.io/tunnel_interface=ethx
```

7.8.2 [illegible]

[illegible]Kube-OVN [illegible] IP [illegible] tunnel [illegible]

[illegible]

- [illegible] IP [illegible]
- [illegible]
- [illegible] Overlay [illegible]

[illegible]

1. [illegible] Node [illegible] IP[illegible]
2. [illegible] Subnet [illegible] `nodeNetwork` [illegible]
3. kube-ovn-daemon [illegible] IP [illegible] OVS[illegible]
4. [illegible] Pod [illegible] `nodeNetwork` [illegible] Pod [illegible] OVS [illegible] IP[illegible]

[illegible]

[illegible] `ovn.kubernetes.io/node_networks` [illegible] JSON [illegible]key [illegible]value [illegible] IP[illegible]

```
kubectl annotate node <node-name> ovn.kubernetes.io/node_networks='{"storage": "192.168.100.10", "app": "172.16.0.10"}'
```

[illegible]

- `storage` [illegible] IP `192.168.100.10` [illegible] IP [illegible]
- `app` [illegible] IP `172.16.0.10` [illegible] IP [illegible]

[illegible] IP[illegible]

[illegible]

[illegible] `spec.nodeNetwork` [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: storage-subnet
spec:
  protocol: IPv4
  cidrBlock: 10.100.0.0/16
  gateway: 10.100.0.1
  nodeNetwork: storage
```

[illegible] `storage-subnet` [illegible] Pod [illegible] `storage` [illegible] tunnel [illegible]

[illegible] `nodeNetwork` [illegible] IP[illegible] `IFACE` [illegible] IP[illegible]

[illegible]

[illegible] OVS [illegible] IP [illegible]

[illegible] OVS [illegible] IP[illegible]

```
ovs-vsctl get open . external-ids:ovn-encap-ip
```

[illegible] IP[illegible]

```
"192.168.1.10,192.168.100.10,172.16.0.10"
```

[illegible] IP[illegible]

```
ovs-vsctl get open . external-ids:ovn-encap-ip-default
```

[illegible] POD [illegible] IP

[illegible] Pod [illegible] Pod [illegible] OVS [illegible] IP [illegible]

```
ovs-vsctl --columns=external_ids find interface external-ids:iface-id="<pod-name>.<namespace>"
```

[illegible] `nodeNetwork` [illegible] `encap-ip` [illegible]

```
external_ids        : {encap-ip="192.168.100.10", iface-id="test-pod.default", ...}
```

[illegible]

[illegible]

1. [illegible]

[illegible]

- `eth0` [illegible]IP [illegible] `192.168.1.10` [illegible] `192.168.1.11`
- `eth1` [illegible]IP [illegible] `10.10.10.10` [illegible] `10.10.10.11`

```
kubectl annotate node node1 ovn.kubernetes.io/node_networks='{"storage": "10.10.10.10"}'
kubectl annotate node node2 ovn.kubernetes.io/node_networks='{"storage": "10.10.10.11"}'
```

2. [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: storage-net
spec:
  protocol: IPv4
  cidrBlock: 10.200.0.0/16
  gateway: 10.200.0.1
  nodeNetwork: storage
  namespaces:
  - storage-namespace
```

3. [illegible] POD

```
apiVersion: v1
kind: Pod
metadata:
  name: storage-pod
  namespace: storage-namespace
spec:
  containers:
  - name: app
    image: docker.io/library/nginx:alpine
```

[illegible] Pod [illegible] `eth1` [illegible]

[illegible]

1. [illegible]
2. [illegible] IP [illegible] IP [illegible]
3. [illegible] Overlay [illegible]Underlay [illegible] Underlay [illegible] Underlay [illegible]
4. [illegible] `nodeNetwork` [illegible] IP[illegible]
5. [illegible] Pod [illegible] `nodeNetwork` [illegible]Pod [illegible]
6. [illegible] `ovn.kubernetes.io/node_networks` [illegible] Pod [illegible]

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2025[illegible]11[illegible]28[illegible]

2025[illegible]11[illegible]28[illegible]

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7.8.3 [illegible]

7.9 BGP 支持

Kube-OVN 支持将 Pods、Subnets、Services 和 EIPs 的 IP 地址通过 BGP 协议向外部进行路由广播，从而使外部可以直接访问到这些 IP 地址。 为了使用该功能，需要在特定节点上部署 `kube-ovn-speaker`，并对需要对外暴露的 Pod 或 Subnet 增加对应的 annotation。

如果要在 EIP 上使用 BGP，需要创建一个 VPC NAT Gateway，该功能请参考发布 EIPs。

7.9.1 安装 kube-ovn-speaker

`kube-ovn-speaker` 内使用 GoBGP 对外发布路由信息，并将访问暴露地址的下一跳路由指向自身。

由于部署 `kube-ovn-speaker` 的节点需要承担回程流量，因此需要选择特定节点进行部署：

```
kubectl label nodes speaker-node-1 ovn.kubernetes.io/bgp=true
kubectl label nodes speaker-node-2 ovn.kubernetes.io/bgp=true
```

> 当存在多个 kube-ovn-speaker 实例时，每个实例都会对外发布路由，上游路由器需要支持多路径 ECMP。

下载对应 yaml:

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/yamls/speaker.yaml
```

修改 yaml 内相应配置：

如果只有一个交换机：

```
- --neighbor-address=10.32.32.254
- --neighbor-ipv6-address=2409:AB00:AB00:2000::AFB:8AFE
- --neighbor-as=65030
- --cluster-as=65000
#        IP
# - --allowed-source-addresses=10.32.32.2,10.32.32.3,10.32.32.4,10.32.32.5
```

如果有一对交换机：

```
- --neighbor-address=10.32.32.252,10.32.32.253
- --neighbor-ipv6-address=2409:AB00:AB00:2000::AFB:8AFC,2409:AB00:AB00:2000::AFB:8AFD
- --neighbor-as=65030
- --cluster-as=65000
#        IP
# - --allowed-source-addresses=10.32.32.2,10.32.32.3,10.32.32.4,10.32.32.5
```

- `neighbor-address`: BGP Peer 的地址，通常为路由器网关地址。
- `neighbor-as`: BGP Peer 的 AS 号。
- `cluster-as`: 容器网络的 AS 号。
- `allowed-source-addresses`: Speaker 允许使用的源 IP 地址列表，用于多网卡的场景。Speaker 会通过 `ip route get` 获取到达 BGP Peer 的路由，并检查路由的源 IP 是否在允许列表中。如果不在，Speaker 将跳过该路由，避免在 ECMP 场景下使用错误的 IP 作为下一跳。该 IP 列表需要包含所有节点。

部署 yaml:

```
kubectl apply -f speaker.yaml
```

按节点配置 YAML 文件

如果需要在不同节点上为 `kube-ovn-speaker` 使用不同 YAML 配置，可以设置参数 `--config=/etc/kube-ovn/$(NODE_NAME).yaml` 对应节点的配置文件，也可以通过 `KUBE_OVN_BGP_SPEAKER_CONFIG_FILE` 环境变量来代替参数 `--config` 的设置。

如果使用配置文件，可以创建一个 ConfigMap 并使用 key 挂载，key 名称需要与节点名称一致，格式为 `<spec.nodeName>.yaml` 例如： `.yaml` 前面的名称为 Pod 的 `spec.nodeName`，以下示例为 `speaker-node-1` 和 `speaker-node-2` 节点的配置：

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: kube-ovn-bgp-speaker-config
  namespace: kube-system
data:
  speaker-node-1.yaml: |
    cluster-as: 65000
    neighbor-address:
      - 10.32.32.254
    neighbor-as: 65030
    router-id: 10.32.32.2
  speaker-node-2.yaml: |
    cluster-as: 65001
    neighbor-address:
      - 10.32.32.254
    neighbor-as: 65030
    router-id: 10.32.32.3
```

□□ v2 Helm Chart □□□□ `bgpSpeaker.configMap` □□ ConfigMap□□□□ `bgpSpeaker.extraEnv` □□□ Pod □□ Downward API □ `NODE_NAME` □□□□□□□

```
bgpSpeaker:
  enabled: true
  configMap: kube-ovn-bgp-speaker-config
  configMapMountPath: /etc/kube-ovn/
  extraEnv:
    - name: KUBE_OVN_BGP_SPEAKER_CONFIG_FILE
      value: /etc/kube-ovn/$(NODE_NAME).yaml
```

□□□□□□□□□□□□□□□□□□□□ ConfigMap □□□ speaker □□□□□□□□□□ `NODE_NAME` □□□□□□□□□□□

```
env:
  - name: NODE_NAME
    valueFrom:
      fieldRef:
        fieldPath: spec.nodeName
  - name: KUBE_OVN_BGP_SPEAKER_CONFIG_FILE
    value: /etc/kube-ovn/$(NODE_NAME).yaml
```

□□□□ `--config` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ YAML □□□□□□□□□□□ speaker □□□□□

□□□□□□□□□□□□□□□□□□□□ kebab-case □□□□□□□□□□□□□

- `neighbor-address` □ `neighbor-ipv6-address` □IPv4 □ IPv6 □□□□□
- `allowed-source-addresses` □ `allowed-source-ipv6-addresses` □□□□□□□□□□□
- `cluster-as` □ `neighbor-as` □AS □□□□□□□□
- `grpc-host` □IPv4 □ IPv6 □□□□□□□
- `router-id` □IPv4 □□□□□□BGP router ID □□□ IPv6□
- `node-ips` □□□□□□□□□ `IPv4` □ `IPv6` □□□□□□□□□□□□□□
- `holdtime` □ `graceful-restart-time` □ `graceful-restart-deferral-time` □□□□□□□□□□□ `90s` □ `5m` □ `1h` □□□□□□□□□□□□□
- `announce-cluster-ip` □ `graceful-restart` □ `passivemode` □ `extended-nexthop` □ `nat-gw-mode` □ `enable-metrics` □ `enable-bfd` □□□□□
- `bfd-min-tx` □ `bfd-min-rx` □ `bfd-detection-multiplier` □□□□

□□□□□□□□□□□□□□□□□□□□ CLI □□□□□□□□□□□□□□□□□□□□ CLI □□□□□□□□□□□□ CLI □□□□□□□□□□□□□□□□□□□□

7.9.2 □□ Pod/Subnet □□

□□□□ BGP □□□□□□□□□□□□□□□□ Subnet □ `natOutgoing` □□□ `false` □□□ Pod IP □□□□□□□□□□□□□

□□ annotation □□□□□□

```
kubectl annotate pod sample ovn.kubernetes.io/bgp=true
kubectl annotate subnet ovn-default ovn.kubernetes.io/bgp=true
```

□□ annotation □□□□□□

```
kubectl annotate pod sample ovn.kubernetes.io/bgp-
kubectl annotate subnet ovn-default ovn.kubernetes.io/bgp-
```

□□□□□□□□□□□□□□□□□□□□□□□□ BGP □□□□□□□□

7.9.3 □□ `ClusterIP` □□ Service □□

□□ Service □ ClusterIP □□□□□□□□□ `kube-ovn-speaker` □□ `--announce-cluster-ip` □□□ `true` □ □□□□□□□□□□□□ BGP □□□□□

□□ annotation □□□□□

```
kubectl annotate service sample ovn.kubernetes.io/bgp=true
```

□□ annotation □□□□□

```
kubectl annotate service sample ovn.kubernetes.io/bgp-
```

7.9.4 □□ EIPs

EIPs □□□□□□□□ VPC NAT Gateway □□□□□□□ `VpcNatGateway` □□□ BGP □□□□□□□□□□□ BGP Sidecar□

□□□□ VPC NAT Gateway □ BGP □□□□□□□□□□□ BGP Speaker Sidecar □□□□ `NetworkAttachmentDefinition` □□□ NAD □□□□□□□□ VPC □□ Subnet □□□□□ Sidecar □□□□□□□□ Kubernetes API □□□□□□□ EIPs □□□ □□□□□□□□□□□ VPC □ CoreDNS □□□□□□□□□□□ NAD□

□□ `NetworkAttachmentDefinition` □ `Subnet` □□ `provider` □□□ `{nadName}.{nadNamespace}.ovn` □

```yaml
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: api-ovn-nad
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "api-ovn-nad.default.ovn"
    }'
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vpc-apiserver-subnet
spec:
  protocol: IPv4
  cidrBlock: 100.100.100.0/24
  provider: api-ovn-nad.default.ovn
```

□ `ovn-vpc-nat-config` ConfigMap □ □□□□ `apiNadProvider` □ BGP Speaker □□□□□□:

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
data:
  apiNadProvider: api-ovn-nad.default.ovn              # What NetworkAttachmentDefinition provider to use so that the sidecar
                                                       # can access the K8S API, as it can't by default due to VPC segmentation
  bgpSpeakerImage: docker.io/kubeovn/kube-ovn:v1.13.0  # Sets the BGP speaker image used
  image: docker.io/kubeovn/vpc-nat-gateway:v1.13.0
```

□□ `ovn-default` □□□□□□□ `provider` □

```yaml
provider: api-ovn-nad.default.ovn
```

□ VPC NAT Gateway □□□□□□ BGP□

```yaml
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-natgw
spec:
  vpc: vpc1
  subnet: net1
  lanIp: 10.0.1.10
  bgpSpeaker:
    enabled: true
    asn: 65500
    remoteAsn: 65000
    neighbors:
      - 100.127.4.161
      - fd:01::1
    enableGracefulRestart: true # Optional
    routerId: 1.1.1.1           # Optional
    holdTime: 1m                # Optional
    password: "password123"     # Optional
    extraArgs:                  # Optional, passed directly to the BGP speaker
      - -v5                     # Enables verbose debugging of the BGP speaker sidecar
  selector:
    - "kubernetes.io/os: linux"
  externalSubnets:
  - ovn-vpc-external-network # Network on which we'll speak BGP and receive/send traffic to the outside world
                             # BGP neighbors need to be on that network
```

⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ BGP ⌧⌧⌧⌧ EIP⌧

```
kubectl annotate eip sample ovn.kubernetes.io/bgp=true
```

7.9.5 ⌧⌧⌧⌧

`kube-ovn-speaker` ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧:

- **Cluster**: ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ speaker ⌧⌧⌧⌧⌧⌧ Pod IPs/Subnet CIDRs⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ IP ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ CIDR ⌧ Pod⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ speaker ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ Pod⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ Pod ⌧ Subnet ⌧⌧⌧⌧⌧⌧⌧⌧⌧
- **Local**: ⌧⌧⌧⌧⌧⌧⌧Pod IPs ⌧⌧⌧⌧⌧⌧ Pod ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ Cluster ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ Pod ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧

> ⌧⌧: ⌧⌧⌧⌧ `Local` ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ `kube-ovn-speaker` ⌧⌧⌧ Pod ⌧⌧⌧⌧⌧⌧⌧ speaker ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧

⌧⌧⌧⌧⌧⌧⌧⌧ `Cluster` ⌧⌧⌧⌧⌧ Pod/Subnet ⌧ annotation `ovn.kubernetes.io/bgp` ⌧⌧⌧⌧⌧⌧

- `ovn.kubernetes.io/bgp=cluster` ⌧ `ovn.kubernetes.io/bgp=true` ⌧⌧⌧⌧ `Cluster` ⌧⌧⌧
- `ovn.kubernetes.io/bgp=local` ⌧⌧⌧⌧ `Local` ⌧⌧⌧

> ⌧⌧⌧⌧⌧ Service ⌧⌧⌧⌧⌧⌧⌧⌧ `kube-proxy` ⌧⌧⌧⌧⌧ `ClusterIP` ⌧⌧ Service ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ `Cluster` ⌧⌧⌧

7.9.6 BGP ⌧⌧⌧⌧

`kube-ovn-speaker` ⌧⌧⌧⌧ BGP ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧

- `announce-cluster-ip` : ⌧⌧⌧⌧⌧⌧ Service ⌧⌧⌧⌧⌧⌧ `false` ⌧
- `auth-password` : BGP peer ⌧⌧⌧⌧⌧⌧
- `holdtime` : BGP ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ 90 ⌧⌧
- `graceful-restart` : ⌧⌧⌧⌧ BGP Graceful Restart⌧
- `graceful-restart-time` : BGP Graceful restart time ⌧⌧⌧ RFC4724 3⌧
- `graceful-restart-deferral-time` : BGP Graceful restart deferral time ⌧⌧⌧ RFC4724 4.1⌧
- `passivemode` : Speaker ⌧⌧⌧ passive ⌧⌧⌧⌧⌧⌧⌧⌧ peer⌧
- `ebgp-multihop` : ebgp ttl ⌧⌧⌧⌧ 1⌧
- `allowed-source-addresses` : ⌧ IP ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧Speaker ⌧⌧⌧⌧⌧ `ip route get` ⌧⌧⌧⌧ BGP Peer ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ IP ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ECMP ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ ⌧ IP ⌧⌧⌧⌧⌧
- `enable-bfd` : ⌧⌧ `kube-ovn-speaker` ⌧⌧ BFD⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ `false` ⌧⌧⌧⌧⌧“⌧⌧⌧⌧ `kube-ovn-speaker` ⌧⌧ BFD”⌧
- `bfd-min-tx` : `kube-ovn-speaker` ⌧⌧ BFD ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ `1000` ⌧1 ⌧⌧⌧
- `bfd-min-rx` : `kube-ovn-speaker` ⌧⌧ BFD ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ `1000` ⌧1 ⌧⌧⌧
- `bfd-detection-multiplier` : `kube-ovn-speaker` ⌧⌧ BFD ⌧⌧⌧⌧⌧⌧⌧⌧⌧ `3` ⌧⌧⌧⌧⌧⌧⌧⌧ = `bfd-min-rx * bfd-detection-multiplier` ⌧

7.9.7 BFD ⌧⌧⌧⌧⌧⌧⌧

⌧ BGP ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ BFD⌧Bidirectional Forwarding Detection⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ BGP ECMP ⌧⌧⌧⌧⌧⌧⌧⌧⌧

⌧ Kube-OVN ⌧⌧⌧⌧BFD ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧

- ⌧⌧⌧⌧⌧⌧⌧ **host network** ⌧⌧⌧⌧⌧⌧ Kubernetes Node ⌧ host network ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ BFD ⌧⌧⌧⌧⌧⌧ BGP ECMP ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧

- [illegible]**VPC** [illegible] `OVN logical router BFD` [illegible] `VPC` CRD [illegible] VPC [illegible] kube-ovn controller [illegible] VPC [illegible] BFD[illegible]

[illegible]

```
flowchart TD
    A[BFD     ] --> B[   :     host network]
    A --> C[   : VPC   ]
    B --> D[   : kube-ovn-speaker    BFD]
    B --> E[   : openbfdd     ]
    C --> F[OVN logical router BFD]
```

[illegible] host network [illegible]

- [illegible] **`kube-ovn-speaker`** [illegible] **BFD**[illegible] `kube-ovn-speaker` [illegible] GoBGP [illegible] BFD [illegible] node host network [illegible] BFD[illegible]
- [illegible] **`openbfdd`** [illegible] BFD DaemonSet[illegible] node host network [illegible] BFD[illegible]

> [illegible] BFD [illegible] `OVN logical router BFD` [illegible] VPC [illegible] host network [illegible] BGP ECMP [illegible]

[illegible] `kube-ovn-speaker` [illegible] BFD

`kube-ovn-speaker` [illegible] GoBGP [illegible] BFD [illegible]RFC 5880 / RFC 5881[illegible] BGP Speaker [illegible] BFD[illegible] `kube-ovn-speaker` [illegible] GoBGP [illegible] node host network [illegible] BFD [illegible]

[illegible]

[illegible] `kube-ovn-speaker` [illegible] BFD [illegible]

```
args:
  - --enable-bfd=true
  #
  - --bfd-min-tx=1000
  - --bfd-min-rx=1000
  - --bfd-detection-multiplier=3
```

[illegible] `1000ms × 3 = 3  ` [illegible]

[illegible] 300ms[illegible]

```
args:
  - --enable-bfd=true
  - --bfd-min-tx=100
  - --bfd-min-rx=100
  - --bfd-detection-multiplier=3
```

[illegible]

- [illegible] **BFD**[illegible]BFD [illegible] BFD [illegible]
- [illegible] UDP [illegible] `3784` [illegible]BFD [illegible]
- [illegible] BFD [illegible] CPU [illegible] BGP [illegible]

[illegible]

[illegible] `gobgp` [illegible] BFD [illegible] JSON [illegible]

```
gobgp -j neighbor <peer-address> | jq '{bfd_config: .bfd, bfd_state: .state.bfd_state}'
```

[illegible]

```
{
  "bfd_config": {
    "enabled": true,
    "port": 3784,
    "desired_minimum_tx_interval": 1000000,
    "required_minimum_receive": 1000000,
    "detection_multiplier": 3
  },
  "bfd_state": {
    "session_state": 1,
    "bfd_async": {
      "transmitted_packets": 100,
      "received_packets": 99
    }
  }
}
```

 `session_state` `1` = UP `2` = DOWN `3` = ADMIN_DOWN `4` = INIT

####

 BFD GoBGP BGP hard reset `BFD is down` BGP

openbfdd

 BFD `kube-ovn-speaker` openbfdd BFD DaemonSet node host network BGP BFD `BFD_MULTI * max(BFD_MIN_TX, BFD_MIN_RX) = 3 * 1000ms = 3 `

DaemonSet `hostNetwork: true` Pod

- **init-peer**: `ip route get` IP Volume
- **bfdd**: openbfdd init IP BFD `start-bfdd.sh` `bfdd-prestart.sh` startupProbe
- **reconcile**: 5 BFD peer

 yaml

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/yamls/bfdd-daemonset.yaml
```

 `GATEWAY_ADDRESS` `ALLOWED_SOURCE_ADDRESSES` Speaker `--allowed-source-addresses`

```
env:
  - name: GATEWAY_ADDRESS
    value: "10.32.32.1"                            #
  - name: ALLOWED_SOURCE_ADDRESSES
    value: "10.32.32.2,10.32.32.3,10.32.32.4,10.32.32.5"  #   IP
  - name: BFD_MIN_TX
    value: "1000"                                  #
  - name: BFD_MIN_RX
    value: "1000"                                  #
  - name: BFD_MULTI
    value: "3"                                     #
```

 DaemonSet

```
kubectl apply -f bfdd-daemonset.yaml
```

BFD

```
#    BFD daemon
bfdd-control status

#      BFD    remote=  , local=    IP
bfdd-control status remote <gateway-ip> local <local-ip>

#    BFD
bfdd-control allow <gateway-ip>

#
bfdd-control session new set mintx <ms> ms   #
bfdd-control session new set minrx <ms> ms   #
bfdd-control session new set multi <n>       #

#
bfdd-control log type command no

#    init         IP
cat /bfdd-peer/local-ip

#    reconcile sidecar
kubectl logs -n kube-system ds/openbfdd -c reconcile

#    init
kubectl logs -n kube-system ds/openbfdd -c init-peer
```

OVN BFD

OVN BFD kube-ovn controller

```
#    OVN BFD
kubectl ko nbctl list bfd

#            BFD
kubectl ko nbctl find bfd logical_port=<lrp-name>

#    OVN BFD
kubectl ko nbctl destroy bfd <uuid>
```

7.9.8 BGP routes debug

```
# show peer neighbor
gobgp neighbor

# show announced routes to one peer
gobgp neighbor 10.32.32.254 adj-out
```

PDF **Slack** **Support**

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GitHub

7.9.9 ▯▯

7.10 [illegible] MetalLB [illegible] Kube-OVN Underlay [illegible]

MetalLB [illegible] Kubernetes [illegible] Kubernetes [illegible] Service [illegible]

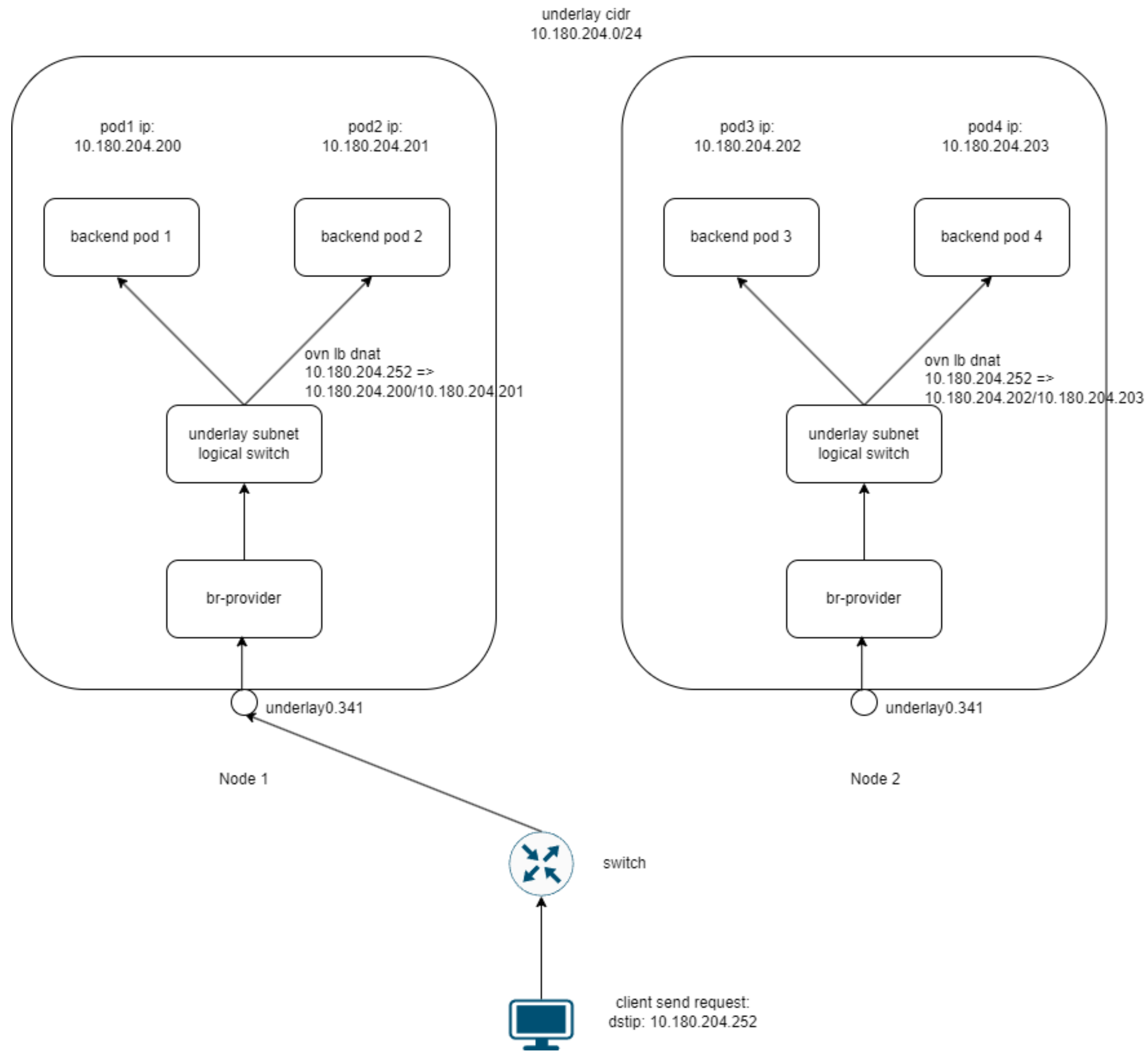
[illegible] MetalLB [illegible] Kube-OVN [illegible] Underlay [illegible]

7.10.1 [illegible]

Kube-OVN [illegible] 1.14.0 [illegible] MetalLB [illegible] Underlay [illegible]

- [illegible] MetalLB [illegible] IP [illegible]
- [illegible] Pod [illegible] MetalLB VIP [illegible] Underlay [illegible]
- [illegible] IP [illegible] SNAT

7.10.2 [illegible]


underlay cidr
10.180.204.0/24
pod1 ip:
10.180.204.200
pod2 ip:
10.180.204.201
pod3 ip:
10.180.204.202
pod4 ip:
10.180.204.203
backend pod 1
backend pod 2
backend pod 3
backend pod 4
ovn lb dnat
10.180.204.252 =>
10.180.204.200/10.180.204.201
ovn lb dnat
10.180.204.252 =>
10.180.204.202/10.180.204.203
underlay subnet
logical switch
underlay subnet
logical switch
br-provider
br-provider
underlay0.341
underlay0.341
Node 1
Node 2
switch
client send request:
dstip: 10.180.204.252

[illegible] 1[illegible]MetalLB VIP [illegible] Kube-OVN Underlay [illegible]

MetalLB [illegible] Kube-OVN Underlay [illegible]

1. [illegible] VIP[illegible] 10.180.204.252[illegible] IP [illegible] MetalLB [illegible] L2 [illegible] Node1 [illegible] metallb [illegible] VIP
2. [illegible] VIP [illegible] `underlay0.341` [illegible]
3. [illegible] `br-provider` [illegible] Underlay [illegible]
4. `br-provider` [illegible] OpenFlow [illegible] OVN [illegible]
5. [illegible] `underlay subnet` [illegible] OVN [illegible] `ovn lb dnat` [illegible]
6. OVN [illegible] Pod[illegible]

[illegible] 10.180.204.0/24[illegible] VIP [illegible] Pod [illegible] IP [illegible]

7.10.3 [illegible]

- Kube-OVN [illegible] `--enable-ovn-lb-prefer-local=true` [illegible]
- Underlay [illegible] `enableExternalLBAddress=true`
- [illegible] Underlay [illegible] `excludeIps` [illegible] MetalLB [illegible] IP [illegible]

7.10.4 [illegible]

1. [illegible] Kube-OVN

[illegible] Kube-OVN[illegible] Kube-OVN [illegible] `--enable-ovn-lb-prefer-local=true` [illegible] `--ls-ct-skip-dst-lport-ips=false` [illegible]

```
#   kube-ovn-controller   Deployment
kubectl edit deployment -n kube-system kube-ovn-controller
```

[illegible]

```
--enable-ovn-lb-prefer-local=true
--ls-ct-skip-dst-lport-ips=false
```

2. [illegible] Underlay [illegible]

[illegible] Underlay [illegible] LoadBalancer [illegible] excludeIps [illegible] MetalLB [illegible] IP [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: underlay-subnet
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 10.180.204.0/24  #
  gateway: 10.180.204.1
  excludeIps:
  - 10.180.204.250
  - 10.180.204.251
  - 10.180.204.252  # MetalLB             VIP 10.180.204.252
  natOutgoing: false
  enableExternalLBAddress: true   #     subnet   cidr    ip      metallb vip
```

3. [illegible] MetalLB

[illegible] MetalLB [illegible] [illegible] MetalLB[illegible]

```
kubectl apply -f https://raw.githubusercontent.com/metallb/metallb/v0.13.7/config/manifests/metallb-native.yaml
```

[illegible] MetalLB [illegible] L2 [illegible]

```yaml
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
  name: underlay-pool
  namespace: metallb-system
spec:
  addresses:
  - 10.180.204.250-10.180.204.254  #       VIP 10.180.204.252
---
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: l2-advert
  namespace: metallb-system
spec:
  ipAddressPools:
  - underlay-pool
```

4. [illegible] LoadBalancer [illegible] Service

[illegible] LoadBalancer [illegible] Service[illegible] Underlay [illegible] Pod[illegible]

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: nginx
  name: deploy-169402624
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      annotations:
        ovn.kubernetes.io/logical_switch: underlay-subnet
      labels:
        app: nginx
    spec:
      containers:
      - args:
        - netexec
        - --http-port
        - "80"
        image: kubeovn/agnhost:2.47
        imagePullPolicy: IfNotPresent
        name: nginx
---
apiVersion: v1
kind: Service
metadata:
  name: nginx-lb
spec:
  externalTrafficPolicy: Local
  ipFamilies:
  - IPv4
  ipFamilyPolicy: PreferDualStack
  ports:
  - port: 80
    protocol: TCP
    targetPort: 80
  selector:
    app: nginx
  type: LoadBalancer
```

7.10.5 [illegible]

1. [illegible] Service [illegible] MetalLB [illegible] IP [illegible]

```
kubectl get svc nginx-lb
```

[illegible] EXTERNAL-IP [illegible] IP [illegible] 10.180.204.252[illegible]

1. [illegible] Service [illegible] IP [illegible]

```
curl http://10.180.204.252
```

1. [illegible] Pod[illegible]

[illegible] Service [illegible] endpoints [illegible] Pod [illegible]

```
#    Service   endpoints
kubectl get endpoints nginx-lb
```

```
#    Pod
kubectl get pods -l app=nginx -o wide
```

1. □□□□□ IP □□□□□

□ nginx Pod □□□□□□□□□□□□□□□□□□□ IP □□□□□□□□□ IP□□□□□□ SNAT □□ IP□

```
kubectl exec -it $(kubectl get pods -l app=nginx -o name | head -n1) -- cat /var/log/nginx/access.log
```

7.10.6 □□□□

IP □□□□□

MetalLB □□□□□□□□□□ Underlay □□ CIDR □□□□□□□□□□ Underlay □□ `excludeIps` □□□□□□□□□□□□ IP □□□□□

□□□□□□

MetalLB □□□□□ Kube-OVN Underlay □□□□□□□□□□□□□□□□ `underlay0.341` □□□□□□□□□□ VLAN □□□□□□□□□□□□□□□□□ VLAN □□□ ARP □□□□□□□□□□ MetalLB VIP□

□□□□□□

□□□□□□□□□□□□□□□□□□□□ - Kube-OVN □□□□ `--enable-ovn-lb-prefer-local=true` □□ - Service □□□□□ `externalTrafficPolicy: Local`

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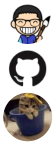
GitHub



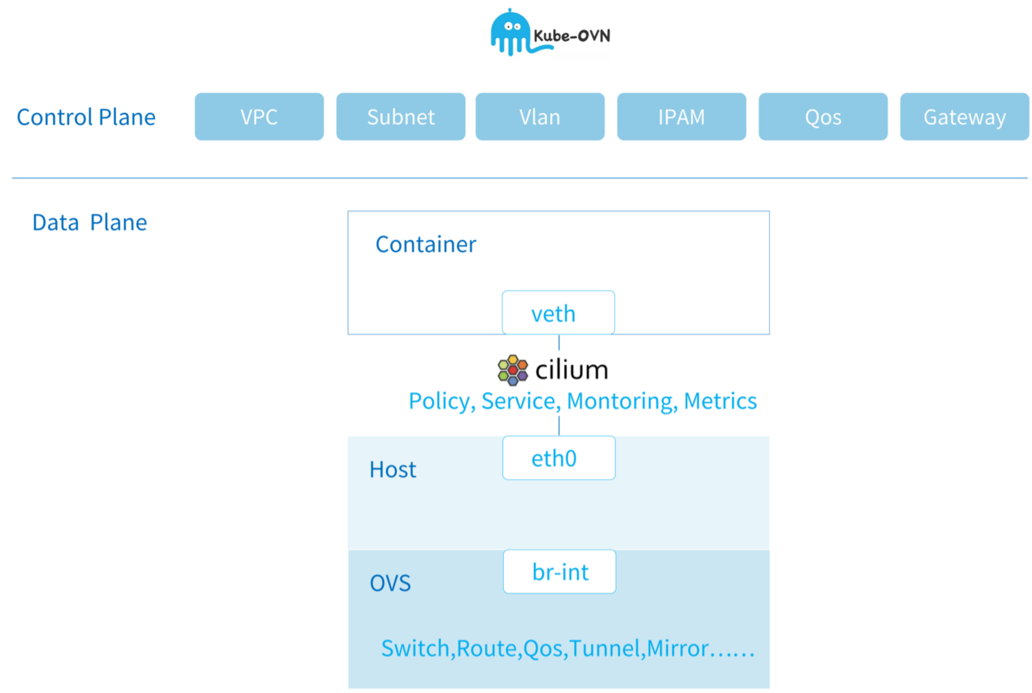
7.10.7 □□

7.11 Cilium □□

Cilium □□□□□ eBPF □□□□□□□□□Kube-OVN □□□□□ CNI Chaining □□□□□□□□□□□□□ □□□□□□□□ Kube-OVN □□□□□□□□□□ eBPF □□□□□□□□□□□□

□□□□ Cilium□Kube-OVN □□□□□□□□□□□□

- □□□□□□□□□□□□
- □□ Hubble □□□□□□□


Kube-OVN
Control Plane
VPC
Subnet
Vlan
IPAM
Qos
Gateway
Data Plane
Container
veth
cilium
Policy, Service, Montoring, Metrics
Host
eth0
OVS
br-int
Switch,Route,Qos,Tunnel,Mirror......

7.11.1 □□□□

1. Linux □□□□□□□ 4.19 □□□□□□□□□□□□□ eBPF □□□□□□
2. □□□□□ Helm □□□ Cilium □□□□□□□ Helm □□□ Installing Helm□

7.11.2 □□ Kube-OVN

□□□□□□□ Cilium □□□□□□□□□□□□ Kube-OVN □□ `networkpolicy` □□□□□□□ CNI □□□□□□□

□ `install.sh` □□□□□□□□□□□

```
ENABLE_NP=false
CNI_CONFIG_PRIORITY=10
```

□□□□□□□□□□□□□ `kube-ovn-controller` □□□□□□□□□□ `networkpolicy`□

```
args:
- --enable-np=false
```

□□ `kube-ovn-cni` □□□□□□□ CNI □□□□□□□

```
args:
- --cni-conf-name=10-kube-ovn.conflist
```

□□□□□□□□ Kube-OVN □□□□□□□□□□□□□□□ Cilium □□□□□□

```
mv /etc/cni/net.d/01-kube-ovn.conflist /etc/cni/net.d/10-kube-ovn.conflist
```

7.11.3 □□ Cilium

□□ `chaining.yaml` □□□□□□□□ Cilium □ generic-veth □□□

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: cni-configuration
  namespace: kube-system
data:
  cni-config: |-
    {
      "name": "generic-veth",
      "cniVersion": "0.3.1",
      "plugins": [
        {
          "type": "kube-ovn",
          "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
          "ipam": {
              "type": "kube-ovn",
              "server_socket": "/run/openvswitch/kube-ovn-daemon.sock"
          }
        },
        {
          "type": "portmap",
          "snat": true,
          "capabilities": {"portMappings": true}
        },
        {
          "type": "cilium-cni"
        }
      ]
    }
```

□□□□□□

```
kubectl apply -f chaining.yaml
```

□□ Helm □□ Cilium□

```
helm repo add cilium https://helm.cilium.io/
helm install cilium cilium/cilium --version 1.11.6 \
    --namespace kube-system \
    --set cni.chainingMode=generic-veth \
    --set cni.customConf=true \
    --set cni.configMap=cni-configuration \
    --set tunnel=disabled \
    --set enableIPv4Masquerade=false \
    --set devices="eth+ ovn0 genev_sys_6081 vxlan_sys_4789" \
    --set enableIdentityMark=false
```

□□ Cilium □□□□□

```
# cilium  status
    /¯¯\
 /¯¯\__/¯¯\    Cilium:         OK
 \__/¯¯\__/    Operator:       OK
 /¯¯\__/¯¯\    Hubble:         disabled
 \__/¯¯\__/    ClusterMesh:    disabled
    \__/

DaemonSet         cilium             Desired: 2, Ready: 2/2, Available: 2/2
Deployment        cilium-operator    Desired: 2, Ready: 2/2, Available: 2/2
Containers:       cilium             Running: 2
                  cilium-operator    Running: 2
Cluster Pods:     8/11 managed by Cilium
Image versions    cilium             quay.io/cilium/cilium:v1.11.6: 2
                  cilium-operator    quay.io/cilium/operator-generic:v1.11.6: 2
```

7.11.4 □ Cilium □□□□□ Kube-OVN NAT □□

Cilium □□□□□□□□□□□□□ NAT □□ Pod □□□□□□□□□□□□□□□□□□ □□ NAT □□□□□ VPC □ Pod □□□□□ SNAT/DNAT □□□□□□□□□□□□□ IP □□ Pod □□□ IP□

Cilium □□ **SIP**□Source IP□□□□□□□□□□□□□□□——□□□□□□□ Pod IP □□□□□□□□□□□□ Hubble □□□ **DROPPED**□□

[illegible] IP [illegible] NAT [illegible]

[illegible]/Pod [illegible]Cilium ≥ 1.20[illegible]

Cilium 1.20 [illegible]cilium/cilium#43505[illegible] [illegible] **annotation** [illegible] Pod [illegible] SIP [illegible]

1. [illegible] annotation[illegible]

```
kubectl annotate namespace <nat-gateway-namespace> \
  config.cilium.io/delegate-source-ip-verification=true
```

2. NAT [illegible] Pod [illegible] SIP [illegible] annotation[illegible]

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    config.cilium.io/disable-source-ip-verification: "true"
```

[illegible] NAT [illegible] Pod [illegible]

[illegible]Cilium < 1.20[illegible]

[illegible] Cilium[illegible] Helm Chart [illegible] `enableSourceIPVerification` [illegible] `false` [illegible] SIP [illegible]

```
enableSourceIPVerification: false
```

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2026[illegible]4[illegible]28[illegible]

2022[illegible]5[illegible]24[illegible]

GitHub

7.11.5 [illegible]

7.12 Cilium NetworkPolicy □□

Kube-OVN □□□□□□□ Cilium □□□□□□□□□□□ Cilium □□□

□□□ Cilium □□□□□□□□ Cilium □□□□□□□□□□□□□□□□□□□□□□□□□□□□□ Cilium L3 □ L4 □□□□□□□□□□□□

7.12.1 □□□□

□□□□ Pod

□□ namespace `test` □□□□□ yaml□□ test namespace □□□□□ label `app=test` □ Pod□□□□□□□□□□ Pod□

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: test
  name: test
  namespace: test
spec:
  replicas: 1
  selector:
    matchLabels:
      app: test
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      labels:
        app: test
    spec:
      containers:
      - image: docker.io/library/nginx:alpine
        imagePullPolicy: IfNotPresent
        name: nginx
```

□□□□□□ yaml□□ default namespace □□□□□ label `app=dynamic` □ Pod □□□□□□□□ Pod□

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: dynamic
  name: dynamic
  namespace: default
spec:
  replicas: 2
  selector:
    matchLabels:
      app: dynamic
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: dynamic
    spec:
      containers:
      - image: docker.io/library/nginx:alpine
        imagePullPolicy: IfNotPresent
        name: nginx
```

□□□□ Pod □□ Label □□:

```
# kubectl get pod -o wide --show-labels
NAME                           READY   STATUS    RESTARTS   AGE   IP            NODE                     NOMINATED NODE   READINESS GATES   LABELS
dynamic-7d8d7874f5-9v5c4       1/1     Running   0          28h   10.16.0.35    kube-ovn-worker          <none>           <none>            app=dynamic,pod-
template-hash=7d8d7874f5
dynamic-7d8d7874f5-s8z2n       1/1     Running   0          28h   10.16.0.36    kube-ovn-control-plane   <none>           <none>            app=dynamic,pod-
template-hash=7d8d7874f5
# kubectl get pod -o wide -n test --show-labels
NAME                           READY   STATUS    RESTARTS   AGE     IP            NODE                     NOMINATED NODE   READINESS GATES   LABELS
dynamic-7d8d7874f5-6dsg6       1/1     Running   0          7h20m   10.16.0.2     kube-ovn-control-plane   <none>           <none>            app=dynamic,pod-
template-hash=7d8d7874f5
```

```
dynamic-7d8d7874f5-tjgtp        1/1     Running   0          7h46m   10.16.0.42   kube-ovn-worker          <none>           <none>            app=dynamic,pod-
template-hash=7d8d7874f5
label-test1-77b6764857-swq4k    1/1     Running   0          3h43m   10.16.0.12   kube-ovn-worker          <none>           <none>            app=test1,pod-
template-hash=77b6764857

#           Pod
test-54c98bc466-mft5s           1/1     Running   0          8h      10.16.0.41   kube-ovn-worker          <none>           <none>            app=test,pod-
template-hash=54c98bc466
```

L3 □□□□□□

□□□□ yaml□□□ `CiliumNetworkPolicy` □□:

```yaml
apiVersion: "cilium.io/v2"
kind: CiliumNetworkPolicy
metadata:
  name: "l3-rule"
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
  - fromEndpoints:
    - matchLabels:
        app: dynamic
```

□ default namespace □□□□ Pod □□□□□□□ Pod □□□□□□□□□□□ □□□ test namespace □□□□□□□ Pod □□□□□□□□□□

default namespace □□□□□:

```
# kubectl exec -it dynamic-7d8d7874f5-9v5c4 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
```

test namespace □ Pod □□□□□□□□□:

```
# kubectl exec -it -n test dynamic-7d8d7874f5-6dsg6 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes
64 bytes from 10.16.0.41: seq=0 ttl=64 time=2.558 ms
64 bytes from 10.16.0.41: seq=1 ttl=64 time=0.223 ms
64 bytes from 10.16.0.41: seq=2 ttl=64 time=0.304 ms

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.223/1.028/2.558 ms
```

□□ Cilium □□□□□□□ `CiliumNetworkPolicy` □□□□□□□□□ `Namespace` □□□□□□□□□□□ Cilium □□□

□□□□□□□□□□□□□□□□□□□ **Namespace** □ Pod □□□□□□□□□□□□□□□□□□ **Namespace** □ Pod □□□□□

□□□□□□ Namespace □□□□□□□□□□□□□ Namespace □□□

□□□□□□□ `CiliumNetworkPolicy` □□□□□ namespace □□:

```yaml
  ingress:
  - fromEndpoints:
    - matchLabels:
        app: dynamic
        k8s:io.kubernetes.pod.namespace: default    #      Namespace    Pod
```

□□□□□□ `CiliumNetworkPolicy` □□□□:

```
# kubectl get cnp -n test  -o yaml l3-rule
apiVersion: cilium.io/v2
kind: CiliumNetworkPolicy
metadata:
  name: l3-rule
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
  - fromEndpoints:
    - matchLabels:
        app: dynamic
    - matchLabels:
        app: dynamic
        k8s:io.kubernetes.pod.namespace: default
```

□□□□ default namespace □□ Pod □□□□□ Pod □□□□:

```
# kubectl exec -it dynamic-7d8d7874f5-9v5c4 -n test -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes
64 bytes from 10.16.0.41: seq=0 ttl=64 time=2.383 ms
64 bytes from 10.16.0.41: seq=1 ttl=64 time=0.115 ms
64 bytes from 10.16.0.41: seq=2 ttl=64 time=0.142 ms

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.115/0.880/2.383 ms
```

□□□□□ Kubernetes □□□□ networkpolicy□□□□□□□ Cilium □□□□□□□□□□□ Namespace □□□ Namespace □□□□□□□□□

□□□ Kube-OVN □□□□□□□Kube-OVN □□□□□ k8s □□□□□□□□□□□ Namespace □□□□ **Pod**□□□□□□□□ Pod□□□□ Namespace □□□□□□ Namespace □□□□□□□□□ Pod□□□□□□□□□□ Pod □□□□

L4 □□□□□□

□□□□ yaml□□□ L4 □□□□□□□□□:

```yaml
apiVersion: "cilium.io/v2"
kind: CiliumNetworkPolicy
metadata:
  name: "l4-rule"
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
  - fromEndpoints:
    - matchLabels:
        app: dynamic
    toPorts:
    - ports:
      - port: "80"
        protocol: TCP
```

□□□□ Namespace □□□□□□□□□□ Pod □□□

```
# kubectl exec -it -n test dynamic-7d8d7874f5-6dsg6 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
bash-5.0#
bash-5.0# curl 10.16.0.41:80
<html>
<head>
        <title>Hello World!</title>
        <link href='//fonts.googleapis.com/css?family=Open+Sans:400,700' rel='stylesheet' type='text/css'>
        <style>
        body {
                background-color: white;
                text-align: center;
                padding: 50px;
                font-family: "Open Sans","Helvetica Neue",Helvetica,Arial,sans-serif;
        }
        #logo {
                margin-bottom: 40px;
        }
        </style>
</head>
<body>
                <h1>Hello World!</h1>
                                <h3>Links found</h3>
        <h3>I am on  test-54c98bc466-mft5s</h3>
        <h3>Cookie                  =</h3>
                                        <b>KUBERNETES</b> listening in 443 available at tcp://10.96.0.1:443<br />
                                                <h3>my name is hanhouchao!</h3>
                        <h3> RequestURI='/'</h3>
</body>
</html>
```

[illegible] Namespace [illegible] Pod [illegible]

```
# kubectl exec -it -n test label-test1-77b6764857-swq4k -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
bash-5.0#
bash-5.0# curl -v 10.16.0.41:80 --connect-timeout 10
*   Trying 10.16.0.41:80...
* After 10000ms connect time, move on!
* connect to 10.16.0.41 port 80 failed: Operation timed out
* Connection timeout after 10001 ms
* Closing connection 0
curl: (28) Connection timeout after 10001 ms
```

[illegible] Namespace [illegible] L3 [illegible]

[illegible] L4 [illegible]ping [illegible] TCP [illegible]

[illegible] ICMP [illegible] L4 [illegible]

L7 [illegible]

chaining [illegible]L7 [illegible] Cilium [illegible] Generic Veth Chaining[illegible]

[illegible] issue 12454 [illegible]

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2026[illegible]7[illegible]4[illegible]

2022[illegible]8[illegible]2[illegible]

GitHub

7.12.2 [illegible]

7.13 Cilium □□□□□□

Kube-OVN □□□□□□□ Cilium □□□□□□□□□□□ Cilium □□□

Cilium □□□□□□□□□□□□□□□□□□□□□□□ Hubble □□□□Hubble □□□□□□□□□□□□□□□□□□□□□□□□

7.13.1 □□ Hubble

□□□ Cilium □□□□□□□□□□□ Hubble □□□□□□□□□□□□□□□□□□□□□□□□□□ Hubble□

□□□□□□□□□ helm □□ Hubble□

```
helm upgrade cilium cilium/cilium --version 1.11.6 \
   --namespace kube-system \
   --reuse-values \
   --set hubble.relay.enabled=true \
   --set hubble.ui.enabled=true
```

□□□□ Hubble □□□□□ `cilium status` □□□□□□□□□ Hubble □□□□□

```
# cilium status
    /¯¯\
 /¯¯\__/¯¯\    Cilium:         OK
 \__/¯¯\__/    Operator:       OK
 /¯¯\__/¯¯\    Hubble:         OK
 \__/¯¯\__/    ClusterMesh:    disabled
    \__/

Deployment        hubble-relay       Desired: 1, Ready: 1/1, Available: 1/1
Deployment        cilium-operator    Desired: 2, Ready: 2/2, Available: 2/2
DaemonSet         cilium             Desired: 2, Ready: 2/2, Available: 2/2
Deployment        hubble-ui          Desired: 1, Ready: 1/1, Available: 1/1
Containers:       cilium             Running: 2
                  hubble-ui          Running: 1
                  hubble-relay       Running: 1
                  cilium-operator    Running: 2
Cluster Pods:     16/17 managed by Cilium
Image versions    hubble-relay       quay.io/cilium/hubble-relay:v1.11.6@sha256:fd9034a2d04d5b973f1e8ed44f230ea195b89c37955ff32e34e5aa68f3ed675a: 1
                  cilium-operator    quay.io/cilium/operator-generic:v1.11.6@sha256:9f6063c7bcaede801a39315ec7c166309f6a6783e98665f6693939cf1701bc17: 2
                  cilium             quay.io/cilium/cilium:v1.11.6@sha256:f7f93c26739b6641a3fa3d76b1e1605b15989f25d06625260099e01c8243f54c: 2
                  hubble-ui          quay.io/cilium/hubble-ui:v0.9.0@sha256:0ef04e9a29212925da6bdfd0ba5b581765e41a01f1cc30563cef9b30b457fea0: 1
                  hubble-ui          quay.io/cilium/hubble-ui-backend:v0.9.0@sha256:000df6b76719f607a9edefb9af94dfd1811a6f1b6a8a9c537cba90bf12df474b: 1
apple@bogon cilium %
```

□□ Hubble □□□□□□□□□□□□□□□□□□□□□□□□□□□ □□□□□□□□□ Hubble CLI :

```
curl -L --fail --remote-name-all https://github.com/cilium/hubble/releases/download/v0.10.0/hubble-linux-amd64.tar.gz
sudo tar xzvfC hubble-linux-amd64.tar.gz /usr/local/bin
```

7.13.2 □□□□□□

Cilium □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

□□□□ `cilium connectivity test` □Cilium □□□□□ `cilium-test` □ Namespace□□□□ cilium-test □□□□□□□□

□□□□□□□□□□ cilium-test namespace □□□□□□□□□□□

```
# kubectl get all -n cilium-test
NAME                                  READY   STATUS    RESTARTS   AGE
pod/client-7df6cfbf7b-z5t2j           1/1     Running   0          21s
pod/client2-547996d7d8-nvgxg          1/1     Running   0          21s
pod/echo-other-node-d79544ccf-hl4gg   2/2     Running   0          21s
pod/echo-same-node-5d466d5444-ml7tc   2/2     Running   0          21s

NAME                      TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
service/echo-other-node   NodePort   10.109.58.126    <none>        8080:32269/TCP   21s
service/echo-same-node    NodePort   10.108.70.32     <none>        8080:32490/TCP   21s

NAME                              READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/client            1/1     1            1           21s
deployment.apps/client2           1/1     1            1           21s
deployment.apps/echo-other-node   1/1     1            1           21s
deployment.apps/echo-same-node    1/1     1            1           21s

NAME                                        DESIRED   CURRENT   READY   AGE
```

```
replicaset.apps/client-7df6cfbf7b           1         1         1       21s
replicaset.apps/client2-547996d7d8          1         1         1       21s
replicaset.apps/echo-other-node-d79544ccf   1         1         1       21s
replicaset.apps/echo-same-node-5d466d5444   1         1         1       21s
```

7.13.3 [illegible]

[illegible] Cilium [illegible] `kube-system namespace` [illegible] Cilium [illegible] pod [illegible] `hubble observe` [illegible]

```
# kubectl get pod -n kube-system -o wide
NAME                                             READY   STATUS    RESTARTS   AGE     IP           NODE                     NOMINATED NODE   READINESS GATES
cilium-d6h56                                     1/1     Running   0          2d20h   172.18.0.2   kube-ovn-worker          <none>           <none>
cilium-operator-5887f78bbb-c7sb2                 1/1     Running   0          2d20h   172.18.0.2   kube-ovn-worker          <none>           <none>
cilium-operator-5887f78bbb-wj8gt                 1/1     Running   0          2d20h   172.18.0.3   kube-ovn-control-plane   <none>           <none>
cilium-tq5xb                                     1/1     Running   0          2d20h   172.18.0.3   kube-ovn-control-plane   <none>           <none>
kube-ovn-pinger-7lgk8                            1/1     Running   0          21h     10.16.0.19   kube-ovn-control-plane   <none>           <none>
kube-ovn-pinger-msvcn                            1/1     Running   0          21h     10.16.0.18   kube-ovn-worker          <none>           <none>

# kubectl exec -it -n kube-system cilium-d6h56 -- bash
root@kube-ovn-worker:/home/cilium# hubble observe --from-namespace kube-system
Jul 29 03:24:25.551: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.561: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: RST)
Jul 29 03:24:25.561: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, RST)
Jul 29 03:24:25.572: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: SYN)
Jul 29 03:24:25.572: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.651: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.661: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: RST)
Jul 29 03:24:25.661: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, RST)
Jul 29 03:24:25.761: kube-system/kube-ovn-pinger-msvcn:52004 -> 172.18.0.3:6443 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.779: kube-system/kube-ovn-pinger-msvcn -> kube-system/kube-ovn-pinger-7lgk8 to-stack FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:25.779: kube-system/kube-ovn-pinger-msvcn <- kube-system/kube-ovn-pinger-7lgk8 to-endpoint FORWARDED (ICMPv4 EchoReply)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 <- kube-system/hubble-relay-959988db5-zc5vv:4245 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 <- kube-system/hubble-relay-959988db5-zc5vv:80 to-endpoint FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 -> kube-system/hubble-relay-959988db5-zc5vv:4245 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 -> kube-system/hubble-relay-959988db5-zc5vv:4245 to-endpoint FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.975: kube-system/kube-ovn-pinger-7lgk8 -> kube-system/kube-ovn-pinger-msvcn to-endpoint FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:25.975: kube-system/kube-ovn-pinger-7lgk8 <- kube-system/kube-ovn-pinger-msvcn to-stack FORWARDED (ICMPv4 EchoReply)
Jul 29 03:24:25.979: kube-system/kube-ovn-pinger-msvcn -> 172.18.0.3 to-stack FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:26.037: kube-system/coredns-6d4b75cb6d-lbgjg:36430 -> 172.18.0.3:6443 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:26.282: kube-system/kube-ovn-pinger-msvcn -> 172.18.0.2 to-stack FORWARDED (ICMPv4 EchoRequest)
```

[illegible] Hubble Relay [illegible]Hubble [illegible]

[illegible]

[illegible] Hubble API[illegible] Hubble Service[illegible] `kubectl port-forward deployment/hubble-relay -n kube-system 4245:4245` [illegible]

[illegible] [illegible]

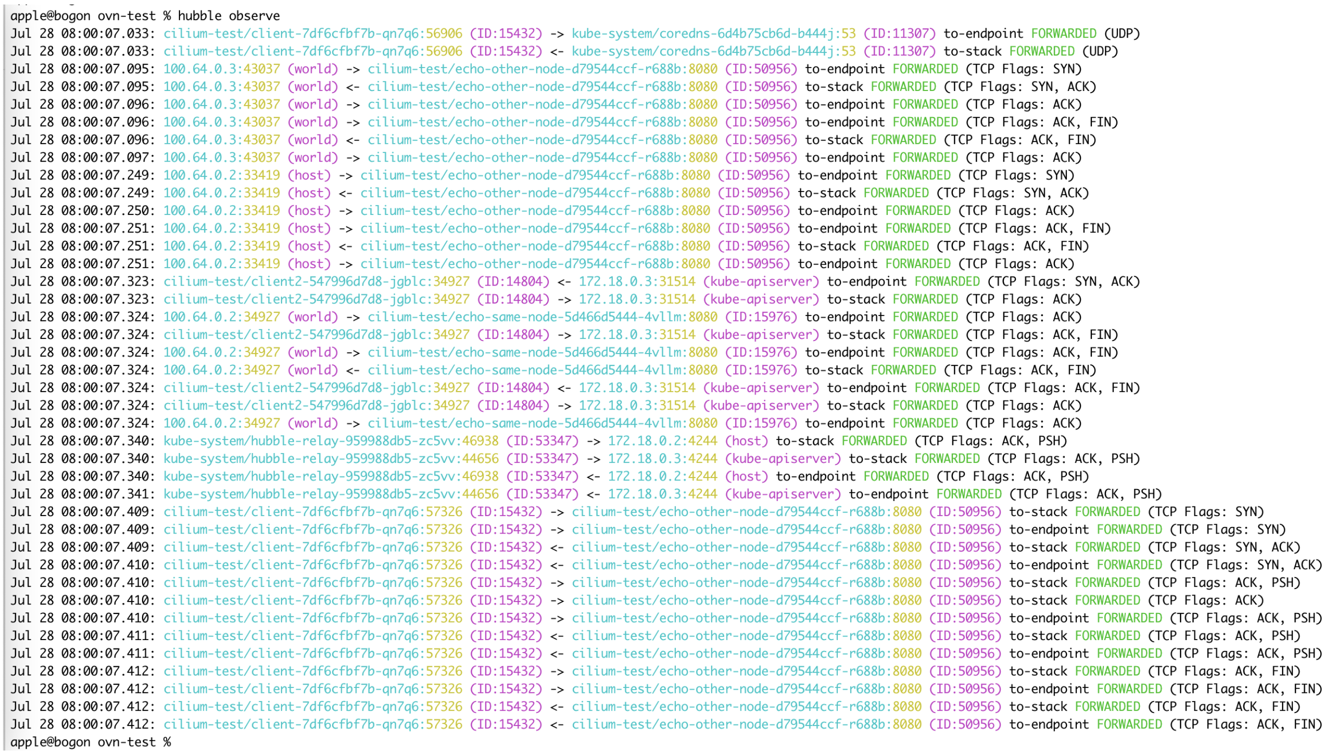
`kubectl port-forward` [illegible]

[illegible] `hubble status` [illegible]

```
# hubble status
Healthcheck (via localhost:4245): Ok
Current/Max Flows: 8,190/8,190 (100.00%)
Flows/s: 22.86
Connected Nodes: 2/2
```

□□□□□

□□□□□□ `hubble observe` □□□□□□□□□□□□□ □□□□ cilium-test □□□□□□□□□□□□



```
apple@bogon ovn-test % hubble observe
Jul 28 08:00:07.033: cilium-test/client-7df6cfbf7b-qn7q6:56906 (ID:15432) -> kube-system/coredns-6d4b75cb6d-b444j:53 (ID:11307) to-endpoint FORWARDED (UDP)
Jul 28 08:00:07.033: cilium-test/client-7df6cfbf7b-qn7q6:56906 (ID:15432) <- kube-system/coredns-6d4b75cb6d-b444j:53 (ID:11307) to-stack FORWARDED (UDP)
Jul 28 08:00:07.095: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.095: 100.64.0.3:43037 (world) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.097: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.249: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.249: 100.64.0.2:33419 (host) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.250: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.323: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) <- 172.18.0.3:31514 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.323: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) <- cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) <- 172.18.0.3:31514 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) -> 172.18.0.2:4244 (host) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:44656 (ID:53347) -> 172.18.0.3:4244 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) <- 172.18.0.2:4244 (host) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.341: kube-system/hubble-relay-959988db5-zc5vv:44656 (ID:53347) <- 172.18.0.3:4244 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.411: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.411: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
apple@bogon ovn-test %
```

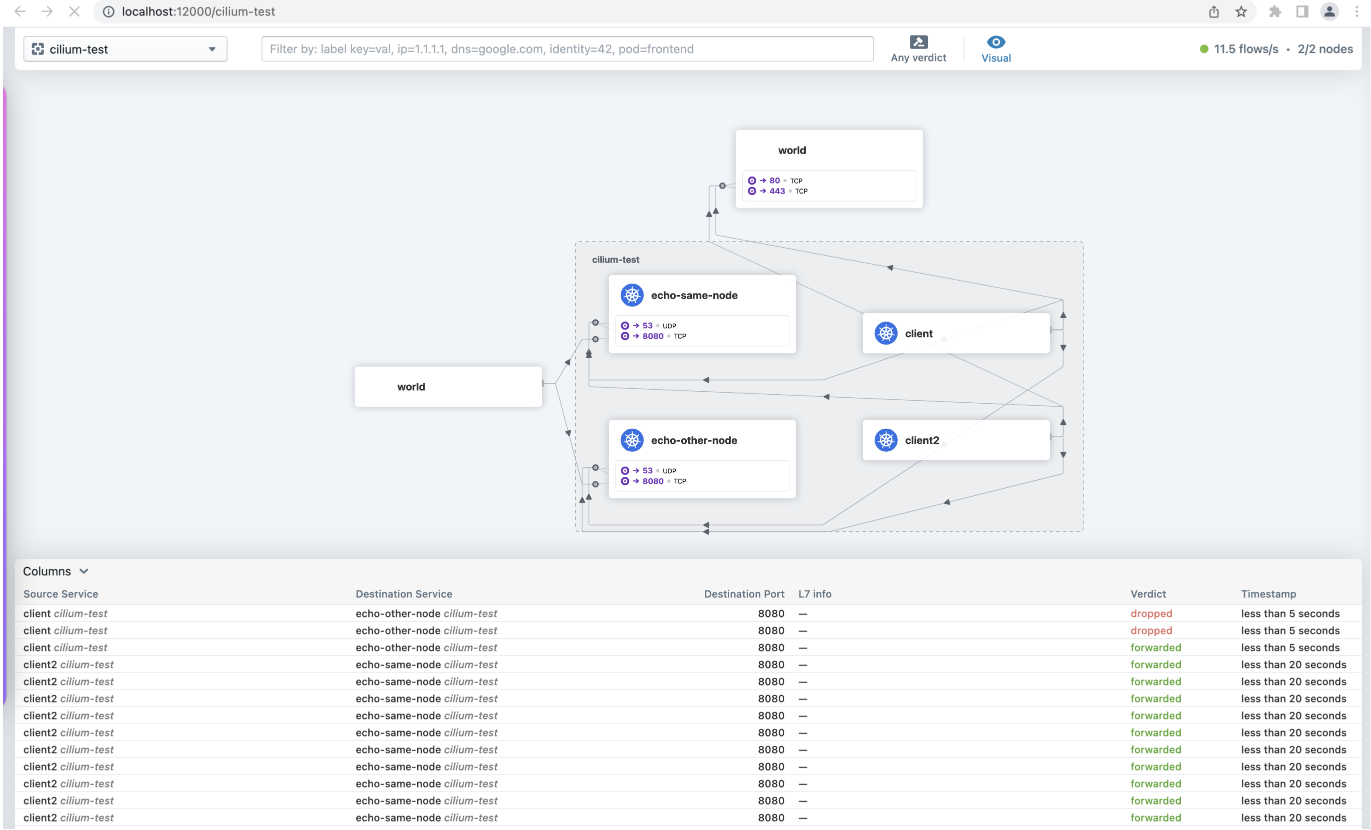
□□□□□□□ `hubble observe` □□ □□□□□□□□□□□□□□ `hubble help observe` □□□□ Hubble CLI □□□□□□□□

7.13.4 □□ UI □□□□□□

□□ `cilium status` □□□□□ Hubble UI □□□□□□□□□□□□ Hubble □□□□□□□□□ UI □□□□

□□□□ `cilium hubble ui` □□□□□□□□□□□□ `hubble-ui service` □□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ Hubble UI □□□□□□□□□□□□□□□□□□□□ `http://localhost:12000` □□ UI □□□□□

□□□□□□□□□ `cilium-test` namespace□□□ Cilium □□□□□□□□□□



Source Service	Destination Service	Destination Port	L7 info	Verdict	Timestamp
client *cilium-test*	echo-other-node *cilium-test*	8080	—	dropped	less than 5 seconds
client *cilium-test*	echo-other-node *cilium-test*	8080	—	dropped	less than 5 seconds
client *cilium-test*	echo-other-node *cilium-test*	8080	—	forwarded	less than 5 seconds
client2 *cilium-test*	echo-same-node *cilium-test*	8080	—	forwarded	less than 20 seconds
client2 *cilium-test*	echo-same-node *cilium-test*	8080	—	forwarded	less than 20 seconds
client2 *cilium-test*	echo-same-node *cilium-test*	8080	—	forwarded	less than 20 seconds
client2 *cilium-test*	echo-same-node *cilium-test*	8080	—	forwarded	less than 20 seconds
client2 *cilium-test*	echo-same-node *cilium-test*	8080	—	forwarded	less than 20 seconds
client2 *cilium-test*	echo-same-node *cilium-test*	8080	—	forwarded	less than 20 seconds
client2 *cilium-test*	echo-same-node *cilium-test*	8080	—	forwarded	less than 20 seconds
client2 *cilium-test*	echo-same-node *cilium-test*	8080	—	forwarded	less than 20 seconds
client2 *cilium-test*	echo-same-node *cilium-test*	8080	—	forwarded	less than 20 seconds
client2 *cilium-test*	echo-same-node *cilium-test*	8080	—	forwarded	less than 20 seconds

7.13.5 Hubble □□□□

Hubble □□□□□□□□ Pod □□□□□□□□□□□□□□ Hubble □□□□□□□□□□□□□□□□□

□□□□□□□□□ `hubble.metrics.enabled` □□□:

```
helm upgrade cilium cilium/cilium --version 1.11.6 \
   --namespace kube-system \
   --reuse-values \
   --set hubble.relay.enabled=true \
   --set hubble.ui.enabled=true \
   --set hubble.metrics.enabled="{dns,drop,tcp,flow,icmp,http}"
```

□□□□□□□ kube-system namespace □□□□□ `hubble-metrics` □□□□□□□□ Endpoints □□ Hubble □□□□□□□□□□□□:

```
# curl 172.18.0.2:9091/metrics
# HELP hubble_drop_total Number of drops
# TYPE hubble_drop_total counter
hubble_drop_total{protocol="ICMPv6",reason="Unsupported L3 protocol"} 2
# HELP hubble_flows_processed_total Total number of flows processed
# TYPE hubble_flows_processed_total counter
hubble_flows_processed_total{protocol="ICMPv4",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 335
hubble_flows_processed_total{protocol="ICMPv4",subtype="to-stack",type="Trace",verdict="FORWARDED"} 335
hubble_flows_processed_total{protocol="ICMPv6",subtype="",type="Drop",verdict="DROPPED"} 2
hubble_flows_processed_total{protocol="TCP",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 8282
hubble_flows_processed_total{protocol="TCP",subtype="to-stack",type="Trace",verdict="FORWARDED"} 6767
hubble_flows_processed_total{protocol="UDP",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 1642
hubble_flows_processed_total{protocol="UDP",subtype="to-stack",type="Trace",verdict="FORWARDED"} 1642
# HELP hubble_icmp_total Number of ICMP messages
# TYPE hubble_icmp_total counter
hubble_icmp_total{family="IPv4",type="EchoReply"} 335
hubble_icmp_total{family="IPv4",type="EchoRequest"} 335
hubble_icmp_total{family="IPv4",type="RouterSolicitation"} 2
# HELP hubble_tcp_flags_total TCP flag occurrences
# TYPE hubble_tcp_flags_total counter
hubble_tcp_flags_total{family="IPv4",flag="FIN"} 2043
hubble_tcp_flags_total{family="IPv4",flag="RST"} 301
hubble_tcp_flags_total{family="IPv4",flag="SYN"} 1169
hubble_tcp_flags_total{family="IPv4",flag="SYN-ACK"} 1169
```

PDF Slack Support

2023□3□26□

2022□8□2□

GitHub

7.13.6 □□

7.14 [illegible]

[illegible] Kube-OVN [illegible]

7.14.1 [illegible]

```yaml
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: external
spec:
  cidrBlock: 172.31.0.0/16
  gatewayType: centralized
  natOutgoing: false
  externalEgressGateway: 192.168.0.1
  policyRoutingTableID: 1000
  policyRoutingPriority: 1500
```

- `natOutgoing`: [illegible] `false` [illegible]
- `externalEgressGateway` [illegible]
- `policyRoutingTableID` [illegible] TableID [illegible]
- `policyRoutingPriority` [illegible]

PDF **Slack** **Support**

2022[illegible]6[illegible]8[illegible]

2022[illegible]5[illegible]24[illegible]

GitHub

7.14.2 [illegible]

7.15 VIP [illegible] IP

VIP [illegible] IP[illegible] IP [illegible]VIP [illegible] OpenStack Neutron Allowed-Address-Pairs[illegible]AAP[illegible] OVN Switch LB Rule [illegible] VIP[illegible] Pod [illegible] `IP` [illegible] IPAM [illegible]VIP [illegible] Pod [illegible] VIP [illegible]

- Allowed-Address-Pairs VIP
- Switch LB rule VIP

7.15.1 1. Allowed-Address-Pairs VIP

[illegible] IP [illegible] Pod [illegible]

- Kubernetes [illegible] Kubernetes [illegible] Kubernetes [illegible] Underlay [illegible] Subnet [illegible]
- LB [illegible] Subnet [illegible] IP[illegible] Pod[illegible]

[illegible]VIP [illegible] Allowed-Address-Pairs [illegible] IP [illegible] IP [illegible]

- Keepalived [illegible] IP [illegible]

1.1 [illegible] VIP

[illegible] IP [illegible] IP [illegible] yaml [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: vip-dynamic-01
spec:
  subnet: ovn-default
  type: ""
```

- `subnet`: [illegible] Subnet [illegible] IP[illegible]
- `type`: [illegible] ipam ip [illegible] `switch_lb_vip` [illegible] vip [illegible] switch lb [illegible] vip [illegible] ip [illegible]

[illegible] VIP[illegible]

```
# kubectl get vip
NAME             V4IP         PV4IP   MAC                  PMAC   V6IP   PV6IP   SUBNET        READY
vip-dynamic-01   10.16.0.12           00:00:00:F0:DB:25                          ovn-default   true
```

[illegible] VIP [illegible] `10.16.0.12` [illegible] IP [illegible]

1.2 [illegible] VIP

[illegible] VIP [illegible] IP [illegible] yaml [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: static-vip01
spec:
  subnet: ovn-default
  v4ip: "10.16.0.121"
```

- `subnet`: [illegible] Subnet [illegible] IP[illegible]
- `v4ip`: [illegible] IP [illegible] `subnet` [illegible] CIDR [illegible]

[illegible] VIP[illegible]

```
# kubectl get vip
NAME             V4IP         PV4IP   MAC                  PMAC   V6IP   PV6IP   SUBNET        READY
static-vip01   10.16.0.121           00:00:00:F0:DB:26                          ovn-default   true
```

[illegible] VIP [illegible] IP [illegible]

1.3 Pod □□ VIP □□ AAP

Pod □□□□ annotation □□ VIP □□ AAP □□□Pod □ labels □□□□ VIP □ `selector` □□□□□□□□□

Pod annotation □□□□□□ VIP□□□□□□□ovn.kubernetes.io/aaps: vip-aap,vip-aap2,vip-aap3

AAP □□□□□□□□□□□ Pod □□□□□□□AAP □□ Pod □□ VIP □□ subnet □□□ Port □□□□

1.3.1 □□ VIP □□ AAP

```yaml
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: vip-aap
spec:
  subnet: ovn-default
  namespace: default
  selector:
    - "app: aap1"
```

VIP □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

- `namespace` : AAP □□□□VIP □□□□□□□□□□VIP □□□□□□□□□□□□□□□ AAP □□□
- `selector` : AAP □□□□□□□□□ VIP □□□□ Pod □□□□□□□□□□□□□ `namespace` □□ Pod□□□□□□□□□□□□□□ `key: value` □□□□□□□□□□□□□□□□□□□□□□□□□□□ AND □□□

□□□□□□□□□ VIP □□□ Port□

```
# kubectl ko nbctl show ovn-default
switch e32e1d3b-c539-45f4-ab19-be4e33a061f6 (ovn-default)
    port aap-vip
        type: virtual
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: busybox
  annotations:
    ovn.kubernetes.io/aaps: vip-aap
  labels:
    app: aap1
spec:
  containers:
    - name: busybox
      image: busybox
      command: ["sleep", "3600"]
      securityContext:
        capabilities:
          add:
            - NET_ADMIN
```

□□□□□□□□□□ AAP □□□□□□

```
# kubectl ko nbctl list logical_switch_port aap-vip
_uuid               : cd930750-0533-4f06-a6c0-217ddac73272
addresses           : []
dhcpv4_options      : []
dhcpv6_options      : []
dynamic_addresses   : []
enabled             : []
external_ids        : {ls=ovn-default, vendor=kube-ovn}
ha_chassis_group    : []
mirror_rules        : []
name                : aap-vip
options             : {virtual-ip="10.16.0.100", virtual-parents="busybox.default"}
parent_name         : []
port_security       : []
tag                 : []
tag_request         : []
type                : virtual
up                  : false
```

virtual-ip □□□□ VIP □□□ IP□virtual-parents □□□□□ AAP □□□ Pod □□□ Port□

□□□□□□□□□□ Pod □□□□□□

```
# kubectl exec -it busybox -- ip addr add 10.16.0.100/16 dev eth0
# kubectl exec -it busybox01 -- ip addr show eth0
35: eth0@if36: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 1400 qdisc noqueue
    link/ether 00:00:00:e2:ab:0c brd ff:ff:ff:ff:ff:ff
    inet 10.16.0.7/16 brd 10.16.255.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 10.16.0.100/16 scope global secondary eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fee2:ab0c/64 scope link
       valid_lft forever preferred_lft forever
```

 Pod IPVIP IP subnet Pod IP

7.15.2 2. Switch LB rule vip

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: slr-01
spec:
  subnet: ovn-default
  type: switch_lb_vip
```

- `subnet` : Subnet IP
- `type` : ipam ip `switch_lb_vip` vip switch lb vip ip

7.15.3 3. v1.17

v1.17 `ovn.kubernetes.io/vip` Pod IP `kube_host_vm_vip` annotation Pod VIP VIP Pod IP

- `ovn.kubernetes.io/vip` Pod IPAM annotation
- Pod IPAM VIP VIP Pod controller VIP
- `kube_host_vm_vip`
- v1.17 VIP `ip_reserved` Pod Subnet `UsingIPs` Pod IP VIP

 Pod IPPool Pod annotationAAP Switch LB VIP

7.15.4 4. StatefulSet Kubevirt VM IP

StatefulSet VM IP VIP

 `StatefulSet` `VM` Pod IP

VM IP `kube-ovn-controller` `keep-vm-ip` `true` Kubevirt VM

PDF Slack Support

20261028

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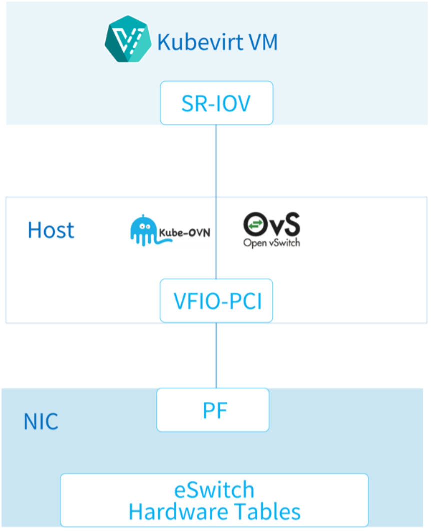
GitHub +3

7.15.5

7.16 Mellanox [illegible] Offload [illegible]

Kube-OVN [illegible] OVS [illegible] CPU [illegible] CPU [illegible]Mellanox [illegible] Accelerated Switching And Packet Processing (ASAP²) [illegible] OVS [illegible] eSwitch [illegible] OVS [illegible] CPU [illegible]


Kubevirt VM
SR-IOV
Host
Kube-OVN
OvS
Open vSwitch
VFIO-PCI
NIC
PF
eSwitch
Hardware Tables

Note

[illegible] 2022 [illegible]

7.16.1 [illegible]

- Mellanox CX5/CX6/CX7/BlueField [illegible] ASAP² [illegible]
- CentOS 8 Stream [illegible] Linux 5.7 [illegible]
- [illegible] `dp_hash` [illegible] `hash` [illegible] OVN LB [illegible]
- [illegible] bond[illegible]

7.16.2 [illegible] SR-IOV [illegible] Device Plugin

Mellanox [illegible] offload [illegible] SR-IOV [illegible] Device Plugin[illegible] sriov-network-operator [illegible]

[illegible] SR-IOV [illegible] Device Plugin

[illegible] SR-IOV

[illegible] ID[illegible] `84:00.0` [illegible] `84.00.1` [illegible]

```
# lspci -nn | grep ConnectX-5
84:00.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
84:00.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
```

□□□□ ID □□□□□□□

```
# ls -l /sys/class/net/ | grep 84:00.0
lrwxrwxrwx 1 root root 0 Feb 4 16:16 enp132s0f0np0 -> ../../devices/pci0000:80/0000:80:08.0/0000:84:00.0/net/enp132s0f0np0
# ls -l /sys/class/net/ | grep 84:00.1
lrwxrwxrwx 1 root root 0 Feb 4 16:16 enp132s0f1np1 -> ../../devices/pci0000:80/0000:80:08.0/0000:84:00.1/net/enp132s0f1np1
```

□□□□□□□□□ bond□

> □□□□□□ enp132s0f0np0 □ enp132s0f1np1 □□ bond1

```
# ip link show enp132s0f0np0 | grep bond
160: enp132s0f0np0: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bond1 state UP mode DEFAULT group default qlen 1000
# ip link show enp132s0f1np1 | grep bond
169: enp132s0f1np1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bond1 state UP mode DEFAULT group default qlen 1000
```

□□ bond □□□□ VF□

```
ifenslave -d bond1 enp132s0f0np0
ifenslave -d bond1 enp132s0f1np1
echo 0 > /sys/class/net/enp132s0f0np0/device/sriov_numvfs
echo 0 > /sys/class/net/enp132s0f1np1/device/sriov_numvfs
ip link set enp132s0f0np0 down
ip link set enp132s0f1np1 down
```

□□□□□□□□□

> OVS □□□□□□□□□□□□□□□□SMFS □ DMFS

- SMFS (software-managed flow steering)□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
- DMFS (device-managed flow steering)□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

□□□□□□□□□□□□□□ sysfs □ devlink API □□□□□

```
#    sysfs
echo <smfs|dmfs> > /sys/class/net/enp132s0f0np0/compat/devlink/steering_mode
echo <smfs|dmfs> > /sys/class/net/enp132s0f1np1/compat/devlink/steering_mode
#    devlink
devlink dev param set pci/84:00.0 name flow_steering_mode value smfs cmode runtime
devlink dev param set pci/84:00.1 name flow_steering_mode value smfs cmode runtime
```

> □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

□□□□ VF □□□

```
# cat /sys/class/net/enp132s0f0np0/device/sriov_totalvfs
127
# cat /sys/class/net/enp132s1f0np1/device/sriov_totalvfs
127
```

□□ VF□□□□□□□□□□□□□□□□□

```
# echo '4' > /sys/class/net/enp132s0f0np0/device/sriov_numvfs
# echo '4' > /sys/class/net/enp132s1f0np1/device/sriov_numvfs
# ip link show enp132s0f0np0
160: enp132s0f0np0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:c0:eb:74:c3:4a brd ff:ff:ff:ff:ff:ff
    vf 0 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 1 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 2 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 3 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
# ip link show enp132s0f1np1
169: enp132s0f1np1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000
    link/ether 08:c0:eb:74:c3:4b brd ff:ff:ff:ff:ff:ff
    vf 0 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 1 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 2 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
    vf 3 link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff, spoof checking off, link-state disable, trust off, query_rss off
# ip link set enp132s0f0np0 up
# ip link set enp132s0f1np1 up
```

□□□□ VF □□□□□ ID□

```
# lspci -nn | grep ConnectX-5 | grep Virtual
84:00.2 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.3 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.4 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

```
84:00.5 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.6 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.7 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:01.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:01.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

[illegible] VF [illegible]

```
echo 0000:84:00.2 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.3 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.4 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.5 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.6 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.7 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:01.0 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:01.1 > /sys/bus/pci/drivers/mlx5_core/unbind
```

[illegible] eSwitch [illegible]

```
devlink dev eswitch set pci/0000:84:00.0 mode switchdev
devlink dev eswitch set pci/0000:84:00.1 mode switchdev
ethtool -K enp132s0f0np0 hw-tc-offload on
ethtool -K enp132s0f1np1 hw-tc-offload on
```

SR-IOV VF [illegible]

SR-IOV VF LAG [illegible] PF [illegible] OVS [illegible] e-switch [illegible] bond [illegible]

- Active-backup
- XOR
- LACP

> SR-IOV VF LAG [illegible] LAG [illegible]bond [illegible] bond PF [illegible] VF[illegible] VF [illegible] bond [illegible] PF [illegible] VF [illegible] PF[illegible] XOR [illegible] LACP [illegible] PF [illegible] VF [illegible] PF [illegible]

[illegible] LACP [illegible]

```
modprobe bonding mode=802.3ad
ip link set enp132s0f0np0 master bond1
ip link set enp132s0f1np1 master bond1
ip link set enp132s0f0np0 up
ip link set enp132s0f1np1 up
ip link set bond1 up
```

> [illegible] bond[illegible]

[illegible] VF [illegible]

```
echo 0000:84:00.2 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.3 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.4 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.5 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.6 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.7 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:01.0 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:01.1 > /sys/bus/pci/drivers/mlx5_core/bind
```

`NetworkManager` [illegible] `NetworkManager` [illegible]

```
systemctl stop NetworkManager
systemctl disable NetworkManager
```

[illegible] DEVICE PLUGIN

[illegible] VF [illegible] Pod [illegible] VF [illegible] SR-IOV Device Plugin [illegible]

[illegible] SR-IOV [illegible] Configmap[illegible]

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
      "resourceList": [{
          "resourcePrefix": "mellanox.com",
          "resourceName": "cx5_sriov_switchdev",
          "selectors": {
                  "vendors": ["15b3"],
                  "devices": ["1018"],
                  "drivers": ["mlx5_core"]
              }
      }
      ]
    }
```

SR-IOV Device Plugin [illegible] ConfigMap [illegible]ConfigMap [illegible] name [illegible] sriovdp-config[illegible]

- `selectors`: VF [illegible]
- `vendors`: [illegible]
- `devices`: [illegible]
- `drivers`: [illegible]

`selectors` [illegible] `pciAddresses` [illegible] `acpiIndexes` [illegible] VF [illegible]SR-IOV ConfigMap [illegible]

[illegible] SR-IOV [illegible]

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/v3.6.2/deployments/sriovdp-daemonset.yaml
```

[illegible] SR-IOV [illegible] Kubernetes Node [illegible]

```
kubectl describe node kube-ovn-01  | grep mellanox

mellanox.com/cx5_sriov_switchdev:  8
mellanox.com/cx5_sriov_switchdev:  8
mellanox.com/cx5_sriov_switchdev  0           0
```

[illegible] sriov-network-operator [illegible] SR-IOV [illegible] Device Plugin

[illegible] node-feature-discovery[illegible]

```
kubectl apply -k https://github.com/kubernetes-sigs/node-feature-discovery/deployment/overlays/default?ref=v0.11.3
```

[illegible] offload [illegible] annotation:

```
kubectl label nodes [offloadNicNode] feature.node.kubernetes.io/network-sriov.capable=true
```

[illegible] Operator[illegible]

```
git clone --depth=1 https://github.com/kubeovn/sriov-network-operator.git
kubectl apply -k sriov-network-operator/deploy
```

[illegible] Operator [illegible]

```
# kubectl get -n kube-system all | grep sriov
NAME                                          READY   STATUS    RESTARTS   AGE
pod/sriov-network-config-daemon-bf9nt         1/1     Running   0          8s
pod/sriov-network-operator-54d7545f65-296gb   1/1     Running   0          10s

NAME                                         DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE
SELECTOR                                                  AGE
daemonset.apps/sriov-network-config-daemon   1         1         1       1            1           beta.kubernetes.io/os=linux,feature.node.kubernetes.io/
network-sriov.capable=true   8s

NAME                                     READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/sriov-network-operator   1/1     1            1           10s

NAME                                                DESIRED   CURRENT   READY   AGE
replicaset.apps/sriov-network-operator-54d7545f65   1         1         1       10s
```

[illegible] `SriovNetworkNodeState` [illegible] `node1` [illegible] Mellanox [illegible]

```
# kubectl get sriovnetworknodestates.sriovnetwork.openshift.io -n kube-system node1 -o yaml
apiVersion: sriovnetwork.openshift.io/v1
kind: SriovNetworkNodeState
spec: ...
```

```
status:
  interfaces:
  - deviceID: "1017"
    driver: mlx5_core
    mtu: 1500
    pciAddress: "0000:5f:00.0"
    totalvfs: 8
    vendor: "15b3"
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    name: ens41f0np0
  - deviceID: "1017"
    driver: mlx5_core
    mtu: 1500
    pciAddress: "0000:5f:00.1"
    totalvfs: 8
    vendor: "15b3"
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    name: ens41f1np1
```

[illegible] `SriovNetworkNodePolicy` [illegible] `nicSelector` [illegible]

```
apiVersion: sriovnetwork.openshift.io/v1
kind: SriovNetworkNodePolicy
metadata:
  name: policy
  namespace: kube-system
spec:
  nodeSelector:
    feature.node.kubernetes.io/network-sriov.capable: "true"
  eSwitchMode: switchdev
  numVfs: 3
  nicSelector:
    pfNames:
    - ens41f0np0
    - ens41f1np1
  resourceName: cx_sriov_switchdev
```

[illegible] `SriovNetworkNodeState` [illegible] `status` [illegible]

```
# kubectl get sriovnetworknodestates.sriovnetwork.openshift.io -n kube-system node1 -o yaml

...
spec:
  interfaces:
  - eSwitchMode: switchdev
    name: ens41f0np0
    numVfs: 3
    pciAddress: 0000:5f:00.0
    vfGroups:
    - policyName: policy
      vfRange: 0-2
      resourceName: cx_sriov_switchdev
  - eSwitchMode: switchdev
    name: ens41f1np1
    numVfs: 3
    pciAddress: 0000:5f:00.1
    vfGroups:
    - policyName: policy
      vfRange: 0-2
      resourceName: cx_sriov_switchdev
status:
  interfaces
  - Vfs:
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.2
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.3
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.4
      vendor: "15b3"
    deviceID: "1017"
    driver: mlx5_core
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:ab
    mtu: 1500
    name: ens41f0np0
    numVfs: 3
    pciAddress: 0000:5f:00.0
    totalvfs: 3
    vendor: "15b3"
  - Vfs:
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.5
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.6
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.7
      vendor: "15b3"
    deviceID: "1017"
    driver: mlx5_core
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    mtu: 1500
    name: ens41f1np1
    numVfs: 3
    pciAddress: 0000:5f:00.1
    totalvfs: 3
    vendor: "15b3"
```

□□ VF □□□□

```
# lspci -nn | grep ConnectX
5f:00.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
5f:00.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
5f:00.2 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.3 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.4 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.5 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.6 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.7 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

□□ PF □□□□□

```
# cat /sys/class/net/ens41f0np0/compat/devlink/mode
switchdev
```

7.16.3 □□ Multus-CNI

SR-IOV Device Plugin □□□□□□□□ ID □□□□ Multus-CNI □□□ Kube-OVN□□□□□□□ Multus-CNI □□□□□□□□□

□□ Multus-CNI □□□□□□□

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/v4.0.2/deployments/multus-daemonset-thick.yml
```

> □□□multus □□□ Thin □ Thick □□□□□□□□□□ SR-IOV □□□□□ Thick □□□

□□ `NetworkAttachmentDefinition`□

```yaml
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: sriov
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: mellanox.com/cx5_sriov_switchdev
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "sriov.default.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

- `provider`: □□□□□ `NetworkAttachmentDefinition` □ {name}.{namespace}.ovn□

7.16.4 Overlay □□

Kube-OVN □□□□□□□□

□□□□□□□□

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

□□□□□□□□ `IFACE` □□□□□□□□□□□□□□□□□□□□ IP□

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="bond1"
#       SR-IOV   Device Plugin           bond    IFACE     bond1      bond     IFACE     enp132s0f0np0   enp132s0f1np1
```

□□ Kube-OVN□

```
bash install.sh
```

□□□□ VF □□□ Pod

□□□□□□□ yaml □□□□□□□ VF □□□□□□□□□□ Pod:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-overlay
  annotations:
    v1.multus-cni.io/default-network: default/sriov
    sriov.default.ovn.kubernetes.io/logical_switch: ovn-default
spec:
  containers:
  - name: nginx-overlay
    image: docker.io/library/nginx:alpine
    resources:
      requests:
        mellanox.com/cx5_sriov_switchdev: '1'
      limits:
        mellanox.com/cx5_sriov_switchdev: '1'
```

- `v1.multus-cni.io/default-network`: □□□□□□□ `NetworkAttachmentDefinition` □ {namespace}/{name}□
- `sriov.default.ovn.kubernetes.io/logical_switch`: □□ Pod □□□□□□□□□□□ Pod □□□□□□□□□□□□□□□□□□□□□□□□□

7.16.5 Underlay □□

Kube-OVN □□□□□□□□

□□□□□□□□

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

□□□□□□□□ `IFACE` □□□□□□□□□□□□□□□□□□□□ IP□

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE=""
#     Underlay    IFACE         PF      IFACE         K8s                  PF
```

□□ Kube-OVN□

```
bash install.sh
```

□□□□ VF □□□ Pod

□□□□□□□ yaml □□□□□□□ VF □□□□□□□□□□ Pod:

```
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: underlay-offload
spec:
  defaultInterface: bond1

---
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan0
spec:
  id: 0
  provider: underlay-offload

---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vlan0
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 10.10.204.0/24
  gateway: 10.10.204.254
  vlan: vlan0
  excludeIps:
  - 10.10.204.1..10.10.204.100

---
apiVersion: v1
kind: Pod
metadata:
  name: nginx-underlay
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{
      "name": "sriov",
      "namespace": "default",
      "default-route": ["10.10.204.254"]
    }]'
    sriov.default.ovn.kubernetes.io/logical_switch: vlan0
spec:
  containers:
  - name: nginx-underlay
    image: docker.io/library/nginx:alpine
    resources:
      requests:
        mellanox.com/cx5_sriov_switchdev: '1'
      limits:
        mellanox.com/cx5_sriov_switchdev: '1'
```

- `v1.multus-cni.io/default-network`: [illegible] `NetworkAttachmentDefinition` [illegible] {namespace}/{name}[illegible]

> [illegible] multus [illegible] VF [illegible] Pod[illegible] VF [illegible] Pod [illegible] VF [illegible] Pod [illegible] multus [illegible]

[illegible] yaml [illegible] VF [illegible] Pod:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-underlay-noVF
  annotations:
    ovn.kubernetes.io/logical_switch: vlan0
spec:
  containers:
  - name: nginx-underlay-noVF
    image: docker.io/library/nginx:alpine
```

[illegible] VF [illegible] Pod[illegible] ovs-kernel [illegible] e-switch [illegible]

7.16.6 [illegible]

[illegible] Pod [illegible] `ovs-ovn` [illegible]

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:91cc45de-e7e9-4935-8f82-1890430b0f66, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(5b45c61b307e_h),packet_type(ns=0/0,id=0/
0),eth(src=00:00:00:c5:6d:4e,dst=00:00:00:e7:16:ce),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no),
packets:941539, bytes:62142230, used:0.260s, offloaded:yes, dp:tc, actions:54235e5753b8_h
ufid:e00768d7-e652-4d79-8182-3291d852b791, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(54235e5753b8_h),packet_type(ns=0/0,id=0/
0),eth(src=00:00:00:e7:16:ce,dst=00:00:00:c5:6d:4e),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no),
packets:82386659, bytes:115944854173, used:0.260s, offloaded:yes, dp:tc, actions:5b45c61b307e_h
```

[illegible] `offloaded:yes, dp:tc` [illegible]

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2025[illegible]9[illegible]10[illegible]

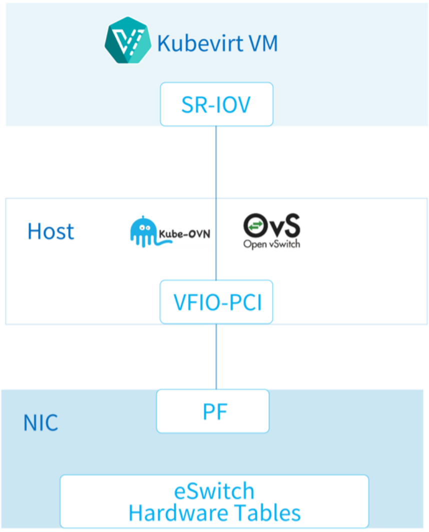
2022[illegible]5[illegible]24[illegible]

GitHub

7.16.7 [illegible]

7.17 [illegible] Offload [illegible]

Kube-OVN [illegible] OVS [illegible] CPU [illegible] CPU [illegible] Agilio CX [illegible] OVS [illegible] OVS [illegible] CPU [illegible]


Kubevirt VM
SR-IOV
Host
Kube-OVN
OvS Open vSwitch
VFIO-PCI
NIC
PF
eSwitch Hardware Tables

Note

[illegible] 2022 [illegible]

7.17.1 [illegible]

- [illegible] Agilio CX [illegible]
- CentOS 8 Stream [illegible] Linux 5.7 [illegible]
- [illegible] `dp_hash` [illegible] `hash` [illegible] OVN LB [illegible]

7.17.2 [illegible] SR-IOV [illegible]

[illegible] Agilio Open vSwitch TC User Guide [illegible]

[illegible]

```
#!/bin/bash
DEVICE=${1}
DEFAULT_ASSY=scan
ASSY=${2:-${DEFAULT_ASSY}}
APP=${3:-flower}

if [ "x${DEVICE}" = "x" -o ! -e /sys/class/net/${DEVICE} ]; then
    echo Syntax: ${0} device [ASSY] [APP]
    echo
    echo This script associates the TC Offload firmware
    echo with a Netronome SmartNIC.
    echo
    echo device: is the network device associated with the SmartNIC
    echo ASSY: defaults to ${DEFAULT_ASSY}
    echo APP: defaults to flower. flower-next is supported if updated
    echo      firmware has been installed.
    exit 1
fi

# It is recommended that the assembly be determined by inspection
# The following code determines the value via the debug interface
if [ "${ASSY}x" = "scanx" ]; then
    ethtool -W ${DEVICE} 0
    DEBUG=$(ethtool -w ${DEVICE} data /dev/stdout | strings)
    SERIAL=$(echo "${DEBUG}" | grep "^SN:")
    ASSY=$(echo ${SERIAL} | grep -oE AMDA[0-9]{4})
fi

PCIADDR=$(basename $(readlink -e /sys/class/net/${DEVICE}/device))
FWDIR="/lib/firmware/netronome"

# AMDA0081 and AMDA0097 uses the same firmware
if [ "${ASSY}" = "AMDA0081" ]; then
    if [ ! -e ${FWDIR}/${APP}/nic_AMDA0081.nffw ]; then
       ln -sf nic_AMDA0097.nffw ${FWDIR}/${APP}/nic_AMDA0081.nffw
   fi
fi

FW="${FWDIR}/pci-${PCIADDR}.nffw"
ln -sf "${APP}/nic_${ASSY}.nffw" "${FW}"

# insert distro-specific initramfs section here...
```

□□□□□□□□□□□□□□

```
./agilio-tc-fw-select.sh ens47np0 scan
rmmod nfp
modprobe nfp
```

□□□□ VF □□□□□□□ VF□

```
# cat /sys/class/net/ens3/device/sriov_totalvfs
65

# echo 4 > /sys/class/net/ens47/device/sriov_numvfs
```

7.17.3 □□ SR-IOV Device Plugin

□□□□□□□□ VF □□□□□□□□□□□□□□ Pod □□□ VF □□□□□□□□□□ SR-IOV Device Plugin □□□□□□□□□□□□□□□□□□□□□ □□□□□□□□

□□ SR-IOV □□ Configmap□

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
      "resourceList": [{
          "resourcePrefix": "corigine.com",
          "resourceName": "agilio_sriov",
          "selectors": {
                  "vendors": ["19ee"],
                  "devices": ["6003"],
                  "drivers": ["nfp_netvf"]
              }
      }
      ]
    }
```

□□ SR-IOV Network Device Plugin □□□□□□□□□□□ intel/sriov-network-device-plugin □□□ k8snetworkplumbingwg□□

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/master/deployments/sriovdp-daemonset.yaml
```

□□ SR-IOV □□□□□□□□□□ Kubernetes Node □□

```
kubectl describe no containerserver  | grep corigine

corigine.com/agilio_sriov:  4
corigine.com/agilio_sriov:  4
corigine.com/agilio_sriov  0           0
```

7.17.4 □□ Multus-CNI

SR-IOV Device Plugin □□□□□□□□□ ID □□□□ Multus-CNI □□□ Kube-OVN□□□□□□□□ Multus-CNI □□□□□□□□□□

□□ Multus-CNI □□□□□□□□

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/master/deployments/multus-daemonset.yml
```

□□ `NetworkAttachmentDefinition` □

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: default
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: corigine.com/agilio_sriov
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "default.default.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

- `provider` : □□□□□□ `NetworkAttachmentDefinition` □ {name}.{namespace}.ovn□

7.17.5 Kube-OVN □□□□□□□□

□□□□□□□

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

□□□□□□□ `IFACE` □□□□□□□□□□□□□□□□□□ IP□

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="ensp01"
```

□□ Kube-OVN□

```
bash install.sh
```

7.17.6 □□□□□ VF □□□□ Pod

□□□□□□□□ yaml □□□□□□□□ VF □□□□□□□□□□□□ Pod:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: default/default
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          corigine.com/agilio_sriov: '1'
        limits:
          corigine.com/agilio_sriov: '1'
```

- `v1.multus-cni.io/default-network` : [illegible] `NetworkAttachmentDefinition` [illegible] {namespace}/{name}[illegible]

[illegible] Pod [illegible] `ovs-ovn` [illegible]

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:91cc45de-e7e9-4935-8f82-1890430b0f66, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(5b45c61b307e_h),packet_type(ns=0/0,id=0/
0),eth(src=00:00:00:c5:6d:4e,dst=00:00:00:e7:16:ce),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no),
packets:941539, bytes:62142230, used:0.260s, offloaded:yes, dp:tc, actions:54235e5753b8_h
ufid:e00768d7-e652-4d79-8182-3291d852b791, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(54235e5753b8_h),packet_type(ns=0/0,id=0/
0),eth(src=00:00:00:e7:16:ce,dst=00:00:00:c5:6d:4e),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no),
packets:82386659, bytes:115944854173, used:0.260s, offloaded:yes, dp:tc, actions:5b45c61b307e_h
```

[illegible] `offloaded:yes, dp:tc` [illegible]

PDF Slack Support

2026[illegible]4[illegible]28[illegible]

2022[illegible]5[illegible]24[illegible]

GitHub

7.17.7 [illegible]

7.18 [illegible] Offload [illegible]

Kube-OVN [illegible] OVS [illegible] CPU [illegible] CPU [illegible] metaScale [illegible] OVS [illegible] OVS [illegible] CPU [illegible]

Note

1. [illegible] 2024 [illegible]
2. [illegible] 1.11 [illegible] Kube-OVN[illegible]

7.18.1 [illegible]

- [illegible] metaScale [illegible]
- MCR [illegible]
- BIOS [illegible] SR-IOV [illegible] VT-d[illegible]

7.18.2 [illegible]

[illegible] hw-offload [illegible] Kube-OVN

1. [illegible]

```
wget https://raw.githubusercontent.com/yunsilicon/kube-ovn/release-1.11/dist/images/install.sh
```

1. [illegible]

[illegible] `/opt/ovs-config/ovs-dpdk-config` [illegible]

```
# specify log level for ovs dpdk, the value is info or dbg, default is info
VLOG=info
# specify nic offload, the value is true or false, default is true
HW_OFFLOAD=true
# specify cpu mask for ovs dpdk, not specified by default
CPU_MASK=0x02
# specify socket memory, not specified by default
SOCKET_MEM="2048,2048"
# specify encap IP
ENCAP_IP=6.6.6.208/24
# specify pci device
DPDK_DEV=0000:b3:00.0
# specify mtu, default is 1500
PF_MTU=1500
# specify bond name if bond enabled, not specified by default
BR_PHY_BOND_NAME=bond0
```

1. [illegible] Kube-OVN

```
bash install.sh
```

[illegible] SR-IOV

1. [illegible] metaScale [illegible] ID[illegible] `b3:00.0` :

```
[root@k8s-master ~]# lspci -d 1f67:
b3:00.0 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.1 Ethernet controller: Device 1f67:1111 (rev 02)
```

1. [illegible] ID [illegible] `p3p1` [illegible]

```
ls -l /sys/class/net/ | grep b3:00.0
lrwxrwxrwx 1 root root 0 May  7 16:30 p3p1 -> ../../devices/pci0000:b2/0000:b2:00.0/0000:b3:00.0/net/p3p1
```

1. [illegible] VF [illegible]

```
cat /sys/class/net/p3p1/device/sriov_totalvfs
512
```

1. [illegible] VF[illegible]

```
echo '10' > /sys/class/net/p3p1/device/sriov_numvfs
```

1. [illegible] VF [illegible]

```
lspci -d 1f67:
b3:00.0 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.1 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.2 Ethernet controller: Device 1f67:1112
b3:00.3 Ethernet controller: Device 1f67:1112
b3:00.4 Ethernet controller: Device 1f67:1112
b3:00.5 Ethernet controller: Device 1f67:1112
b3:00.6 Ethernet controller: Device 1f67:1112
b3:00.7 Ethernet controller: Device 1f67:1112
b3:01.0 Ethernet controller: Device 1f67:1112
b3:01.1 Ethernet controller: Device 1f67:1112
b3:01.2 Ethernet controller: Device 1f67:1112
b3:01.3 Ethernet controller: Device 1f67:1112
```

1. [illegible] switchdev [illegible]

```
devlink dev eswitch set pci/0000:b3:00.0 mode switchdev
```

[illegible] SR-IOV Device Plugin

1. [illegible] SR-IOV [illegible] ConfigMap[illegible]

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
        "resourceList": [{
                "resourceName": "xsc_sriov",
                "resourcePrefix": "yunsilicon.com",
                "selectors": {
                    "vendors": ["1f67"],
                    "devices": ["1012", "1112"]
                }}
        ]
    }
```

1. [illegible] SR-IOV Device Plugin [illegible] DevicePlugin[illegible]
2. [illegible] SR-IOV [illegible]

```
# kubectl describe node <node name> | grep yunsilicon.com/xsc_sriov
  yunsilicon.com/xsc_sriov:  10
  yunsilicon.com/xsc_sriov:  10
  yunsilicon.com/xsc_sriov  0             0
```

[illegible] Multus-CNI

1. [illegible] Multus-CNI [illegible] Multus-CNI

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/master/deployments/multus-daemonset.yml
```

1. [illegible] NetworkAttachmentDefinition

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
```

```
metadata:
  name: sriov-net1
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: yunsilicon.com/xsc_sriov
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "sriov-net1.default.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

□□□□ SR-IOV □ Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  annotations:
    v1.multus-cni.io/default-network: default/sriov-net1
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          yunsilicon.com/xsc_sriov: '1'
        limits:
          yunsilicon.com/xsc_sriov: '1'
```

Offload □□

□□□□ Pod □□□□□ `ovs-ovn` □□□□□□□□□□□□□□□□□□□□

```
ovs-appctl dpctl/dump-flows type=offloaded
flow-dump from pmd on cpu core: 9
ct_state(-new+est-rel+rpl+trk),ct_mark(0/
0x3),recirc_id(0x2d277),in_port(15),packet_type(ns=0,id=0),eth(src=00:00:00:9d:fb:1a,dst=00:00:00:ce:cf:b9),eth_type(0x0800),ipv4(dst=10.16.0.14,frag=no),
packets:6, bytes:588, used:7.276s, actions:ct(zone=4,nat),recirc(0x2d278)
ct_state(-new+est-rel-rpl+trk),ct_mark(0/
0x3),recirc_id(0x2d275),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9,dst=00:00:00:9d:fb:1a),eth_type(0x0800),ipv4(dst=10.16.0.18,frag=no),
packets:5, bytes:490, used:7.434s, actions:ct(zone=6,nat),recirc(0x2d276)
ct_state(-new+est-rel-rpl+trk),ct_mark(0/0x1),recirc_id(0x2d276),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9,dst=00:00:00:9d:fb:1a/
01:00:00:00:00:00),eth_type(0x0800),ipv4(frag=no), packets:5, bytes:490, used:7.434s, actions:15
recirc_id(0),in_port(15),packet_type(ns=0,id=0),eth(src=00:00:00:9d:fb:1a/
01:00:00:00:00:00,dst=00:00:00:ce:cf:b9),eth_type(0x0800),ipv4(dst=10.16.0.14/255.192.0.0,frag=no), packets:6, bytes:588, used:7.277s,
actions:ct(zone=6,nat),recirc(0x2d277)
recirc_id(0),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9/01:00:00:00:00:00,dst=00:00:00:9d:fb:1a),eth_type(0x0800),ipv4(dst=10.16.0.18/255.192.0
packets:6, bytes:588, used:7.434s, actions:ct(zone=4,nat),recirc(0x2d275)
ct_state(-new+est-rel+rpl+trk),ct_mark(0/
0x1),recirc_id(0x2d278),in_port(15),packet_type(ns=0,id=0),eth(dst=00:00:00:ce:cf:b9/01:00:00:00:00:00),eth_type(0x0800),ipv4(frag=no), packets:6, bytes:588,
used:7.277s, actions:8
```

□□□□□□□□□□□□□□

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7.18.3 [illegible]

7.19 [illegible] Offload [illegible]

> **Note**
>
> [illegible] 2024 [illegible]

7.19.1 [illegible]

- [illegible] 2200E [illegible]
- [illegible] HADOS [illegible]
- BIOS [illegible] SR-IOV

7.19.2 [illegible]

[illegible] SR-IOV

1. [illegible] 2200E [illegible] vendor ID[illegible]1f47[illegible] ID[illegible]00:0a.0[illegible]00:0b.0[illegible] 2200E [illegible]

```
lspci | grep 1f47
00:0a.0 Ethernet controller: Device 1f47:1001 (rev 10)
00:0b.0 Ethernet controller: Device 1f47:1001 (rev 10)
```

1. [illegible] VF [illegible]

```
cat /sys/bus/pci/devices/0000\:00\:0a.0/sriov_totalvfs
256
```

1. [illegible] VF[illegible] VF [illegible]

```
echo 7 > /sys/bus/pci/devices/0000\:00\:0a.0/sriov_numvfs
```

1. [illegible] VF [illegible]

```
lspci | grep 1f47
00:0a.0 Ethernet controller: Device 1f47:1001 (rev 10)
00:0a.1 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.2 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.3 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.4 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.5 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.6 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.7 Ethernet controller: Device 1f47:110f (rev 10)
00:0b.0 Ethernet controller: Device 1f47:1001 (rev 10)
```

[illegible] SR-IOV Device Plugin

1. [illegible] SR-IOV [illegible] Configmap[illegible] SR-IOV Device Plugin [illegible] VF [illegible] Pod [illegible]

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
        "resourceList": [{
                "resourceName": "sriov_dpu",
                "resourcePrefix": "yusur.tech",
                "selectors": {
                    "vendors": ["1f47"],
                    "devices": ["110f"]
                }}
        ]
    }
```

1. □□□□□ SR-IOV Device Plugin

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/v3.6.2/deployments/sriovdp-daemonset.yaml
```

1. □□ SR-IOV □□□□□□□□□□□ kubernetes Node □□□□□□

```
kubectl describe node node1 | grep yusur
  yusur.tech/sriov_dpu:  7
  yusur.tech/sriov_dpu:  7
  yusur.tech/sriov_dpu  0            0
```

7.19.3 □□ Multus-CNI

□□ Multus-CNI□□□□ Kube-OVN □□□□ SRIOV □□□ Device ID□

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/v4.0.2/deployments/multus-daemonset-thick.yml
```

□□ `NetworkAttachmentDefinition`□

```
apiVersion:
  "k8s.cni.cncf.io/v1"
kind:
  NetworkAttachmentDefinition
metadata:
  name: test
  namespace: kube-system
  annotations:
    k8s.v1.cni.cncf.io/resourceName: yusur.tech/sriov_dpu
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "test.kube-system.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

- `provider`: □□□□□ NetworkAttachmentDefinition □ `{name}.{namespace}.ovn`□

7.19.4 Kube-OVN □□□□□□□□

1. □□ □□□□

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.12/dist/images/install.sh
```

1. □□□□□□□IFACE □□□□□□□□□□□□□□□□□□ IP□

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="p0"
```

1. □□ kube-ovn

```
bash install.sh
```

□□□□□ VF □□□ pod

□□□□□□□ yaml □□□□□□□ VF □□□□□□□□□□ Pod□

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: kube-system/test
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          yusur.tech/sriov_dpu: '1'
        limits:
          yusur.tech/sriov_dpu: '1'
```

- `v1.multus-cni.io/default-network`：[illegible] `NetworkAttachmentDefinition` [illegible] `{namespace}/{name}` [illegible]

Offload [illegible]

[illegible] Pod [illegible] `ovs-ovn` [illegible]

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:67c2e10f-92d4-4574-be70-d072815ff166, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0),recirc_id(0),dp_hash(0/
0),in_port(d85b161b6840_h),packet_type(ns=0/0,id=0/
0),eth(src=0a:c9:1c:70:01:09,dst=8a:18:a4:22:b7:7d),eth_type(0x0800),ipv4(src=10.0.1.10,dst=10.0.1.6,proto=6,tos=0/0x3,ttl=0/
0,frag=no),tcp(src=60774,dst=9001), packets:75021, bytes:109521630, offload_packets:75019, offload_bytes:109521498, used:3.990s,offloaded:yes,dp:tc,
actions:set(tunnel(tun_id=0x5,dst=192.168.201.12,ttl=64,tp_dst=6081,geneve({class=0x102,type=0x80,len=4,0xa0006}),flags(csum|key))),genev_sys_6081
ufid:7940666e-a0bd-42a5-8116-1e84e81bb338, skb_priority(0/0),tunnel(tun_id=0x5,src=192.168.201.12,dst=192.168.201.11,ttl=0/
0,tp_dst=6081,geneve({class=0x102,type=0x80,len=4,0x6000a}),flags(+key)),skb_mark(0/0),ct_state(0/0),ct_zone(0/0),ct_mark(0/0),ct_label(0/
0),recirc_id(0),dp_hash(0/0),in_port(genev_sys_6081),packet_type(ns=0/0,id=0/
0),eth(src=8a:18:a4:22:b7:7d,dst=0a:c9:1c:70:01:09),eth_type(0x0800),ipv4(src=10.0.1.6,dst=10.0.1.10,proto=6,tos=0/0,ttl=0/
0,frag=no),tcp(src=9001,dst=60774), packets:6946, bytes:459664, offload_packets:6944, offload_bytes:459532, used:4.170s,
dp:tc,offloaded:yes,actions:d85b161b6840_h
```

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7.19.5 [illegible]

7.20 DPDK ⌧⌧

⌧⌧⌧⌧⌧ Kube-OVN ⌧⌧⌧ OVS-DPDK ⌧⌧⌧⌧ KubeVirt ⌧⌧⌧⌧⌧ DPDK ⌧⌧⌧⌧⌧⌧⌧⌧

> ⌧⌧⌧ KubeVirt ⌧⌧⌧⌧⌧⌧ OVS-DPDK⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ patch Vhostuser implementation ⌧⌧ KubeVirt ⌧ KVM Device Plugin ⌧⌧⌧ OVS-DPDK⌧

7.20.1 ⌧⌧⌧⌧

- ⌧⌧⌧⌧⌧⌧⌧⌧ DPDK ⌧⌧⌧⌧⌧⌧⌧⌧
- ⌧⌧⌧⌧⌧ Hugepages⌧

7.20.2 ⌧⌧⌧⌧ DPDK ⌧⌧

⌧⌧⌧⌧⌧⌧ `driverctl` ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ DPDK ⌧⌧⌧⌧⌧⌧⌧

```
driverctl set-override 0000:00:0b.0 uio_pci_generic
```

7.20.3 ⌧⌧⌧⌧

⌧⌧⌧ OVS-DPDK ⌧⌧⌧⌧⌧⌧⌧⌧⌧ Kube-OVN ⌧⌧⌧⌧⌧⌧⌧

```
kubectl label nodes <node> ovn.kubernetes.io/ovs_dp_type="userspace"
```

⌧⌧⌧ OVS-DPDK ⌧⌧⌧ `/opt/ovs-config` ⌧⌧⌧⌧⌧⌧⌧⌧⌧ `ovs-dpdk-config` ⌧

```
ENCAP_IP=192.168.122.193/24
DPDK_DEV=0000:00:0b.0
```

- `ENCAP_IP` : ⌧⌧⌧⌧⌧⌧⌧
- `DPDK_DEV` : ⌧⌧⌧ PCI ID⌧

7.20.4 ⌧⌧ Kube-OVN

⌧⌧⌧⌧⌧⌧⌧

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

⌧⌧ DPDK ⌧⌧⌧⌧⌧⌧⌧⌧⌧

```
bash install.sh --with-hybrid-dpdk
```

7.20.5 ⌧⌧⌧⌧

⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ vhostuser ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ OVS-DPDK ⌧⌧⌧

⌧⌧ KVM Device Plugin ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ KVM Device Plugin⌧

```
kubectl apply -f https://raw.githubusercontent.com/kubevirt/kubernetes-device-plugins/master/manifests/kvm-ds.yml
```

⌧⌧ NetworkAttachmentDefinition⌧

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-dpdk
  namespace: default
spec:
  config: >-
    {
        "cniVersion": "0.3.0",
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "ovn-dpdk.default.ovn",
        "vhost_user_socket_volume_name": "vhostuser-sockets",
        "vhost_user_socket_name": "sock"
    }
```

[illegible] Dockerfile [illegible] VM [illegible]

```
FROM quay.io/kubevirt/virt-launcher:v0.46.1

# wget http://cloud.centos.org/centos/7/images/CentOS-7-x86_64-GenericCloud.qcow2
COPY CentOS-7-x86_64-GenericCloud.qcow2 /var/lib/libvirt/images/CentOS-7-x86_64-GenericCloud.qcow2
```

[illegible]

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: vm-config
data:
  start.sh: |
    chmod u+w /etc/libvirt/qemu.conf
    echo "hugetlbfs_mount = \"/dev/hugepages\"" >> /etc/libvirt/qemu.conf
    virtlogd &
    libvirtd &

    mkdir /var/lock

    sleep 5

    virsh define /root/vm/vm.xml
    virsh start vm

    tail -f /dev/null
  vm.xml: |
    <domain type='kvm'>
      <name>vm</name>
      <uuid>4a9b3f53-fa2a-47f3-a757-dd87720d9d1d</uuid>
      <memory unit='KiB'>2097152</memory>
      <currentMemory unit='KiB'>2097152</currentMemory>
      <memoryBacking>
        <hugepages>
          <page size='2' unit='M' nodeset='0'/>
        </hugepages>
      </memoryBacking>
      <vcpu placement='static'>2</vcpu>
      <cputune>
        <shares>4096</shares>
        <vcpupin vcpu='0' cpuset='4'/>
        <vcpupin vcpu='1' cpuset='5'/>
        <emulatorpin cpuset='1,3'/>
      </cputune>
      <os>
        <type arch='x86_64' machine='pc'>hvm</type>
        <boot dev='hd'/>
      </os>
      <features>
        <acpi/>
        <apic/>
      </features>
      <cpu mode='host-model'>
        <model fallback='allow'/>
        <topology sockets='1' cores='2' threads='1'/>
        <numa>
          <cell id='0' cpus='0-1' memory='2097152' unit='KiB' memAccess='shared'/>
        </numa>
      </cpu>
      <on_reboot>restart</on_reboot>
      <devices>
        <emulator>/usr/libexec/qemu-kvm</emulator>
        <disk type='file' device='disk'>
          <driver name='qemu' type='qcow2' cache='none'/>
          <source file='/var/lib/libvirt/images/CentOS-7-x86_64-GenericCloud.qcow2'/>
          <target dev='vda' bus='virtio'/>
        </disk>

        <interface type='vhostuser'>
          <mac address='00:00:00:0A:30:89'/>
          <source type='unix' path='/var/run/vm/sock' mode='server'/>
           <model type='virtio'/>
          <driver queues='2'>
            <host mrg_rxbuf='off'/>
          </driver>
        </interface>
        <serial type='pty'>
          <target type='isa-serial' port='0'>
            <model name='isa-serial'/>
          </target>
        </serial>
        <console type='pty'>
          <target type='serial' port='0'/>
        </console>
        <channel type='unix'>
          <source mode='bind' path='/var/lib/libvirt/qemu/channel/target/domain-1-vm/org.qemu.guest_agent.0'/>
          <target type='virtio' name='org.qemu.guest_agent.0' state='connected'/>
          <alias name='channel0'/>
          <address type='virtio-serial' controller='0' bus='0' port='1'/>
        </channel>

      </devices>
    </domain>
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: vm-deployment
  labels:
    app: vm
spec:
  replicas: 1
  selector:
    matchLabels:
      app: vm
  template:
    metadata:
      labels:
        app: vm
      annotations:
        k8s.v1.cni.cncf.io/networks: default/ovn-dpdk
        ovn-dpdk.default.ovn.kubernetes.io/ip_address: 10.16.0.96
        ovn-dpdk.default.ovn.kubernetes.io/mac_address: 00:00:00:0A:30:89
    spec:
      nodeSelector:
        ovn.kubernetes.io/ovs_dp_type: userspace
      securityContext:
        runAsUser: 0
      volumes:
        - name: vhostuser-sockets
          emptyDir: {}
        - name: xml
          configMap:
            name: vm-config
        - name: hugepage
          emptyDir:
            medium: HugePages-2Mi
        - name: libvirt-runtime
          emptyDir: {}
      containers:
        - name: vm
          image: vm-vhostuser:latest
          command: ["bash", "/root/vm/start.sh"]
          securityContext:
            capabilities:
              add:
                - NET_BIND_SERVICE
                - SYS_NICE
                - NET_RAW
                - NET_ADMIN
            privileged: false
            runAsUser: 0
          resources:
            limits:
              cpu: '2'
              devices.kubevirt.io/kvm: '1'
              memory: '8784969729'
              hugepages-2Mi: 2Gi
            requests:
              cpu: 666m
              devices.kubevirt.io/kvm: '1'
              ephemeral-storage: 50M
              memory: '4490002433'
          volumeMounts:
            - name: vhostuser-sockets
              mountPath: /var/run/vm
            - name: xml
              mountPath: /root/vm/
            - mountPath: /dev/hugepages
              name: hugepage
            - name: libvirt-runtime
              mountPath: /var/run/libvirt
```

[illegible] Pod [illegible]

```
# virsh set-user-password vm root 12345
Password set successfully for root in vm

# virsh console vm
Connected to domain 'vm'
Escape character is ^] (Ctrl + ])

CentOS Linux 7 (Core)
Kernel 3.10.0-1127.el7.x86_64 on an x86_64
```

```
localhost login: root
Password:
Last login: Fri Feb 25 09:52:54 on ttyS0
```

[illegible]

```
ip link set eth0 mtu 1400
ip addr add 10.16.0.96/16 dev eth0
ip ro add default via 10.16.0.1
ping 114.114.114.114
```

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7.20.6 [illegible]

7.21 OpenStack [illegible]

[illegible] OpenStack [illegible] Kubernetes [illegible] OpenStack Neutron [illegible] OVN [illegible] Kube-OVN [illegible] OVN [illegible] OpenStack [illegible] Kubernetes [illegible]

7.21.1 [illegible]

[illegible] OVN-IC [illegible] Kubernetes [illegible] OpenStack [illegible] Kubernetes[illegible]

[illegible]

1. [illegible] OpenStack [illegible] Kubernetes [illegible] CIDR [illegible]
2. [illegible]
3. [illegible] IP [illegible]
4. [illegible] Kubernetes [illegible] OpenStack [illegible] VPC[illegible]

[illegible] OVN-IC [illegible]

[illegible] `OVN-IC` [illegible]

```
docker run --name=ovn-ic-db -d --network=host -v /etc/ovn/:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn kubeovn/kube-ovn:v1.17.0 bash
start-ic-db.sh
```

Kubernetes [illegible]

[illegible] `kube-system` Namespace [illegible] `ovn-ic-config` ConfigMap[illegible]

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

- `enable-ic`: [illegible]
- `az-name`: [illegible]
- `ic-db-host`: [illegible] `OVN-IC` [illegible]
- `ic-nb-port`: `OVN-IC` [illegible] 6645[illegible]
- `ic-sb-port`: `OVN-IC` [illegible] 6646[illegible]
- `gw-nodes`: [illegible]
- `auto-route`: [illegible]

OpenStack [illegible]

[illegible] Kubernetes [illegible]

```
# openstack router create router0
# openstack router list
+--------------------------------------+---------+--------+-------+----------------------------------+
| ID                                   | Name    | Status | State | Project                          |
+--------------------------------------+---------+--------+-------+----------------------------------+
| d5b38655-249a-4192-8046-71aa4d2b4af1 | router0 | ACTIVE | UP    | 98a29ab7388347e7b5ff8bdd181ba4f9 |
+--------------------------------------+---------+--------+-------+----------------------------------+
```

[illegible] OpenStack [illegible] OVN [illegible]

```
ovn-nbctl set NB_Global . name=op-az
```

[illegible] `OVN-IC` [illegible] `OVN-IC` [illegible]

```
/usr/share/ovn/scripts/ovn-ctl --ovn-ic-nb-db=tcp:192.168.65.3:6645 \
  --ovn-ic-sb-db=tcp:192.168.65.3:6646 \
  --ovn-northd-nb-db=unix:/run/ovn/ovnnb_db.sock \
  --ovn-northd-sb-db=unix:/run/ovn/ovnsb_db.sock \
  start_ic
```

- `ovn-ic-nb-db` [illegible] `ovn-ic-sb-db` : OVN-IC [illegible]
- `ovn-northd-nb-db` [illegible] `ovn-northd-sb-db` : [illegible] OVN [illegible]

[illegible]

```
ovs-vsctl set open_vswitch . external_ids:ovn-is-interconn=true
```

[illegible] OpenStack [illegible] OVN [illegible]

[illegible] `ts` [illegible] `router0` [illegible]

```
ovn-nbctl lrp-add router0 lrp-router0-ts 00:02:ef:11:39:4f 169.254.100.73/24
ovn-nbctl lsp-add ts lsp-ts-router0 -- lsp-set-addresses lsp-ts-router0 router \
  -- lsp-set-type lsp-ts-router0 router \
  -- lsp-set-options lsp-ts-router0  router-port=lrp-router0-ts
ovn-nbctl lrp-set-gateway-chassis lrp-router0-ts {gateway chassis} 1000
ovn-nbctl set NB_Global . options:ic-route-adv=true options:ic-route-learn=true
```

[illegible] Kubernetes [illegible]

```
# ovn-nbctl lr-route-list router0
IPv4 Routes
                10.0.0.22            169.254.100.34 dst-ip (learned)
             10.16.0.0/16            169.254.100.34 dst-ip (learned)
```

[illegible] `router0` [illegible] Kubernetes [illegible] Pod [illegible]

7.21.2 [illegible] OVN

[illegible]OpenStack [illegible] Kubernetes [illegible] OVN[illegible] VPC [illegible] Subnet [illegible]

[illegible] Kube-OVN [illegible] OVN[illegible]OpenStack [illegible] Neutron [illegible] OVN [illegible]OpenStack [illegible] networking-ovn [illegible] Neutron [illegible]

Neutron [illegible]

[illegible] Neutron [illegible] `/etc/neutron/plugins/ml2/ml2_conf.ini` [illegible]

```
[ovn]
...
ovn_nb_connection = tcp:[192.168.137.176]:6641,tcp:[192.168.137.177]:6641,tcp:[192.168.137.178]:6641
ovn_sb_connection = tcp:[192.168.137.176]:6642,tcp:[192.168.137.177]:6642,tcp:[192.168.137.178]:6642
ovn_l3_scheduler = OVN_L3_SCHEDULER
```

- `ovn_nb_connection` [illegible] `ovn_sb_connection` : [illegible] Kube-OVN [illegible] `ovn-central` [illegible]

[illegible] OVS [illegible]

```
ovs-vsctl set open . external-ids:ovn-remote=tcp:[192.168.137.176]:6642,tcp:[192.168.137.177]:6642,tcp:[192.168.137.178]:6642
ovs-vsctl set open . external-ids:ovn-encap-type=geneve
ovs-vsctl set open . external-ids:ovn-encap-ip=192.168.137.200
```

- `external-ids:ovn-remote` : [illegible] Kube-OVN [illegible] `ovn-central` [illegible]
- `ovn-encap-ip` : [illegible] IP [illegible]

□ Kubernetes □□□ OpenStack □□□

□□□□□□□□□ Kubernetes □□□ OpenStack □□□□□□□□ OpenStack □□□□□□□ Pod□

> **Note**
>
> □□□□□□□□□ `kube-ovn-controller` □ args □□□□□□□□ `--enable-external-vpc=true` □

□□ OpenStack □□□□□□□□□□□□□□□□□□□□□□□□□

```
# openstack router list
+--------------------------------------+---------+--------+-------+----------------------------------+
| ID                                   | Name    | Status | State | Project                          |
+--------------------------------------+---------+--------+-------+----------------------------------+
| 22040ed5-0598-4f77-bffd-e7fd4db47e93 | router0 | ACTIVE | UP    | 62381a21d569404aa236a5dd8712449c |
+--------------------------------------+---------+--------+-------+----------------------------------+
# openstack network list
+--------------------------------------+----------+--------------------------------------+
| ID                                   | Name     | Subnets                              |
+--------------------------------------+----------+--------------------------------------+
| cd59e36a-37db-4c27-b709-d35379a7920f | provider | 01d73d9f-fdaa-426c-9b60-aa34abbfacae |
+--------------------------------------+----------+--------------------------------------+
# openstack subnet list
+--------------------------------------+-------------+--------------------------------------+----------------+
| ID                                   | Name        | Network                              | Subnet         |
+--------------------------------------+-------------+--------------------------------------+----------------+
| 01d73d9f-fdaa-426c-9b60-aa34abbfacae | provider-v4 | cd59e36a-37db-4c27-b709-d35379a7920f | 192.168.1.0/24 |
+--------------------------------------+-------------+--------------------------------------+----------------+
# openstack server list
+--------------------------------------+-------------------+--------+-----------------------+--------+--------+
| ID                                   | Name              | Status | Networks              | Image  | Flavor |
+--------------------------------------+-------------------+--------+-----------------------+--------+--------+
| 8433d622-a8d6-41a7-8b31-49abfd64f639 | provider-instance | ACTIVE | provider=192.168.1.61 | ubuntu | m1     |
+--------------------------------------+-------------------+--------+-----------------------+--------+--------+
```

□ Kubernetes □□□□ VPC □□□

```
# kubectl get vpc
NAME                                           STANDBY   SUBNETS
neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93   true      ["neutron-cd59e36a-37db-4c27-b709-d35379a7920f"]
ovn-cluster                                    true      ["join","ovn-default"]
```

`neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93` □□ OpenStack □□□□□ VPC □□□

□□□□□□□□ Kube-OVN □□□ VPC □ Subnet □□□□ Pod □□□□

VPC, Subnet □□ Namespace `net2` □□□□ Pod:

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: net2
---
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  creationTimestamp: "2021-06-20T13:34:11Z"
  generation: 2
  labels:
    ovn.kubernetes.io/vpc_external: "true"
  name: neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93
  resourceVersion: "583728"
  uid: 18d4c654-f511-4def-a3a0-a6434d237c1e
spec:
  namespaces:
  - net2
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93
  namespaces:
    - net2
  cidrBlock: 12.0.1.0/24
  natOutgoing: false
---
apiVersion: v1
kind: Pod
metadata:
  name: ubuntu
  namespace: net2
spec:
  containers:
    - image: docker.io/kubeovn/kube-ovn:v1.8.0
      command:
        - "sleep"
        - "604800"
      imagePullPolicy: IfNotPresent
      name: ubuntu
  restartPolicy: Always
```

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7.21.3 □□

7.22 [illegible] IPsec [illegible]

Kube-OVN [illegible] v1.13.0 [illegible] OVN/OVS [illegible] IPsec [illegible]Geneve/Vxlan/STT[illegible]

7.22.1 [illegible]

- [illegible] UDP 500[illegible]IKE[illegible] UDP 4500[illegible]NAT-T[illegible]
- [illegible] `xfrm` / `af_key` [illegible] strongSwan/libreswan [illegible]

7.22.2 [illegible]

`kube-ovn-cni` [illegible] CertificateSigningRequest[illegible] `kube-ovn-controller` [illegible] approve [illegible] `kube-ovn-cni` [illegible] IPsec [illegible] ipsec [illegible]

7.22.3 [illegible] IPsec

[illegible] `kube-ovn-controller` Deployment [illegible] `kube-ovn-cni` DaemonSet [illegible] `--enable-ovn-ipsec=false` [illegible] `--enable-ovn-ipsec=true` [illegible]

```
ENABLE_OVN_IPSEC=true
```

7.22.4 [illegible] cert-manager [illegible]

[illegible] cert-manager[illegible] IPsec [illegible] `--cert-manager-ipsec-cert=true` [illegible] kube-ovn-cni [illegible] cert-manager [illegible] IPsec [illegible] Kube-OVN [illegible] CA[illegible]

```
args:
- --enable-ovn-ipsec=true
- --cert-manager-ipsec-cert=true
```

7.22.5 [illegible]

- [illegible] `kube-ovn-cni` [illegible] `ovs-appctl ipsec/show` [illegible] IPsec [illegible]SA[illegible]SPI [illegible]
- [illegible] `kubectl get csr` [illegible] kube-ovn-cni [illegible] CSR[illegible] IPsec [illegible] CSR [illegible] approve[illegible] controller [illegible] `signer.go` [illegible]
- [illegible] IPsec [illegible] `false` [illegible] IPsec [illegible] SA[illegible] `kube-ovn-cni` Pod [illegible]

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2023[illegible]4[illegible]18[illegible]

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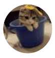

7.22.6 [illegible]

7.23 OVN □□□□

□□□□□□□□ Pod□□□□□□□□□□□ GRE/ERSPAN □□□□□□□□□□

> □□□□□ Kube-OVN □□□□□ v1.12□

7.23.1 □□ Multus-CNI

□□ Multus-CNI □□ □□ Multus□

7.23.2 □□□□□□

□□□□□□□□□□□□□

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: |
    {
      "cniVersion": "0.3.1",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "attachnet.default.ovn"
    }
```

□□ `provider` □□□□□ `<NAME>.<NAMESPACE>.ovn` □

7.23.3 □□ Underlay □□

□□□□□□□□□□□□□□□□□□□□□□□□MTU □□□□□□□□ LSP/Pod□□□□□□□ Underlay □□□□□□□

□□□□□□□□ Underlay □□□

```
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: net1
spec:
  defaultInterface: eth1
---
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan1
spec:
  id: 0
  provider: net1
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.19.0.0/16
  excludeIps:
  - 172.19.0.2..172.19.0.20
  gateway: 172.19.0.1
  vlan: vlan1
  provider: attachnet.default.ovn
```

□□□□□□ `provider` □□□□□□□□ `provider` □□□

7.23.4 □□□□□□□ Pod

□□□□□□□□□□□□□□□□□□□ Pod□

```
apiVersion: v1
kind: Pod
```

```
metadata:
  name: pod1
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
spec:
  containers:
  - name: bash
    image: docker.io/kubeovn/kube-ovn:v1.17.0
    args:
    - bash
    - -c
    - sleep infinity
    securityContext:
      privileged: true
```

[illegible] Pod [illegible] IP [illegible]

```
$ kubectl get ips | grep pod1
pod1.default                        10.16.0.12   00:00:00:FF:34:24  kube-ovn-worker  ovn-default
pod1.default.attachnet.default.ovn  172.19.0.21  00:00:00:A0:30:68  kube-ovn-worker  subnet1
```

[illegible] IP [illegible] `172.19.0.21` [illegible]

7.23.5 [illegible] OVN [illegible]

[illegible] OVN [illegible]

```
kubectl ko nbctl mirror-add mirror1 gre 99 from-lport 172.19.0.21
kubectl ko nbctl lsp-attach-mirror coredns-787d4945fb-gpnkb.kube-system mirror1
```

[illegible] `coredns-787d4945fb-gpnkb.kube-system` [illegible] OVN LSP [illegible] `<POD_NAME>.<POD_NAMESPACE>` [illegible]

[illegible] OVN [illegible]

```
ovn-nbctl mirror-add <NAME> <TYPE> <INDEX> <FILTER> <IP>

NAME   - add a mirror with given name
TYPE   - specify TYPE 'gre' or 'erspan'
INDEX  - specify the tunnel INDEX value
         (indicates key if GRE, erpsan_idx if ERSPAN)
FILTER - specify FILTER for mirroring selection
         ('to-lport' / 'from-lport')
IP     - specify Sink / Destination i.e. Remote IP

ovn-nbctl mirror-del [NAME]         remove mirrors
ovn-nbctl mirror-list               print mirrors

ovn-nbctl lsp-attach-mirror PORT MIRROR   attach source PORT to MIRROR
ovn-nbctl lsp-detach-mirror PORT MIRROR   detach source PORT from MIRROR
```

7.23.6 [illegible] Pod

[illegible] Pod [illegible]

```
root@pod1:/kube-ovn# ip link add mirror1 type gretap local 172.19.0.21 key 99 dev net1
root@pod1:/kube-ovn# ip link set mirror1 up
```

[illegible] Pod [illegible]

```
root@pod1:/kube-ovn# tcpdump -i mirror1 -nnve
tcpdump: listening on mirror1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
05:13:30.328808 00:00:00:a3:f5:e2 > 00:00:00:97:0f:6e, ethertype ARP (0x0806), length 42: Ethernet (len 6), IPv4 (len 4), Request who-has 10.16.0.7 tell
10.16.0.4, length 28
05:13:30.559167 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 212: (tos 0x0, ttl 64, id 57364, offset 0, flags [DF], proto UDP (17),
length 198)
    10.16.0.4.53 > 10.16.0.6.50472: 34511 NXDomain*- 0/1/1 (170)
05:13:30.559343 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 212: (tos 0x0, ttl 64, id 57365, offset 0, flags [DF], proto UDP (17),
length 198)
    10.16.0.4.53 > 10.16.0.6.45177: 1659 NXDomain*- 0/1/1 (170)
05:13:30.560625 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 200: (tos 0x0, ttl 64, id 57367, offset 0, flags [DF], proto UDP (17),
length 186)
    10.16.0.4.53 > 10.16.0.6.43848: 2636*- 0/1/1 (158)
05:13:30.562774 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 191: (tos 0x0, ttl 64, id 57368, offset 0, flags [DF], proto UDP (17),
length 177)
    10.16.0.4.53 > 10.16.0.6.37755: 48737 NXDomain*- 0/1/1 (149)
05:13:30.563523 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 187: (tos 0x0, ttl 64, id 57369, offset 0, flags [DF], proto UDP (17),
length 173)
    10.16.0.4.53 > 10.16.0.6.53887: 45519 NXDomain*- 0/1/1 (145)
05:13:30.564940 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 201: (tos 0x0, ttl 64, id 57370, offset 0, flags [DF], proto UDP (17),
length 187)
    10.16.0.4.53 > 10.16.0.6.40846: 25745 NXDomain*- 0/1/1 (159)
05:13:30.565140 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 201: (tos 0x0, ttl 64, id 57371, offset 0, flags [DF], proto UDP (17),
length 187)
    10.16.0.4.53 > 10.16.0.6.45214: 61875 NXDomain*- 0/1/1 (159)
05:13:30.566023 00:00:00:a3:f5:e2 > 00:00:00:55:e4:4e, ethertype IPv4 (0x0800), length 80: (tos 0x0, ttl 64, id 45937, offset 0, flags [DF], proto UDP (17),
length 66)
    10.16.0.4.44116 > 172.18.0.1.53: 16025+ [1au] AAAA? kube-ovn.io. (38)
```

7.23.7 [illegible]

1. [illegible] ERSPAN [illegible]OVN [illegible] Linux [illegible] 4.14[illegible] ERSPAN [illegible] IPv6 [illegible]Linux [illegible] 4.16[illegible]
2. [illegible] OVN [illegible]

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7.23.8 [illegible]

7.24 [illegible] DNS [illegible] Kube-OVN [illegible]

NodeLocal DNSCache [illegible] DaemonSet [illegible] DNS [illegible] DNS [illegible] Kube-OVN [illegible]

7.24.1 [illegible] DNS [illegible]

[illegible] **Kubernetes** [illegible] **DNS** [illegible]

[illegible] Kubernetes [illegible] Nodelocaldnscache[illegible]

[illegible]

```
#!bin/bash

localdns=169.254.20.10
domain=cluster.local
kubedns=10.96.0.10

wget https://raw.githubusercontent.com/kubernetes/kubernetes/master/cluster/addons/dns/nodelocaldns/nodelocaldns.yaml
sed -i "s/__PILLAR__LOCAL__DNS__/$localdns/g; s/__PILLAR__DNS__DOMAIN__/$domain/g; s/,__PILLAR__DNS__SERVER__//g; s/__PILLAR__CLUSTER__DNS__/$kubedns/g"
nodelocaldns.yaml

kubectl apply -f nodelocaldns.yaml
```

[illegible] kubelet [illegible] `/var/lib/kubelet/config.yaml` [illegible] clusterDNS [illegible] DNS IP 169.254.20.10[illegible] kubelet [illegible]

Kube-OVN [illegible] **DNS** [illegible]

[illegible] Kubernetes [illegible] Nodelocal DNScache [illegible]Kube-OVN [illegible]

UNDERLAY SUBNET [illegible] U2O [illegible]

[illegible] Underlay Subnet [illegible] DNS [illegible] U2O [illegible] `kubectl edit subnet {your subnet}` [illegible] `spec.u2oInterconnection = true` , [illegible] Overlay Subnet [illegible]

[illegible] KUBE-OVN-CONTROLLER [illegible] DNS IP

```
kubectl edit deployment kube-ovn-controller -n kube-system
```

[illegible] `spec.template.spec.containers.args` [illegible] `--node-local-dns-ip=169.254.20.10`

[illegible] POD

[illegible] Pod [illegible] `/etc/resolv.conf` [illegible] nameserver [illegible] DNS IP[illegible] Pod [illegible] nameserver [illegible] DNS ClusterIP[illegible] u2o [illegible] Pod [illegible] Pod [illegible]

7.24.2 [illegible] DNS [illegible]

[illegible] Pod [illegible] Pod [illegible] DNS [illegible] 169.254.20.10[illegible]

```
# kubectl exec -it pod1 -- nslookup github.com
Server:         169.254.20.10
Address:        169.254.20.10:53


Name:   github.com
Address: 20.205.243.166
```

[illegible] DNS [illegible] ovn0 [illegible] DNS [illegible]DNS [illegible]

```
# tcpdump -i any port 53

06:20:00.441889 659246098c56_h P   ifindex 17 00:00:00:73:f1:06 ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1291+ A? baidu.com.
(27)
06:20:00.441889 ovn0  In  ifindex 7 00:00:00:50:32:cd ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1291+ A? baidu.com. (27)
06:20:00.441950 659246098c56_h P   ifindex 17 00:00:00:73:f1:06 ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1611+ AAAA?
baidu.com. (27)
06:20:00.441950 ovn0  In  ifindex 7 00:00:00:50:32:cd ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1611+ AAAA? baidu.com. (27)
06:20:00.442203 ovn0  Out ifindex 7 00:00:00:52:99:d8 ethertype IPv4 (0x0800), length 145: 169.254.20.10.53 > 10.16.0.2.40230: 1611* 0/1/0 (97)
```

```
06:20:00.442219 659246098c56_h Out ifindex 17 00:00:00:ea:b3:5e ethertype IPv4 (0x0800), length 145: 169.254.20.10.53 > 10.16.0.2.40230: 1611* 0/1/0 (97)
06:20:00.442273 ovn0  Out ifindex 7 00:00:00:52:99:d8 ethertype IPv4 (0x0800), length 125: 169.254.20.10.53 > 10.16.0.2.40230: 1291* 2/0/0 A 39.156.66.10, A
110.242.68.66 (77)
06:20:00.442278 659246098c56_h Out ifindex 17 00:00:00:ea:b3:5e ethertype IPv4 (0x0800), length 125: 169.254.20.10.53 > 10.16.0.2.40230: 1291* 2/0/0 A
39.156.66.10, A 110.242.68.66 (77)
```

7.24.3 ⌧⌧⌧⌧

⚠️ ⌧⌧⌧

⌧⌧⌧⌧⌧⌧⌧⌧ NetworkPolicy⌧⌧⌧⌧⌧⌧ NetworkPolicy ⌧⌧⌧⌧⌧⌧⌧ DNS IP⌧⌧ 169.254.20.10⌧⌧⌧⌧⌧ CIDR⌧⌧⌧⌧ NetworkPolicy ⌧⌧ DNS ⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧ Pod ⌧⌧⌧⌧⌧⌧⌧⌧⌧

NetworkPolicy ⌧⌧⌧⌧

⌧⌧⌧⌧⌧⌧⌧⌧ Pod ⌧⌧⌧⌧ DNS ⌧⌧⌧⌧⌧⌧⌧⌧⌧ NetworkPolicy ⌧⌧⌧⌧⌧⌧

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-local-dns-and-node-cidr
  namespace: default  #
spec:
  podSelector: {}  #       Pod
  policyTypes:
  - Ingress
  - Egress
  egress:
  #        DNS
  - to:
    - ipBlock:
        cidr: 169.254.20.10/32
  #        CIDR           CIDR
  - to:
    - ipBlock:
        cidr: 10.0.0.0/8  #      CIDR
  ingress:
  #        DNS
  - from:
    - ipBlock:
        cidr: 169.254.20.10/32
  #        CIDR               CIDR
  - from:
    - ipBlock:
        cidr: 10.0.0.0/8  #      CIDR
```

⌧⌧⌧⌧⌧

- `169.254.20.10/32`⌧⌧⌧ DNS ⌧⌧⌧ IP ⌧⌧
- `10.0.0.0/8`⌧⌧⌧⌧⌧ CIDR⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧⌧

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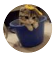

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7.24.4 ⌧⌧

7.25 [illegible] VPC NAT [illegible]

7.25.1 [illegible]

[illegible] VPC [illegible] Overlay [illegible] `natOutgoing` [illegible]Subnet [illegible] Pod [illegible] SNAT [illegible] IP[illegible] Pod [illegible] SNAT[illegible]

[illegible] NAT [illegible] CIDR [illegible] IP [illegible] SNAT[illegible]

7.25.2 [illegible]

[illegible] `subnet.Spec` [illegible] `natOutgoing` [illegible] `natOutgoingPolicyRules` [illegible]

```yaml
spec:
  natOutgoing: true
  natOutgoingPolicyRules:
    - action: forward
      match:
        srcIPs: 10.0.11.0/30,10.0.11.254
    - action: nat
      match:
        srcIPs: 10.0.11.128/26
        dstIPs: 114.114.114.114,8.8.8.8
```

[illegible] NAT [illegible]

1. [illegible] IP [illegible] 10.0.11.0/30 [illegible] 10.0.11.254 [illegible] SNAT[illegible]
2. [illegible] IP [illegible] 10.0.11.128/26 [illegible] IP [illegible] 114.114.114.114 [illegible] 8.8.8.8 [illegible] SNAT[illegible]

[illegible]

`action` [illegible] `match` [illegible] action, action [illegible] `forward` [illegible] `nat` [illegible] `forward` [illegible] SNAT, `nat` [illegible] SNAT[illegible] natOutgoingPolicyRules [illegible] SNAT[illegible]

`match` [illegible] `srcIPs` [illegible] `dstIPs` [illegible] IP [illegible] IP[illegible] `match.srcIPs` [illegible] `match.dstIPs` [illegible] CIDR [illegible] IP[illegible]

[illegible] match [illegible] `natOutgoingPolicyRules` [illegible] action [illegible]

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2023[illegible]10[illegible]25[illegible]

2023[illegible]6[illegible]5[illegible]

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7.25.3 [illegible]

8. [illegible]

8.1 [illegible]

[illegible] Kube-OVN [illegible]

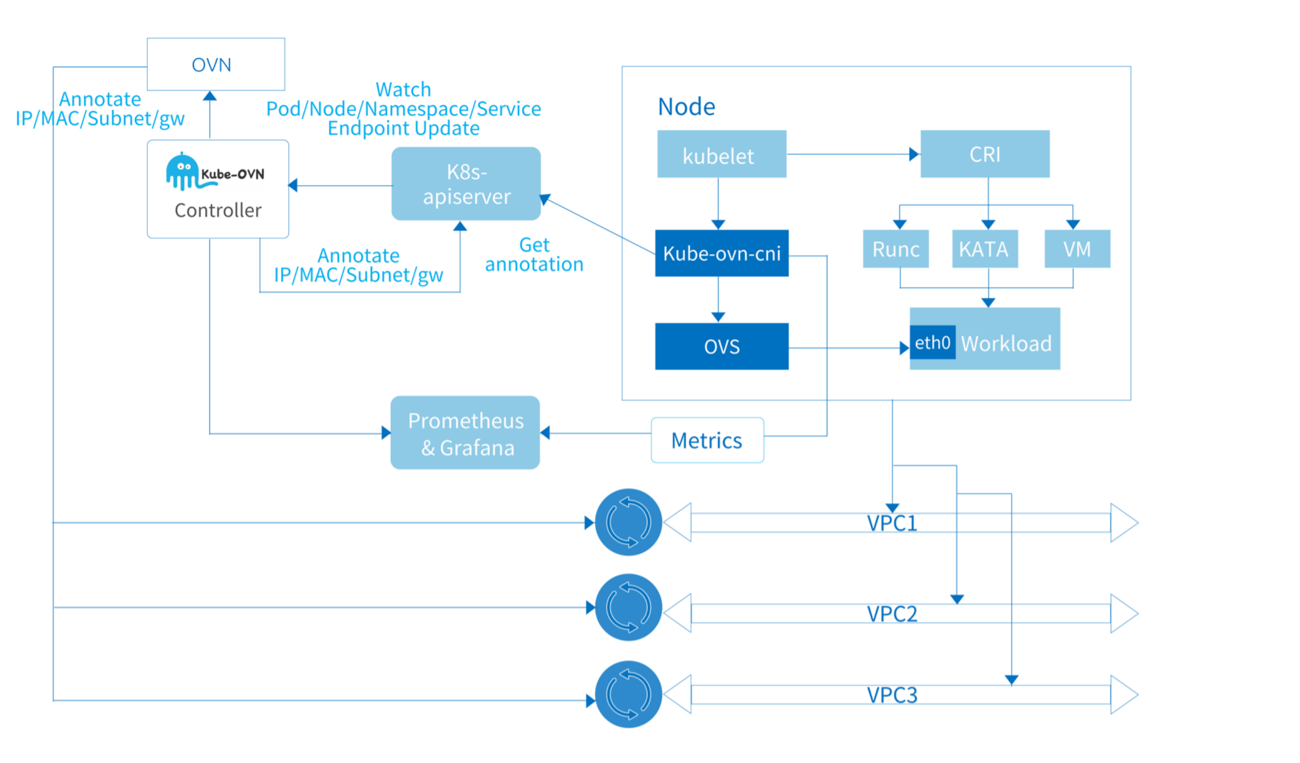
[illegible]Kube-OVN [illegible] Kubernetes [illegible] OVN [illegible] SDN [illegible] Kube-OVN [illegible] OVN [illegible] Kubernetes [illegible] CNI[illegible]Service [illegible] Networkpolicy[illegible] SDN [illegible]VPC[illegible]QoS[illegible]ACL [illegible]

[illegible] Kube-OVN [illegible] Cilium[illegible]Submariner[illegible]Prometheus[illegible]KubeVirt [illegible]

8.1.1 [illegible]

Kube-OVN [illegible]

- [illegible] OVN/OVS [illegible]
- [illegible] Agent[illegible]
- [illegible]


OVN
Annotate
IP/MAC/Subnet/gw
Watch
Pod/Node/Namespace/Service
Endpoint Update
Kube-OVN
Controller
K8s-
apiserver
Annotate
IP/MAC/Subnet/gw
Get
annotation
Node
kubelet
CRI
Kube-ovn-cni
Runc
KATA
VM
OVS
eth0
Workload
Prometheus
& Grafana
Metrics
VPC1
VPC2
VPC3

[illegible] OVN/OVS [illegible]

[illegible] OVN/OVS [illegible] Kube-OVN [illegible]OVN/OVS [illegible] SDN [illegible] Kube-OVN [illegible] ovn-architecture(7) [illegible] OVN [illegible]Kube-OVN [illegible] OVN [illegible] Kubernetes [illegible]

[illegible] OVN/OVS [illegible] Kubernetes [illegible]

OVN-CENTRAL

`ovn-central` Deployment [illegible] OVN [illegible] `ovn-nb` , `ovn-sb` , [illegible] `ovn-northd` [illegible]

- `ovn-nb` [illegible] API [illegible] `kube-ovn-controller` [illegible] `ovn-nb` [illegible]

- `ovn-sb` [illegible] `ovn-nb` [illegible]
- `ovn-northd` [illegible] `ovn-nb` [illegible] `ovn-sb` [illegible]

[illegible] `ovn-central` [illegible] Raft [illegible]

OVS-OVN

`ovs-ovn` [illegible] DaemonSet [illegible] Pod [illegible] `openvswitch`, `ovsdb`, [illegible] `ovn-controller` [illegible] `ovn-central` [illegible] Agent [illegible]

[illegible] Agent

[illegible] Kube-OVN [illegible] OVN [illegible] Kubernetes [illegible]

KUBE-OVN-CONTROLLER

[illegible] Deployment [illegible] Kubernetes [illegible] OVN [illegible] Kube-OVN [illegible] `kube-ovn-controller` [illegible] OVN [illegible] Pod[illegible]Service[illegible]Endpoint[illegible]Node[illegible]NetworkPolicy[illegible]VPC[illegible]Subnet[illegible]Vlan[illegible]ProviderNetwork[illegible]

[illegible] Pod [illegible] `kube-ovn-controller` [illegible] Pod [illegible] IPAM [illegible] `ovn-central` [illegible] ACL [illegible] `kube-ovn-controller` [illegible] CIDR[illegible] Pod [illegible] annotation [illegible] annotation [illegible] `kube-ovn-cni` [illegible]

KUBE-OVN-CNI

[illegible] DaemonSet [illegible] CNI [illegible] OVS [illegible]

[illegible] DaemonSet [illegible] `kube-ovn` [illegible] `kubelet` [illegible] `kube-ovn-cni` [illegible] CNI [illegible] `kube-ovn-cni` [illegible] `/opt/cni/bin` [illegible]

`kube-ovn-cni` [illegible]

1. [illegible] `ovn-controller` [illegible] `vswitchd` [illegible]
2. [illegible] CNI add/del [illegible]
    a. [illegible] veth [illegible] OVS [illegible]
    b. [illegible] OVS [illegible]
    c. [illegible] iptables/ipset/route [illegible]
3. [illegible] QoS.
4. [illegible] `ovn0` [illegible]
5. [illegible] Vlan/Underlay/EIP [illegible]
6. [illegible]

[illegible]

[illegible] Kube-OVN [illegible]

KUBE-OVN-SPEAKER

[illegible] DaemonSet [illegible] Pod IP [illegible]

[illegible] BGP [illegible]

KUBE-OVN-PINGER

[illegible] DaemonSet [illegible] OVS [illegible] Kube-OVN [illegible]

KUBE-OVN-MONITOR

[illegible] Deployment [illegible] OVN [illegible] Kube-OVN [illegible]

KUBECTL-KO

[illegible] kubectl [illegible] kubectl [illegible]

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2022[illegible]5[illegible]24[illegible]

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8.1.2 [illegible]

8.2 What's Next

This document lists the features merged into the master branch for the next minor release.

8.2.1 Post-v1.16.0

- Add routed subnet mode with `/32` IPv4 or `/128` IPv6 Pod addresses and gateway hairpinning through the OVN logical router. #7294
- QoS: support configurable ingress/egress burst limits via per-NIC annotations (`ovn.kubernetes.io/ingress_burst`). #6628
- VPC Egress Gateway: accept integer Mbps values and `M` , `Mi` , `G` , or `Gi` bit-rate quantities for per-replica bandwidth limits. #7115
- VPC NAT Gateway: allow placing the NAT gateway StatefulSet and its iptables-EIP custom resources in a user-specified namespace. #6595
- iptables-EIP: surface the bound external subnet in the `kubectl get iptables-eip` printer. #6672
- Helm Chart (kube-ovn-v2): ship PrometheusRule and Grafana dashboards out of the box. #6630

8.2.2 Post-v1.15.0

- NetworkPolicy now supports provider-scoped policies for multi-network pods using the `ovn.kubernetes.io/policy-for` annotation. #6223
- Support static IP/MAC for multiple interfaces on the same logical switch. #6060
- MetalLB underlay integration now supports IPv6 and dual-stack. #6159
- KubeVirt live-migration multi-chassis options now apply to all VM NICs, not just the primary one. #6241
- Add human-readable descriptions to all Kube-OVN CRD fields for better `kubectl explain` output. #6133 #6147
- Support BGP and EVPN (L3VPN) for VPC Egress Gateway, running FRR in the egress gateway Pod. #6224
- Support per-pod NIC granular DHCP control via annotations, overriding subnet-level DHCP settings. #6475
- Extend SecurityGroup API with tiers, larger priority range (1-16384), localAddress and port matches. #6330
- Use BigInt for SubnetStatus IP count fields to support large address ranges. #6403
- Improve hairpin logic to support FIP, SNAT from any CIDR within the VPC and LoadBalancers. #6445
- VPC NAT Gateway
- Support user-defined annotations on NAT gateway Pod template. #6256
- Allow SNAT EIP to FIP EIP traffic. #6329
- Allow any EIP to share external subnet gateway in one native VLAN. #6426
- VPC Egress Gateway
- Support custom resources and bandwidth limits. #6407
- Interconnection
- Add vendor ID to transit switches to avoid conflicts with other OVN controllers. #6186
- Reliability
- OpenFlow synchronization: detect and recover from stale or missing OVS flows automatically. #6117
- OVN DB: back up Raft header before rejoining cluster to improve recovery. #6106
- Performance
- Strip `managedFields` from informer cache to reduce memory usage. #6119
- Add field selectors to informer factory to reduce API server load. #6091
- Security
- Replace wildcard RBAC verbs with explicit verb lists. #6233
- Specify ephemeral storage limits for containers. #6259
- Helm Chart
- Make DaemonSet update strategy configurable via `values.yaml` . #6136

- Introduce `extraEnv` variable for all components. #6142
- Add `affinity` and `nodeSelector` support for ovs-ovn and ovs-ovn-dpdk DaemonSets. #6308
- Add `external-gateway-config-ns` option for controller. #6211
- Introduce ServiceMonitor and labels to services. #6340

8.2.3 Post-v1.14.0

- ACL log supports ratelimiting. #5938
- Subnet with centralized gateway now supports nodeSelectors. #5956
- Overlay encapsulation NIC selection. #5946
- Performace: skip conntrack for specific dst CIDRs. #5821
- NetworkPolicy supports `lax` mode which only deny traffic type of TCP, UDP and SCTP. That means ARP, ICMP and DHCP traffic are alaways allowed. #5745
- Remove internal-port type interface code. #5794
- IPPool
- Multiple IPPools now can bind to the same Namespace. #5731
- Pods in a bound namespace will only get IPs from the bound pool(s), not other ranges in the subnet. #5731
- IPPool will create an AddressSet that can be work with VPC Policy Route and ACL. #5920
- `AdminNetworkPolicy` now supports specify egress peers using FQDNs. #5703
- Using ARP for IPv4 network ready check: now you don't need ACL allow rules for gateway to make Pod running. #5716
- Non-primary CNI mode: you can run Kube-OVN as the secondary only network, without annoying unused annotations and logical switch port allocations. #5618
- VPC NAT Gateway:
- No default EIP mode: the secondary interface can initialize without a default EIP to avoid the waste. #5605
- Custom routes: you can control the route rules within the vpc-nat-gateway Pods to control traffic paths. #5608
- Gratuitous ARP: VPC NAT Gateway automatically sends gratuitous ARP packets during initialization to accelerate network convergence. #5607
- Healthchecks for static endpoints in `SwitchLBRules` : SLR with both selector or endpoints key can support healthchecks. #5435
- Underlay
- Node Selectors for `ProviderNetwork` : instead of adding/removing nodes to the `ProviderNetwork` one by one, you can use node selectors to simplify the workflow. #5518
- Different `NetworkProvider` s can now share the same VLAN. #5471
- Auto create VLAN sub-interfaces. #5966
- Auto move VLAN sub-interfaces to OVS bridges. #5949
- Adding `pod_name` and `pod_namespace` labels to interface metrics. #5463
- IPSec
- Support `cert-manager` to issue certificates. #5365
- Request new certificate if current certificate is not trusted. #5710
- kubectl-ko
- Collect IPSec and xFRM information. #5472
- Replace `Endpoint` with `EndpointSlice` . #5425
- NetworkAttachment caching: reduce APIServer load in large-scale deployments with Multus. #5386
- Upgrade `OVS` to 3.5 and `OVN` to 25.03. #5537

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8.2.4 □□

8.3 [illegible]

[illegible]Kube-OVN [illegible] [illegible] [illegible]

8.3.1 [illegible]

Kube-OVN [illegible]

1. `default` [illegible] Pod [illegible] IP [illegible] CIDR [illegible] `10.16.0.0/16` [illegible] `10.16.0.1` [illegible]
2. `join` [illegible] Node [illegible] Pod [illegible], [illegible] CIDR [illegible] `100.64.0.0/16` [illegible] `100.64.0.1` [illegible]

[illegible]

```
POD_CIDR="10.16.0.0/16"
POD_GATEWAY="10.16.0.1"
JOIN_CIDR="100.64.0.0/16"
EXCLUDE_IPS=""
```

`EXCLUDE_IP` [illegible] `POD_CIDR` [illegible] `192.168.10.20..192.168.10.30` [illegible]

[illegible] Overlay [illegible] Service CIDR [illegible]

[illegible] Join [illegible]

8.3.2 Service [illegible]

[illegible] `kube-proxy` [illegible] iptables [illegible] Kube-OVN [illegible] Kube-OVN [illegible] Service [illegible] CIDR [illegible]

[illegible]

```
SVC_CIDR="10.96.0.0/12"
```

[illegible]

[illegible] `kube-ovn-controller` Deployment [illegible]

```
args:
- --service-cluster-ip-range=10.96.0.0/12
```

[illegible]

8.3.3 Overlay [illegible]

[illegible]Kube-OVN [illegible] Kubernetes Node IP [illegible]

[illegible]

```
IFACE=eth1
```

[illegible] `ens[a-z0-9]*,eth[a-z0-9]*` [illegible]

[illegible] `kube-ovn-cni` DaemonSet [illegible]

```
args:
- --iface=eth1
```

[illegible] annotation `ovn.kubernetes.io/tunnel_interface` [illegible] annotation [illegible] `iface` [illegible] annotation[illegible]

```
kubectl annotate node no1 ovn.kubernetes.io/tunnel_interface=ethx
```

[illegible] Overlay [illegible]

[illegible] `/32` [illegible]

[illegible] `/32` [illegible] `/128` [illegible]Kube-OVN [illegible] IP [illegible] `HOST_TUNNEL_SRC=true` [illegible]

```
HOST_TUNNEL_SRC=true
```

[illegible] `HOST_TUNNEL_SRC=true` [illegible] Kube-OVN daemon[illegible] Helm [illegible] chart [illegible]

```
# charts/kube-ovn
func:
  HOST_TUNNEL_SRC: true

# charts/kube-ovn-v2
features:
  enableHostTunnelSrc: true
```

[illegible] Helm [illegible] `false` [illegible]daemon [illegible] Overlay [illegible]

8.3.4 MTU [illegible]

[illegible] Overlay [illegible]Kube-OVN [illegible] MTU [illegible] MTU [illegible] Overlay [illegible] Pod [illegible] MTU [illegible] MTU - 100[illegible]Underlay [illegible]Pod [illegible] MTU[illegible]

[illegible] Overlay [illegible] MTU [illegible] `kube-ovn-cni` DaemonSet [illegible]

```
args:
- --mtu=1333
```

8.3.5 [illegible]

[illegible]Kube-OVN [illegible] `mirror0` [illegible] tcpdump [illegible]

```
ENABLE_MIRROR=true
```

[illegible] `kube-ovn-cni` DaemonSet [illegible]:

```
args:
- --enable-mirror=true
```

[illegible]

8.3.6 LB [illegible]

[illegible] Underlay [illegible]kube-proxy [illegible] Service [illegible] OVN [illegible] L2 LB [illegible] ClusterIP [illegible] Service [illegible] LB [illegible] ClusterIP [illegible] kube-proxy [illegible] kube-proxy[illegible]

[illegible]

```
ENABLE_LB=false
```

[illegible] `kube-ovn-controller` Deployment [illegible]

```
args:
- --enable-lb=false
```

LB [illegible]

[illegible] Kube-OVN v1.12.0 [illegible] subnet crd [illegible] spec [illegible] `enableLb` [illegible] Kube-OVN [illegible] LB [illegible] LB [illegible] `kube-ovn-controller` Deployment [illegible] `enable-lb` [illegible] load-balancer [illegible] `enableLb` [illegible] load-balancer [illegible] v1.12.0 [illegible] `enableLb` [illegible]

8.3.7 NetworkPolicy [illegible]

Kube-OVN [illegible] OVN [illegible] ACL [illegible] NetworkPolicy[illegible] NetworkPolicy [illegible] Cilium Chain [illegible] eBPF [illegible] NetworkPolicy[illegible] Kube-OVN [illegible] NetworkPolicy [illegible]

[illegible]

```
ENABLE_NP=false
```

[illegible] `kube-ovn-controller` Deployment [illegible]

```
args:
- --enable-np=false
```

NetworkPolicy [illegible]

8.3.8 EIP [illegible] SNAT [illegible]

[illegible] EIP [illegible] SNAT [illegible] `kube-ovn-controller` [illegible]

[illegible]

```
ENABLE_EIP_SNAT=false
```

[illegible] `kube-ovn-controller` Deployment [illegible]

```
args:
- --enable-eip-snat=false
```

EIP [illegible] SNAT [illegible] EIP [illegible] SNAT [illegible]

8.3.9 Load Balancer [illegible] Service [illegible]

[illegible] VPC [illegible] Load Balancer [illegible] Service[illegible] LoadBalancer [illegible] Service[illegible]

[illegible]

```
ENABLE_LB_SVC=true
```

[illegible] `kube-ovn-controller` Deployment [illegible]

```
args:
- --enable-lb-svc=true
```

8.3.10 [illegible] ECMP [illegible]

[illegible] ECMP [illegible] ECMP [illegible] `kube-ovn-controller` Deployment [illegible]:

```
args:
- --enable-ecmp=true
```

[illegible] Kube-OVN v1.12.0 [illegible] subnet crd [illegible] spec [illegible] `enableEcmp` [illegible] ECMP [illegible] ECMP [illegible] `kube-ovn-controller` Deployment [illegible] `enable-ecmp` [illegible] v1.12.0 [illegible]

[illegible]

8.3.11 Kubevirt VM [illegible]

[illegible] Kubevirt [illegible] VM [illegible] `kube-ovn-controller` [illegible] StatefulSet Pod [illegible] IP [illegible] VM [illegible]

[illegible] 1.10.6 [illegible] `kube-ovn-controller` Deployment [illegible]

```yaml
args:
- --keep-vm-ip=false
```

8.3.12 CNI [illegible]

Kube-OVN [illegible] `/opt/cni/bin` [illegible] CNI [illegible] `/etc/cni/net.d` [illegible] CNI [illegible] `01-kube-ovn.conflist` [illegible] CNI [illegible]

```
CNI_CONF_DIR="/etc/cni/net.d"
CNI_BIN_DIR="/opt/cni/bin"
CNI_CONFIG_PRIORITY="01"
```

[illegible] `kube-ovn-cni` DaemonSet [illegible] Volume [illegible]

```yaml
volumes:
- name: cni-conf
  hostPath:
    path: "/etc/cni/net.d"
- name: cni-bin
  hostPath:
    path:"/opt/cni/bin"
...
args:
- --cni-conf-name=01-kube-ovn.conflist
```

[illegible] Kube-OVN [illegible] `loopback` [illegible] `portmap` [illegible] `macvlan` [illegible] `ipvlan` [illegible] CNI [illegible]

```
INSTALL_CNI_PLUGINS=false
```

[illegible] `false` [illegible]Kube-OVN [illegible] CNI [illegible] Helm v1 [illegible] `cni_conf.INSTALL_CNI_PLUGINS: false` [illegible] Helm v2 [illegible] `cni.installCNIPlugins: false` [illegible]

8.3.13 CNI [illegible] recompute

`kube-ovn-cni` DaemonSet [illegible] OVN controller recompute [illegible]

```
ENABLE_RECOMPUTE=true
```

Helm v1 [illegible] Helm v2 [illegible]

```yaml
# Helm v1
func:
  ENABLE_RECOMPUTE: true

# Helm v2
features:
  enableRecompute: true
```

[illegible] `kube-ovn-cni` DaemonSet [illegible] `--enable-recompute=true` [illegible]daemon [illegible] 10 [illegible] recompute[illegible] `false` [illegible]

8.3.14 [illegible]

Kube-OVN [illegible] Overlay [illegible] Geneve[illegible] Vxlan [illegible] STT[illegible]

```
TUNNEL_TYPE="vxlan"
```

[illegible] `ovs-ovn` DaemonSet [illegible]

```yaml
env:
- name: TUNNEL_TYPE
  value: "vxlan"
```

[illegible] STT [illegible] ovs [illegible]

[illegible]

8.3.15 SSL [illegible]

OVN NB/SB [illegible] API [illegible] Kube-OVN secure-serving [illegible] TLS[illegible]TLS [illegible] [illegible] OVN central/DB [illegible] TLS[illegible] secure-serving [illegible] TLS [illegible] cipher suites[illegible]

```
ENABLE_SSL=true
TLS_MIN_VERSION="TLS12"
TLS_MAX_VERSION="TLS13"
TLS_CIPHER_SUITES="TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,TLS_AES_128_GCM_SHA256"
```

`TLS_MIN_VERSION` [illegible] `TLS_MAX_VERSION` [illegible] `TLS10` [illegible] `TLS11` [illegible] `TLS12` [illegible] `TLS13` [illegible] [illegible] OVN NB/SB [illegible]

`TLS_CIPHER_SUITES` [illegible] Go TLS cipher suite [illegible]OVN DB [illegible] [illegible] OpenSSL [illegible]

```
TLS_AES_128_GCM_SHA256
TLS_AES_256_GCM_SHA384
TLS_CHACHA20_POLY1305_SHA256
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256
TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256
```

[illegible] cipher suites [illegible] `kube-ovn-tls` Secret [illegible] `kube-ovn-tls` [illegible] RSA [illegible] ECDSA cipher suites [illegible] RSA [illegible] TLS [illegible] ECDSA-only suites[illegible] ECDSA [illegible]

[illegible] Helm [illegible] chart [illegible] `componentTLSArgs` [illegible] `--tls-min-version` [illegible] `--tls-max-version` [illegible] `--tls-cipher-suites` [illegible] secure-serving [illegible] `TLS_*` [illegible] TLS [illegible] secure-serving[illegible] chart [illegible] classic chart [illegible] `func.SECURE_SERVING` [illegible]v2 chart [illegible] `features.enableSecureServing` [illegible] TLS [illegible] cipher suites [illegible] `networking` [illegible]

classic Helm chart [illegible]

```yaml
func:
  SECURE_SERVING: true
networking:
  ENABLE_SSL: true
  TLS_MIN_VERSION: TLS12
  TLS_MAX_VERSION: TLS13
  TLS_CIPHER_SUITES:
    - TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
    - TLS_AES_128_GCM_SHA256
```

v2 Helm chart [illegible]

```yaml
features:
  enableSecureServing: true
networking:
  enableSsl: true
  tlsMinVersion: TLS12
  tlsMaxVersion: TLS13
  tlsCipherSuites:
    - TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
    - TLS_AES_128_GCM_SHA256
```

[illegible] `ENABLE_SSL=true` [illegible] `networking.enableSsl=true` [illegible]Kube-OVN [illegible] `kube-ovn-tls` Secret[illegible] Secret [illegible] `cacert` [illegible] `cert` [illegible] `key` [illegible] `/var/run/tls` [illegible] classic Helm chart [illegible] v2 Helm chart [illegible] Secret [illegible] classic Helm chart [illegible] `installMode: dataPlaneOnly` [illegible] `networking.ENABLE_SSL=true` [illegible] `kube-ovn-tls` [illegible] OVN central [illegible] CA[illegible]classic chart [illegible] Secret [illegible] Secret[illegible]v2 Helm chart [illegible] `installMode: dataPlaneOnly` [illegible] v2 chart[illegible]

8.3.16 [illegible] conntrack [illegible] CIDR [illegible]

[illegible] conntrack [illegible] NAT/[illegible] `kube-ovn-controller` [illegible] CIDR [illegible]

```
SKIP_CONNTRACK_DST_CIDRS="10.10.0.0/16,fd00:10::/64"
```

```yaml
args:
- --skip-conntrack-dst-cidrs=10.10.0.0/16,fd00:10::/64
```

8.3.17 OVN IPSec

 Kube-OVN OVN IPSec Geneve/Vxlan/STT

```
ENABLE_OVN_IPSEC=true
```

 daemon

```
args:
- --enable-ovn-ipsec=true
```

 OVN IPSec

8.3.18 AdminNetworkPolicy / DNS

 AdminNetworkPolicyANP/CNP DNS `DNSNameResolver` CRD

```
ENABLE_ANP=true
ENABLE_DNS_NAME_RESOLVER=true
```

```
args:
- --enable-anp=true
- --enable-dns-name-resolver=true
```

8.3.19 VPC NAT Gateway

 VPC NAT Gateway controller

```
ENABLE_NAT_GW=true
```

 `ovn-vpc-nat-gw-config` ConfigMap `enable-vpc-nat-gw`

8.3.20 ovn-northd

 `ovn-northd`

```
OVN_NORTHD_N_THREADS=4
```

 1 CPU

8.3.21 NetworkPolicy

Kube-OVN `standard` / `lax` NetworkPolicy

```
NP_ENFORCEMENT=standard
```

```
args:
- --np-enforcement=standard
```

 NetworkPolicy `ovn.kubernetes.io/network_policy_enforcement` annotation NetworkPolicy

8.3.22 VXLAN TX checksum offload

 VXLAN checksum offload

```
SET_VXLAN_TX_OFF=true
```

```yaml
args:
- --set-vxlan-tx-off=true
```

8.3.23 KubeVirt [illegible]

[illegible] KubeVirt [illegible]

```
ENABLE_LIVE_MIGRATION_OPTIMIZE=true
```

8.3.24 OVN LB [illegible] endpoint

[illegible] OVN [illegible] LoadBalancer [illegible] endpoint[illegible] `externalTrafficPolicy=Local` [illegible]

```
ENABLE_OVN_LB_PREFER_LOCAL=true
```

```yaml
args:
- --enable-ovn-lb-prefer-local=true
```

8.3.25 [illegible] ConfigMap [illegible]

[illegible] ConfigMap[illegible] `external-gw-config` [illegible] `kube-system` [illegible]

```
EXTERNAL_GATEWAY_CONFIG_NS=kube-system
```

8.3.26 Non-Primary CNI [illegible]

[illegible] Kube-OVN [illegible] attachment network [illegible] Pod [illegible] IP[illegible] CNI [illegible]controller [illegible] daemon [illegible]

```yaml
args:
- --non-primary-cni-mode=true
```

[illegible] Helm `cni_conf.NON_PRIMARY_CNI=true` [illegible]chart v1[illegible]/ `cni.nonPrimaryCNI=true` [illegible]chart v2[illegible] Non-Primary CNI [illegible]

8.3.27 Controller Leader Election [illegible]

controller [illegible] Kubernetes leader election[illegible] duration [illegible] leader [illegible] controller [illegible]

```yaml
args:
- --leader-elect-lease-duration=30s
- --leader-elect-renew-deadline=20s
- --leader-elect-retry-period=6s
```

[illegible] `30s` [illegible] `20s` [illegible] `6s` [illegible] duration[illegible]controller [illegible] Kubernetes client [illegible]

- `lease-duration` [illegible] `renew-deadline` [illegible]
- `renew-deadline` [illegible] `retry-period` [illegible] Kubernetes client-go leader election [illegible] jitter factor[illegible] `1.2` [illegible]

[illegible]

```bash
export LEADER_ELECT_LEASE_DURATION=1m
export LEADER_ELECT_RENEW_DEADLINE=40s
export LEADER_ELECT_RETRY_PERIOD=10s
bash install.sh
```

Helm [illegible] chart [illegible]classic chart [illegible] `kube-ovn-controller.leaderElection` [illegible]

```yaml
kube-ovn-controller:
  leaderElection:
    leaseDuration: "30s"
    renewDeadline: "20s"
    retryPeriod: "6s"
```

v2 chart [illegible] `controller.leaderElection` [illegible]

```
controller:
  leaderElection:
    leaseDuration: "30s"
    renewDeadline: "20s"
    retryPeriod: "6s"
```

8.3.28 kube-ovn-monitor [illegible]

`kube-ovn-monitor` [illegible] Helm [illegible] Deployment [illegible]

```
# classic chart
kube-ovn-monitor:
  replicas: 1

# v2 chart
monitor:
  replicas: 1
```

classic chart [illegible] `kube-ovn-monitor.replicas` [illegible]v2 chart [illegible] `monitor.replicas` [illegible] `1` [illegible] OVN central [illegible] central [illegible] OVN Raft [illegible] `kube-ovn-monitor` [illegible]

[illegible] Deployment [illegible] master [illegible] Pod [illegible] master [illegible] central [illegible] Pod [illegible] `Pending` [illegible]

[illegible] Helm [illegible] Deployment[illegible] chart [illegible] `1` [illegible] Helm values [illegible] `kubectl scale` [illegible] Deployment[illegible]

8.3.29 [illegible] ip

kube-ovn-controller/kube-ovn-cni/kube-ovn-monitor [illegible] ip[illegible] 0.0.0.0 [illegible] ip [illegible]

```
ENABLE_BIND_LOCAL_IP=true
```

[illegible] kube-ovn-monitor [illegible] pod ip [illegible]

```
# netstat -tunlp |grep kube-ovn
tcp        0      0 172.18.0.5:10661        0.0.0.0:*               LISTEN      2612/./kube-ovn-mon
```

[illegible] deployment [illegible] daemonSet [illegible]

```
env:
- name: ENABLE_BIND_LOCAL_IP
  value: "false"
```

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2022[illegible]5[illegible]24[illegible]

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8.3.30 [illegible]

8.4 [illegible]

Kube-OVN [illegible] Minor [illegible] Patch [illegible]Minor [illegible]OVN/OVS [illegible]API [illegible]Patch [illegible] Bug [illegible] API[illegible]

8.4.1 [illegible]

Kube-OVN [illegible] `master`, `release-1.12` [illegible] `release-1.11` [illegible] `release-1.12` [illegible] Bug [illegible] backport [illegible]

[illegible] `release-1.11` [illegible] backport [illegible] Bug [illegible]

8.4.2 [illegible]

Minor [illegible]Patch [illegible] Bug [illegible] Bug [illegible]

8.4.3 Patch [illegible]

[illegible] Patch [illegible] hack/release.sh [illegible]

1. [illegible] Build [illegible]
2. [illegible] tag [illegible] Docker Hub[illegible]
3. [illegible] tag [illegible] Github[illegible]
4. [illegible]
5. [illegible]
6. [illegible] Release Note PR[illegible]
7. [illegible] Release Note ([illegible])
8. [illegible] Merge github action [illegible] Release Note PR
9. [illegible] Github Release [illegible]
10. [illegible] Github Release [illegible] Release[illegible] `v1.12.12` [illegible] Release Note [illegible] Release [illegible]

8.4.4 Minor [illegible]

[illegible] Minor [illegible]

1. [illegible] Github [illegible] `release-1.13` ([illegible])
2. [illegible] `VERSION`, `dist/images/install.sh`, `charts/kube-ovn/values.yaml` [illegible] `charts/kube-ovn/Chart.yaml` [illegible] Minor [illegible] `v1.14.0` ([illegible])
3. [illegible] tag [illegible] Docker Hub ([illegible])
4. [illegible] tag [illegible] Github ([illegible])
5. [illegible] `v1.13` [illegible] `mkdocs.yml` [illegible] `version` [illegible] `branch` [illegible] ([illegible])
6. [illegible] Release Note PR[illegible]
7. [illegible] Release Note ([illegible])
8. [illegible] Merge github action [illegible] Release Note PR
9. [illegible] Github Release [illegible]
10. [illegible] Github Release [illegible] Release[illegible] `v1.13.0` [illegible] Release Note [illegible] Release [illegible]
11. [illegible] `VERSION` [illegible] Patch [illegible] `v1.13.1`

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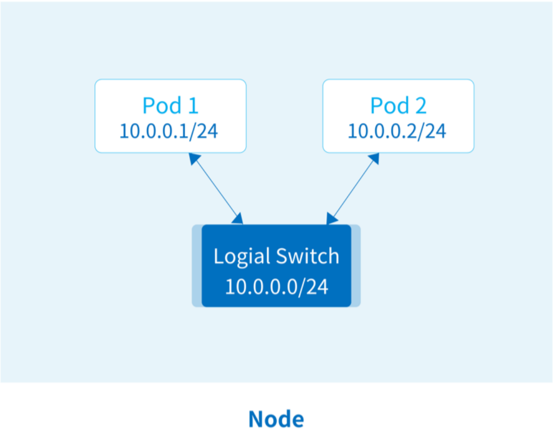
8.4.5 □□

8.5 Underlay [illegible]

[illegible] Underlay [illegible]

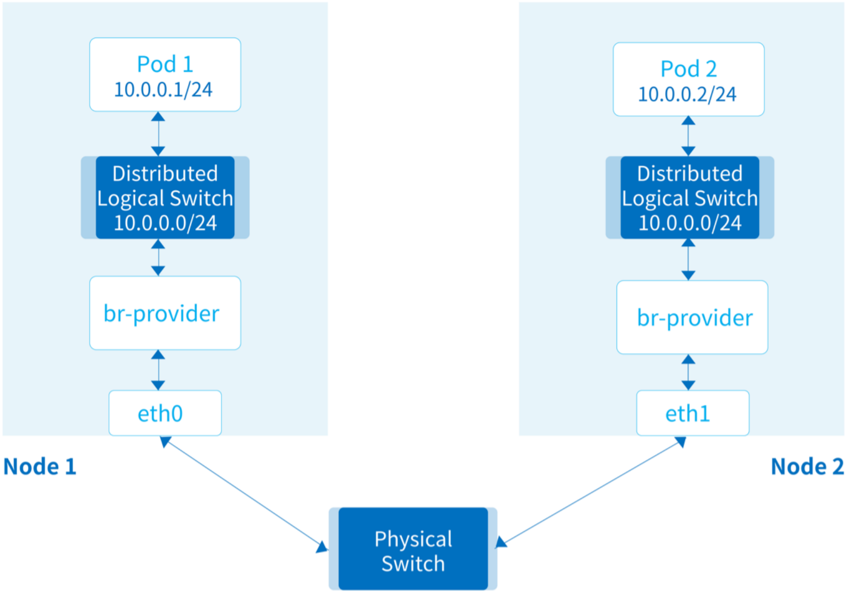
8.5.1 [illegible]

[illegible]


Pod 1
10.0.0.1/24
Pod 2
10.0.0.2/24
Logial Switch
10.0.0.0/24
Node

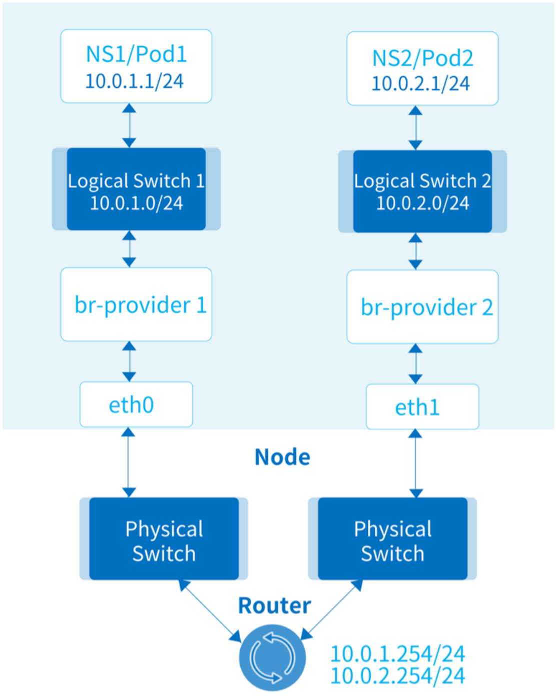
8.5.2 [illegible]

[illegible]


Pod 1
10.0.0.1/24
Distributed Logical Switch
10.0.0.0/24
br-provider
eth0
Node 1
Pod 2
10.0.0.2/24
Distributed Logical Switch
10.0.0.0/24
br-provider
eth1
Node 2
Physical Switch

8.5.3 [illegible]

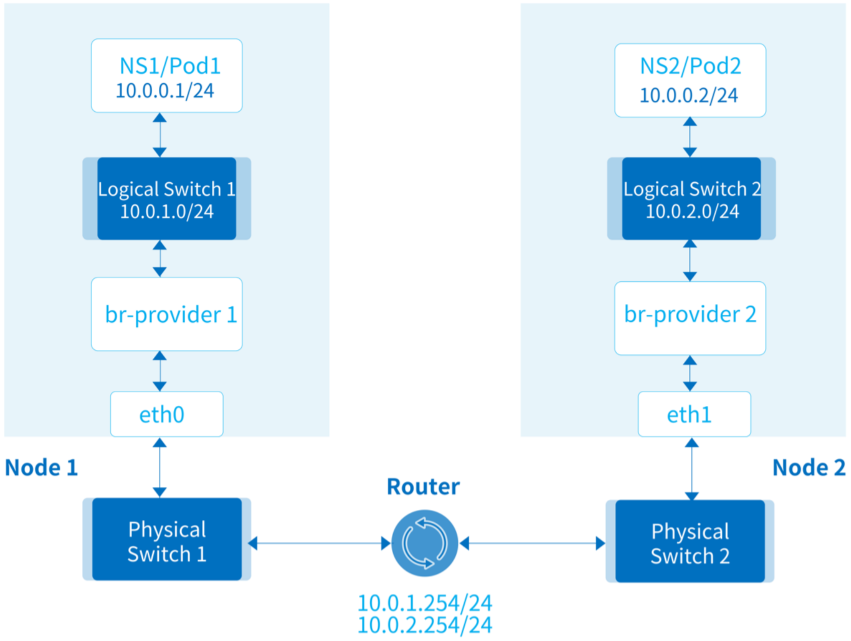
[illegible]


NS1/Pod1
10.0.1.1/24
Logical Switch 1
10.0.1.0/24
br-provider 1
eth0
NS2/Pod2
10.0.2.1/24
Logical Switch 2
10.0.2.0/24
br-provider 2
eth1
Node
Physical Switch
Physical Switch
Router
10.0.1.254/24
10.0.2.254/24

> [illegible] br-provider-1 [illegible] br-provider-2 [illegible] OVS [illegible] Provider Network[illegible]

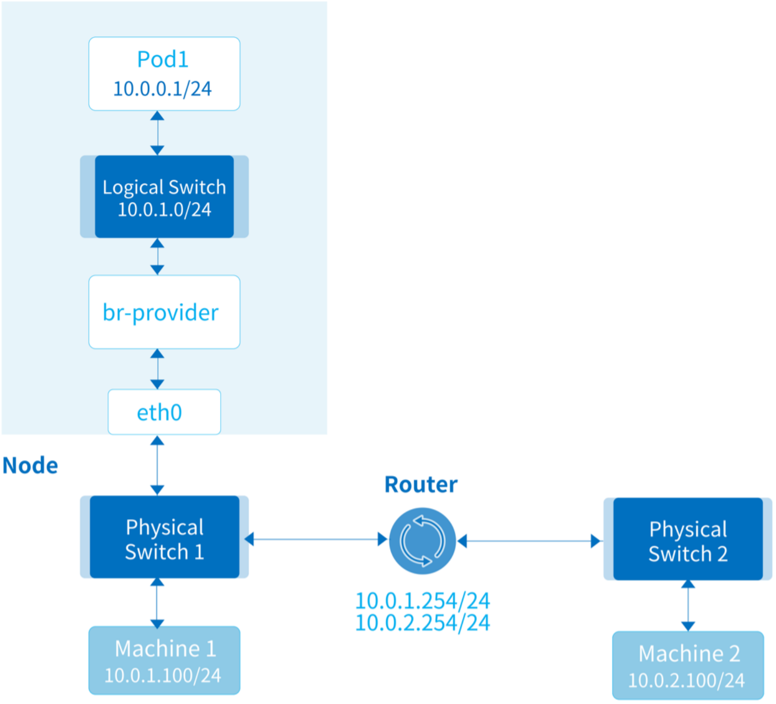
8.5.4 [illegible]

[illegible]


NS1/Pod1
10.0.0.1/24
Logical Switch 1
10.0.1.0/24
br-provider 1
eth0
Node 1
Physical Switch 1
NS2/Pod2
10.0.0.2/24
Logical Switch 2
10.0.2.0/24
br-provider 2
eth1
Node 2
Physical Switch 2
Router
10.0.1.254/24
10.0.2.254/24

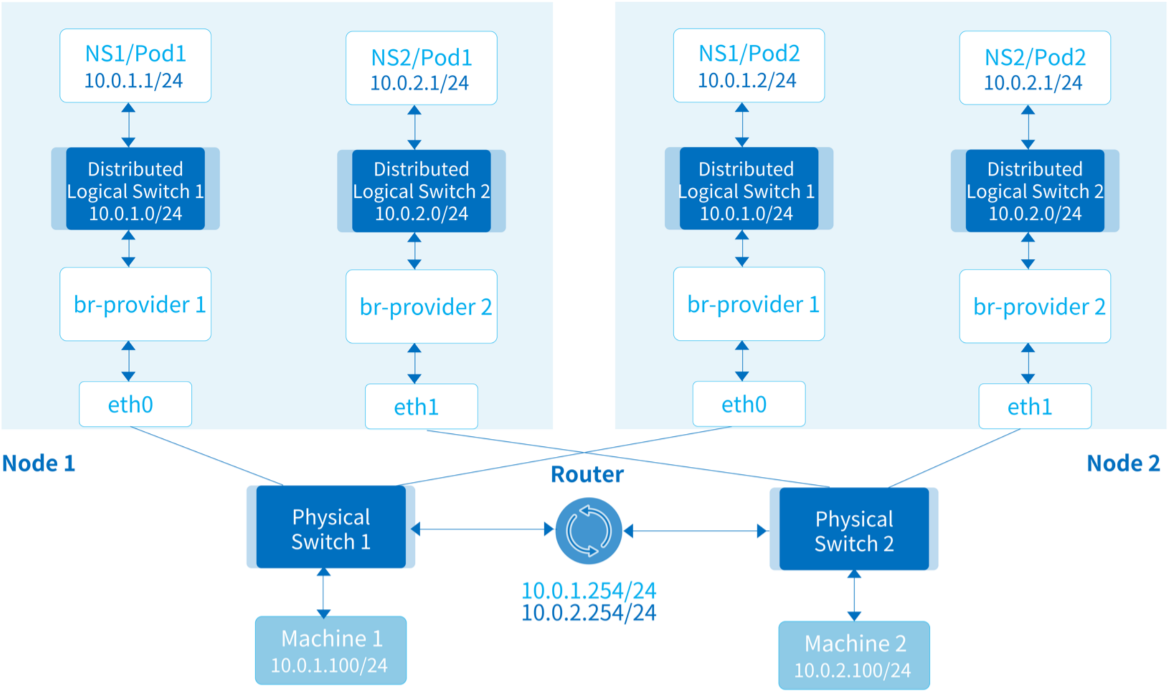
8.5.5 [illegible]

[illegible]

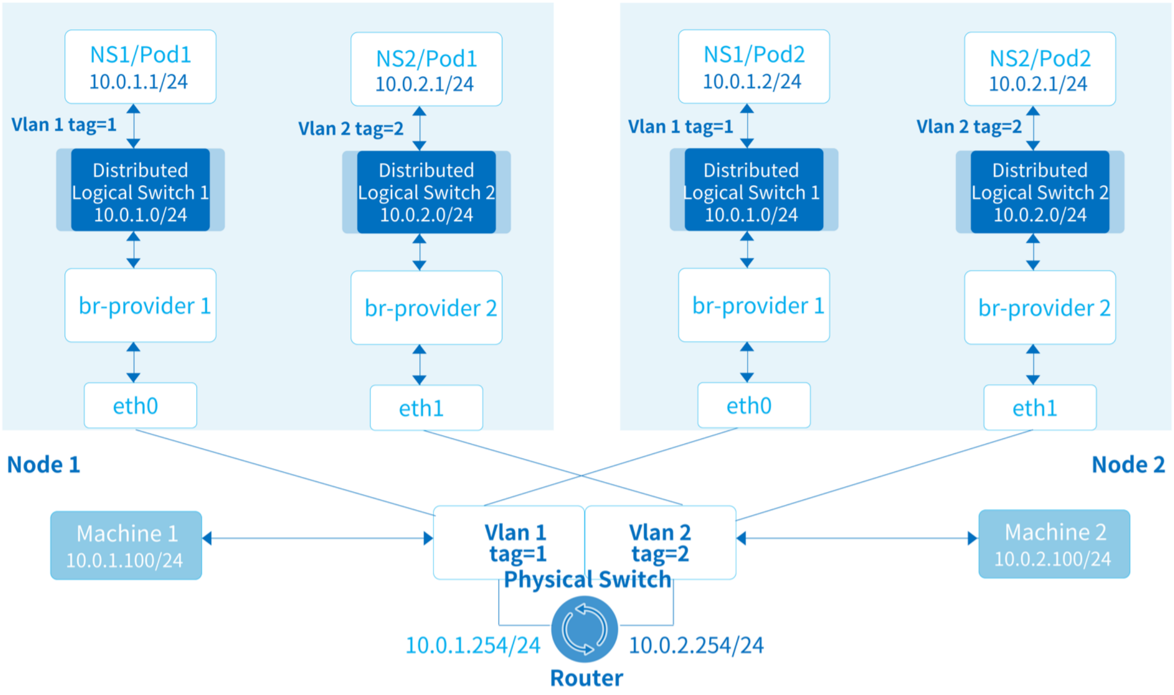

Pod1
10.0.0.1/24
Logical Switch
10.0.1.0/24
br-provider
eth0
Node
Router
Physical Switch 1
Physical Switch 2
10.0.1.254/24
10.0.2.254/24
Machine 1
10.0.1.100/24
Machine 2
10.0.2.100/24

> [illegible] Pod [illegible]

8.5.6 [illegible] Vlan Tag [illegible]


NS1/Pod1
10.0.1.1/24
NS2/Pod1
10.0.2.1/24
NS1/Pod2
10.0.1.2/24
NS2/Pod2
10.0.2.1/24
Distributed Logical Switch 1
10.0.1.0/24
Distributed Logical Switch 2
10.0.2.0/24
Distributed Logical Switch 1
10.0.1.0/24
Distributed Logical Switch 2
10.0.2.0/24
br-provider 1
br-provider 2
br-provider 1
br-provider 2
eth0
eth1
eth0
eth1
Node 1
Router
Node 2
Physical Switch 1
Physical Switch 2
10.0.1.254/24
10.0.2.254/24
Machine 1
10.0.1.100/24
Machine 2
10.0.2.100/24

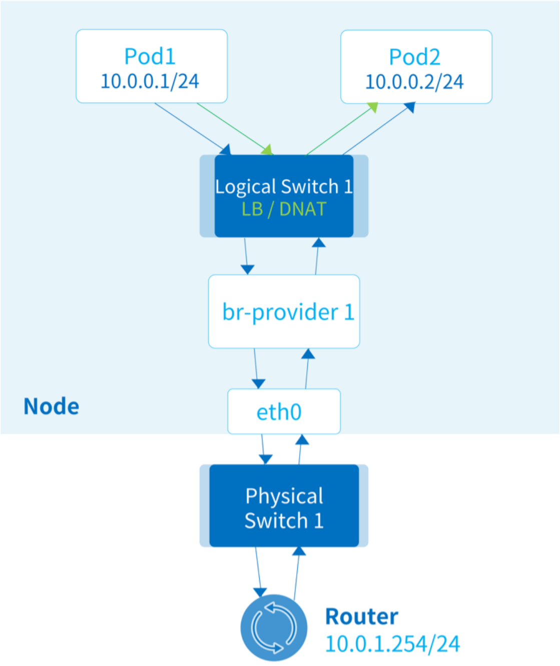
8.5.7 □ VLAN □□


NS1/Pod1
10.0.1.1/24
Vlan 1 tag=1
Distributed
Logical Switch 1
10.0.1.0/24
br-provider 1
eth0
NS2/Pod1
10.0.2.1/24
Vlan 2 tag=2
Distributed
Logical Switch 2
10.0.2.0/24
br-provider 2
eth1
NS1/Pod2
10.0.1.2/24
Vlan 1 tag=1
Distributed
Logical Switch 1
10.0.1.0/24
br-provider 1
eth0
NS2/Pod2
10.0.2.1/24
Vlan 2 tag=2
Distributed
Logical Switch 2
10.0.2.0/24
br-provider 2
eth1
Node 1
Node 2
Machine 1
10.0.1.100/24
Vlan 1
tag=1
Vlan 2
tag=2
Physical Switch
Machine 2
10.0.2.100/24
10.0.1.254/24
10.0.2.254/24
Router

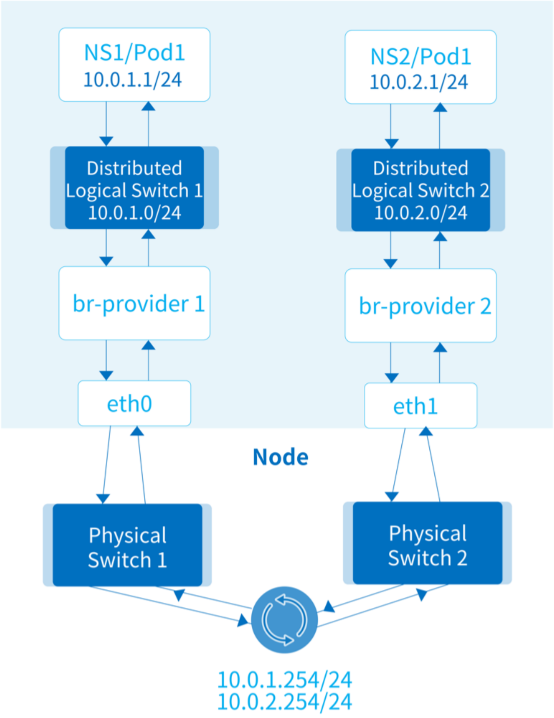
8.5.8 Pod □□ Service IP

Kube-OVN □□□ Kubernetes Service □□□□□□□□□□□□□□□□□□□□□ □ Pod □□□□ Service IP □□□□ Pod □□□□□□□□□□□□ Service IP□□□ MAC □□□□□ MAC □□□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□ DNAT □□□□□□ IP □□□□□□ Service □□□□□ Endpoint □ IP □□□□ □□□□□□□□□□□□□□□□□□□□ MAC □□

Service [illegible] Pod


Pod1
10.0.0.1/24
Pod2
10.0.0.2/24
Logical Switch 1
LB / DNAT
br-provider 1
Node
eth0
Physical Switch 1
Router
10.0.1.254/24

Service [illegible] Pod


NS1/Pod1
10.0.1.1/24
NS2/Pod1
10.0.2.1/24
Distributed Logical Switch 1
10.0.1.0/24
Distributed Logical Switch 2
10.0.2.0/24
br-provider 1
br-provider 2
eth0
eth1
Node
Physical Switch 1
Physical Switch 2
10.0.1.254/24
10.0.2.254/24

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2022年5月20日

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8.5.9 [illegible]

8.6 Iptables [illegible]

Kube-OVN [illegible] ipset [illegible] iptables [illegible] VPC [illegible]Overlay[illegible] NAT [illegible]

[illegible] ipset [illegible]

[illegible]**IPv4/IPv6**[illegible]	[illegible]	[illegible]
ovn40services/ovn60services	hash:net	Service [illegible]
ovn40subnets/ovn60subnets	hash:net	Overlay [illegible] NodeLocal DNS IP [illegible]
ovn40subnets-nat/ovn60subnets-nat	hash:net	[illegible] NatOutgoing [illegible] Overlay [illegible]
ovn40subnets-distributed-gw/ovn60subnets-distributed-gw	hash:net	[illegible] Overlay [illegible]
ovn40other-node/ovn60other-node	hash:net	[illegible] IP [illegible]
ovn40local-pod-ip-nat/ovn60local-pod-ip-nat	hash:ip	[illegible]
ovn40subnets-nat-policy	hash:net	[illegible] natOutgoingPolicyRules [illegible]
ovn40natpr-418e79269dc5-dst	hash:net	natOutgoingPolicyRules [illegible] rule [illegible] dstIPs
ovn40natpr-418e79269dc5-src	hash:net	natOutgoingPolicyRules [illegible] rule [illegible] srcIPs

[illegible] iptables [illegible]IPv4[illegible]

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
filter	INPUT	-m set --match-set ovn40services src -j ACCEPT	[illegible] k8s Service [illegible] Pod [illegible]	--
filter	INPUT	-m set --match-set ovn40services dst -j ACCEPT	[illegible]	--
filter	INPUT	-m set --match-set ovn40subnets src -j ACCEPT	[illegible]	--
filter	INPUT	-m set --match-set ovn40subnets dst -j ACCEPT	[illegible]	--
filter	FORWARD	-m set --match-set ovn40services src -j ACCEPT	[illegible]	--
filter	FORWARD	-m set --match-set ovn40services dst -j ACCEPT	[illegible]	--
filter	FORWARD	-m set --match-set ovn40subnets src -j ACCEPT	[illegible]	--
filter	FORWARD	-m set --match-set ovn40subnets dst -j ACCEPT	[illegible]	--
filter	FORWARD	-s 10.16.0.0/16 -m comment --comment "ovn-subnet-gateway,ovn-default"	[illegible] subnet [illegible]	10.16.0.0/16 [illegible] subnet [illegible] cidr [illegible]comment [illegible] ovn-subnet-gateway [illegible] iptables [illegible] subnet [illegible] ovn-default [illegible] subne [illegible]
filter	FORWARD	-d 10.16.0.0/16 -m comment --comment "ovn-subnet-gateway,ovn-default"	[illegible] subnet [illegible]	[illegible]
filter	OUTPUT	-p udp -m udp --dport 6081 -j MARK --set-xmark 0x0	[illegible] SNAT	UDP: bad checksum on VXLAN interface
nat	PREROUTING	-m comment --comment "kube-ovn prerouting rules" -j OVN-PREROUTING	[illegible] OVN-PREROUTING [illegible]	--
nat	POSTROUTING	-m comment --comment "kube-ovn postrouting rules" -j OVN-POSTROUTING	[illegible] OVN-POSTROUTING [illegible]	--
nat	OVN-PREROUTING	-i ovn0 -m set --match-set ovn40subnets src -m set --match-set ovn40services dst -j MARK --set-xmark 0x4000/0x4000	[illegible] Pod [illegible] Service [illegible] masquerade [illegible]	[illegible] LB [illegible]
nat	OVN-PREROUTING	-p tcp -m addrtype --dst-type LOCAL -m set --match-		[illegible] kube-proxy [illegible] ipvs [illegible]

□	□	□□ set KUBE-NODE-PORT-LOCAL-TCP dst -j MARK --set-xmark 0x80000/0x80000	□□ □ ExternalTrafficPolicy □ Local □ Service □□□TCP□□□□□□□	□□
nat	OVN-PREROUTING	-p udp -m addrtype --dst-type LOCAL -m set --match-set KUBE-NODE-PORT-LOCAL-UDP dst -j MARK --set-xmark 0x80000/0x80000	□ ExternalTrafficPolicy □ Local □ Service □□□UDP□□□□□□□	□□
nat	OVN-POSTROUTING	-m set --match-set ovn40services src -m set --match-set ovn40subnets dst -m mark --mark 0x4000/0x4000 -j SNAT --to-source	□□□□□ Service IP □□ Overlay Pod □□□□□ IP □□□ IP□	□ kube-proxy □□ ipvs □□□□□
nat	OVN-POSTROUTING	-m mark --mark 0x4000/0x4000 -j MASQUERADE	□□□□□□□□□□ SNAT	--
nat	OVN-POSTROUTING	-m set --match-set ovn40subnets src -m set --match-set ovn40subnets dst -j MASQUERADE	□□□□□□ Pod □□□ Service □□□□ SNAT	--
nat	OVN-POSTROUTING	-m mark --mark 0x80000/0x80000 -m set --match-set ovn40subnets-distributed-gw dst -j RETURN	□□ ExternalTrafficPolicy □ Local □ Service □□□□ Endpoint □□□□□□□□□□□□ SNAT	--
nat	OVN-POSTROUTING	-m mark --mark 0x80000/0x80000 -j MASQUERADE	□□ ExternalTrafficPolicy □ Local □ Service □□□□ Endpoint □□□□□□□□□□ SNAT	--
nat	OVN-POSTROUTING	-p tcp -m tcp --tcp-flags SYN NONE -m conntrack --ctstate NEW -j RETURN	Pod IP □□□□□□□□□ SNAT	--
nat	OVN-POSTROUTING	-s 10.16.0.0/16 -m set ! --match-set ovn40subnets dst -j SNAT --to-source 192.168.0.101	Pod □□□□□□□□□□□□□□ NatOutgoing □□□□□ IP □□□□□□□□□ SNAT	10.16.0.0/16 □□□□□□192.168.0.101 □□□□□□□ IP
nat	OVN-POSTROUTING	-m set --match-set ovn40subnets-nat src -m set ! --match-set ovn40subnets dst -j MASQUERADE	Pod □□□□□□□□□□□□□□ NatOutgoing□□□ SNAT	--
nat	OVN-POSTROUTING	-m set --match-set ovn40subnets-nat-policy src -m set ! --match-set ovn40subnets dst -j OVN-NAT-POLICY	Pod □□□□□□□□□□□□□□□ natOutgoingPolicyRules□□□□□□□□□□ SNAT	□□□ natOutgoingPolicyRules □□□□□□□□□□ OV NAT-POLICY □
nat			□□	

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
	OVN-POSTROUTING	-m mark --mark 0x90001/0x90001 -j MASQUERADE --random-fully		[illegible] OVN-NAT-POLICY [illegible] tag 0x90001/0x90001 [illegible] SNAT
nat	OVN-POSTROUTING	-m mark --mark 0x90002/0x90002 -j RETURN	[illegible]	[illegible] OVN-NAT-POLICY [illegible], [illegible] tag 0x90002/0x90002 [illegible] SNAT
nat	OVN-NAT-POLICY	-s 10.0.11.0/24 -m comment --comment natPolicySubnet-net1 -j OVN-NAT-PSUBNET-aa98851157c5	[illegible]	10.0.11.0/24 [illegible] net1 [illegible] CIDR[illegible] OVN-NAT-PSUBNET-aa98851157c5 [illegible] natOutgoingPolicyRules [illegible]
nat	OVN-NAT-PSUBNET-xxxxxxxxxxxx	-m set --match-set ovn40natpr-418e79269dc5-src src -m set --match-set ovn40natpr-418e79269dc5-dst dst -j MARK --set-xmark 0x90002/0x90002	[illegible]	418e79269dc5 [illegible] natOutgoingPolicyRules [illegible] ID[illegible] status.natOutgoingPolicyRules[index].Rule [illegible] srcIPs [illegible] ovn40natpr-418e79269dc src[illegible] dstIPS [illegible] ovn40natpr-418e79269dc5-dst [illegible] tag 0x90002
mangle	OVN-OUTPUT	-d 10.241.39.2/32 -p tcp -m tcp --dport 80 -j MARK --set-xmark 0x90003/0x90003	[illegible] kubelet [illegible] [illegible] tproxy	
mangle	OVN-PREROUTING	-d 10.241.39.2/32 -p tcp -m tcp --dport 80 -j TPROXY --on-port 8102 --on-ip 172.18.0.3 --tproxy-mark 0x90004/0x90004	[illegible] kubelet [illegible] [illegible] tproxy	

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2022[illegible]9[illegible]6[illegible]

GitHub

8.6.1 [illegible]

8.7 ⊠⊠⊠⊠⊠⊠⊠

8.7.1 ⊠⊠⊠⊠

Kube-OVN ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ Github ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠/⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ Github ⊠⊠⊠ Issue ⊠ PR⊠⊠⊠ Maintainer ⊠ Review ⊠⊠⊠ Github Action ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

8.7.2 ⊠⊠⊠⊠

Kube-OVN ⊠⊠ Go ⊠⊠⊠⊠⊠ Go Modules ⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠ `GO111MODULE="on"` ⊠

golangci-lint ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠local-installation⊠

⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠Kube-OVN ⊠⊠⊠⊠⊠ Docker buildx ⊠⊠⊠⊠⊠⊠⊠⊠ Docker ⊠⊠⊠⊠⊠ ⊠⊠⊠ buildx:

```
docker buildx create --use
```

8.7.3 ⊠⊠⊠⊠

⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ Kube-OVN ⊠⊠⊠⊠⊠

```
git clone https://github.com/kubeovn/kube-ovn.git
cd kube-ovn
make release
```

⊠⊠⊠⊠⊠ ARM ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

```
make release-arm
```

8.7.4 ⊠⊠ base ⊠⊠

⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠OVS/OVN ⊠⊠⊠⊠⊠⊠⊠ base ⊠⊠⊠⊠⊠⊠⊠⊠⊠

base ⊠⊠⊠⊠⊠ Dockerfile ⊠ `dist/images/Dockerfile.base` ⊠

⊠⊠⊠⊠⊠

```
# build x86 base image
make base-amd64

# build arm base image
make base-arm64
```

8.7.5 ⊠⊠ E2E

Kube-OVN ⊠⊠:

- KIND ⊠⊠⊠⊠⊠ Kubernetes ⊠⊠⊠ `go install sigs.k8s.io/kind@latest`
- jinjanator ⊠⊠⊠⊠⊠: `pip install jinjanator`
- Ginkgo ⊠⊠⊠⊠⊠⊠⊠⊠ `go install github.com/onsi/ginkgo/v2/ginkgo; go get github.com/onsi/gomega/...`

> ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

⊠⊠⊠⊠ E2E ⊠⊠⊠

```
make kind-init
make kind-install
make e2e
```

⊠⊠⊠⊠ Underlay E2E ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

```
make kind-init
make kind-install-underlay
make e2e-underlay-single-nic
```

□□□□ ovn vpc nat gw eip, fip, snat, dnat □□□□□□□□□□□

```
make kind-init
make kind-install
make ovn-vpc-nat-gw-conformance-e2e
```

□□□□ iptables vpc nat gw eip, fip, snat, dnat □□□□□□□□□□□

```
make kind-init
make kind-install-vpc-nat-gw
make iptables-vpc-nat-gw-conformance-e2e
```

□□□□ loadbalancer service □□□□□□□□□□□

```
make kind-init
make kind-install-lb-svc
make kube-ovn-lb-svc-conformance-e2e
```

□□□□□□□□□□□□□

```
make kind-clean
```

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8.7.6 □□

8.8 [illegible] OVS/OVN [illegible]

[illegible] OVN/OVS [illegible] SDN [illegible] Kubernetes [illegible] Kube-OVN [illegible]Kube-OVN [illegible] OVN/OVS [illegible] OVN/OVS [illegible] Kube-OVN [illegible] [illegible]

[illegible]

- 4228eab1d7 [illegible] vswitchd ofport_usage [illegible]
- 54056ea65d [illegible] timer[illegible]
- 6b4dcb311f [illegible] fdb [illegible]
- f627b7721e [illegible] hairpin [illegible] fdb [illegible]
- 3f3e3a436f [illegible] ovsdb-tool [illegible] join-cluster [illegible] Server ID [illegible]
- a6cb8215a8 [illegible] QoS [illegible]
- d4d76ddb2e ovsdb-tool[illegible] fix-cluster [illegible]
- ffd2328d4a netdev[illegible] CPU [illegible]
- d088c5d8c2 ovs-router[illegible] kube-ipvs0 [illegible]
- 1b31f07dc6 [illegible]
- 54b7678229 [illegible] ovs-sandbox [illegible] docker run[illegible]
- 9ee66bd91b [illegible]
- e889d46924 [illegible] Underlay [illegible] resubmit [illegible]
- f9e97031b5 ovn-controller[illegible] Kube-OVN [illegible] localnet [illegible] GARP[illegible]
- 78cade0187 [illegible] conntrack [illegible]
- 85aa6263ad northd[illegible] DNS IP [illegible] conntrack[illegible]
- 34dc3e3fcf lflow[illegible] lport [illegible] conntrack[illegible]
- a297b840c2 [illegible] DNAT [illegible] lsp[illegible]
- 03e35ed9c5 ovn-controller[illegible]
- e7d3ba53cd [illegible] ACL [illegible] DNS IP [illegible] conntrack[illegible]
- 9286e1fd57 [illegible]
- e5916eb53a [illegible] lr-lb DNAT [illegible]
- e4e6ea9c5f [illegible] BFD LRP[illegible]
- e76880e792 northd[illegible] nb [illegible] version_compatibility[illegible]
- 477695a010 northd[illegible] localnet [illegible] lrp [illegible] arp/nd [illegible]
- 2209d9a335 [illegible] ovn-controller [illegible] LSP [illegible]
- 52b727b331 [illegible] ovn-northd [illegible]
- 53b9837f7f [illegible] LB [illegible] ACL [illegible]
- 737d9f932e [illegible] GW [illegible]
- sbrec-delete-null-check [illegible] ovn-controller [illegible] sbrec [illegible]

[illegible]

- 20626ea909 [illegible] LB [illegible] ACL [illegible]
- a2d9ff3ccd Deb [illegible]

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2026[illegible]4[illegible]28[illegible]

2022[illegible]5[illegible]24[illegible]

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8.8.1 [illegible]

8.9 [illegible]

Kube-OVN [illegible] OVN/OVS [illegible] `Geneve` [illegible] `Vxlan` [illegible] `STT` [illegible] OVN [illegible] OVN Architecture Design Decision[illegible]

8.9.1 Geneve

`Geneve` [illegible] Kube-OVN [illegible] OVN [illegible] Offload [illegible] `Geneve` [illegible] 24bit [illegible] datapath [illegible] datapath [illegible] 32768 [illegible]

[illegible] Mellanox [illegible] OVS [illegible] `Geneve` [illegible] 5.4 [illegible] backport [illegible]

[illegible] UDP [illegible] TCP over UDP [illegible] TCP [illegible] CPU [illegible]

8.9.2 Vxlan

`Vxlan` [illegible] OVN [illegible] Offload [illegible] OVN [illegible]datapath [illegible] 4096 [illegible] datapath[illegible] datapath [illegible] 4096 [illegible] `inport` [illegible] ACL [illegible]

[illegible] Mellanox [illegible] OVS [illegible] `Vxlan` [illegible]

[illegible] UDP [illegible] TCP over UDP [illegible] TCP [illegible] CPU [illegible]

8.9.3 STT

Warning

OpenVswitch [illegible] 3.6 [illegible] STT [illegible] Tunnel [illegible]

`STT` [illegible] OVN [illegible] TCP [illegible] TCP [illegible] TCP [illegible] OVN [illegible] datapath[illegible]

[illegible] OVS [illegible]

[illegible] OVS [illegible]

8.9.4 [illegible]

- VXLAN vs GENEVE: Understand The Difference
- OVN FAQ
- What is Geneve

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2022[illegible]6[illegible]17[illegible]

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8.9.5 [illegible]

8.10 Kube-OVN [illegible]

[illegible] Kube-OVN [illegible]

8.10.1 ovn-monitor

OVN [illegible]

[illegible]	[illegible]	[illegible]
Gauge	kube_ovn_ovn_status	OVN [illegible] (2) [illegible] follower[illegible] (1) [illegible] leader, (0) [illegible]
Gauge	kube_ovn_failed_req_count	OVN [illegible]
Gauge	kube_ovn_log_file_size_bytes	OVN [illegible]
Gauge	kube_ovn_db_file_size_bytes	OVN [illegible]
Gauge	kube_ovn_chassis_info	OVN chassis [illegible] (1) [illegible](0) [illegible]
Gauge	kube_ovn_db_status	OVN [illegible], (1) [illegible] (0) [illegible]
Gauge	kube_ovn_logical_switch_info	OVN logical switch [illegible] (1)[illegible] logical switch [illegible]
Gauge	kube_ovn_logical_switch_external_id	OVN logical switch external_id [illegible] (1)[illegible] external-id [illegible]
Gauge	kube_ovn_logical_switch_port_binding	OVN logical switch [illegible] logical switch port [illegible] (1)[illegible]
Gauge	kube_ovn_logical_switch_tunnel_key	[illegible] OVN logical switch [illegible] tunnel key [illegible]
Gauge	kube_ovn_logical_switch_ports_num	OVN logical switch [illegible] logical port [illegible]
Gauge	kube_ovn_logical_switch_port_info	OVN logical switch port [illegible] (1)[illegible]
Gauge	kube_ovn_logical_switch_port_tunnel_key	[illegible] OVN logical switch port [illegible] tunnel key [illegible]
Gauge	kube_ovn_cluster_enabled	(1) OVN [illegible] (0) OVN [illegible]
Gauge	kube_ovn_cluster_role	[illegible] (1)[illegible]
Gauge	kube_ovn_cluster_status	[illegible] (1)[illegible]
Gauge	kube_ovn_cluster_term	RAFT term [illegible]
Gauge	kube_ovn_cluster_leader_self	[illegible] leader (1) [illegible] (0) [illegible]
Gauge	kube_ovn_cluster_vote_self	[illegible] leader (1) [illegible] (0) [illegible]
Gauge	kube_ovn_cluster_election_timer	[illegible] election timer [illegible]
Gauge	kube_ovn_cluster_log_not_committed	[illegible] commit [illegible] RAFT [illegible]
Gauge	kube_ovn_cluster_log_not_applied	[illegible] apply [illegible] RAFT [illegible]
Gauge	kube_ovn_cluster_log_index_start	[illegible] RAFT [illegible]
Gauge	kube_ovn_cluster_log_index_next	RAFT [illegible]
Gauge	kube_ovn_cluster_inbound_connections_total	[illegible]
Gauge	kube_ovn_cluster_outbound_connections_total	[illegible]
Gauge	kube_ovn_cluster_inbound_connections_error_total	[illegible]
Gauge	kube_ovn_cluster_outbound_connections_error_total	[illegible]

8.10.2 ovs-monitor

`ovsdb` [illegible] `vswitchd` [illegible]

[illegible]	[illegible]	[illegible]
Gauge	ovs_status	OVS [illegible] (1) [illegible](0) [illegible]
Gauge	ovs_info	OVS [illegible] (1)[illegible]
Gauge	failed_req_count	OVS [illegible]
Gauge	log_file_size_bytes	OVS [illegible]
Gauge	db_file_size_bytes	OVS [illegible]
Gauge	datapath	Datapath [illegible] (1)[illegible]
Gauge	dp_total	[illegible] OVS [illegible] datapath [illegible]
Gauge	dp_if	Datapath [illegible] (1)[illegible]
Gauge	dp_if_total	[illegible] datapath [illegible] port [illegible]
Gauge	dp_flows_total	Datapath [illegible] flow [illegible]
Gauge	dp_flows_lookup_hit	Datapath [illegible] flow [illegible]
Gauge	dp_flows_lookup_missed	Datapath [illegible] flow [illegible]
Gauge	dp_flows_lookup_lost	Datapath [illegible] userspace [illegible]
Gauge	dp_masks_hit	Datapath [illegible] mask [illegible]
Gauge	dp_masks_total	Datapath [illegible] mask [illegible]
Gauge	dp_masks_hit_ratio	Datapath [illegible] [illegible] mask [illegible]
Gauge	interface	OVS [illegible] (1)[illegible]
Gauge	interface_admin_state	[illegible] (0) [illegible] down, (1) [illegible] up, (2) [illegible]
Gauge	interface_link_state	[illegible] (0) [illegible] down, (1) [illegible] up, (2) [illegible]
Gauge	interface_mac_in_use	OVS Interface [illegible] MAC [illegible]
Gauge	interface_mtu	OVS Interface [illegible] MTU[illegible]
Gauge	interface_of_port	OVS Interface [illegible] OpenFlow Port ID[illegible]
Gauge	interface_if_index	OVS Interface [illegible] Index[illegible]
Gauge	interface_tx_packets	OVS Interface [illegible]
Gauge	interface_tx_bytes	OVS Interface [illegible]
Gauge	interface_rx_packets	OVS Interface [illegible]
Gauge	interface_rx_bytes	OVS Interface [illegible]
Gauge	interface_rx_crc_err	OVS Interface [illegible]
Gauge	interface_rx_dropped	OVS Interface [illegible]
Gauge	interface_rx_errors	OVS Interface [illegible]
Gauge	interface_rx_frame_err	OVS Interface [illegible]
Gauge	interface_rx_missed_err	OVS Interface [illegible] miss [illegible]
Gauge	interface_rx_over_err	OVS Interface [illegible] overrun [illegible]
Gauge	interface_tx_dropped	OVS Interface [illegible]

[illegible]	[illegible]	[illegible]
Gauge	interface_tx_errors	OVS Interface [illegible]
Gauge	interface_collisions	OVS interface [illegible]

8.10.3 kube-ovn-pinger

[illegible]

[illegible]	[illegible]	[illegible]
Gauge	pinger_ovs_up	[illegible] OVS [illegible]
Gauge	pinger_ovs_down	[illegible] OVS [illegible]
Gauge	pinger_ovn_controller_up	[illegible] ovn-controller [illegible]
Gauge	pinger_ovn_controller_down	[illegible] ovn-controller [illegible]
Gauge	pinger_inconsistent_port_binding	OVN-SB [illegible] portbinding [illegible] OVS interface [illegible]
Gauge	pinger_apiserver_healthy	kube-ovn-pinger [illegible] apiserver[illegible]
Gauge	pinger_apiserver_unhealthy	kube-ovn-pinger [illegible] apiserver[illegible]
Histogram	pinger_apiserver_latency_ms	kube-ovn-pinger [illegible] apiserver [illegible]
Gauge	pinger_internal_dns_healthy	kube-ovn-pinger [illegible]
Gauge	pinger_internal_dns_unhealthy	kube-ovn-pinger [illegible]
Histogram	pinger_internal_dns_latency_ms	kube-ovn-pinger [illegible]
Gauge	pinger_external_dns_healthy	kube-ovn-pinger [illegible]
Gauge	pinger_external_dns_unhealthy	kube-ovn-pinger [illegible]
Histogram	pinger_external_dns_latency_ms	kube-ovn-pinger [illegible]
Histogram	pinger_pod_ping_latency_ms	kube-ovn-pinger ping Pod [illegible]
Gauge	pinger_pod_ping_lost_total	kube-ovn-pinger ping Pod [illegible]
Gauge	pinger_pod_ping_count_total	kube-ovn-pinger ping Pod [illegible]
Histogram	pinger_node_ping_latency_ms	kube-ovn-pinger ping Node [illegible]
Gauge	pinger_node_ping_lost_total	kube-ovn-pinger ping Node [illegible]
Gauge	pinger_node_ping_count_total	kube-ovn-pinger ping Node [illegible]
Histogram	pinger_external_ping_latency_ms	kube-ovn-pinger ping [illegible] [illegible]
Gauge	pinger_external_ping_lost_total	kube-ovn-pinger ping [illegible]

8.10.4 kube-ovn-controller

`kube-ovn-controller` [illegible]

[illegible]	[illegible]	[illegible]
Histogram	rest_client_request_latency_seconds	[illegible] apiserver [illegible]
Counter	rest_client_requests_total	[illegible] apiserver [illegible]
Counter	lists_total	API list [illegible]
Summary	list_duration_seconds	API list [illegible]
Summary	items_per_list	API list [illegible]
Counter	watches_total	API watch [illegible]
Counter	short_watches_total	[illegible] API watch [illegible]
Summary	watch_duration_seconds	API watch [illegible]
Summary	items_per_watch	API watch [illegible]
Gauge	last_resource_version	[illegible] resource version[illegible]
Histogram	ovs_client_request_latency_milliseconds	[illegible] OVN [illegible]
Gauge	subnet_available_ip_count	[illegible] IP [illegible]
Gauge	subnet_used_ip_count	[illegible] IP [illegible]

8.10.5 kube-ovn-cni

`kube-ovn-cni` [illegible]

[illegible]	[illegible]	[illegible]
Histogram	cni_op_latency_seconds	CNI [illegible]
Counter	cni_wait_address_seconds_total	CNI [illegible]
Counter	cni_wait_connectivity_seconds_total	CNI [illegible]
Counter	cni_wait_route_seconds_total	CNI [illegible]
Histogram	rest_client_request_latency_seconds	[illegible] apiserver [illegible]
Counter	rest_client_requests_total	[illegible] apiserver [illegible]
Counter	lists_total	API list [illegible]
Summary	list_duration_seconds	API list [illegible]
Summary	items_per_list	API list [illegible]
Counter	watches_total	API watch [illegible]
Counter	short_watches_total	[illegible] API watch [illegible]
Summary	watch_duration_seconds	API watch [illegible]
Summary	items_per_watch	API watch [illegible]
Gauge	last_resource_version	[illegible] resource version[illegible]
Histogram	ovs_client_request_latency_milliseconds	[illegible] OVN [illegible]

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GitHub

8.10.6 ⌧⌧

8.11 Kube-OVN [illegible]

[illegible] Kube-OVN [illegible] Kube-OVN [illegible] CRD [illegible] CRD [illegible]

8.11.1 [illegible] Condition [illegible]

[illegible]	[illegible]	[illegible]
type	String	[illegible]
status	String	[illegible] `True` [illegible] `False` [illegible] `Unknown`
reason	String	[illegible]
message	String	[illegible]
observedGeneration	Int64	[illegible]
lastUpdateTime	Time	[illegible]
lastTransitionTime	Time	[illegible]

[illegible] CRD [illegible]Status [illegible] Condition [illegible]

8.11.2 [illegible]

Subnet [illegible]

SUBNET

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `Subnet`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	SubnetSpec	Subnet [illegible]
status	SubnetStatus	Subnet [illegible]

SubnetSpec

[illegible]	[illegible]	[illegible]
default	Bool	[illegible]
vpc	String	[illegible] VPC[illegible] ovn-cluster
protocol	String	IP [illegible] `IPv4` [illegible] `IPv6` [illegible] `Dual` [illegible] MAC [illegible] Underlay [illegible] `Mac`
namespaces	[]String	[illegible] namespace [illegible]
cidrBlock	String	[illegible] 10.16.0.0/16
gateway	String	[illegible] CIDRBlock [illegible]
excludeIps	[]String	[illegible]
provider	String	[illegible] OVN[illegible] NetworkAttachmentDefinition [illegible]Kube-OVN [illegible] Subnet [illegible]
gatewayType	String	Overlay [illegible] `distributed` [illegible] `centralized`
gatewayNode	String	[illegible] centralized [illegible]
natOutgoing	Bool	[illegible] NAT[illegible] `externalEgressGateway` [illegible]
externalEgressGateway	String	[illegible] `natOutgoing` [illegible]
policyRoutingPriority	Uint32	[illegible]
policyRoutingTableID	Uint32	[illegible] TableID[illegible]
mtu	Uint32	[illegible] MTU [illegible]
private	Bool	[illegible]
routed	Bool	[illegible] Pod [illegible] `/32` [illegible]IPv4[illegible] `/128` [illegible]IPv6[illegible] OVN [illegible]
allowSubnets	[]String	[illegible]
vlan	String	[illegible] Vlan [illegible]
vips	[]String	[illegible] virtual [illegible] lsp [illegible] virtual-ip [illegible]
logicalGateway	Bool	[illegible]
disableGatewayCheck	Bool	[illegible] Pod [illegible]
disableInterConnection	Bool	[illegible]
enableDHCP	Bool	[illegible] lsp [illegible] dhcp [illegible]
dhcpV4Options	String	[illegible] lsp dhcpv4_options [illegible] DHCP_Options [illegible]
dhcpV6Options	String	[illegible] lsp dhcpv6_options [illegible] DHCP_Options [illegible]
enableIPv6RA	Bool	[illegible] lrp [illegible] ipv6_ra_configs [illegible]
ipv6RAConfigs	String	[illegible] lrp [illegible]ipv6_ra_configs [illegible]
acls	[]Acl	[illegible] logical-switch [illegible] acls [illegible]
allowEWTraffic	Bool	[illegible]
natOutgoingPolicyRules	[]NatOutgoingPolicyRule	NAT [illegible]
u2oInterconnectionIP	String	Underlay/Overlay [illegible] IP [illegible]
u2oInterconnection	Bool	[illegible] Overlay/Underlay [illegible]
enableLb	*Bool	[illegible] logical-switch [illegible] load-balancer [illegible]

[illegible]	[illegible]	[illegible]
enableEcmp	Bool	[illegible] ECMP [illegible]
enableMulticastSnoop	Bool	[illegible]
enableExternalLBAddress	Bool	[illegible]
routeTable	String	[illegible]
namespaceSelectors	[]LabelSelector	[illegible]

Acl

[illegible]	[illegible]	[illegible]
direction	String	Acl [illegible] `from-lport` [illegible] `to-lport`
priority	Int	Acl [illegible] 0 [illegible] 32767
match	String	Acl [illegible]
action	String	Acl [illegible] `allow-related` [illegible] `allow-stateless` [illegible] `allow` [illegible] `drop` [illegible] `reject` [illegible]

NatOutgoingPolicyRule

[illegible]	[illegible]	[illegible]
match	NatOutGoingPolicyMatch	[illegible]
action	String	[illegible]

NatOutGoingPolicyMatch

[illegible]	[illegible]	[illegible]
srcIPs	String	[illegible] IP [illegible]
dstIPs	String	[illegible] IP [illegible]

SubnetStatus

[illegible]	[illegible]	[illegible]
conditions	[]SubnetCondition	[illegible] Condition [illegible]
v4availableIPs	BigInt	[illegible] IPv4 IP [illegible] IPv6 [illegible] float64 [illegible]
v4availableIPrange	String	[illegible] IPv4 [illegible]
v4usingIPs	BigInt	[illegible] IPv4 IP [illegible]
v4usingIPrange	String	[illegible] IPv4 [illegible]
v6availableIPs	BigInt	[illegible] IPv6 IP [illegible]
v6availableIPrange	String	[illegible] IPv6 [illegible]
v6usingIPs	BigInt	[illegible] IPv6 IP [illegible]
v6usingIPrange	String	[illegible] IPv6 [illegible]
activateGateway	String	[illegible]
dhcpV4OptionsUUID	String	[illegible] lsp dhcpv4_options [illegible] DHCP_Options [illegible]
dhcpV6OptionsUUID	String	[illegible] lsp dhcpv6_options [illegible] DHCP_Options [illegible]
u2oInterconnectionIP	String	[illegible] Overlay/Underlay [illegible] IP [illegible]
u2oInterconnectionMAC	String	[illegible] Overlay/Underlay [illegible] MAC [illegible]
u2oInterconnectionVPC	String	[illegible] Overlay/Underlay [illegible] VPC
natOutgoingPolicyRules	[]NatOutgoingPolicyRuleStatus	NAT [illegible]
mcastQuerierIP	String	[illegible] IP [illegible]
mcastQuerierMAC	String	[illegible] MAC [illegible]

IP [illegible]

IP

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `IP`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	IPSpec	IP [illegible]

IPSpec

[illegible]	[illegible]	[illegible]
podName	String	[illegible] Pod [illegible]
namespace	String	[illegible] Pod [illegible] Namespace [illegible]
subnet	String	IP [illegible] Subnet
attachSubnets	[]String	[illegible] IP [illegible]
nodeName	String	[illegible] Pod [illegible]
ipAddress	String	IP [illegible] `v4IP v6IP` [illegible]
v4IpAddress	String	IPv4 IP [illegible]
v6IpAddress	String	IPv6 IP [illegible]
attachIps	[]String	[illegible] IP [illegible] IP [illegible]
macAddress	String	[illegible] Pod [illegible] MAC [illegible]
attachMacs	[]String	[illegible] IP [illegible] MAC [illegible]
containerID	String	[illegible] Pod [illegible] Container ID
podType	String	[illegible] Pod[illegible] `StatefulSet` [illegible] `VirtualMachine` [illegible]

Vpc [illegible]

VPC

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `Vpc`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	VpcSpec	Vpc [illegible]
status	VpcStatus	Vpc [illegible]

VpcSpec

[illegible]	[illegible]	[illegible]
defaultSubnet	String	[illegible]
namespaces	[]String	Vpc [illegible]
staticRoutes	[]StaticRoute	[illegible]
policyRoutes	[]PolicyRoute	[illegible]
vpcPeerings	[]VpcPeering	VPC [illegible]
enableExternal	Bool	[illegible]
extraExternalSubnets	[]String	[illegible]
enableBfd	Bool	[illegible] BFD ([illegible])
bfdPort	BFDPort	BFD [illegible]

StaticRoute

[illegible]	[illegible]	[illegible]
policy	String	[illegible]
cidr	String	[illegible]
nextHopIP	String	[illegible] IP [illegible]
ecmpMode	String	ECMP [illegible]
bfdId	String	BFD ID
routeTable	String	[illegible]

PolicyRoute

[illegible]	[illegible]	[illegible]
priority	Int	[illegible]
match	String	[illegible]
action	String	[illegible] `allow` [illegible] `drop` [illegible] `reroute`
nextHopIP	String	[illegible] IP [illegible] action [illegible] reroute [illegible]

VpcPeering

[illegible]	[illegible]	[illegible]
remoteVpc	String	[illegible] VPC [illegible]
localConnectIP	String	[illegible] IP [illegible]

BFDPort

[illegible]	[illegible]	[illegible]
enabled	Bool	[illegible] BFD
ip	String	BFD [illegible] IP [illegible]
nodeSelector	LabelSelector	[illegible] BFD LRP [illegible]

VpcStatus

[illegible]	[illegible]	[illegible]
conditions	[]VpcCondition	Vpc [illegible] Condition [illegible]
standby	Bool	[illegible] VPC
default	Bool	[illegible] VPC
defaultLogicalSwitch	String	[illegible]
router	String	[illegible]
tcpLoadBalancer	String	TCP [illegible]
udpLoadBalancer	String	UDP [illegible]
sctpLoadBalancer	String	SCTP [illegible]
tcpSessionLoadBalancer	String	TCP [illegible]
udpSessionLoadBalancer	String	UDP [illegible]
sctpSessionLoadBalancer	String	SCTP [illegible]
subnets	[]String	VPC [illegible]
vpcPeerings	[]String	VPC [illegible]
enableExternal	Bool	[illegible]
extraExternalSubnets	[]String	[illegible]
enableBfd	Bool	[illegible] BFD

8.11.3 Underlay [illegible]

Vlan

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `Vlan`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	VlanSpec	Vlan [illegible]
status	VlanStatus	Vlan [illegible]

VLANSPEC

[illegible]	[illegible]	[illegible]
id	Int	Vlan tag [illegible] 0~4096
provider	String	Vlan [illegible] ProviderNetwork [illegible]

VLANSTATUS

[illegible]	[illegible]	[illegible]
subnets	[]String	Vlan [illegible]
conflict	Bool	[illegible]
conditions	[]VlanCondition	Vlan [illegible] Condition [illegible]

ProviderNetwork

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `ProviderNetwork`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	ProviderNetworkSpec	ProviderNetwork [illegible]
status	ProviderNetworkStatus	ProviderNetwork [illegible]

PROVIDERNETWORKSPEC

[illegible]	[illegible]	[illegible]
defaultInterface	String	[illegible]
customInterfaces	[]CustomInterface	[illegible]
nodeSelector	LabelSelector	[illegible] OVS [illegible] matchLabels [illegible] matchExpressions[illegible] nodeSelector[illegible]excludeNodes [illegible]
excludeNodes	[]String	[illegible]
exchangeLinkName	Bool	[illegible] OVS [illegible]

CustomInterface

[illegible]	[illegible]	[illegible]
interface	String	Underlay [illegible]
nodes	[]String	[illegible]

PROVIDERNETWORKSTATUS

[illegible]	[illegible]	[illegible]
ready	Bool	[illegible]
readyNodes	[]String	[illegible]
notReadyNodes	[]String	[illegible]
vlans	[]String	[illegible] Vlan [illegible]
conditions	[]ProviderNetworkCondition	ProviderNetwork [illegible] Condition [illegible]

8.11.4 [illegible]

SecurityGroup

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `SecurityGroup`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	SecurityGroupSpec	SecurityGroup [illegible]
status	SecurityGroupStatus	SecurityGroup [illegible]

SECURITYGROUPSPEC

[illegible]	[illegible]	[illegible]
ingressRules	[]SecurityGroupRule	[illegible]
egressRules	[]SecurityGroupRule	[illegible]
allowSameGroupTraffic	Bool	[illegible]

SecurityGroupRule

[illegible]	[illegible]	[illegible]
ipVersion	String	IP [illegible] `ipv4` [illegible] `ipv6`
protocol	SgProtocol	[illegible] `all` [illegible] `icmp` [illegible] `tcp` [illegible] `udp`
priority	Int	[illegible] 1-200[illegible]
remoteType	SgRemoteType	[illegible] `address` [illegible] `securityGroup`
remoteAddress	String	[illegible]
remoteSecurityGroup	String	[illegible]
portRangeMin	Int	[illegible] 1
portRangeMax	Int	[illegible] 65535
policy	SgPolicy	[illegible] `allow` [illegible] `drop`

SECURITYGROUPSTATUS

[illegible]	[illegible]	[illegible]
portGroup	String	[illegible]
allowSameGroupTraffic	Bool	[illegible]
ingressMd5	String	[illegible] MD5 [illegible]
egressMd5	String	[illegible] MD5 [illegible]
ingressLastSyncSuccess	Bool	[illegible]
egressLastSyncSuccess	Bool	[illegible]

8.11.5 [illegible] IP

Vip

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `Vip`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	VipSpec	Vip [illegible]
status	VipStatus	Vip [illegible]

VIPSPEC

[illegible]	[illegible]	[illegible]
namespace	String	VIP [illegible]
subnet	String	VIP [illegible]
type	String	VIP [illegible]
v4ip	String	IPv4 [illegible]
v6ip	String	IPv6 [illegible]
macAddress	String	MAC [illegible]
selector	[]String	[illegible]
attachSubnets	[]String	[illegible]

VIPSTATUS

[illegible]	[illegible]	[illegible]
conditions	[]VipCondition	VIP [illegible] Condition [illegible]
type	String	VIP [illegible]
v4ip	String	IPv4 [illegible]
v6ip	String	IPv6 [illegible]
mac	String	MAC [illegible]

SwitchLBRule

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `SwitchLBRule`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	SwitchLBRuleSpec	SwitchLBRule [illegible]
status	SwitchLBRuleStatus	SwitchLBRule [illegible]

SWITCHLBRULESPEC

[illegible]	[illegible]	[illegible]
vip	String	[illegible] IP [illegible]
namespace	String	[illegible]
selector	[]String	[illegible]
endpoints	[]String	[illegible]
sessionAffinity	String	[illegible]
ports	[]SwitchLBRulePort	[illegible]

SwitchLBRulePort

[illegible]	[illegible]	[illegible]
name	String	[illegible]
port	Int32	[illegible]
targetPort	Int32	[illegible]
protocol	String	[illegible]

SWITCHLBRULESTATUS

[illegible]	[illegible]	[illegible]
conditions	[]SwitchLBRuleCondition	SwitchLBRule [illegible] Condition [illegible]
ports	String	SwitchLBRule [illegible]
service	String	SwitchLBRule [illegible] service [illegible]

8.11.6 QoS [illegible] IP [illegible]

QoSPolicy

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `QoSPolicy`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	QoSPolicySpec	QoSPolicy [illegible]

QOSPOLICYSPEC

[illegible]	[illegible]	[illegible]
bandwidthLimitRules	QoSPolicyBandwidthLimitRules	[illegible]
shared	Bool	[illegible]
bindingType	QoSPolicyBindingType	[illegible]

IPPool

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `IPPool`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	IPPoolSpec	IPPool [illegible]
status	IPPoolStatus	IPPool [illegible]

IPPOOLSPEC

[illegible]	[illegible]	[illegible]
subnet	String	[illegible]
namespaces	[]String	[illegible] IPPool [illegible] IPPool [illegible] Pod
ips	[]String	IP [illegible]

IPPOOLSTATUS

[illegible]	[illegible]	[illegible]
v4AvailableIPs	BigInt	IPv4 [illegible] IP [illegible]
v4AvailableIPRange	String	IPv4 [illegible] IP [illegible]
v4UsingIPs	BigInt	IPv4 [illegible] IP [illegible]
v4UsingIPRange	String	IPv4 [illegible] IP [illegible]
v6AvailableIPs	BigInt	IPv6 [illegible] IP [illegible]
v6AvailableIPRange	String	IPv6 [illegible] IP [illegible]
v6UsingIPs	BigInt	IPv6 [illegible] IP [illegible]
v6UsingIPRange	String	IPv6 [illegible] IP [illegible]
conditions	[]IPPoolCondition	IP [illegible] Condition [illegible]

8.11.7 NAT [illegible] IP [illegible]

IptablesEIP

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `IptablesEIP`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	IptablesEIPSpec	IptablesEIP [illegible]
status	IptablesEIPStatus	IptablesEIP [illegible]

IPTABLESEIPSPEC

□□□□	□□	□□
v4ip	String	IPv4 □□
v6ip	String	IPv6 □□
macAddress	String	MAC □□
natGwDp	String	NAT □□□□□□
qosPolicy	String	QoS □□
externalSubnet	String	□□□□

IPTABLESEIPSTATUS

□□□□	□□	□□
ready	Bool	IptablesEIP □□□□□□□
ip	String	IptablesEIP □□□ IP □□□□□□□□□ IPv4 □□
redo	String	IptablesEIP CRD □□□□□□□□□
nat	String	IptablesEIP □□□□□□□□□ `fip` □ `snat` □□ `dnat`
qosPolicy	String	QoS □□□□
conditions	[]IptablesEIPCondition	IptablesEIP □□□□□□□□□□□□□□□□□□ Condition □□

OvnEip

□□□□	□□	□□
apiVersion	String	□□ Kubernetes □□□□□□□□□□□□□□□□□□□ kubeovn.io/v1
kind	String	□□ Kubernetes □□□□□□□□□□□□□□□□□□ `OvnEip`
metadata	ObjectMeta	□□ Kubernetes □□□□□□□□
spec	OvnEipSpec	OvnEip □□□□□□□□□
status	OvnEipStatus	OvnEip □□□□□□□

OVNEIPSPEC

□□□□	□□	□□
externalSubnet	String	□□□□
v4Ip	String	IPv4 □□
v6Ip	String	IPv6 □□
macAddress	String	MAC □□
type	String	□□□□□□ lrp□lsp □ nat

IptablesFIPRule

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `IptablesFIPRule`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	IptablesFIPRuleSpec	IptablesFIPRule [illegible]

IPTABLESFIPRULESPEC

[illegible]	[illegible]	[illegible]
eip	String	[illegible] IP [illegible]
internalIP	String	[illegible] IP [illegible]

OvnFip

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `OvnFip`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	OvnFipSpec	OvnFip [illegible]
status	OvnFipStatus	OvnFip [illegible]

OVNFIPSPEC

[illegible]	[illegible]	[illegible]
ovnEip	String	[illegible] OVN EIP
ipType	String	IP [illegible] vip [illegible] ip
ipName	String	IP [illegible]
vpc	String	[illegible] VPC
v4Ip	String	IPv4 [illegible]
v6Ip	String	IPv6 [illegible]
type	String	[illegible] distributed [illegible] centralized

IptablesDnatRule

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `IptablesDnatRule`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	IptablesDnatRuleSpec	IptablesDnatRule [illegible]

IPTABLESDNATRULESPEC

[illegible]	[illegible]	[illegible]
eip	String	[illegible] IP [illegible]
externalPort	String	[illegible]
protocol	String	[illegible]
internalIP	String	[illegible] IP [illegible]
internalPort	String	[illegible]

OvnDnatRule

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `OvnDnatRule`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	OvnDnatRuleSpec	OvnDnatRule [illegible]
status	OvnDnatRuleStatus	OvnDnatRule [illegible]

OVNDNATRULESPEC

[illegible]	[illegible]	[illegible]
ovnEip	String	[illegible] OVN EIP
ipType	String	IP [illegible] vip [illegible] ip
ipName	String	IP [illegible]
internalPort	String	[illegible]
externalPort	String	[illegible]
protocol	String	[illegible]
vpc	String	[illegible] VPC
v4Ip	String	IPv4 [illegible]
v6Ip	String	IPv6 [illegible]

OVNDNATRULESTATUS

[illegible]	[illegible]	[illegible]
vpc	String	[illegible] VPC
v4Eip	String	IPv4 EIP [illegible]
v6Eip	String	IPv6 EIP [illegible]
externalPort	String	[illegible]
v4Ip	String	IPv4 [illegible]
v6Ip	String	IPv6 [illegible]
internalPort	String	[illegible]
protocol	String	[illegible]
ipName	String	IP [illegible]
ready	Bool	DNAT [illegible]
conditions	[]OvnDnatRuleCondition	OVN DNAT [illegible] Condition [illegible]

IptablesSnatRule

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `IptablesSnatRule`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	IptablesSnatRuleSpec	IptablesSnatRule [illegible]

IPTABLESSNATRULESPEC

[illegible]	[illegible]	[illegible]
eip	String	[illegible] IP [illegible]
internalCIDR	String	[illegible] CIDR [illegible]

OvnSnatRule

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `OvnSnatRule`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	OvnSnatRuleSpec	OvnSnatRule [illegible]
status	OvnSnatRuleStatus	OvnSnatRule [illegible]

OVNSNATRULESPEC

[illegible]	[illegible]	[illegible]
ovnEip	String	[illegible] OVN EIP
vpcSubnet	String	VPC [illegible]
ipName	String	IP [illegible]
vpc	String	[illegible] VPC
v4IpCidr	String	IPv4 CIDR [illegible]
v6IpCidr	String	IPv6 CIDR [illegible]

8.11.8 VPC [illegible]

VpcNatGateway

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `VpcNatGateway`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	VpcNatGatewaySpec	VpcNatGateway [illegible]
status	VpcNatGatewayStatus	VpcNatGateway [illegible]

VPCNATGATEWAYSPEC

[illegible]	[illegible]	[illegible]
vpc	String	VPC [illegible] Pod [illegible] VPC [illegible]
subnet	String	VPC [illegible] Pod [illegible]
externalSubnets	[]String	[illegible]
lanIp	String	VPC [illegible] Pod [illegible] IP [illegible]
selector	[]String	[illegible] Kubernetes Selector [illegible]
tolerations	[]Toleration	[illegible] Kubernetes [illegible]
affinity	Affinity	[illegible] Kubernetes [illegible]
qosPolicy	String	QoS [illegible]
bgpSpeaker	VpcBgpSpeaker	BGP speaker [illegible]
routes	[]Route	[illegible]

VpcBgpSpeaker

[illegible]	[illegible]	[illegible]
enabled	Bool	[illegible] BGP speaker
asn	Uint32	[illegible]
remoteAsn	Uint32	[illegible]
neighbors	[]String	BGP [illegible]
holdTime	Duration	BGP [illegible]
routerId	String	BGP [illegible] ID
password	String	BGP [illegible]
enableGracefulRestart	Bool	[illegible]
extraArgs	[]String	[illegible]

Route

[illegible]	[illegible]	[illegible]
cidr	String	[illegible] CIDR
nextHopIP	String	[illegible] IP [illegible]

VPCNATGATEWAYSTATUS

[illegible]	[illegible]	[illegible]
qosPolicy	String	QoS [illegible]
externalSubnets	[]String	[illegible]
selector	[]String	[illegible] Kubernetes Selector [illegible]
tolerations	[]Toleration	[illegible] Kubernetes [illegible]
affinity	Affinity	[illegible] Kubernetes [illegible]

VpcEgressGateway

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `VpcEgressGateway`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	VpcEgressGatewaySpec	VpcEgressGateway [illegible]
status	VpcEgressGatewayStatus	VpcEgressGateway [illegible]

VPCEGRESSGATEWAYSPEC

[illegible]	[illegible]	[illegible]
vpc	String	[illegible] VPC[illegible] VPC
bgpConf	String	[illegible] `BgpConf` [illegible]
evpnConf	String	[illegible] `EvpnConf` [illegible] EVPN [illegible]
replicas	Int32	[illegible] 1
prefix	String	[illegible] `<prefix><veg-name>`
image	String	[illegible] kube-ovn-controller [illegible]
internalSubnet	String	[illegible] VPC [illegible]
externalSubnet	String	[illegible]
internalIPs	[]String	[illegible] IP [illegible] `v4,v6` [illegible]
externalIPs	[]String	[illegible] IP [illegible] `v4,v6` [illegible]
internalIPPool	String	[illegible] `IPPool` [illegible] `internalSubnet` [illegible] `internalIPs` [illegible]
externalIPPool	String	[illegible] `IPPool` [illegible] `externalSubnet` [illegible] `externalIPs` [illegible]
selectors	[]VpcEgressGatewaySelector	[illegible]/Pod [illegible]
trafficPolicy	String	[illegible] `Cluster` [illegible] `Local` [illegible] VPC [illegible]
bfd	VpcEgressGatewayBFDConfig	BFD [illegible]
policies	[]VpcEgressGatewayPolicy	[illegible]
nodeSelector	[]VpcEgressGatewayNodeSelector	[illegible]
tolerations	[]Toleration	[illegible] Kubernetes [illegible]
resources	ResourceRequirements	[illegible]
bandwidth	BandwidthLimit	[illegible]/[illegible] Mbps [illegible] bit-rate quantity
observability	VpcEgressGatewayObservability	[illegible]conntrack [illegible]JSON [illegible] ServiceMonitor [illegible]

VpcEgressGatewaySelector

[illegible]	[illegible]	[illegible]
namespaceSelector	LabelSelector	[illegible]
podSelector	LabelSelector	Pod [illegible]

VpcEgressGatewayBFDConfig

[illegible]	[illegible]	[illegible]
enabled	Boolean	[illegible] BFD[illegible] VPC [illegible] `.spec.bfd.enabled=true` [illegible] false
minRX	Int32	BFD [illegible]ms[illegible] 1000
minTX	Int32	BFD [illegible]ms[illegible] 1000
multiplier	Int32	BFD [illegible] 3

VpcEgressGatewayPolicy

[illegible]	[illegible]	[illegible]
snat	Boolean	[illegible] SNAT/MASQUERADE[illegible] false
ipBlocks	[]String	[illegible] CIDR [illegible]
subnets	[]String	[illegible]

VpcEgressGatewayNodeSelector

[illegible]	[illegible]	[illegible]
matchLabels	map[String]String	[illegible]
matchExpressions	[]NodeSelectorRequirement	[illegible]
matchFields	[]NodeSelectorRequirement	[illegible]

BandwidthLimit

[illegible]	[illegible]	[illegible]
ingress	Int64 [illegible] String	[illegible] VPC [illegible] Mbps[illegible]quantity [illegible] `M` [illegible] `Mi` [illegible] `G` [illegible] `Gi` bit/s
egress	Int64 [illegible] String	[illegible] VPC [illegible] Mbps[illegible]quantity [illegible] `M` [illegible] `Mi` [illegible] `G` [illegible] `Gi` bit/s

VpcEgressGatewayObservability

[illegible] Kubernetes 1.29 [illegible] schema [illegible] VPC Egress Gateway [illegible]

[illegible]	[illegible]	[illegible]
resources	ResourceRequirements	[illegible]observer sidecar [illegible] request [illegible] `20m` CPU [illegible] `64Mi` [illegible]limit [illegible] `200m` CPU [illegible] `256Mi` [illegible]
interfaceMetrics	VpcEgressGatewayObservabilityFeature	[illegible]
conntrack	VpcEgressGatewayConntrackObservability	conntrack [illegible]
serviceMonitor	VpcEgressGatewayServiceMonitor	[illegible] ServiceMonitor [illegible]

VpcEgressGatewayObservabilityFeature

[illegible]	[illegible]	[illegible]
enabled	Boolean	[illegible] false

VpcEgressGatewayConntrackObservability

[illegible]	[illegible]	[illegible]
metrics	VpcEgressGatewayObservabilityFeature	[illegible] NAT conntrack [illegible]
log	VpcEgressGatewayConntrackLog	JSON Lines NAT [illegible]

VpcEgressGatewayConntrackLog

[illegible]	[illegible]	[illegible]
enabled	Boolean	[illegible] JSON [illegible] observer [illegible] false
events	[]String	[illegible] `start` [illegible]/[illegible] `end` [illegible]
rateLimit	VpcEgressGatewayConntrackLogRateLimit	[illegible] Pod [illegible]
filters	VpcEgressGatewayConntrackLogFilters	[illegible]include [illegible] exclude [illegible]

VpcEgressGatewayConntrackLogRateLimit

[illegible]	[illegible]	[illegible]
recordsPerSecond	Int32	[illegible] 1 [illegible] 100000[illegible] 100
burst	Int32	[illegible] 1 [illegible] 1000000[illegible] 1000

VpcEgressGatewayConntrackLogFilters

[illegible]	[illegible]	[illegible]
include	[]VpcEgressGatewayConntrackLogFilter	[illegible] 64 [illegible] include [illegible] NAT [illegible]
exclude	[]VpcEgressGatewayConntrackLogFilter	[illegible] 64 [illegible] exclude [illegible]exclude [illegible]

VpcEgressGatewayConntrackLogFilter

[illegible] AND [illegible] OR [illegible]

[illegible]	[illegible]	[illegible]
addressFamilies	[]String	[illegible] `ipv4` [illegible]/[illegible] `ipv6`
protocols	[]String	[illegible] `tcp` [illegible] `udp` [illegible] `sctp` [illegible] `icmp` [illegible] `icmpv6` [illegible]/[illegible] `other`
natTypes	[]String	[illegible] `snat` [illegible] `dnat` [illegible]/[illegible] `snat_dnat`
original	VpcEgressGatewayConntrackTupleFilter	[illegible] tuple [illegible]
translated	VpcEgressGatewayConntrackTupleFilter	[illegible] tuple [illegible]

VpcEgressGatewayConntrackTupleFilter

[illegible]	[illegible]	[illegible]
sourceCIDRs	[]String	[illegible] CIDR[illegible] 64 [illegible]
destinationCIDRs	[]String	[illegible] CIDR[illegible] 64 [illegible]
sourcePorts	[]VpcEgressGatewayPortRange	[illegible] 64 [illegible]
destinationPorts	[]VpcEgressGatewayPortRange	[illegible] 64 [illegible]

VpcEgressGatewayPortRange

[illegible]	[illegible]	[illegible]
start	Int32	[illegible] 0 [illegible] 65535
end	Int32	[illegible] `start` [illegible] 65535

VpcEgressGatewayServiceMonitor

[illegible]	[illegible]	[illegible]
labels	map[String]String	[illegible]ServiceMonitor [illegible] selector [illegible]
annotations	map[String]String	[illegible]ServiceMonitor annotations

VPCEGRESSGATEWAYSTATUS

[illegible]	[illegible]	[illegible]
replicas	Int32	[illegible] scale [illegible]
labelSelector	String	[illegible] scale [illegible]
ready	Boolean	[illegible]
phase	String	[illegible] `Pending` [illegible] `Processing` [illegible] `Completed`
internalIPs	[]String	[illegible] IP [illegible]
externalIPs	[]String	[illegible] IP [illegible]
conditions	[]Condition	[illegible]
workload	VpcEgressWorkload	[illegible]

VpcEgressWorkload

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] API [illegible]
kind	String	[illegible] `Deployment`
name	String	[illegible]
nodes	[]String	[illegible]

BgpConf

[illegible]cluster-scoped[illegible] `VpcEgressGateway.spec.bgpConf` [illegible]

[illegible]	[illegible]	[illegible]
apiVersion	String	[illegible] Kubernetes [illegible] kubeovn.io/v1
kind	String	[illegible] Kubernetes [illegible] `BgpConf`
metadata	ObjectMeta	[illegible] Kubernetes [illegible]
spec	BgpConfSpec	BgpConf [illegible]

BGPCONFSPEC

□□□□	□□	□□
localASN	UInt32	□□ BGP AS Number
peerASN	UInt32	□□ BGP AS Number
routerId	String	□□□BGP Router ID
neighbours	[]String	BGP □□ IP □□
password	String	□□□TCP MD5 □□□□
holdTime	Duration	□□□BGP Hold time
keepaliveTime	Duration	□□□BGP Keepalive □□
connectTime	Duration	□□□BGP Connect □□
ebgpMultiHop	Boolean	□□□□□□□ eBGP MultiHop
gracefulRestart	Boolean	□□□□□□□ BGP Graceful Restart

EvpnConf

□□□□cluster-scoped□□□□□ `VpcEgressGateway.spec.evpnConf` □□□□□□ EVPN □□□□□ VNI □ Route Target□

□□□□	□□	□□
apiVersion	String	□□ Kubernetes □□□□□□□□□□□□□□□□□□ kubeovn.io/v1
kind	String	□□ Kubernetes □□□□□□□□□□□□□□□□□□ `EvpnConf`
metadata	ObjectMeta	□□ Kubernetes □□□□□□□
spec	EvpnConfSpec	EvpnConf □□□□□□□□

EVPNCONFSPEC

□□□□	□□	□□
vni	UInt32	EVPN VNI
routeTargets	[]String	Route Target □□

DNSNameResolver

□□□□cluster-scoped□□□□□ `enable-dns-name-resolver` □□□ ANP/CNP □□□□□□□□□□□ DNS □□□□□

□□□□	□□	□□
apiVersion	String	□□ Kubernetes □□□□□□□□□□□□□□□□□□ kubeovn.io/v1
kind	String	□□ Kubernetes □□□□□□□□□□□□□□□□□□ `DNSNameResolver`
metadata	ObjectMeta	□□ Kubernetes □□□□□□□
spec	DNSNameResolverSpec	DNSNameResolver □□□□□□□□
status	DNSNameResolverStatus	DNSNameResolver □□□□□□

DNSNAMERESOLVERSPEC

□□□□	□□	□□
name	String	□□□□□ DNS □□□□□□□□□ `*.` □□□□□□□□□□□□□□□

DNSNAMERESOLVERSTATUS

□□□□	□□	□□
resolvedNames	[]DNSNameResolverResolvedName	□□□□□□□ DNS □□□□ IP □□

DNSNameResolverResolvedName

□□□□	□□	□□
dnsName	String	□□□□□□□ DNS □□
resolvedAddresses	[]DNSNameResolverResolvedAddress	□□□□ IP□TTL □□□□□□□
conditions	[]Condition	□□□□□□□□□ `Degraded` □□□□□□□□□□
resolutionFailures	Int32	□□□□□□□□□□□□□□□□□ 5 □□ TTL □□□□□□□□□

DNSNameResolverResolvedAddress

□□□□	□□	□□
ip	String	□□□□ IP □□
ttlSeconds	Int32	TTL □□□ `lastLookupTime + ttlSeconds` □□□□
lastLookupTime	Time	□□□□□□□□□□□□

VpcDns

□□□□	□□	□□
apiVersion	String	□□ Kubernetes □□□□□□□□□□□□□□□□□□ kubeovn.io/v1
kind	String	□□ Kubernetes □□□□□□□□□□□□□□□□□ `VpcDns`
metadata	ObjectMeta	□□ Kubernetes □□□□□□□
spec	VpcDNSSpec	VpcDns □□□□□□□□
status	VpcDNSStatus	VpcDns □□□□□□

VPCDNSSPEC

□□□□	□□	□□
replicas	Int32	□□□□
vpc	String	□□ VPC
subnet	String	□□□□

PDF Slack Support

2026[illegible]10[illegible]28[illegible]

2023[illegible]2[illegible]16[illegible]

GitHub

8.11.9 [illegible]

8.12 Annotation [illegible]

Kube-OVN [illegible] Pod [illegible] Node Annotation [illegible] Annotation [illegible]

> [illegible] Annotation [illegible]

8.12.1 Pod Annotation

Key	Value	Description
ovn.kubernetes.io/allocated	`true` or `false`	Pod [illegible]
ovn.kubernetes.io/routed	`true` or `false`	Pod [illegible] OVN [illegible]
ovn.kubernetes.io/routes	String	Pod [illegible]
ovn.kubernetes.io/mac_address	String	Pod [illegible] Mac [illegible] Pod [illegible] Annotation [illegible] Mac [illegible]
ovn.kubernetes.io/ip_address	String	Pod [illegible] IP [illegible] Pod [illegible] Annotation [illegible] IP [illegible]
ovn.kubernetes.io/ip_family	`ipv4` or `ipv6`	[illegible] Pod [illegible] IP [illegible]
\<nadName>.\<nadNamespace>.kubernetes.io/ip_address.\<interfaceName>	String	[illegible] IP[illegible]interfaceName [illegible] Multus [illegible] interface [illegible] ovn.kubernetes.io/ip_address [illegible]
\<nadName>.\<nadNamespace>.kubernetes.io/mac_address.\<interfaceName>	String	[illegible] MAC[illegible]interfaceName [illegible] Multus [illegible] interface [illegible]
ovn.kubernetes.io/cidr	String	Pod [illegible] CIDR
ovn.kubernetes.io/gateway	String	Pod [illegible] Gateway [illegible]
ovn.kubernetes.io/ip_pool	IP [illegible]	Pod [illegible] Workload [illegible] IP [illegible]
ovn.kubernetes.io/bgp	`true`, `cluster`, `local`	[illegible] BGP [illegible] Pod [illegible]
ovn.kubernetes.io/snat	String	Pod [illegible] SNAT [illegible]
ovn.kubernetes.io/eip	String	Pod [illegible] EIP [illegible]
ovn.kubernetes.io/vip	String	v1.17 [illegible] Annotation [illegible]`ovn.kubernetes.io/ip_address`[illegible]`ovn.kubernetes.io/ip_pool`[illegible] IPPool [illegible] Pod [illegible]
ovn.kubernetes.io/aaps	String	Pod [illegible] AAPs (Additional Allowed Addresses Pairs) [illegible]
ovn.kubernetes.io/virtualmachine	String	Pod [illegible] VirtualMachineInstance
ovn.kubernetes.io/activation_strategy	String	Pod [illegible]
ovn.kubernetes.io/logical_router	String	Pod [illegible] VPC
ovn.kubernetes.io/layer2_forward	`true` or `false`	Pod [illegible] OVN LSP [illegible] `unknown` [illegible]
ovn.kubernetes.io/port_security	`true` or `false`	Pod [illegible] Port Security
ovn.kubernetes.io/logical_switch	String	Pod [illegible]
ovn.kubernetes.io/vlan_id	Int	Pod [illegible] Vlan ID
ovn.kubernetes.io/ingress_rate	Int	Pod [illegible] Mbits/s
ovn.kubernetes.io/egress_rate	Int	Pod [illegible] Mbits/s
ovn.kubernetes.io/security_groups	String [illegible]	Pod [illegible] Security Group
ovn.kubernetes.io/default_route	`true` or `false`	[illegible]
ovn.kubernetes.io/provider_network	String	Pod [illegible] ProviderNetwork
ovn.kubernetes.io/mirror	`true` or `false`	Pod [illegible]
ovn.kubernetes.io/north_gateway	String	Pod [illegible]
ovn.kubernetes.io/latency	Int	Pod [illegible] ms

Key	Value	Description
ovn.kubernetes.io/limit	Int	Pod [illegible] qdisc [illegible]
ovn.kubernetes.io/loss	Float	Pod [illegible]
ovn.kubernetes.io/jitter	Int	Pod [illegible] ms
ovn.kubernetes.io/generate-hash	`true` or `false`	[illegible] Pod [illegible]
ovn.kubernetes.io/attachmentprovider	String	Pod [illegible]
ovn.kubernetes.io/ingress_burst	Int	Pod [illegible] burst [illegible]
ovn.kubernetes.io/vpc_nat_gw	String	VPC NAT Gateway Pod [illegible]
ovn.kubernetes.io/vpc_nat_gw_init	`true` or `false`	VPC NAT Gateway Pod [illegible]
ovn.kubernetes.io/vpc_nat_gw_container_restarted	`true` or `false`	VPC NAT Gateway [illegible]
ovn.kubernetes.io/vpc_nat_gw_activated	`true` or `false`	VPC NAT Gateway [illegible]
\<provider>.kubernetes.io/ip_pool	IP [illegible]	[illegible] provider [illegible] IP [illegible]
\<provider>.kubernetes.io/ip_family	`ipv4` or `ipv6`	[illegible] provider [illegible] IP [illegible]
\<provider>.kubernetes.io/dhcp-v4-options	String	[illegible] provider [illegible] DHCPv4 [illegible]
\<provider>.kubernetes.io/port_security	`true` or `false`	[illegible] provider [illegible]

8.12.2 Node Annotation

Key	Value	Description
ovn.kubernetes.io/allocated	`true` or `false`	[illegible] `ovn0` [illegible] `join` [illegible]
ovn.kubernetes.io/mac_address	String	Node `ovn0` [illegible] Mac [illegible]
ovn.kubernetes.io/ip_address	String	Node `ovn0` [illegible] IP [illegible]
ovn.kubernetes.io/cidr	String	Node `ovn0` [illegible] `join` [illegible] CIDR
ovn.kubernetes.io/gateway	String	Node `ovn0` [illegible] `join` [illegible] Gateway [illegible]
ovn.kubernetes.io/chassis	String	Node [illegible] OVN-SouthBoundDB [illegible] Chassis ID
ovn.kubernetes.io/port_name	String	Node `ovn0` [illegible] OVN-NorthboundDB [illegible] LSP [illegible]
ovn.kubernetes.io/logical_switch	String	Node `ovn0` [illegible]
ovn.kubernetes.io/tunnel_interface	String	[illegible]
ovn.kubernetes.io/node_networks	String	[illegible] Pod [illegible] attachment networks [illegible] NetworkAttachmentDefinition [illegible]

8.12.3 Namespace Annotation

Key	Value	Description
ovn.kubernetes.io/cidr	CIDR [illegible]	[illegible] Namespace [illegible] CIDR
ovn.kubernetes.io/exclude_ips	excludeIPs [illegible]	[illegible] Namespace [illegible] excludeIPs

8.12.4 Subnet Annotation

Key	Value	Description
ovn.kubernetes.io/bgp	`true`, `cluster`, `local`	[illegible] BGP [illegible]
ovn.kubernetes.io/vpc	String	[illegible] VPC[illegible] VPC [illegible]

8.12.5 Service Annotation

Key	Value	Description
ovn.kubernetes.io/bgp	`true`, `cluster`, `local`	[illegible] BGP [illegible] Service [illegible] ClusterIP Service [illegible] Pod/ Subnet [illegible]
ovn.kubernetes.io/switch_lb_vip	String	Service [illegible] Kube-OVN [illegible] VIP [illegible]
ovn.kubernetes.io/vpc	String	Service [illegible] VPC
ovn.kubernetes.io/service_external_ip_from_subnet	`true` or `false`	Service [illegible] IP [illegible]
ovn.kubernetes.io/service_health_check	`true` or `false`	Service [illegible]
ovn.kubernetes.io/lb_svc_img	String	[illegible]

8.12.6 Networkpolicy Annotation

Key	Value	Description
ovn.kubernetes.io/enable_log	`true` or `false`	[illegible] NetworkPolicy [illegible]
ovn.kubernetes.io/log_acl_actions	"allow,drop,pass" [illegible]	[illegible] Action ACL [illegible]
ovn.kubernetes.io/acl_log_meter_rate	Int	NetworkPolicy [illegible]/[illegible]
ovn.kubernetes.io/network_policy_enforcement	`standard` or `lax`	[illegible] NetworkPolicy [illegible]
ovn.kubernetes.io/network_policy_for	String	[illegible] NetworkPolicy [illegible] provider[illegible] `<nadName>.<nadNamespace>.ovn` [illegible]

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2026[illegible]10[illegible]28[illegible]

2024[illegible]6[illegible]12[illegible]

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8.12.7 [illegible]

8.13 [illegible]

[illegible]

8.13.1 [illegible]

[illegible]

Bad	Good
[illegible]	[illegible]

[illegible]

Bad	Good
Kube-OVN [illegible] 1.10 [illegible] Kube-OVN[illegible]	Kube-OVN [illegible] 1.10 [illegible] Kube-OVN[illegible]

[illegible]

Bad	Good
[illegible] `wget 127.0.0.1`	[illegible] `wget 127.0.0.1`

8.13.2 [illegible]

yaml [illegible] yaml[illegible]

<table>
<tr><th>Bad</th><th>Good</th></tr>
<tr><td><pre>````
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
    name: attach-subnet
````</pre></td><td><pre>````yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
    name: attach-subnet
````</pre></td></tr>
</table>

[illegible] bash[illegible]

<table>
<tr><th>Bad</th><th>Good</th></tr>
<tr><td><pre>````
wget 127.0.0.1
````</pre></td><td><pre>````bash
wget 127.0.0.1
````</pre></td></tr>
</table>

[illegible] `#` [illegible]

**Bad**

```
oilbeater@macdeMac-3 ~ ping 114.114.114.114 -c 3
PING 114.114.114.114 (114.114.114.114): 56 data bytes
64 bytes from 114.114.114.114: icmp_seq=0 ttl=83 time=10.429 ms
64 bytes from 114.114.114.114: icmp_seq=1 ttl=79 time=11.360 ms
64 bytes from 114.114.114.114: icmp_seq=2 ttl=76 time=10.794 ms

--- 114.114.114.114 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 10.429/10.861/11.360/0.383 ms
```

**Good**

```
# ping 114.114.114.114 -c 3
PING 114.114.114.114 (114.114.114.114): 56 data bytes
64 bytes from 114.114.114.114: icmp_seq=0 ttl=83 time=10.429 ms
64 bytes from 114.114.114.114: icmp_seq=1 ttl=79 time=11.360 ms
64 bytes from 114.114.114.114: icmp_seq=2 ttl=76 time=10.794 ms

--- 114.114.114.114 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 10.429/10.861/11.360/0.383 ms
```

[illegible] `#` [illegible]

**Bad**

```
# mv /etc/origin/ovn/ovnnb_db.db /tmp
# mv /etc/origin/ovn/ovnsb_db.db /tmp
```

**Good**

```
mv /etc/origin/ovn/ovnnb_db.db /tmp
mv /etc/origin/ovn/ovnsb_db.db /tmp
```

## 8.13.3 [illegible]

[illegible] `md` [illegible]

**Bad**

```
[    ](http://kubeovn.github.io/prepare)
```

**Good**

```
[    ](./prepare.md)
```

**Bad**

```
[Kubernetes    ](http://kubernetes.io)
```

**Good**

```
[Kubernetes    ](http://kubernetes.io){: target="_blank" }
```

## 8.13.4 [illegible]

[illegible]

**Bad**

~~~
```bash
wget 127.0.0.1
```
~~~

**Good**

~~~

```bash
wget 127.0.0.1
```
~~~

[illegible]

**Bad**

~~~



```bash
wget 127.0.0.1
```
~~~

**Good**

~~~

```bash
wget 127.0.0.1
```
~~~

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2025年9月10日

2022年7月19日

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## 8.13.5 [illegible]

# 9. [illegible]

Kube-OVN [illegible]

## 9.1 [illegible]

- Slack[illegible] CNCF Slack [illegible] #kube-ovn [illegible]
- LinkedIn[illegible]Kube-OVN [illegible]
- Issue / [illegible] GitHub Issues [illegible]GitHub Discussions [illegible]

## 9.2 [illegible]

- [illegible]

## 9.3 [illegible]

- [illegible] [illegible] [illegible] [illegible] [illegible]
- [illegible] USERS.md [illegible]

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2026[illegible]4[illegible]28[illegible]

2022[illegible]6[illegible]20[illegible]

GitHub

## 9.4 [illegible]